10

Clinical Technique for Amalgam Restorations

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Dental amalgam (*silver amalgam* or simply *amalgam*) is a metallic, polycrystalline restorative material originally composed of a mixture of silver–tin alloy and mercury. Current alloys that are amalgamated with mercury are silver–tin–copper. The unset silver-colored mixture is pressed (condensed) into a tooth defect (cavitation) that has been specifically prepared to retain the amalgam. The material is then contoured to restore the tooth's form so that, when the material hardens, the tooth is returned to normal function (Fig. 10.1). Amalgam has been the primary direct restorative material in the United States for more than 150 years. It has been the subject of intense research and has been found to be safe and beneficial as a direct restorative material (see Chapter 13).[1-3] The cost-effective nature of amalgam restorations has benefited many people. According to the U.S. Public Health Service, "hundreds of millions of teeth have been retained that otherwise would have been sacrificed because restorative alternatives would have been too expensive for many people."[4]

In addition to being cost effective, amalgam has the unique property of being "self-sealing." Self-sealing occurs when percolation of oral fluids (i.e., microleakage) between the amalgam restoration and the prepared cavity walls results in corrosion of the amalgam and a subsequent accumulation of corrosion products in the microscopic space. Microleakage between the restoration and the adjacent tooth structure is reduced as corrosion products fill the space. The self-sealing process is self-limiting and requires several months. Amalgam is the only restorative material with an interfacial seal that improves over time.[5-7]

Amalgam was introduced to the United States in the 1830s. Initially, amalgam restorations were made by dentists filing silver coins and mixing the filings with mercury, creating a puttylike mass that was placed into the defective tooth. As knowledge increased and research intensified, major advancements in the formulation and use of amalgam occurred. Concerns about mercury toxicity in the use of amalgam were, however, expressed in many countries; concerns reached major proportions in the early 1990s.

Today, the popularity of amalgam as a direct restorative material has decreased.[8,9] The decline is attributed in part to increased interest in tooth-colored composite resin restorations and the minimally invasive nature of their preparation steps. Concerns have been raised that the preparation used when planning for amalgam may result in unnecessary weakening of teeth because of the demand for additional removal of tooth structure so as to accommodate amalgam's strength and retention requirements. While it is true that preparations for composite resin restorations may allow smaller, more conservative preparations for very early caries lesions, treatment of more advanced lesions may result in essentially the same amount of tooth structure loss regardless of the type of restorative material being used. The choice of a composite resin material over amalgam may be based on the assumption that both materials will perform equally well over time for all patients. However, a growing body of evidence suggests that the risk of developing secondary caries adjacent to amalgam restorations is at least two times less likely than that of composite resin restorations in high caries risk patients, and therefore the reduction/phase-out of dental amalgam may have been premature.[10,11] The decline of amalgam use is also due to perceived concerns over individual and environmental safety relative to the presence of elemental mercury in amalgam restorations. Safe, professional handling of mercury in mixing the amalgam mass, removal of old amalgam restorations, and amalgam scrap disposal is certainly appropriate and absolutely essential. Following best management practices for amalgam waste, as presented by the American Dental Association, results in appropriate amalgam use.[12]

Types of Amalgam Restorative Materials

Low-Copper Amalgam

Low-copper amalgams were primarily used before the early 1960s. When the setting reaction occurred, the material was subject to corrosion because of the formation of a tin–mercury phase (gamma-2). The corrosion led to the rapid breakdown of amalgam restorations. Subsequent research led to the development of high-copper amalgam materials. Currently low-copper amalgams are rarely used in the United States.

High-Copper Amalgam

High-copper amalgams are predominantly used today in the United States. In this book, unless otherwise specified, the term *amalgam*



• **Fig. 10.1** Clinical example of an amalgam restoration. (From Hatrick CD, Eakle WS, Bird WF: *Dental materials: Clinical applications for dental assistants and dental hygienists,* ed 2, St. Louis, Saunders, 2011.)

refers to high-copper dental amalgam. The increase in copper content to 12% or greater designates an amalgam as a high-copper type. The advantage of the added copper is that it preferentially reacts with the tin and reduces the formation of the more corrosive phase (gamma-2) within the amalgam mass. This change in composition reduces the deleterious corrosion effects on the restoration. However, enough corrosion occurs at the amalgam–tooth interface to result in the sealing of the restoration, which reduces microleakage.[13,14] These materials may provide satisfactory clinical performance for more than 12 years.[15,16] High-copper amalgams are available with admixed or spherical alloy structure.

Admixed Amalgam

Admixed amalgam contains irregularly shaped and sized alloy particles, sometimes combined with spherical shapes, which are mixed to form a mass that is placed into the tooth preparation. The irregular shape of many of the particles results in a mass that requires greater condensation pressure (which many dentists prefer) and permits the dentist to displace matrix bands to generate proximal contacts more easily.

Spherical Amalgam

A spherical amalgam contains small, round alloy particles that are mixed with mercury to form the mass that is placed into the tooth preparation. Because of the shape of the particles, the material is condensed into the tooth preparation with little condensation pressure. This advantage is combined with its high early strength to provide a material that is well suited for very large amalgam restorations such as complex amalgams or foundations.[17]

New Amalgam Alloys

Because of the concern about mercury toxicity, new compositions of amalgam have been promoted as mercury-free or low-mercury amalgam restorative materials. Alloys with gallium or indium or alloys using cold-welding techniques have been presented as alternatives to mercury-containing amalgams. None of these new alloys have shown sufficient promise to become a universal replacement for current amalgam materials.[18-21]

Important Amalgam Properties

The linear coefficient of the thermal expansion of amalgam is 2.5 times greater than that of tooth structure, but it is closer to that of tooth structure than the linear coefficient of thermal expansion of composite.[22-24] Although the compressive strength of high-copper amalgam is similar to tooth structure, the tensile strength is lower, making amalgam restorations prone to fracture during flexure.[9,25,26] Usually, high-copper amalgam fracture is a bulk fracture, not a marginal fracture. All amalgams are brittle and have low edge strength. As such, the amalgam restoration must have sufficient bulk (usually 1.5–2 mm in any occlusally loaded area, depending on the position within the tooth) and a 90-degree marginal configuration.

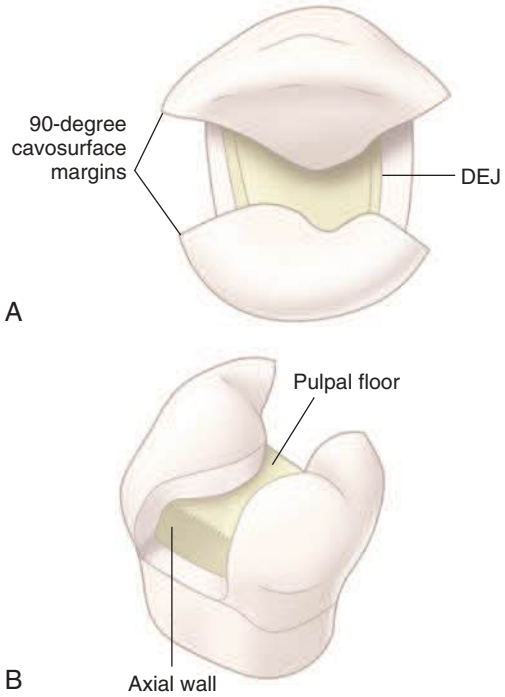
Creep and flow relate to the deformation of a material under load over time. High-copper amalgams exhibit no clinically relevant creep or flow.[27,28] Because amalgam is metallic in structure, it also is a good thermal conductor. At minimum, a dentin desensitizer should be used immediately prior to amalgam placement to limit sensitivity secondary to rapid fluid movement in the dentinal tubules caused by thermal changes. A liner or base should be placed in areas of deep caries removal prior to amalgam placement to limit thermal sensitivity.

General Considerations for Amalgam Restorations

Restoration With Amalgam

Amalgam is effective as a direct restorative material because of its easy insertion into a tooth preparation and, when hardened, its ability to restore the tooth to proper form and function. The tooth preparation not only must remove the fault in the tooth and remove weakened tooth structure, but its form must also allow the amalgam material to function properly. The required tooth preparation form must allow the amalgam to (1) possess a uniform specified minimum thickness for strength (so that it will not flex and fracture under load), (2) produce a 90-degree amalgam angle (butt-joint form for maximum edge thickness) at the margin, and (3) be mechanically retained in the tooth (Fig. 10.2A and B). Amalgam restorations initially leak and therefore require steps to protect from pulpal sensitivity until self-sealing is able to occur (Fig. 10.3A).[13] After desensitizing the prepared tooth structure, mixing, inserting, carving, and finishing the amalgam are relatively fast and easy. The placing and contouring of amalgam restorations are generally easier than those for composite restorations.[29] For these reasons, it is considered a user-friendly material that is less technique (i.e., operator) sensitive as compared with composite.

Some practitioners have continued to use bonded amalgam restorations in their practice (see Fig. 10.3B). As noted in Chapter 4, this book does not promote the use of bonded amalgams.[30-33] The bonding of amalgam requires use of a dual-cure adhesive system to form a hybrid layer with the dentin followed by condensation of amalgam so that amalgam particles become intermingled with the curing adhesive resin. A bonded amalgam restoration, done properly, may seal the prepared tooth structure and may temporarily



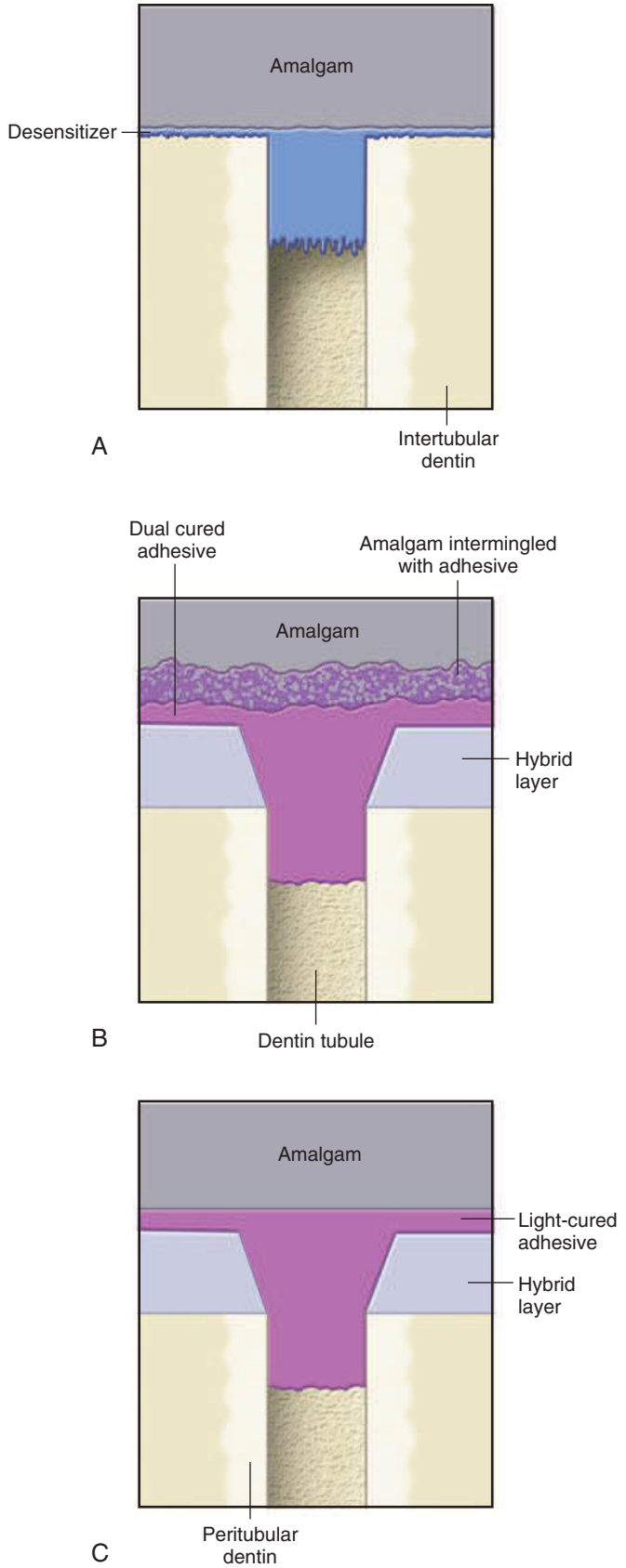
• **Fig. 10.2** A and B, Diagrams of Class II amalgam tooth preparations illustrating uniform pulpal and axial wall depths, 90-degree cavosurface margins, and occlusal convergence of walls.

strengthen adjacent tooth structure. The strengthening and retention gained by bonding, however, is minimal and short term. Consequently, bonded amalgam restorations still require the same tooth preparation retention and resistance form as do nonbonded amalgam restorations.[34-35] Isolation requirements for a bonded amalgam restoration are the same as for a composite restoration.

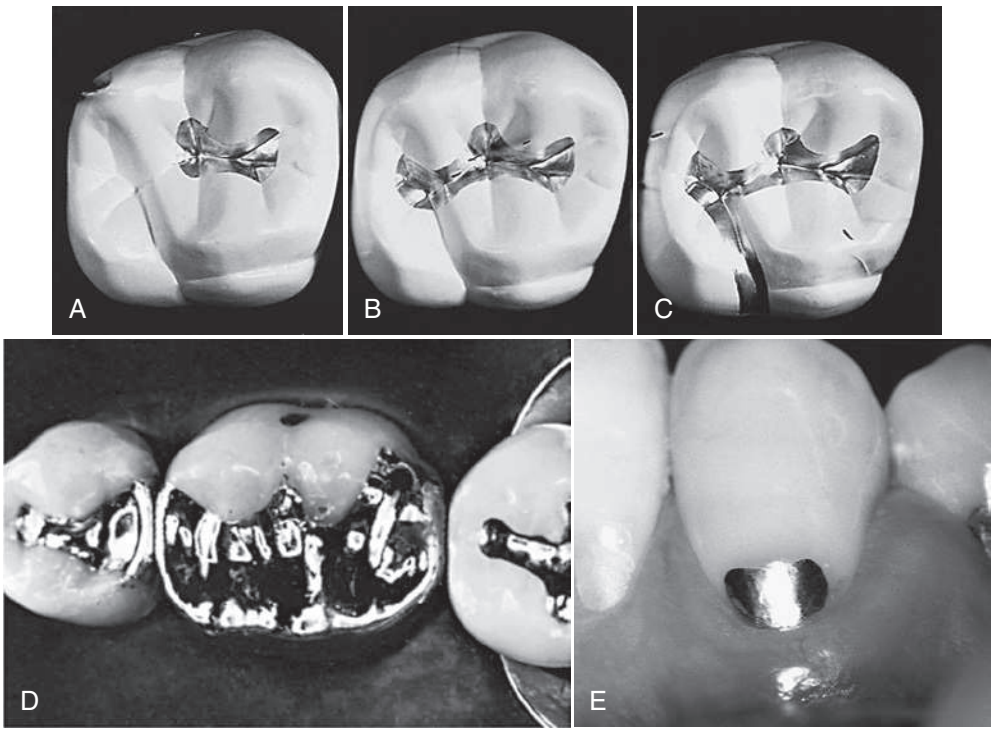
Another amalgam technique uses light-cured adhesive to seal the dentin under the amalgam material (see Fig. 10.3C). This procedure, as is true of all procedures that use adhesive technology, requires proper isolation. The prepared tooth structure is etched (i.e., demineralized), primed, and sealed with adhesive. The adhesive is polymerized before insertion of the amalgam. This technique seals the dentinal tubules effectively.[36,37]

Uses of Amalgam

Amalgam is used for the restoration of many carious or fractured posterior teeth and in the replacement of failed restorations. Understanding the physical properties of amalgam and the principles of tooth preparation is necessary to produce amalgam restorations that provide optimal service. When properly placed, an amalgam restoration is able to provide many years of service.[38-43] Although improved techniques and materials are available, amalgam failures do occur. Much clinical time is spent replacing restorations that fail as a result of recurrent caries lesions, marginal deterioration (i.e., ditching), fractures, or poor contours.[44,45] Attention to detail throughout the procedure may significantly decrease the incidence of failures, however, and extend the life of the restoration.[46-48] Careful evaluation of existing amalgam restorations is important because they have the potential to provide long-term clinical service and should not be removed and replaced unless accurately identified to be defective or undermined by a secondary caries lesion.[49]



• **Fig. 10.3** Methods to limit initial pulpal sensitivity after amalgam placement. A, Dentin desensitization. B, Amalgam bonding (amalgam intermingled with adhesive resin). C, Dentin sealing (adhesive resin placed and cured before amalgam placement).



• **Fig. 10.4** Amalgam restorations. A–C, Class I. D, Complex. E, Class V. (Most practitioners would restore all of these teeth with composite, except tooth No. 30.)

Because of its strength and ease of use, amalgam provides an excellent means for restoring large defects in nonesthetic areas.[50] A review of almost 3500 four-surface and five-surface amalgams revealed successful outcomes at 5 years for 72% of the four-surface and 65% of the five-surface amalgams. When considering complex amalgam restorations and discussing treatment options with patients, this result should be compared with the 5-year success rates for gold and porcelain crowns, which were 84% and 85%, respectively.[51]

Indications

Amalgam may be used for Class I, II, V, and VI restorations; for foundations; and for caries-control restorations (Figs. 10.4 and 10.5; see also Chapter 2). Occasionally amalgam may be utilized for Class III areas if isolation problems exist. Likewise, Class V amalgam restorations may be indicated in anterior areas where esthetics is not an important consideration and the patient has high caries risk.

Contraindications

Amalgams are contraindicated in patients who are allergic to alloy components. The use of amalgam in more prominent esthetic areas of the mouth may represent a relative contraindication. These areas include anterior teeth and, in some patients, premolars and molars. Amalgam should not be used when composite resin would offer more conservation of tooth structure and equal clinical performance.

Occlusal Factors

Amalgam generally has greater wear resistance than does composite in patients that have heavy occlusal function.[2,52,53] Amalgam also may be more appropriate than composite resin when there is little to no natural occlusal tooth structure remaining that requires the restoration to restore most or all of the occlusal contacts.[53]

Isolation Factors

Isolation of the operating field is important for moisture control, access, and visibility, to protect the patient from aspirating or ingesting foreign objects, protecting the pulp in the event of pulpal exposure, and protecting the operator medicolegally. However, minor moisture contamination of amalgam during the insertion procedure may not have as adverse an effect on the final restoration as the same contamination would for a composite restoration.

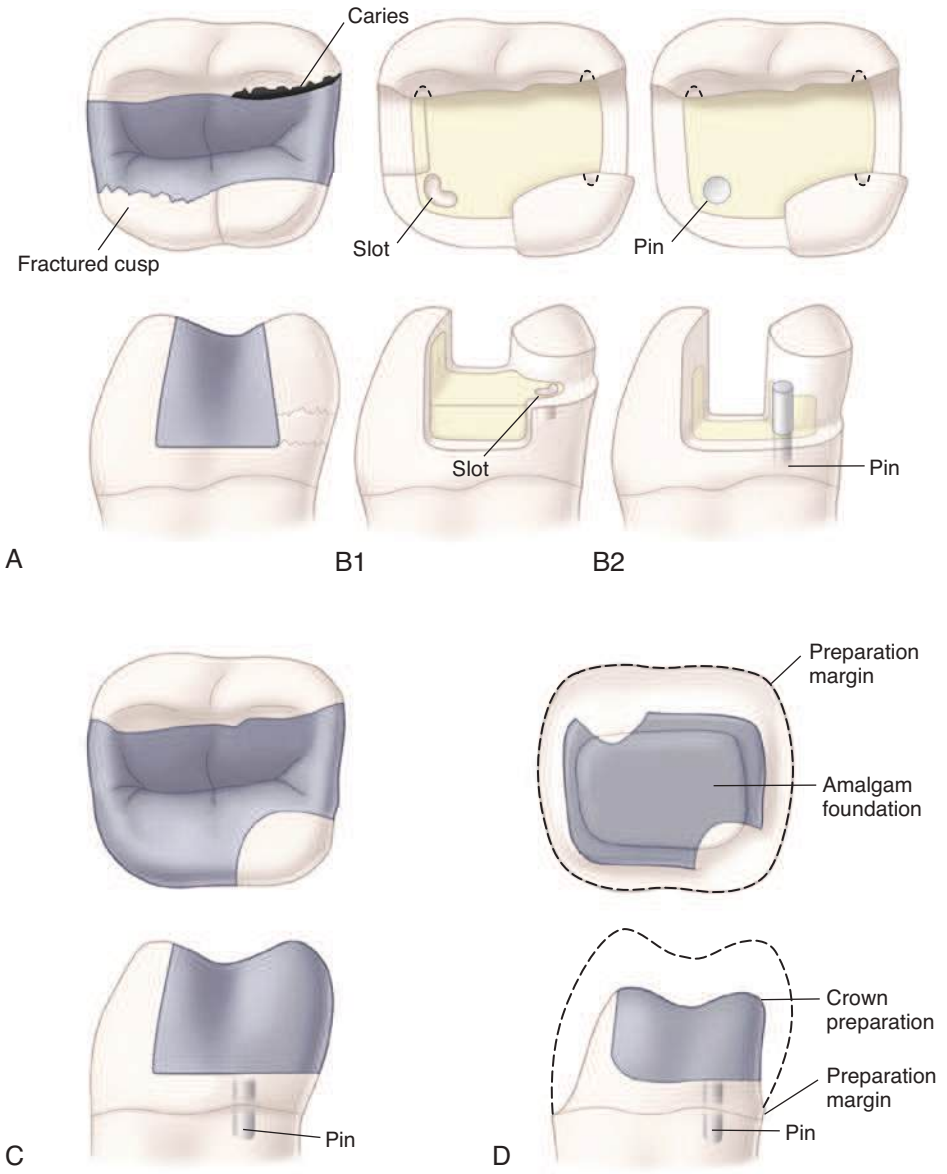
Operator Ability and Commitment Factors

The tooth preparation for an amalgam restoration is very exacting. It requires a specific design with depths that allow appropriate amalgam thickness and a precise marginal form. The failure of amalgam restorations is often related to inappropriate tooth preparation. However, the insertion and finishing procedures for amalgam are much easier than for composite.[29]

Advantages

Some of the advantages of amalgam restorations already have been stated, but the following list presents the primary reasons for the successful use of amalgam restorations over many years.

1. Ease of use
2. High compressive strength
3. Excellent wear resistance
4. Favorable long-term clinical research results
5. Lower cost than for composite restorations



• **Fig. 10.5** Amalgam foundation. A, Defective restoration (defective amalgam, mesiolingual fractured cusp, distofaciocclusal caries). B, Tooth preparation with secondary retention, using slot (B1) and pin (B2). C, Amalgam foundation placed. D, Tooth with amalgam foundation prepared for crown. Note that dashed lines on the occlusal surface in B1 and B2 only indicate the position of the retention grooves on the facial and lingual walls of the proximal preparation. The grooves do not extend onto the occlusal surface.

Disadvantages

The primary disadvantages of amalgam restorations relate to esthetics and increased tooth structure removal during tooth preparation. The following is a list of these and other disadvantages of amalgam restorations.

1. Noninsulating
2. Nonesthetic
3. Less conservative tooth preparation than for composite restorations (more removal of tooth structure during tooth preparation)
4. More difficult tooth preparation than for composite restorations
5. Initial marginal leakage

The following sections begin with the introduction of general clinical techniques and associated concepts to be considered when restoring caries lesions or defects with dental amalgam. The general technique is then followed by specific discussion addressing the creation of Class I, II, III, V, VI, and Complex Amalgam restorations.

General Clinical Technique for Amalgam Restorations

Initial Clinical Procedures

Complete examination, diagnosis, and treatment planning must be completed before a patient is scheduled for operative

appointments (except in emergencies). A brief review of patient's dental record (including medical factors), treatment plan, and radiographs should precede each restorative procedure. At the beginning of each appointment, the dentist should also carefully examine the operating site to confirm the diagnosis of the tooth or teeth scheduled for treatment.

Local Anesthesia

Local anesthesia is recommended for most operative procedures. Profound anesthesia contributes to comfortable and uninterrupted operation and usually results in a marked reduction in salivation.

Isolation of the Operating Site

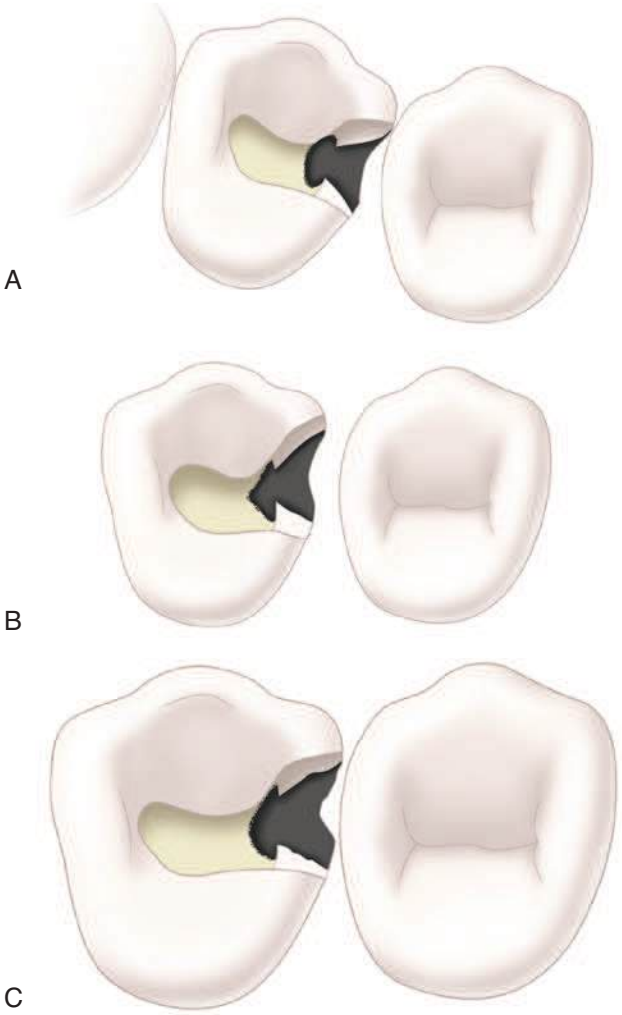
Complete instructions for the control of moisture are given in Online Chapter 15. Isolation for amalgam restorations may be accomplished with a rubber dam, cotton rolls, with or without a retraction cord.

Other Preoperative Considerations

Preoperative assessment of the occlusion should be made. This step should occur before rubber dam placement. The dentist should identify not only the occlusal contacts of the tooth to be restored but also the contacts on the opposing and adjacent teeth. Knowing the preoperative location of occlusal contacts is important in planning the restoration outline form and in establishing the proper occlusal contacts on the restoration. Remembering the location of the contacts on adjacent teeth provides guidance in determining when the restoration contacts have been correctly positioned and adjusted.

A wedge placed preoperatively in the gingival embrasure is useful when restoring a proximal surface. This step causes slight separation of the operated tooth from the adjacent tooth and may help protect the adjacent proximal surface, the rubber dam, and the interdental papilla.

It is important to preoperatively visualize the anticipated extension of the tooth preparation. Because the tooth preparation requires specific depths, extensions, and marginal forms, the connection of the various parts of the tooth preparation should result in minimal tooth structure removal (i.e., as little as is necessary), while maintaining the strength of the cuspal and marginal ridge areas of the tooth as much as possible (Fig. 10.6). The projected facial and lingual extensions of a proximal box should be visualized before preparing the occlusal outline form through the marginal ridge of the tooth, reducing the chance of overextension while maintaining a butt-joint form of the facial or lingual proximal margins.

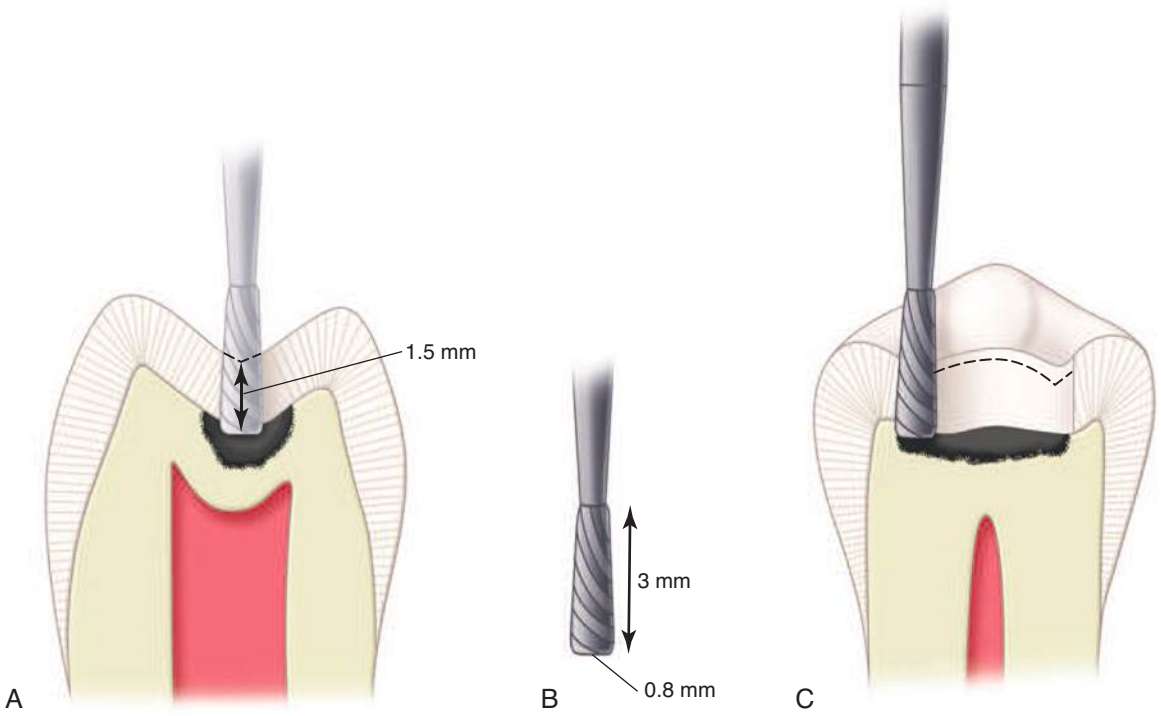


• **Fig. 10.6** Preoperative visualization of tooth preparation extensions when a cavitated caries lesion is present gingival to the proximal contact and in the central groove area. A, Rotated tooth (lingual extension owing to faulty central groove). B, Open proximal contact (preparation extended wider faciolingually to develop a proximal contact with appropriate physiologic proximal contours). C, Normal relationship.

General Concepts Guiding Preparation for Amalgam Restorations

A solid comprehension of concepts presented in Chapter 4 is essential prior to the learning of information presented in the following sections. For an amalgam restoration to be successful, numerous steps must be accomplished correctly. After an accurate diagnosis is made, the dentist must create a tooth preparation that not only removes the defect (e.g., caries lesion, old restorative material, malformed structure) but also leaves remaining tooth structure as strong as possible by leaving as much dentin support as possible. Tooth preparation that conserves tooth structure is strongly recommended because it limits pulpal irritation and preserves the integrity of both the tooth and the subsequent restoration.[54-60]

Discussion in this chapter will include principles, techniques, and procedures using classic examples of amalgam preparations. However, clinically, the outline form of the preparation should always conform to the level of disease in the tooth and not necessarily to the examples presented here. Making the tooth preparation form appropriate for the use of amalgam as the restorative material is equally important. The physical properties of amalgam require that it be placed into a tooth preparation that provides for an approximately 90-degree or slightly greater occlusal cavosurface margin and 90-degree axial cavosurface margin (because of the amalgam's limited edge strength). The preparation must allow a minimum thickness of 1.5 to 2 mm so that the amalgam will not flex and fracture when under occlusal load (most amalgams fail by bulk fracture). The amalgam



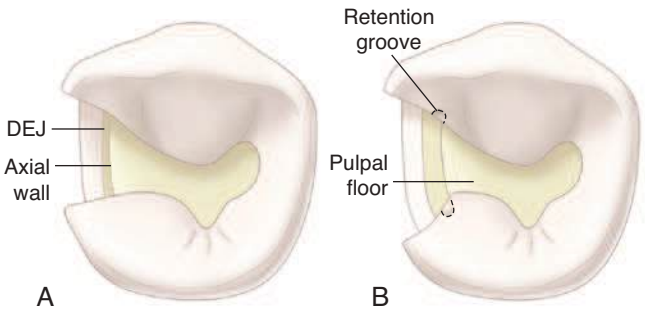
• **Fig. 10.7** Pulpal floor depth. A, Pulpal depth measured from central groove. B, No. 245 bur dimensions. C, Guides to proper pulpal floor depth: (1) one half the length of the No. 245 bur, (2) 1.5 mm, or (3) 0.2 mm inside (internal to) the dentinoenamel junction (DEJ).

must be placed into a prepared undercut form in the tooth so as to be mechanically retained (amalgam does not bond to tooth structure). After appropriate tooth preparation, the success of the final restoration depends on proper insertion, carving, and finishing of the amalgam material.

Principles

The basic principles of tooth preparation must be followed for amalgam tooth preparations to ensure clinical success. As presented in Chapter 4, the preparation for amalgam is discussed in two stages, for academic purposes, to facilitate student understanding of proper extension, form, and caries lesion removal. The initial stage (1) places the tooth preparation extension into sound tooth structure at the marginal areas (not pulpally or axially); (2) extends the depth (pulpally, axially, or both) at the periphery of the preparation to a prescribed, uniform dimension; (3) provides an initial form consistent with amalgam retention in the tooth; and (4) establishes the preparation external walls in a form that will result in a 90-degree amalgam margin upon restoration. The final stage of tooth preparation removes any remaining defect (caries lesion or old restorative material) and incorporates any additional preparation features (grooves, coves, slots, or pins) as needed to achieve the appropriate retention and resistance forms.

In academic institutions, assessing the tooth preparation after the initial preparation stage provides an opportunity to evaluate a student's knowledge and ability to extend the external walls properly and establish proper initial depth. If the student were to remove an extensive caries lesion in its entirety before any evaluation, the attending faculty would not know whether the prepared depths were obtained because of appropriate caries lesion removal or inappropriate overcutting of the tooth.



• **Fig. 10.8** Axial wall depth. A, If no retention grooves needed, axial depth 0.2 mm inside (internal to) the dentinoenamel junction (DEJ). B, If retention grooves needed (i.e., proximal divergence secondary to wide faciolingual caries lesion extension), axial depth 0.5 mm inside (internal to) the DEJ. Note that dashed lines on the occlusal surface only indicate the position of the retention grooves associated with the facial and lingual walls of the proximal preparation. The grooves do not extend onto the occlusal surface.

Initial Tooth Preparation

All initial depths of a tooth preparation for amalgam relate to the dentinoenamel junction (DEJ) except in the following two instances: (1) when the occlusal enamel has become significantly thinner and (2) when the preparation extends onto the root surface. The initial depth pulpally is 0.2 mm inside (internal to) the DEJ or 1.5 mm as measured from the depth of the central groove (Fig. 10.7), whichever results in the greater thickness of amalgam. The initial depth of the axial wall is 0.2 mm inside the DEJ when retention grooves are not used and 0.5 mm inside the DEJ when retention grooves are used (Fig. 10.8). The deeper extension allows placement of the retention

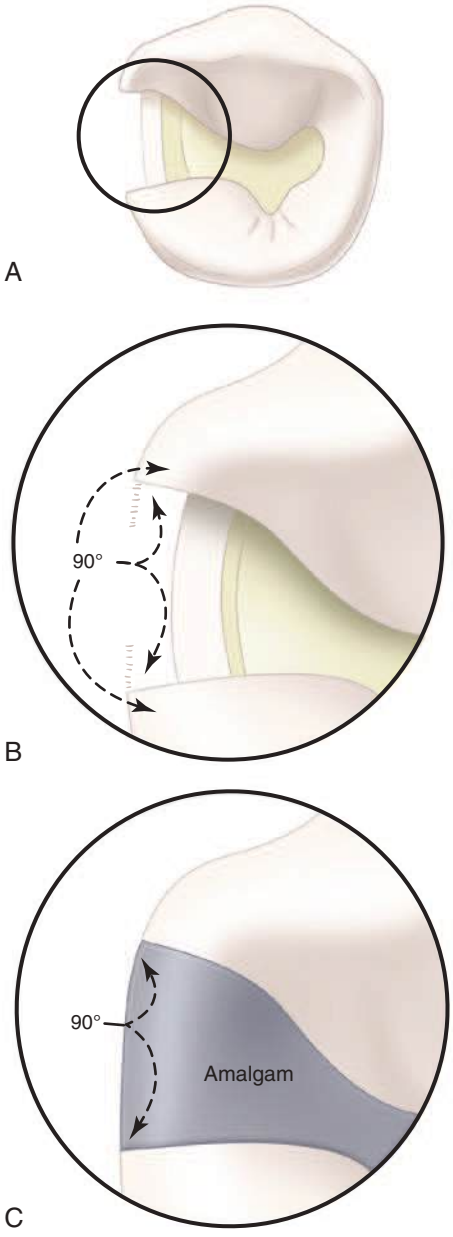
groove without undermining marginal enamel. Axial depths on the root surface should be 0.75 to 1 mm deep so as to provide room for placement of retention grooves or coves.

Outline Form

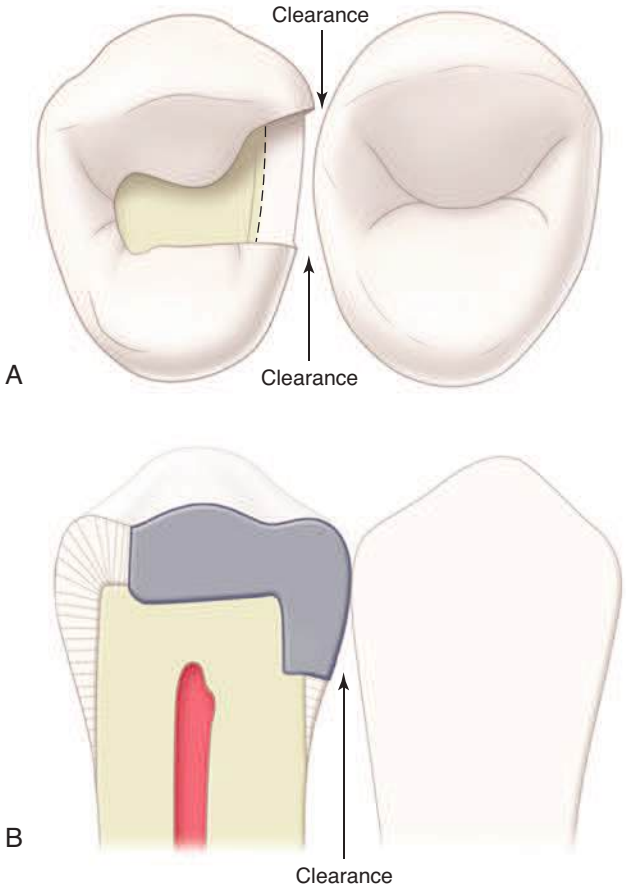
The initial extension of the tooth preparation should be visualized preoperatively by estimating the extent of the defect, the preparation form requirements of the amalgam, and the need for adequate access and visibility to place the amalgam into the tooth. Enamel cavosurface margins must be left at 90 degrees or greater to limit the potential for enamel fracture. For enamel strength, the marginal enamel rods should be supported by sound dentin. These requirements for enamel strength must be combined with marginal requirements for amalgam (90-degree butt joint) when establishing the periphery of the tooth preparation (Fig. 10.9).

The preparation extension is dictated primarily by the existing caries lesion, old restorative material, or defect. Adequate extension to provide access for tooth preparation, caries lesion removal, matrix placement, and amalgam insertion also must be considered. When making the preparation extensions, every effort should be made to preserve the dentinal support (i.e., the strength) of cusps and marginal ridges.

When viewed from the occlusal, the facial and lingual proximal cavosurface margins of a Class II preparation should be 90 degrees (i.e., perpendicular to a tangent drawn through the point of extension facially and lingually) (see Fig. 10.9). In most instances, proximal caries lesions severe enough to require surgical intervention result in facial and lingual proximal walls that must be extended into the facial or lingual embrasure. This extension provides adequate access for performing the preparation (with decreased risk of damaging the adjacent tooth), easier placement of the matrix band, and easier condensation and carving of the amalgam. Such extension provides "clearance" between the cavosurface margin and the adjacent tooth (Fig. 10.10). Clearance also allows the operator to confirm that no voids exist at the proximal margins of the finished restoration. Occasionally the level of disease may not require extension of the proximal margins beyond the proximal contact. This is especially helpful in areas that are more esthetically demanding.



• **Fig. 10.9** Proximal cavosurface margins. A, Facial and lingual proximal cavosurface margins prepared at 90-degree angles to a tangent drawn through the point on the external tooth surface. B, A 90-degree proximal cavosurface margin produces a 90-degree amalgam margin; C, 90-degree amalgam margins.

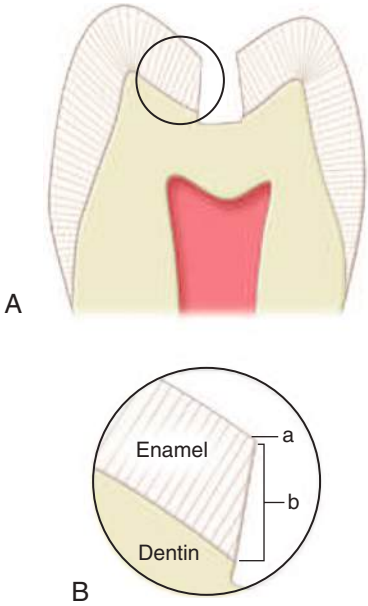


• **Fig. 10.10** Proximal box preparation clearance of adjacent tooth. A, Occlusal view. B, Lingual view of a cross section through the central groove.

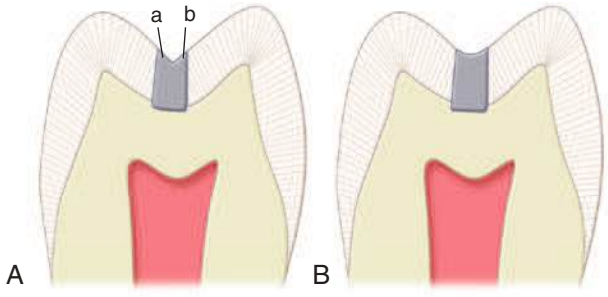
Factors dictating the outline form are presented in greater detail in Chapter 4. They include caries lesion, old restorative material, inclusion of all of the defect, proximal or occlusal contact relationship, and the need for convenience form.

Cavosurface Margin

Enamel must have a marginal configuration of approximately 90 degrees or greater, and amalgam must have a marginal configuration of approximately 90 degrees. Marginal wall configurations with angles less than 90 degrees in enamel or amalgam are subject to fracture, as both of these materials are brittle. Preparation walls on the occlusal surface usually have obtuse enamel margins (representing the strongest enamel margin) and result in amalgam margins that are slightly less than 90 degrees (Figs. 10.11 and 10.12). Rounding of the central groove area when carving the occlusal amalgam enables the marginal configuration to be closer to 90 degrees (see Fig. 10.12). Preparation walls on vertical aspect (in the occlusogingival axis) of the tooth (facial, lingual, mesial, or distal) should result in 90-degree enamel walls (representing a strong enamel margin) (see Fig. 10.9) that meet the inserted amalgam at a butt joint (enamel and amalgam both having 90-degree margins).



• **Fig. 10.11** Occlusal cavosurface margins. A, Tooth preparation. B, Occlusal margin representing the strongest enamel margin. Full-length enamel rods *(a)* and shorter enamel rods *(b)*.



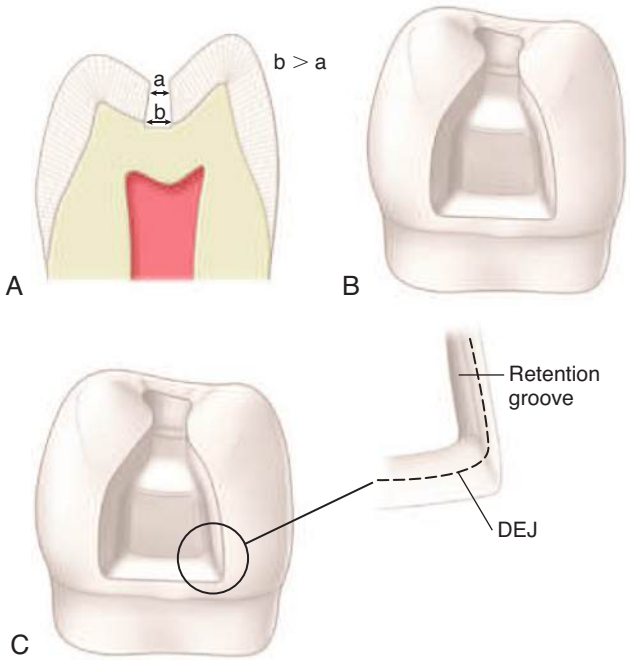
• **Fig. 10.12** Amalgam form at occlusal cavosurface margins. A, Amalgam carved too deep resulting in acute angles *a* and *b* and stress concentrations within the amalgam, increasing the potential for fracture. B, Amalgam carved with appropriate anatomy, resulting in an amalgam margin close to 90 degrees, although the enamel cavosurface margin is obtuse.

Resistance Form

Resistance form preparation features help the tooth and the restoration resist fractures caused by occlusal forces. Resistance features that assist in preventing the tooth from fracturing include (1) maintaining as much tooth structure as possible (preserving the dentin supporting cusps and marginal ridges); (2) having pulpal and gingival walls prepared perpendicular to occlusal forces, when possible; (3) having rounded internal preparation angles; (4) removing unsupported or weakened tooth structure; and (5) placing slots and pins into the tooth as part of the final stage of tooth preparation, when indicated. The placement of slots and pins is considered a secondary resistance form feature and is discussed in the section Clinical Technique for Complex Amalgam Restorations. Resistance form features that assist in preventing the amalgam from fracturing include (1) adequate thickness of amalgam (at least 1.5–2 mm in areas of occlusal contact and 0.75 mm in axial areas); (2) amalgam margin of 90 degrees; (3) boxlike preparation form, which provides uniform amalgam thickness; and (4) rounded axiopulpal line angles in Class II tooth preparations. Many of these resistance form features may be achieved using the No. 330 or No. 245 bur.

Retention Form

Retention form preparation features retain (i.e., "lock") the restorative material in the tooth. For composite restorations, adhesion (bonding) provides most of the needed retention. Amalgam restorations must be mechanically retained in the tooth. Amalgam retention form (Fig. 10.13) is provided by (1) preparation of the



• **Fig. 10.13** Typical amalgam tooth preparation retention form features. A and B, Occlusal convergence of prepared walls (primary retention form). C, Retention grooves in proximal box (secondary retention form) if the proximal preparation is wide faciolingually.

vertical walls (especially facial and lingual walls) that converge occlusally (i.e., *primary retention*); (2) retention features such as grooves, coves, slots, and pins that are placed during the final stage of tooth preparation (i.e., *secondary retention*), and (3) engagement of the inserted amalgam into any surface irregularities in the preparation that may exist. Primary retention form features are achieved by the orientation and type of the preparation instrument such as the No. 330 or No. 245 pear-shaped carbide bur or diamond (see Fig. 10.7B). Secondary retention form features are discussed in subsequent sections.

Convenience Form

Convenience form includes features that allow adequate access and visibility of the operating site to facilitate tooth preparation and restoration. Convenience form includes extension of the outline form so that (1) the caries lesion may be accessed for removal, (2) the matrix may be placed, and (3) amalgam may be inserted, carved, and finished—all while maintaining resistance form.

Final Tooth Preparation

Removal of Defective Restorative Material and/or Soft Dentin

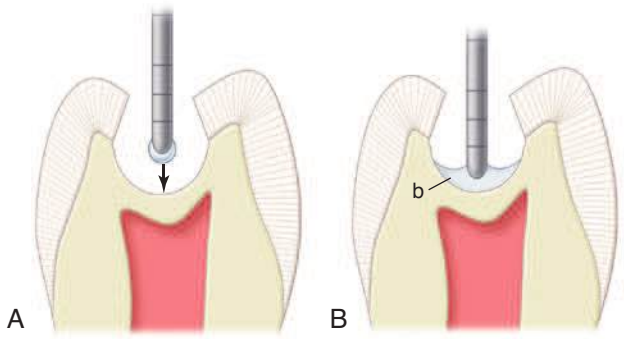
Any old restorative material or soft dentin remaining after the initial preparation will be located in the axial or pulpal walls (the extension of the peripheral preparation margins should have already been to sound tooth structure). Chapters 2 and 4 discuss (1) when to leave or remove old restorative material, (2) how to remove remaining soft dentin, and (3) what should be done to protect the pulp.

The removal of remaining soft dentin should be accomplished using the largest instrument that fits the carious area because it is least likely to penetrate the tooth in an uncontrolled manner. A hand instrument is more reliable than a rotating bur for judging the adequacy of removal of soft dentin. The removal of carious dentin should be stopped when the tooth structure feels firm unless exposure of the pulp becomes likely (see Chapter 2). This situation often occurs before all lightly stained or discolored dentin is removed.[61]

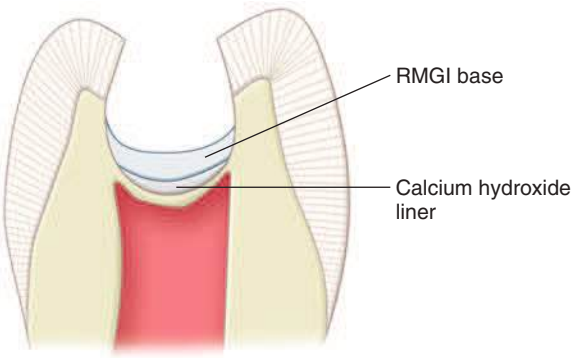
Pulp Protection

If the tooth preparation is of ideal or shallow depth, no liner or base is indicated. In deeper caries removal (where the remaining dentin thickness is judged to be 1–1.5 mm), a layer (i.e., 0.5–0.75 mm) of a resin-modified glass ionomer (RMGI) material should be placed (Fig. 10.14).[62,63] The RMGI insulates the pulp from thermal changes, bonds to dentin, releases fluoride, is strong enough to resist the forces of condensation, and reduces microleakage.[63-65] The RMGI is applied only over the deepest portion of the caries removal. It should be placed in small increments and should flow when it is touched to the dentin surface. The entire dentin surface should not be covered. Dentin peripheral to the liner should be available for support of the restoration.[66]

For pulpal protection in very deep caries removal (where the remaining dentin thickness is judged to be <0.5 mm and suspicion of potential microscopic pulpal exposure is increased), a thin layer (i.e., 0.5–0.75 mm) of a calcium hydroxide liner may be placed (Fig. 10.15). The calcium hydroxide liner may elicit tertiary dentin formation if the original odontoblasts are no longer vital. If the calcium hydroxide liner is used, it is placed by using the same instrument and the same technique as described for the RMGI liner. The calcium hydroxide liner should be placed only over the deepest portion of the caries removal (nearest the pulp). A layer of RMGI liner should be used to cover the calcium hydroxide.[67] The entire dentin surface should not be covered (see Figs. 10.14 and 10.15). The RMGI liner is recommended to cover the calcium hydroxide to resist the forces of condensation and to prevent dissolution over time by sealing the deeply excavated area.[64] Pulp exposure will require a direct pulp cap with calcium hydroxide or mineral trioxide aggregate (MTA), or endodontic treatment.[68]



• **Fig. 10.14** Liner/Base application. A, Inserting resin-modified glass ionomer (RMGI) with periodontal probe. B, In moderately deep caries removal, a base *(b)* thickness of 0.5 to 0.75 mm is indicated.



• **Fig. 10.15** Placement of calcium hydroxide liner and RMGI base.

Secondary Resistance and Retention Forms

Adequate resistance and retention form may no longer be present in the tooth preparation after caries lesion removal has resulted in excessive loss of vertical wall height. Therefore strategic features (such as the placement of grooves, coves, slots, pins) that enhance the resistance and retention form are added. Usually, the larger the tooth preparation, the greater the need for secondary resistance and retention forms.

Final Procedures: Debridement and Inspection

After the previous steps are performed, the completed tooth preparation should be cleaned of all debris (debridement) so that a careful inspection may be accomplished. The usual procedure in cleaning is to free the preparation of visible debris using short bursts of water spray from the air-water syringe followed by short

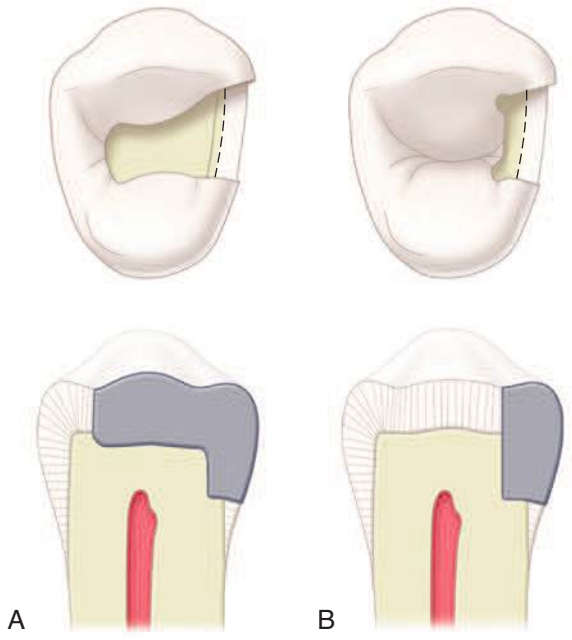
bursts with the air syringe. If debris still clings to the preparation, it may be necessary to dislodge this material with an explorer, a small damp cotton pellet, or a commercially available applicator tip (i.e., a "microbrush") moistened with water. Excess water should be removed so as to allow a clear field for inspection. However, it is important not to desiccate the tooth by overuse of air as this may damage the odontoblasts associated with the desiccated tubules. The preparation should be viewed from all angles. Careful assessment should be made to ensure that the depths are proper, the external walls provide for enamel support and correct orientation for amalgam (i.e., that the preparation is retentive and the resultant restoration margins will not be prone to fracture), and the caries lesion has been removed as indicated. Preparation walls should be smooth and transitions should be gently rounded.[69] The cavosurface margin should be precise and clearly visualized.

Preparation Designs

Typical tooth preparation for amalgam has generally been referred to as *conventional tooth preparation*. A "box-only" tooth preparation is indicated if no occlusal caries lesion is present (i.e., only a proximal caries lesion is present). Fig. 10.16 illustrates various preparation designs. Appropriate details of specific tooth preparations are presented subsequently.

General Concepts Guiding Restoration With Amalgam

After tooth preparation, the tooth must be readied for the insertion of amalgam. Disinfectants may be used, but are not considered essential.[70,71] However, it is highly recommended that a dentin desensitizer (current commercial formulations contain 5% glutaraldehyde and 35% 2-hydroxyethyl methacrylate [HEMA] and water) be placed on the prepared dentin per manufacturer's instructions (see Fig. 10.3). This type of dentin desensitizer cross-links tubular fluid proteins and, in the presence of HEMA, forms lamellar plugs in the dentinal tubules.[72] These plugs are thought to be responsible for reducing the potential for rapid tubular fluid movement and, thereby, the sensitivity of dentin. This step may occur before or after the matrix application.



• **Fig. 10.16** Types of amalgam tooth preparations. A, Conventional. B, Box only.

Matrix Placement

A matrix primarily is used when a proximal surface is to be restored. The objectives of a matrix are to provide proper contact, provide proper contour, confine the restorative material, and reduce the amount of excess material. For a matrix to be effective, it should be easy to apply and remove, extend below the gingival margin enough that it can be engaged by a wedge, extend above the adjacent marginal ridge height so as to allow for proper condensation, and resist deformation during material insertion.

In some clinical circumstances, a matrix may be necessary for Class I or V amalgam restorations. Matrix application, along with wedging to increase interproximal space, may help protect the adjacent tooth from being damaged during preparation.

Mixing (Triturating) the Amalgam

Because of its superior clinical performance, high-copper amalgam is recommended. Preproportioned, disposable capsules are available in sizes ranging from 400 to 800 mg. Some precapsulated brands require activation of the capsules before trituration. The speed and time of trituration are factors that impact the setting reaction of the material. Alterations in either may cause changes in the properties of the inserted amalgam. Amalgam should be triturated (i.e., mixed) according to the manufacturer's directions. Correctly mixed amalgam should not be dry and crumbly; rather, it should retain sufficient "wetness" so as to aid in achieving a homogeneous and well-adapted restoration.[73] It is often necessary to make several mixes to complete the restoration, particularly for large preparations.

Insertion of the Amalgam

An amalgam carrier is used to transfer amalgam to the tooth preparation. Increments extruded from the carrier should be smaller (often only 50% or less of a full-carrier tip) for a small preparation, particularly during the initial insertion. A flat-faced, circular or elliptic condenser may be used to condense amalgam in the deeper areas of the preparation. The principal objectives during the insertion of amalgam are to condense the amalgam mass and to adapt it to the preparation walls and the matrix (when used). Thorough condensation is essential to produce a restoration free of voids, and helps to reduce marginal leakage.[74,75] Optimal condensation is necessary to minimize the mercury content in the restoration, which decreases corrosion and enhances restoration strength and marginal integrity.[75] Spherical amalgam is more easily condensed than admixed (lathe-cut) amalgam, but some practitioners prefer the handling properties of the admixed type. Condensation of amalgam that contains spherical particles requires larger condensers than are commonly used for admixed amalgam. Smaller condensers tend to penetrate a mass of spherical amalgam, resulting in less effective force to compact or adapt the amalgam within the preparation. In contrast, smaller condensers are indicated for the initial increments of admixed amalgam because it is more resistant to condensation pressure. This is because the area of a circular condenser face (nib) increases by the square of the diameter; doubling

the diameter requires four times more force for the same pressure per unit area. Each portion must be thoroughly condensed prior to placement of the next increment. Each condensed increment should fill only one third to one half the preparation depth. Each condensing stroke should overlap the previous condensing stroke to ensure that the entire mass is well condensed.

Lateral condensation (facially, lingually, and proximally directed condensation) is important in the proximal box portions of preparations to ensure confluence of the amalgam with the margins, the elimination of voids, and an adequate proximal contact. Generally, smaller amalgam condensers are used first, which allows the amalgam to be properly condensed into the internal line angles and secondary retention features. Subsequently, larger condensers are used. Amalgam preparations should be somewhat overfilled to ensure adequate condensation on the occlusal surface. The condensation of a mix should be completed within the time specified by the manufacturer (usually 2.5 to 3.5 minutes). Otherwise crystallization of the unused portion will be too advanced to react properly with the condensed portion. The mix should be discarded if it becomes dry, and another mix quickly made to continue the insertion.

Precarve Burnishing

To ensure that the marginal amalgam is well condensed before carving, the overpacked amalgam should be burnished immediately with a large burnisher, using heavy strokes mesiodistally and faciolingually, a procedure referred to as *precarve burnishing*. Precarve burnishing is useful to finalize the condensation, remove excess mercury-rich amalgam, and initiate the carving process. To maximize its effectiveness, the burnisher head should be large enough that in the final strokes it contacts the cusp slopes but not the preparation margins. Precarve burnishing produces denser amalgam at the margins of the occlusal preparations restored with high-copper amalgam alloys and initiates contouring of the restoration.[76,77]

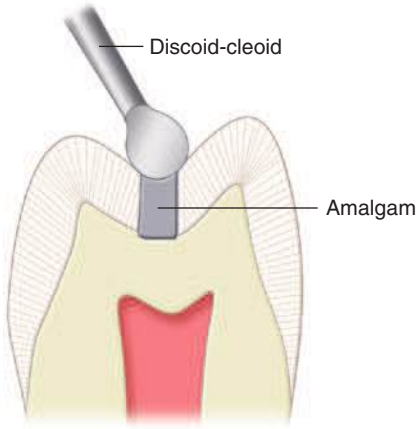
Carving the Amalgam

The following discussion of carving (and shaping) of amalgam assumes the use of *sharp carving instruments.* All carving instruments dull with repeated use and sterilization cycles and, as a result, lose their efficiency. Carving a freshly condensed amalgam, which is setting and steadily getting harder, with a dull instrument requires the use of ever-increasing pressure on the instrument and increases the likelihood of losing control (slipping) and/or increasing the amount of time required to complete the carving. Always use sharp carving instruments!

The amalgam material selected for the restoration has a specific setting time. After precarve burnishing has been accomplished, the remainder of the accessible restoration must be contoured to achieve proper form and, as a result, function. The insertion (condensation) and carving of the material must occur before the material has hardened so much that it becomes uncarvable.

Occlusal Areas

A discoid–cleoid instrument may be used to carve the occlusal surface of an amalgam restoration. The rounded end (discoid) is positioned on the unprepared enamel adjacent to the amalgam margin and pulled parallel to the margin (Fig. 10.17). This removes any excess at the margin while not allowing the marginal amalgam to be carved below the preparation margins (i.e., "submarginated"). The pointed end (cleoid) of the instrument may be used to define



• **Fig. 10.17** Carving the occlusal margins.

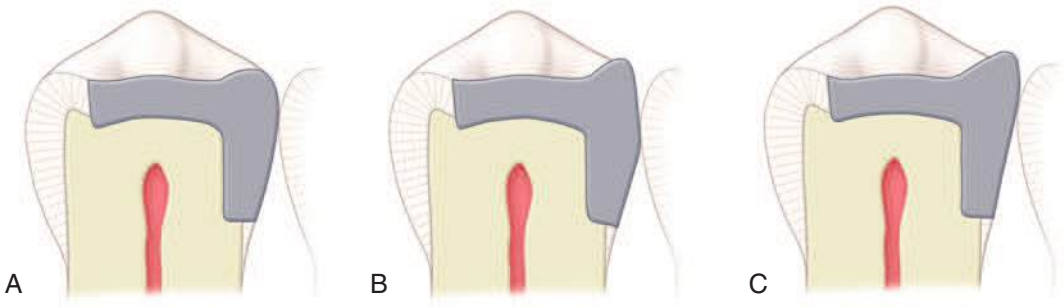


• **Fig. 10.18** Defining the marginal ridge and the occlusal embrasure with an explorer.

the primary grooves, fossae, and cuspal inclines. The Hollenback carver is also useful for carving these areas.

The reproduction of grooves and fossae is necessary to provide appropriate mastication and sluiceways for the escape of food from the occlusal table. The mesial and distal fossae are carved to be inferior to the marginal ridge height, helping limit the potential for food to be wedged into the occlusal embrasure. Having rounded and relatively shallow occlusal anatomy also helps achieve a 90-degree amalgam margin on the occlusal surface and to ensure adequate occlusogingival dimension of the final amalgam restoration for strength (see Fig. 10.12B).

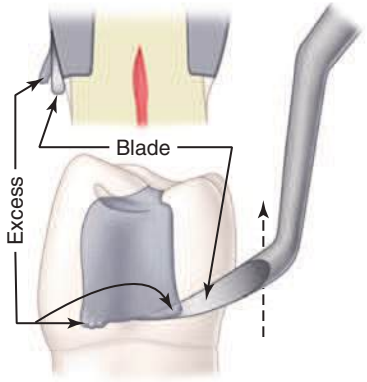
For multiple surface restorations (which require use of a matrix), the initial carving of the occlusal surface should be rapid, concentrating primarily on the marginal ridge height and occlusal embrasure areas. Occlusal embrasure areas are developed with a thin explorer tip or carving instrument by mirroring the contours of the adjacent tooth. The explorer tip is pulled along the inside of the matrix band, creating the occlusal embrasure form. When viewed from the facial or lingual direction, the embrasure form created should be identical to that of the adjacent tooth, assuming that the adjacent tooth has appropriate contour. Likewise, the height of the amalgam marginal ridge should generally be the same as that of the adjacent tooth (Figs. 10.18 and 10.19). If both these areas are developed properly, the potential for fracture of the marginal ridge area of the restoration while checking the occlusion is significantly reduced. Placing the initial carving emphasis on the occlusal areas for a



• **Fig. 10.19** Proximal contour. A, Correct proximal contour. B, Incorrect marginal ridge height and occlusal embrasure form. C, The occlusogingival proximal contour is too straight, the contact is too high, and the occlusal embrasure form is incorrect.



• **Fig. 10.20** Positioning of the carving instrument to prevent overcarving amalgam and to develop the desired gingival contours.



• **Fig. 10.21** Gingival excess may be removed with amalgam knives.

large restoration permits the operator to remove the matrix more quickly so as to carve any extensive axial surfaces of the restoration, especially the proximal areas. Some of these areas may be relatively inaccessible and must be carved while the amalgam material is still not fully set. Remaining necessary carving or contouring on the occlusal surface may be done later, and if the amalgam is too hard to carve, the use of rotary instruments in the handpiece may be required.

Facial and Lingual Areas

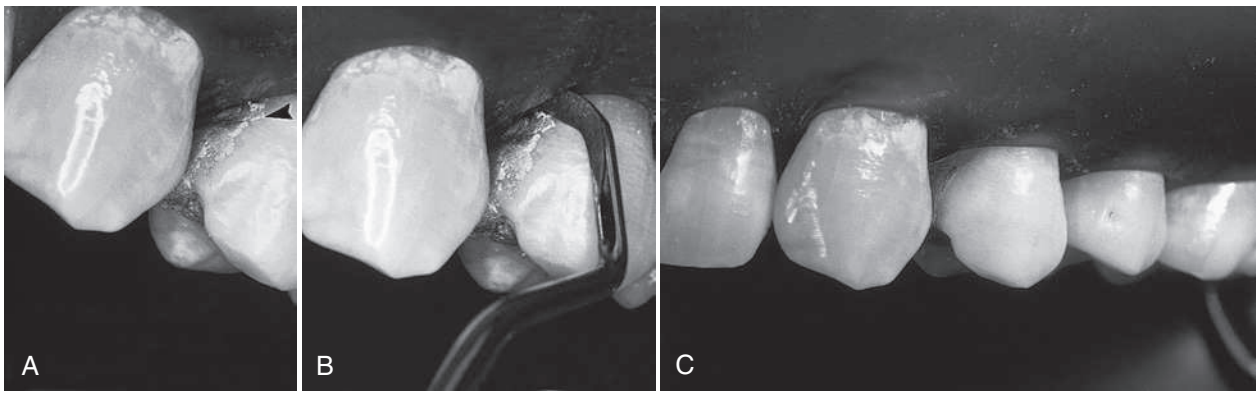
Most facial and lingual areas are accessible and may be carved directly. The side of an explorer tine may be very effective in creating correct contours when the amalgam is early in its setting reaction. A Hollenbeck carver or base of the amalgam knife (scaler 34/35) is also useful for carving these areas. With regard to the cervical areas, it is important to remove any excess and develop the proper contour of the restoration, which is usually convex. The convexity is developed by using the occlusal and gingival unprepared tooth structure as guides for initiating the carving (Fig. 10.20). The marginal areas are blended together, resulting in the desired convexity and providing a physiologic contour that promotes the health of adjacent gingival tissue.

Proximal Embrasure Areas

The initial development of the occlusal embrasure of the proximal surface while the matrix is still in place has already been described. After removal of the matrix, the amalgam knife (or scaler 34/35) is an excellent instrument for removing proximal excess and developing proximal contours and embrasures (Figs. 10.21 and 10.22). The knife is positioned below the gingival margin and drawn occlusally to carefully shave off excess, to refine the proximal contour (below the contact) and the gingival embrasure form. The sharp tip of the knife also is beneficial in developing the facial and lingual embrasure forms. Care must be exercised in not carving away any of the contour necessary to create the desired proximal contact.

Development of proper proximal contour and contact is important for the physiologic health of interproximal soft tissue. Likewise, ensuring a smooth junction between the tooth and the amalgam is important. An amalgam overhang (excess of amalgam) may result in compromised gingival health. Voids at the cavosurface margins may result in recurrent caries lesion formation.

The proximal portion of the carved amalgam is evaluated by visual assessment (reflecting light into the contact area from the lingual and occlusal perspectives to confirm the dimensions and location of the proximal contact) and placement of very thin dental floss through the contact area. If dental floss is used, it must be used judiciously, ensuring that amalgam in the contact area is not inadvertently removed. A piece of floss may be inserted through the contact and into the gingival embrasure area by initially wrapping the floss around the adjacent tooth and exerting pressure on that tooth rather than the restored tooth while moving the floss through the contact area. When the floss is into the gingival embrasure area, it is wrapped around the restored tooth and moved occlusally and gingivally to determine whether excess exists and to smooth the proximal amalgam material. If excess material is detected along the gingival margin, the amalgam knife should again be used until a smooth margin is created.



• **Fig. 10.22** Removal of gingival excess of amalgam. A, Excess of amalgam *(arrowhead)* at the gingival corner of the restoration. B, Use of the amalgam knife for removal of gingival excess. C, Gingival corner of restoration with excess removed.

Postcarve Burnishing

Some operators prefer to *postcarve* burnish the amalgam surface by using a small burnisher. Postcarve burnishing is done by lightly rubbing the carved surface with a burnisher of suitable size and shape to improve smoothness and produce a satin (not shiny) appearance. The surface should not be rubbed hard enough to produce grooves in the amalgam. Postcarve burnishing may improve the marginal integrity of low- and high-copper amalgams and may improve the smoothness of the restoration.[78-80] Postcarve burnishing in conjunction with precarve burnishing may serve as a substitute for conventional polishing.[80]

Evaluation of Occlusal Contact Areas on the Restoration

After completion of the carving and the removal of the rubber dam, the occlusal contacts on the restoration must be evaluated. Even if the carving has been carefully accomplished, the restoration occasionally is "high," indicating premature occlusal contact. A piece of articulating paper is placed over the restoration and adjacent teeth, and the patient is instructed to close gently into occlusion. *It is essential that the teeth be completely dry for proper marking with articulating paper.* The patient is advised not to bite firmly because of the danger of fracturing the restoration, which is weak at this stage. The operator should visually assess the contact potential of the restored tooth and the extent of closure. If local anesthesia is still present, it may be difficult for the patient to tell when the teeth are in contact. After the patient has reopened the mouth and the articulating paper is removed, the following two features of the occlusal relationship suggest that the restoration is high: (1) Cusp tips of adjacent teeth are not in occlusal contact when it is known from the preoperative occlusal assessment that they should be touching, and (2) an opposing cusp prematurely occludes with the new restoration.

Contact areas on the amalgam should be assessed by the color imparted by the articulating paper. Deeply colored areas (heavy contacts) and those with light-colored centers (even heavier contacts) are reduced until all markings are uniformly of a light hue, and contacts are noted on adjacent teeth (Fig. 10.23). Heavy occlusal contacts are commonly referred to as "prematurities" or "high spots." Observing the bite opening between nearby teeth when the restoration is in contact indicates how much occlusal adjustment is indicated. For example, if the adjacent teeth are 0.5 mm apart (by visual estimation), the high spot(s) should be reduced by approximately that amount. This observation may expedite the occlusal adjustment compared with making an insufficient adjustment and having to repeat closure-and-carving numerous times. The sequence of closure, observation, and carving is repeated until the appropriate surfaces of opposing teeth are touching.

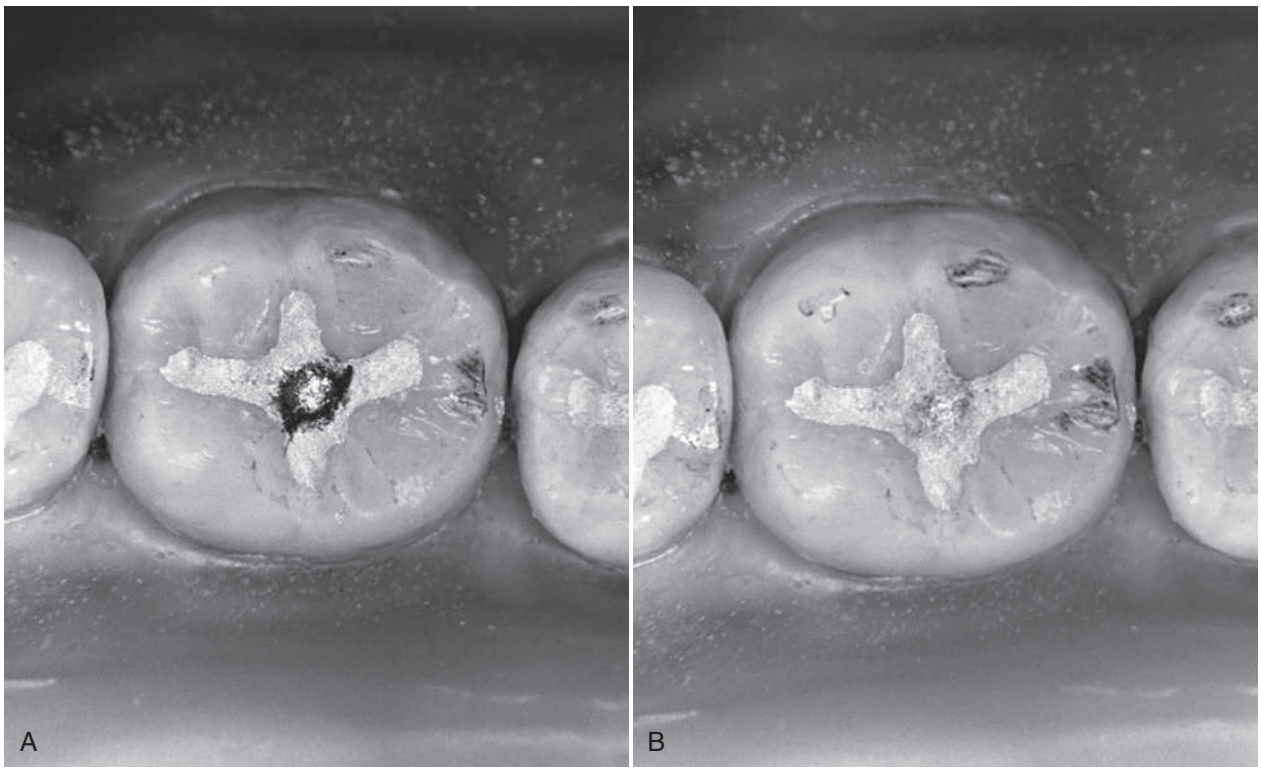
Occlusal contacts on a cuspal incline or ridge slope are undesirable because they cause a deflective lateral force on the tooth. These should be adjusted until the resulting contact is stable (i.e., the occlusal contact should be perpendicular to the occlusal load where possible in maximum intercuspation, or, in other words, the occlusal force should be parallel to the long axis of the tooth).

To this point in the procedure, the patient has been instructed to close vertically into maximum intercuspation (i.e., "bite together on your back teeth"). After replacing the articulating paper over the tooth, the patient is asked to occlude lightly and to slide the teeth lightly from "side to side" to identify excursive interferences and "front to back" to identify protrusive interferences (see Chapter 1). Any additional occlusal marks are evaluated, and undesirable contact areas (interferences) are eliminated. Note that amalgam restorations carved completely out of occlusion (i.e., that are "overcarved") may allow supraeruptive tooth movement resulting in undesirable tooth contacts and this, too, should be avoided. Finally, the patient should be cautioned to protect the newly condensed restoration by not using it to chew firm food for 24 hours.

Finishing and Polishing of the Amalgam

Most amalgams do not require further finishing and polishing. These procedures are occasionally necessary, however, to (1) refine the anatomy, contours, and marginal integrity; and (2) refine the surface texture of the restoration. Additional finishing and polishing procedures for amalgam restorations are not attempted within 24 hours of insertion because crystallization of the restoration is incomplete.[73] If used, these procedures are often delayed until all of the patient's amalgam restorations have been placed, rather than finishing and polishing periodically during the course of treatment. An amalgam restoration is less prone to tarnish and corrosion if a smooth, homogeneous surface is achieved.[81-83] Polishing of high-copper amalgams is less important than it is for low-copper amalgams because high-copper amalgams are less susceptible to tarnishing and marginal breakdown.[56,82-87]

The finishing procedure may be initiated by use of a green carborundum or white alumina stone. (Fig. 10.24A). The green



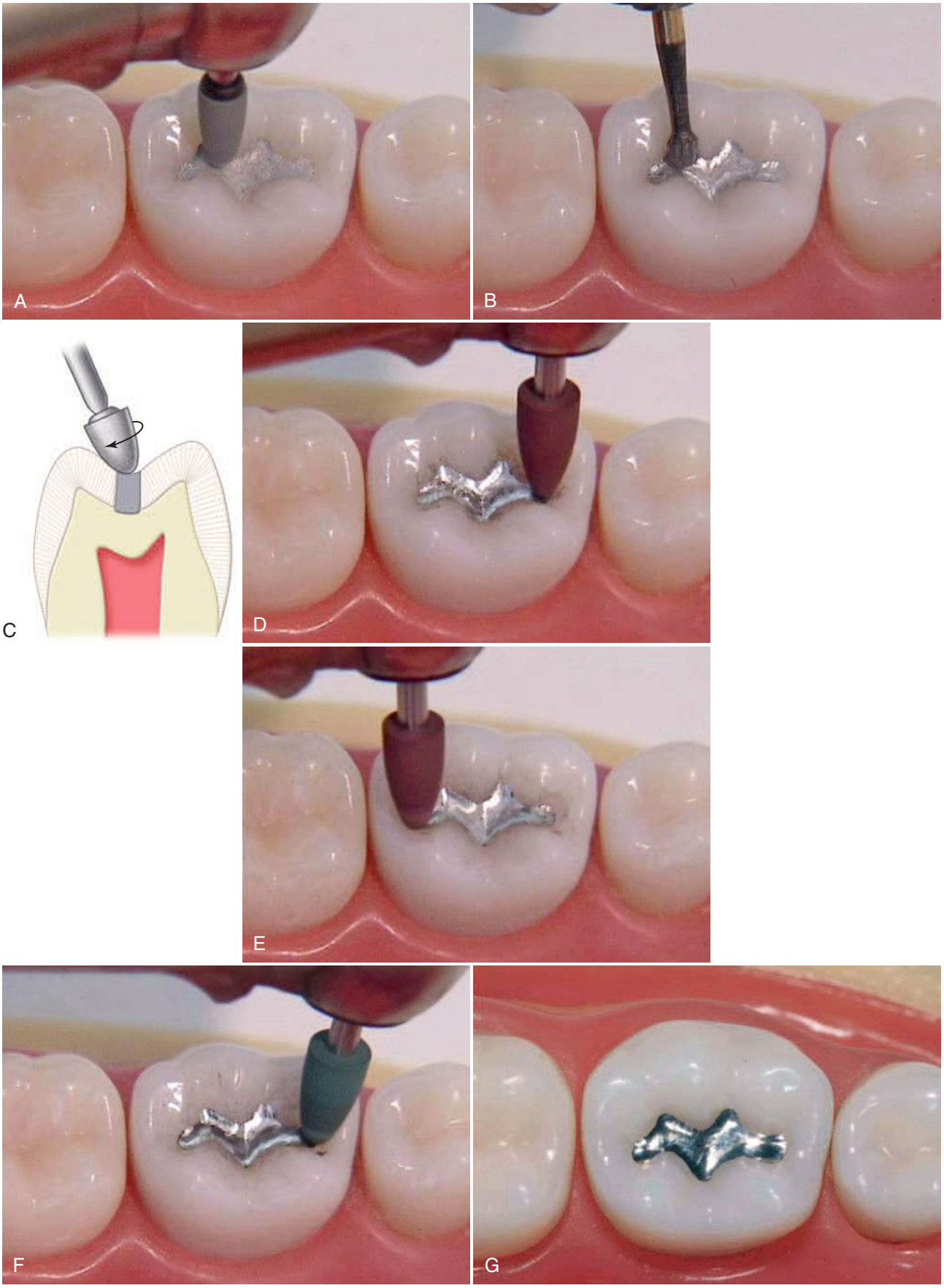
• **Fig. 10.23** Occluding the restoration. A, Heavy occlusal contacts on new amalgam should be adjusted. Articulating paper marks heavy contacts as dark areas, and it marks very heavy contacts as dark areas with light-colored, potentially shiny, centers. B, Amalgam should not be carved out of occlusion. Rather, it should have light occlusal contact or contacts, as indicated by faint markings.

stone is more abrasive than the white stone; the tip of either stone may be blunted on a diamond wheel before use. This helps prevent marring the center of the restoration while the margins are being adjusted. During the surfacing of amalgam, the stone's long axis is held at a 90-degree angle to the margins. Reduction of any occlusal contact should be avoided. After the stone is used, the margins should be reevaluated with the tine of an explorer and any additional discrepancies removed. The surface may be smoothed further using light pressure with an appropriate finishing bur (see Fig. 10.24B). A large, round finishing bur is generally used for this finishing step. If the groove and fossa features are not sufficiently defined, a small round finishing bur also may accentuate them without eliminating the occlusal contact areas. The long axis of the bur or stone should be at a ~45-degree angle to the margin to allow the unprepared tooth structure to guide the bur and prevent unnecessary removal of amalgam (see Fig. 10.24C). A smooth surface should be achieved before the polishing procedure is initiated. The finishing bur should remove the minor scratches that resulted from use of the green or white stone. Often, however, these scratches can be removed only with the use of rubber abrasive points.

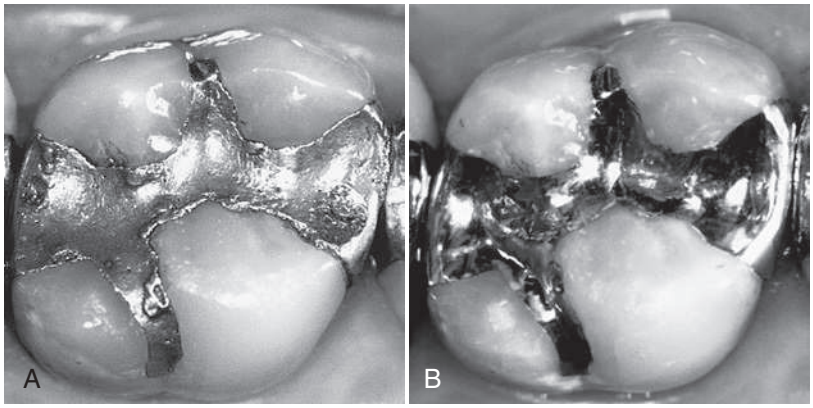
The polishing procedure is initiated by using a coarse, rubber abrasive point at low speed and air-water spray to produce an amalgam surface with a smooth, satin appearance (see Fig. 10.24D and E). If the amalgam surface does not exhibit this appearance after only a few seconds of polishing, the surface was too rough at the start. In this instance, resurfacing with a finishing bur is necessary, followed by the coarse, rubber abrasive point to develop the satiny appearance. It is important that the rubber points be used at low speed (≤6000 revolutions per minute [rpm]) or just above "stall out" speed so as to limit the danger of point disintegration (which may occur at high rotational speeds) and the danger of frictional temperature elevation of the restoration and the tooth. Temperature above 140°F [>60°C] is able to cause irreparable damage to the pulp, the restoration, or both. When overheated, the amalgam surface appears cloudy, even though it may have a high polish. This cloudy appearance indicates that mercury has been brought to the surface, which results in increased corrosion of the amalgam and loss of strength.[73]

Polishing with the coarse abrasive rubber point will result in a moderately polished surface. No deep scratches should remain on the amalgam surface. After the area is washed free of abrasive particles and dried, a high polish may be imparted to the restoration with a series of medium grit and fine-grit abrasive points (see Fig. 10.24F). As with the more abrasive points, the finer abrasive points must be used at a low speed. If a high luster does not appear within a few seconds, the restoration requires additional polishing with the more abrasive points. The system illustrated in Fig. 10.24 includes coarse-grit, medium-grit, and fine-grit rubber abrasive points. Using these points in sequence, from coarse to fine, produces an amalgam surface with a brilliant luster (see Fig. 10.24G). As an alternative to rubber abrasive points, final polishing may be accomplished using a rubber cup with flour of pumice followed by a high-luster agent such as precipitated chalk. Finishing and polishing of older, existing restorations may be performed to improve their contour, margins, surface, or anatomy, when indicated (Fig. 10.25).

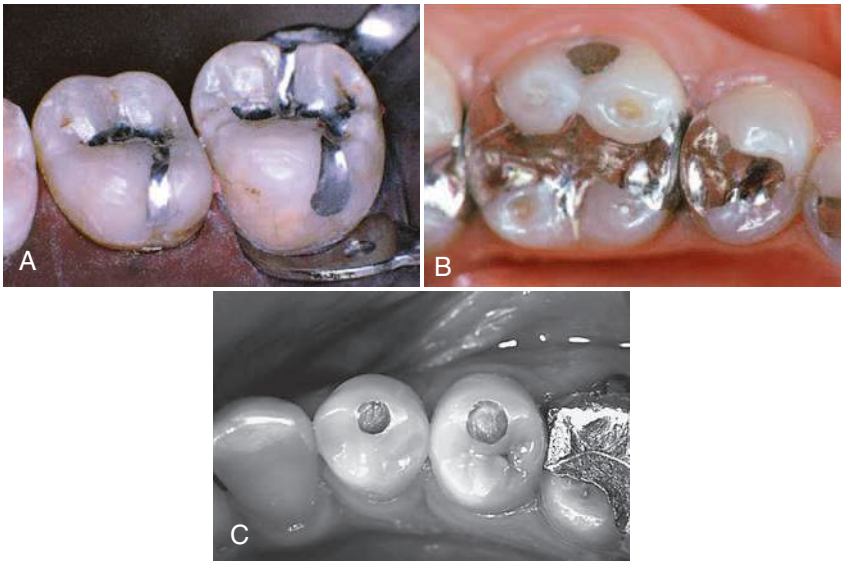
These procedures should not leave the restoration undercontoured and should not alter the carefully designed occlusal contacts. The tip of an explorer should pass from the tooth surface to the restoration surface (and vice versa), without jumping or catching, thus



• **Fig. 10.24** Polishing the amalgam. A, When necessary, a carborundum or fine-grit alumina stone is used to develop continuity of surface from the tooth to the restoration. B, The restoration is surfaced with a round finishing bur. C, The stone's long axis or the bur's long axis is held at a right angle to the margin. D, Polishing is initiated with a coarse rubber abrasive point at low speed. E, The point should produce a smooth, satin appearance. F, A high polish is obtained with medium-grit and fine-grit abrasive points. G, Polished restoration. (Courtesy Aldridge D. Wilder, DDS.)



• **Fig. 10.25** A, Existing amalgam restoration exhibiting marginal deterioration and surface roughness. B, Same restoration after finishing and polishing.



• **Fig. 10.26** Clinical examples of Class I, II, and VI amalgam restorations. A, Class I amalgam in the occlusal surface of the first molar. B, Class II amalgams in a premolar and molar. C, Class VI amalgams in premolars.

verifying continuity of contour across the margin. Every effort to avoid removal of adjacent tooth structure should be made.

Repairing an Amalgam Restoration

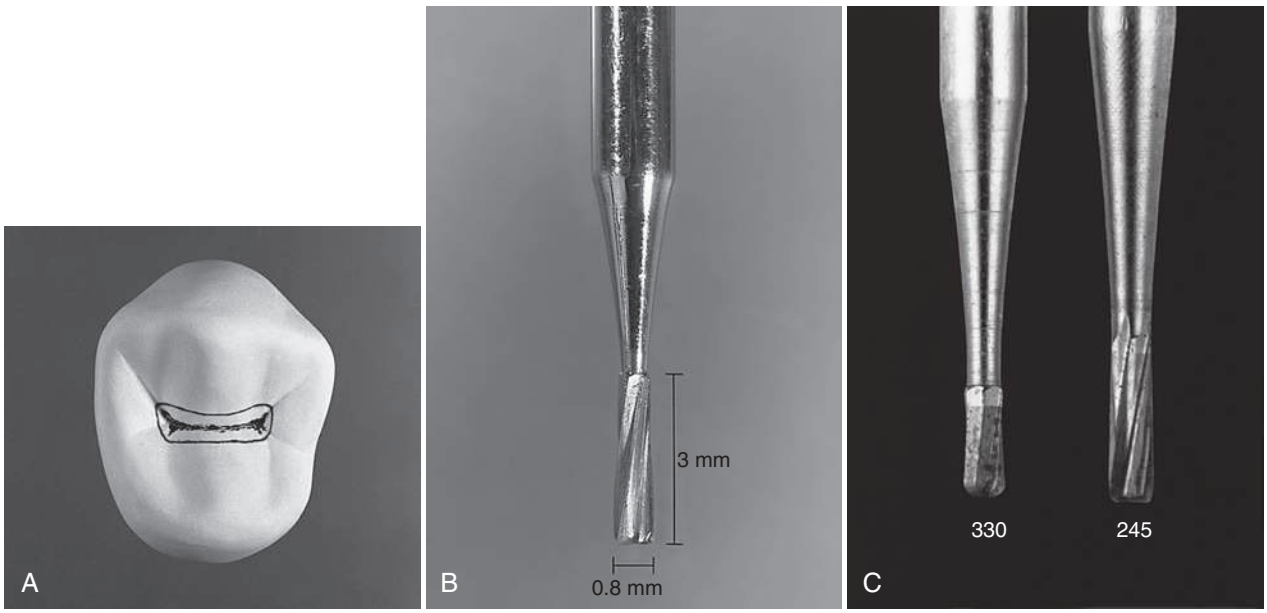
If an amalgam restoration fractures during insertion, generally all of the inserted amalgam must be removed and new amalgam condensed. If a small portion of amalgam fractures during insertion and the amalgam is still carvable, it may be possible to apply and condense newly triturated amalgam to repair the affected area. If a small void in the amalgam is discovered after the matrix is removed, for example, and if the amalgam is still carvable and the area is accessible, any poorly condensed amalgam in the void should be removed with an explorer or other instrument, and the void repaired with newly triturated amalgam. In cases where a new mix of amalgam is added to carvable existing amalgam, the "new" amalgam will adhere to the "old."

If a repair needs to be made to set, uncarvable amalgam, the defective area may be re-prepared as if it were a small restoration. Appropriate depth and retention form must be generated, sometimes entirely within the existing amalgam restoration. If necessary, another matrix must be placed. A new mix of amalgam may be condensed directly into the defect.

Clinical Technique for Class I Amalgam Restorations

This section describes the use of amalgam for Class I restorations. Class I restorations restore defects on the occlusal surfaces of posterior teeth, the occlusal thirds of the facial and lingual surfaces of molars, and the lingual surfaces of maxillary anterior teeth (Fig. 10.26).

The procedural description for a small (i.e., "conservative") Class I amalgam restoration simply and clearly presents the basic information relating to the entire amalgam restoration technique, including tooth preparation and placement and contouring of the restoration. This basic procedural information may be expanded to describe Class I amalgam restoration of lesions/defects of all sizes. The maxillary first premolar is used for illustration in this section.



• **Fig. 10.27** Class I preparation outline. A, Ideal outline includes all carious occlusal pits and fissures. B, Dimensions of head of a No. 245 bur. C, No. 330 and No. 245 burs compared.

Initial Clinical Procedures

A preoperative assessment of the occlusal relationship of the involved and adjacent teeth is necessary. After administration of local anesthetic, isolation of the operating field with the rubber dam is recommended.[73,88] The rubber dam may be applied in the few minutes necessary for onset of profound anesthesia before initiating the tooth preparation. Rubber dam isolation is especially indicated when removing deep caries (judged to be <1 mm from the pulp), during amalgam condensation, and for mercury hygiene. The rubber dam should be used for isolation of the operating site when a caries lesion is extensive. If caries removal exposes the pulp, pulp capping may be more successful if the site is isolated with a properly applied rubber dam. In addition, the dam prevents moisture contamination of the amalgam mix during insertion.[60] For a single maxillary tooth, where caries is not extensive, adequate control of the operating field may also be achieved with cotton rolls and high-volume evacuation.

Tooth Preparation for Class I Amalgam Restorations

This section describes the specific technique for preparing the tooth for a Class I amalgam restoration. It is divided into initial and final stages.

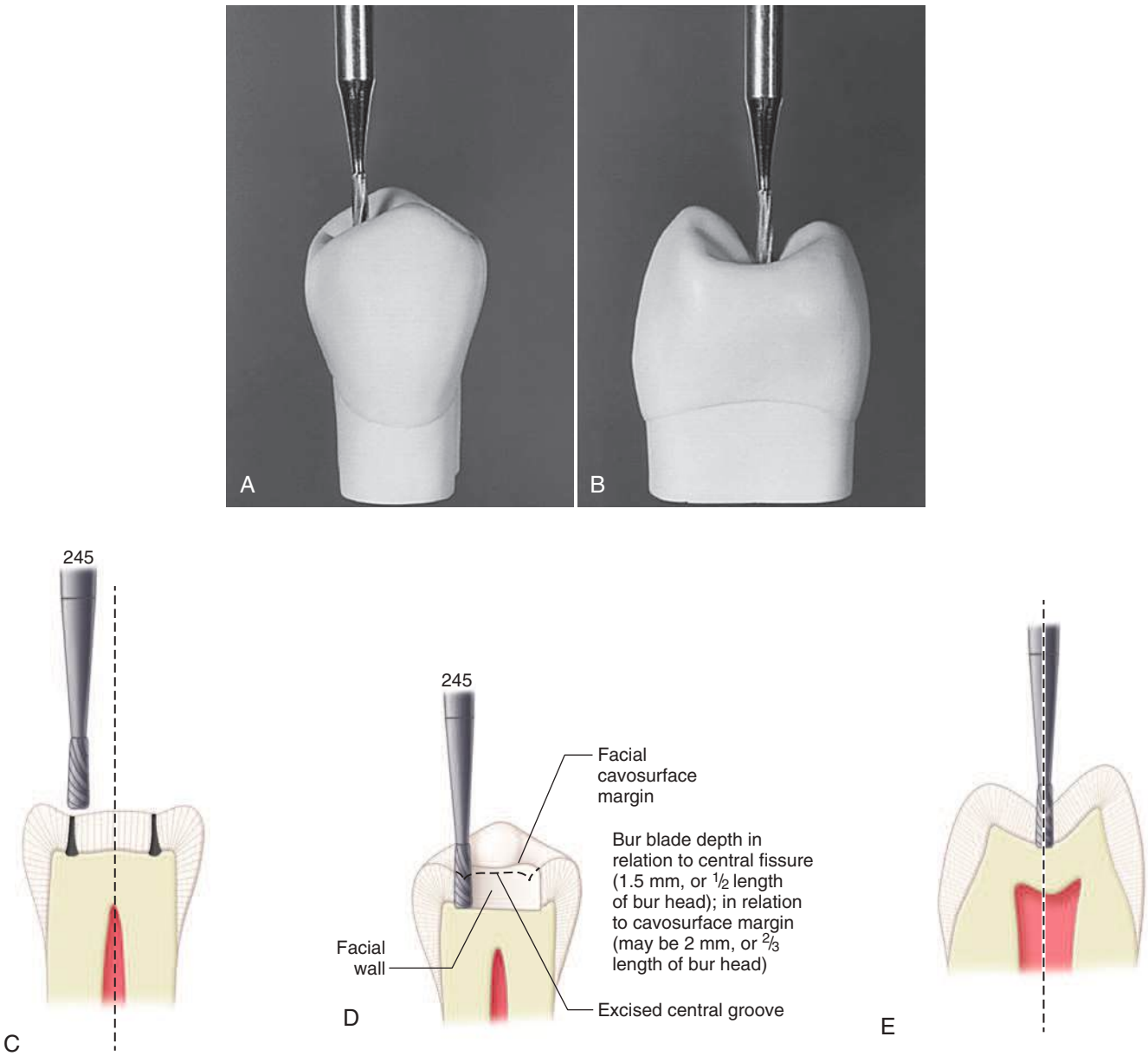
Initial Tooth Preparation

Initial tooth preparation is defined as establishing the outline form by extension of the external walls to sound tooth structure while maintaining a specified, limited depth (usually just inside the DEJ) and providing resistance and retention forms. The outline form for the Class I occlusal amalgam tooth preparation should include only the defective occlusal pits and fissures (in a way that sharp, angular, abrupt transitions in the marginal outline are avoided). Commonly, but not always, the marginal outline for maxillary premolars is butterfly shaped because of extension to include the developmental fissures facially and lingually. The ideal outline form for an amalgam restoration incorporates the following resistance form principles that are basic to all amalgam tooth preparations of occlusal surfaces (Fig. 10.27A). These principles allow margins to be positioned in areas that are sound and subject to minimal occlusal loading while maintaining the strength and health of the tooth by conserving uncompromised tooth structure. The resistance principles are as follows:

1. Extending around the cusps to conserve tooth structure and prevent the internal line angles from approaching the pulp horns
2. Keeping the facial and lingual margin extensions as minimal as possible between the central carious fissure and the cusp tips
3. Extending the outline to include fissures, placing the margins on relatively smooth, sound tooth structure
4. Minimally extending into the marginal ridges (only enough to include the defect) without removing dentinal support
5. Eliminating a weak wall of enamel by joining two outlines that come close together (i.e., <0.5 mm apart)
6. Extending the outline form to include enamel undermined by the caries lesion
7. Using enameloplasty on the terminal ends of shallow fissures to conserve tooth structure
8. Establishing an optimal, conservative depth of the pulpal wall

A No. 245 bur, with a head length of 3 mm and a tip diameter of 0.8 mm, or a smaller No. 330 bur is recommended to prepare the Class I tooth preparation (see Fig. 10.27B and C). The silhouette of the No. 245 bur reveals sides slightly convergent toward the shank. This produces an occlusal convergence of the facial and lingual preparation walls, providing adequate retention form for the tooth preparation. The slightly rounded corners of the end of the No. 245 bur produce slightly rounded internal line angles that render the tooth more resistant to fracture from occlusal force.[89] The No. 330 bur is a smaller and pear-shaped version of the No. 245 bur and is also indicated for amalgam preparations (see Fig. 10.27C).

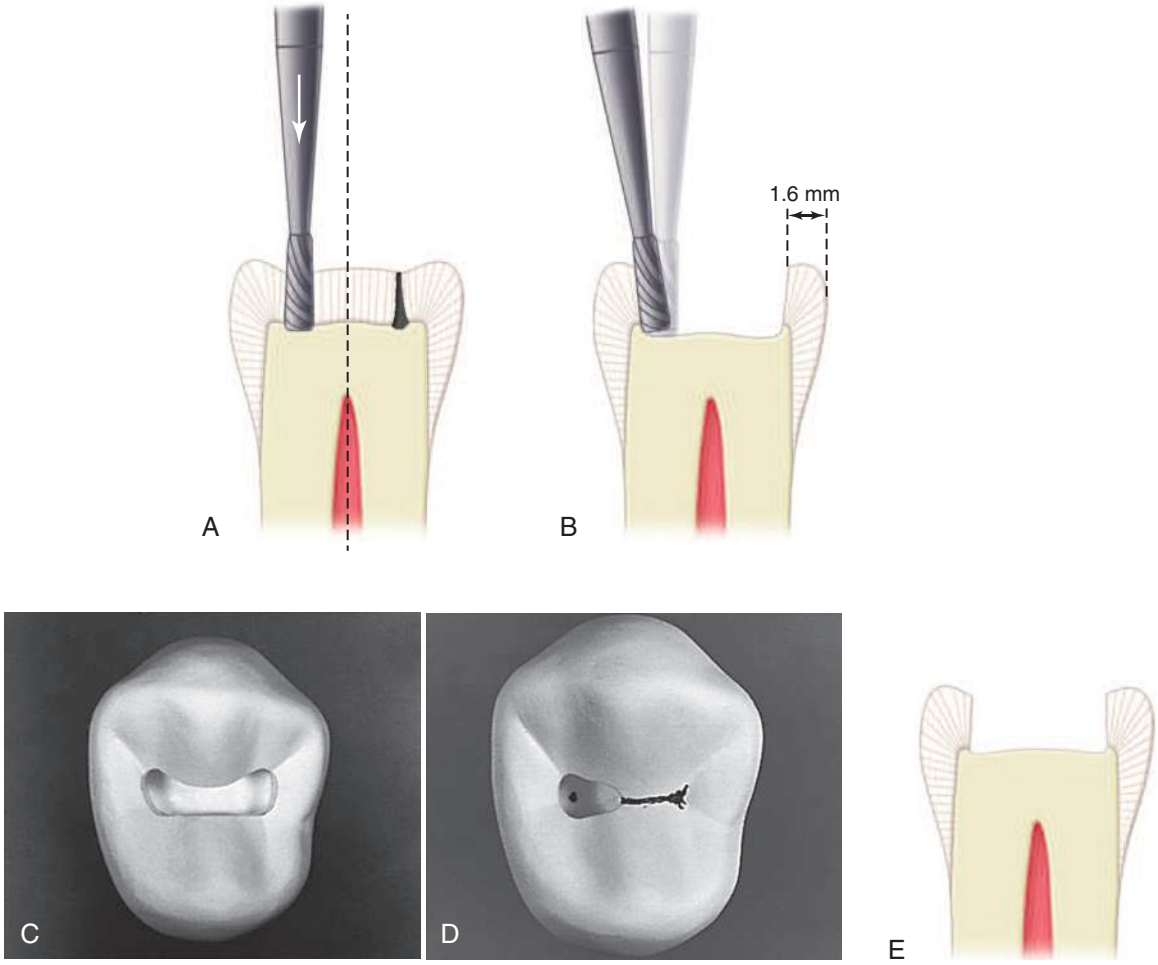
Class I occlusal tooth preparation is begun by entering the deepest or most carious pit with a "punch cut" using the No. 245 carbide bur at high speed with air-water spray. A punch cut is performed by orienting the bur such that its long axis parallels



• **Fig. 10.28** A, No. 245 bur oriented parallel to long axis of tooth crown for entry as viewed from lingual aspect. B, The bur positioned for entry as viewed from the distal aspect. C, The bur is positioned over the most carious pit (distal) for entry. The distal aspect of the bur is positioned over the distal pit. D, Mesiodistal longitudinal section. Relationship of head of No. 245 bur to excised central fissure and cavosurface margin at ideal pulpal floor depth, which is just inside the DEJ. E, Faciolingual longitudinal section. Dotted line indicates the long axis of tooth crown and the direction of the bur.

the long axis of the tooth crown (Fig. 10.28A and B). The bur is inserted directly into the defective pit. When the pits are equally defective, the distal pit should be entered as illustrated. Entering the distal pit first provides increased visibility for the mesial extension. The bur should be positioned such that its distal aspect is directly over the distal pit, minimizing extension into the marginal ridge (see Fig. 10.28C). The bur should be rotating when it is applied to the tooth and should not stop rotating until it is removed from the tooth. On posterior teeth, the approximate depth of the DEJ is located at 1.5 to 2 mm from the occlusal surface. As the bur enters the pit, an initial target depth of 1.5 mm should be established. This is one half the length of the cutting portion of the No. 245 bur. The 1.5-mm pulpal depth is measured at the central fissure (see Fig. 10.28D and E). The depth of the preparation is modified as needed so that the pulpal wall is established 0.1 to 0.2 mm into dentin. In other words, the depth of the prepared external walls should be 1.5 to 2 mm ("just inside the DEJ") (see Fig. 10.28D and E), depending on the cuspal incline. The length of the blades of an unfamiliar bur should be measured before it is used as a depth gauge.

Distal extension into the distal marginal ridge to include a fissure or caries occasionally requires a slight tilting of the bur distally (≤10 degrees). This creates a slight occlusal divergence to the distal wall to prevent undermining the marginal ridge of its dentin support (Fig. 10.29A–C). Because the facial and lingual prepared walls converge, this slight divergence does not compromise



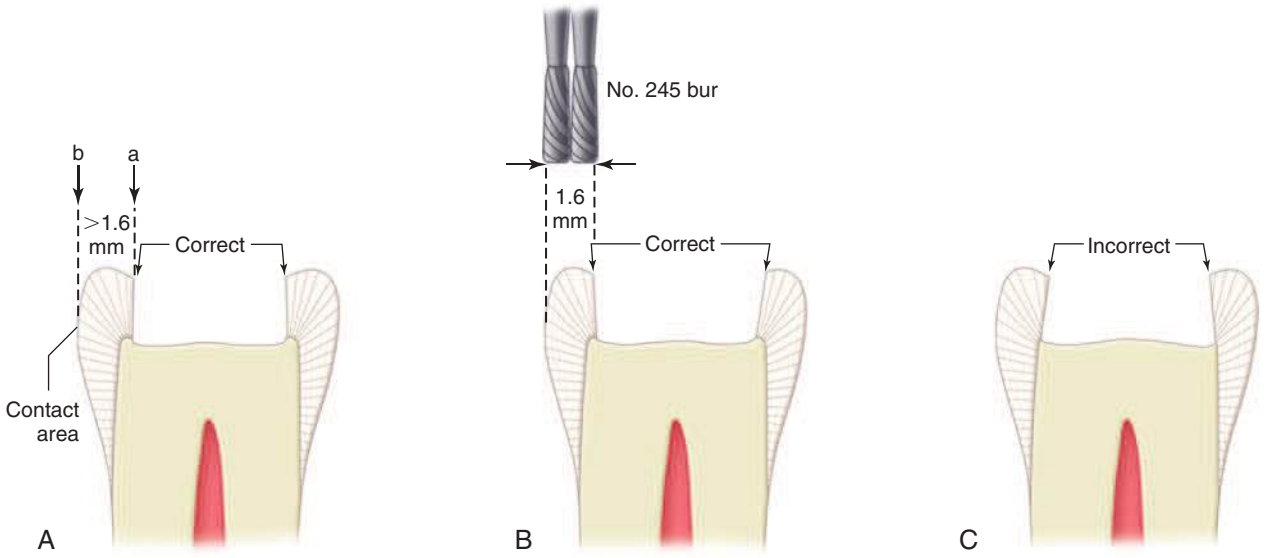
• **Fig. 10.29** A, Enter the pit with a punch cut to just inside the DEJ (depth of 1.5–2 mm or one half to two thirds the head length of the No. 245 bur). The 1.5-mm depth is measured at central fissure; the measurement of same entry cut at the prepared external wall is 2 mm. B, Incline the bur distally to establish proper occlusal divergence to distal wall to prevent removal of the dentin supporting the marginal ridge enamel when the pulpal floor is in dentin, and distal extension is necessary to include a fissure or caries lesion. For such an extension on premolars, the distance from the margin to the proximal surface (i.e., imaginary projection) must not be less than 1.6 mm (i.e., two diameters of bur end). C, Occlusal view of the initial tooth preparation that has mesial and distal walls that diverge occlusally. D, Distofacial and distolingual fissures that radiate from the pit are included before extending along the central fissure. E, Mesiodistal longitudinal section. The pulpal floors are generally flat but may follow the rise and fall of the occlusal surface.

the overall retention form. For premolars, the distance from the margin of such an extension to the proximal surface usually should not be less than ~1.6 mm, or two diameters of the end of the No. 245 bur (see Fig. 10.29B) measured from a tangent to the proximal surface (i.e., the proximal surface height of contour). For molars, this minimal distance is ~2 mm.

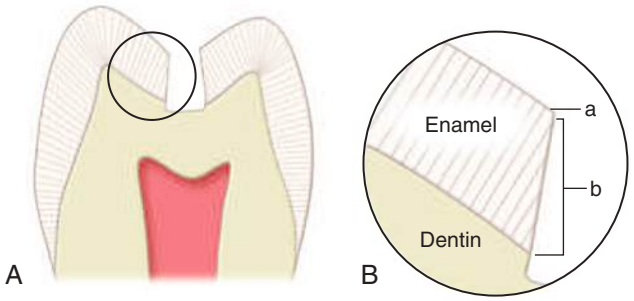
Minimal distal (or mesial) extension often does not require changing the orientation of the bur's axis from being parallel to the long axis of the tooth crown. In this case, the mesial and distal walls are parallel to the long axis of the tooth crown (or slightly convergent occlusally) (see Fig. 10.29D and E).

While maintaining the bur's orientation and depth, the preparation is extended distofacially or distolingually to include any fissures that radiate from the pit (see Fig. 10.29D). When these fissures require extensions of more than a few tenths of a millimeter, however, consideration should be given to changing to a bur of smaller diameter or to using enameloplasty. Both of these approaches conserve the tooth structure and therefore minimize weakening of the tooth.

The bur's orientation and depth are maintained while extending along the central fissure toward the mesial pit, following the DEJ (see Fig. 10.29D and E). When the central fissure has minimal caries, one pass through the fissure at the prescribed depth provides the desired minimal width to the isthmus. Ideally the width of the isthmus should be just wider than the diameter of the bur. It is well established that a tooth preparation with a narrow occlusal isthmus is less prone to fracture.[90,91] As previously described for the distal margin, the orientation of the bur should not change as it approaches the mesial pit if the mesial extension is minimal. If the fissure extends farther onto the marginal ridge, the long axis



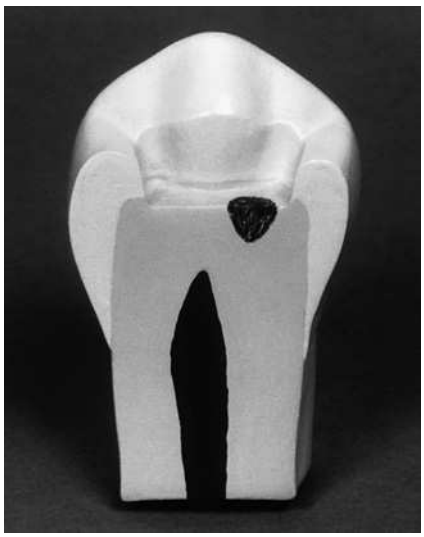
• **Fig. 10.30** The direction of the mesial and distal walls is influenced by the remaining thickness of the marginal ridge as measured from the mesial or distal margin *(a)* to the proximal surface (i.e., imaginary projection of proximal surface) *(b)*. A, Mesial and distal walls should converge occlusally when the distance from *a* to *b* is greater than 1.6 mm. B, When the operator judges that the extension will leave only 1.6-mm thickness (two diameters of No. 245 bur) of marginal ridge (i.e., premolars), the mesial and distal walls must diverge occlusally to conserve ridge-supporting dentin. C, Extending the mesial or distal walls to a two-diameter limit without diverging the wall occlusally undermines the marginal ridge enamel.



• **Fig. 10.31** A and B, The ideal and strongest enamel margin is formed by full-length enamel rods *(a)* resting on sound dentin supported on the preparation side by shorter rods, also resting on sound dentin *(b)*.

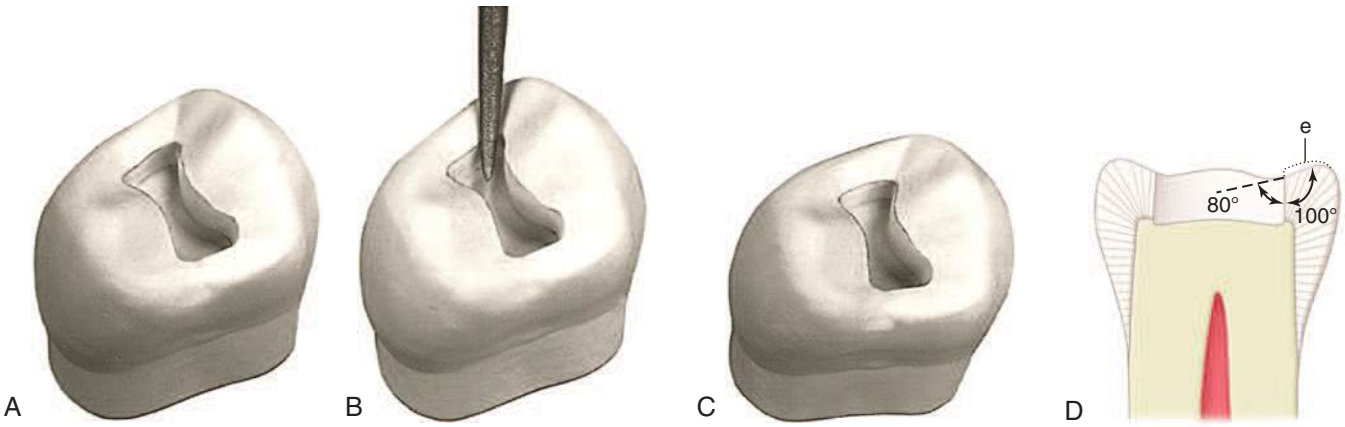
of the bur should be changed to establish a slight occlusal divergence to the mesial wall if the marginal ridge would be otherwise undermined of its dentinal support. Fig. 10.30 illustrates the correct and incorrect preparation of the mesial and distal walls. The remainder of any occlusal enamel defects is included in the outline, and the facial and lingual walls are extended, if necessary, to remove enamel undermined by the caries lesion.[92] The strongest and ideal enamel margin should be composed of full-length enamel rods attached to sound dentin, supported on the preparation side by shorter rods that are also attached to sound dentin (Fig. 10.31).

The Class I tooth preparation should have an outline form with gently flowing curves and distinct cavosurface margins. A faciolingual width of no more than 1 to 1.5 mm and a depth of 1.5 to 2 mm are considered ideal, but this goal is subject to the extension of the caries lesion. The pulpal floor, depending on the enamel thickness, is almost always in dentin (see Fig. 10.29C and E). Although conservation of the tooth structure is important, the convenience form requires that the extent of the preparation be adjusted to provide adequate access and visibility.



• **Fig. 10.32** Mesiodistal longitudinal section showing example of the pulpal floor in dentin and the caries lesion that is exposed after the initial tooth preparation. The caries lesion is surrounded by sound dentin on the pulpal floor for the resistance form.

This completes the initial amalgam preparation for a Class I caries lesion. The initial preparation should ensure that all the caries lesion is removed from the DEJ, resulting in a very narrow peripheral seat of healthy dentin on the pulpal wall surrounding any remaining caries located in the center of the pulpal wall. For the initial tooth preparation, the pulpal wall should remain at the initial ideal depth, even if any restorative material or soft caries lesion remains in the pulpal wall (Fig. 10.32). Remaining caries (and, if present, old restorative material) is removed during the final tooth preparation.



• **Fig. 10.33** Enameloplasty. A, Developmental defect at terminal end of fissure. B, Fine-grit diamond finishing instrument in position to remove the defect. C, Smooth surface after enameloplasty. D, The cavosurface angle should not exceed 100 degrees, and the margin–amalgam angle should not be less than 80 degrees. Enamel external surface *(e)* before enameloplasty.

Primary resistance form is provided by the following:

1. Sufficient area of relatively flat pulpal floor in sound tooth structure (i.e., the "peripheral seat") to resist forces directed in the long axis of the tooth and to provide a horizontal area for the restoration
2. Minimal extension of external walls (to conserve tooth strength)
3. Strong, ideal enamel margins
4. Sufficient depth (i.e., 1.5 mm) for adequate thickness of the restoration, providing resistance to fracture

The slight occlusal convergence of two or more opposing, external walls provides the primary retention form.

Usually, the No. 245 bur is used for extensions into the mesial and distal fissures. During such extensions, the remaining depth of the fissure may be viewed in cross section by looking at the wall being extended. When the remaining fissure is no deeper than one quarter to one third the thickness of enamel, enameloplasty is indicated. Enameloplasty refers to eliminating the developmental fault by removing it with the side of an appropriately shaped rotary finishing instrument, leaving a smooth surface (Fig. 10.33A–C). This procedure frequently reduces the need for further extension. The extent to which enameloplasty should be used cannot be determined exactly until the process of extending into the fissured area occurs, at which time the depth of the fissure into enamel may be observed. The surface left by enameloplasty should meet the tooth preparation wall, preferably with a cavosurface angle no greater than approximately 100 degrees; this would produce a distinct margin for amalgam of no less than 80 degrees (see Fig. 10.33D). If enameloplasty is unsuccessful in eliminating a mesial (or distal) fissure that extends to the crest of a marginal ridge or beyond, three alternatives exist:

1. Make no further change in the outline form.
2. Extend through the marginal ridge when margins would be lingual to the contact (Fig. 10.34).
3. Include the fissure in a conservative Class II tooth preparation.

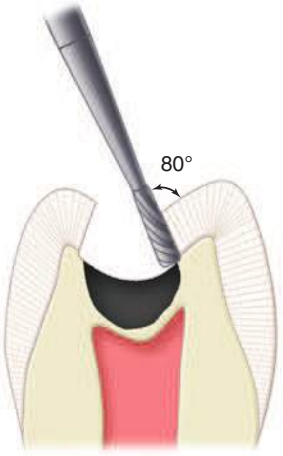
The first alternative usually should be strongly considered except in patients at high risk for caries. Enameloplasty is not indicated if an area of occlusal contact is involved. In this case, the choices are either to consider the preparation completed (an option for patients at low risk for caries) or to extend the preparation to include the fissure as previously described.



• **Fig. 10.34** Mesial fissure that cannot be eliminated by enameloplasty may be included in the preparation if the margins are able to be lingual of the occlusal contact.

A tooth may be found to have a large or "extensive" Class I caries lesion. A lesion is considered extensive if the distance between soft dentin and the pulp is judged to be less than 1 mm or when the faciolingual extent of the defect has involved much of the cuspal inclines. In this case, selective removal of soft dentin and insertion of a liner, if needed, may precede the establishment of outline, resistance, and retention forms. This approach protects the pulp as early as possible from any additional insult that may occur during tooth preparation. Normally, however, the outline, primary resistance, and primary retention forms are established through proper orientation of the No. 245 bur and appropriate extension of the preparation. As is the case for smaller caries lesions, an initial depth to reach the DEJ (measured approximately 1.5 mm at any pit or fissure and 2 mm on the prepared external walls) should still be established and maintained. The preparation is extended laterally at the DEJ to remove all enamel undermined by the caries lesion by alternately cutting and examining the lateral extension of the lesion. For a caries lesion extending up the cuspal inclines, it may be necessary to alter the bur's long axis to prepare a 90- to 100-degree cavosurface angle while maintaining the initial depth (Fig. 10.35). If not, an obtuse cavosurface angle may remain (resulting in an acute, or weak, amalgam margin), or the pulpal floor (over the pulp horns) may be prepared

too deeply. Primary resistance form is obtained by extending the outline of the tooth preparation to include only undermined and defective tooth structure while preparing strong enamel walls and allowing strong cuspal areas to remain. Primary retention form is obtained by the occlusal convergence of the enamel walls. Secondary retention form may result from undercut areas that are occasionally left in dentin (and that are not filled in by a liner or base) after peripheral removal of soft dentin. When extending the outline form, enameloplasty should be used in any indicated area (as described previously and in detail in Chapter 4).



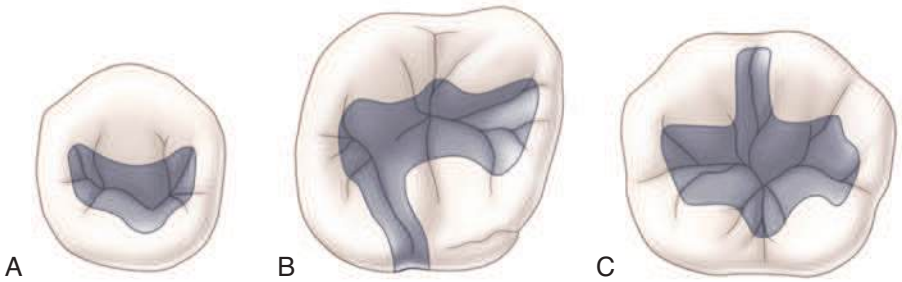
• **Fig. 10.35** Initial tooth preparation with an extensive caries lesion. When extending laterally to remove enamel undermined by the caries lesion, the bur's long axis is altered to prepare a 90- to 100-degree cavosurface angle. A 100-degree cavosurface angle on the cuspal incline results in an 80-degree marginal amalgam angle.

When the defect extends to more than one half the distance between the primary groove and a cusp tip, reducing the cuspal tooth structure and restoring it with amalgam (also referred to as "capping the cusp") may be indicated as discussed in Clinical Technique for Complex Amalgam Restorations. When the distance is two thirds, cusp reduction and coverage is usually required because of the risk of cusp fracture during subsequent functional occlusal loading. Fig. 10.36 illustrates examples of large (extensive) Class I amalgam preparation outlines.

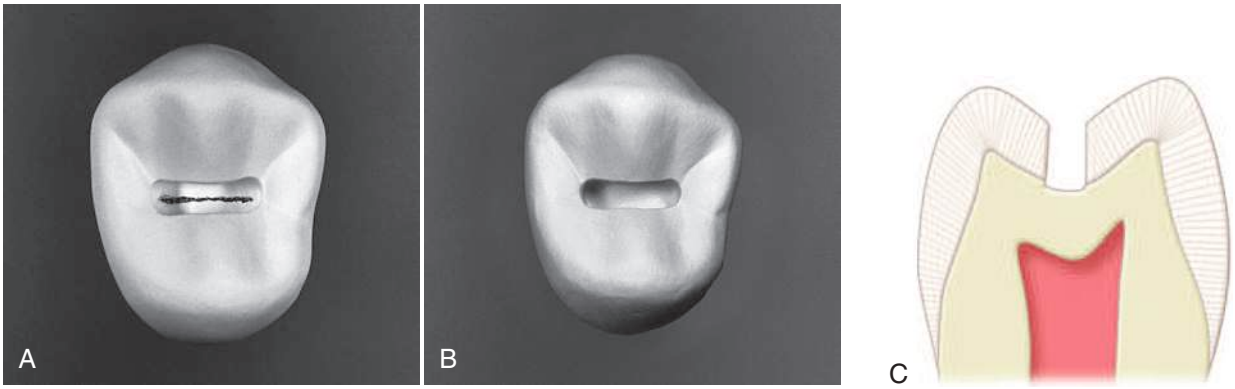
Final Tooth Preparation

The final tooth preparation includes (1) removal of remaining defective enamel and soft dentin on the pulpal floor as indicated; (2) pulp protection, where indicated; (3) procedures for finishing the external walls; and (4) final procedures of debridement (cleaning) and inspecting the preparation.

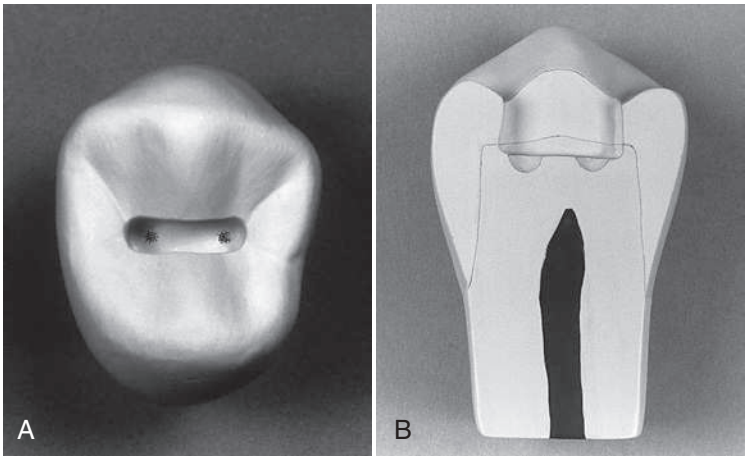
If several enamel pit-and-fissure remnants remain in the floor, or if a central fissure remnant extends over most of the floor, the floor should be deepened with the No. 245 bur to eliminate the defect or to uncover the caries lesion to a maximum preparation depth of 2 mm (Fig. 10.37). If these remnants are few and small, they may be removed with an appropriate carbide bur (Fig. 10.38). Removal of the remaining lesion (i.e., caries lesion that extends pulpally from the established pulpal floor) is best accomplished



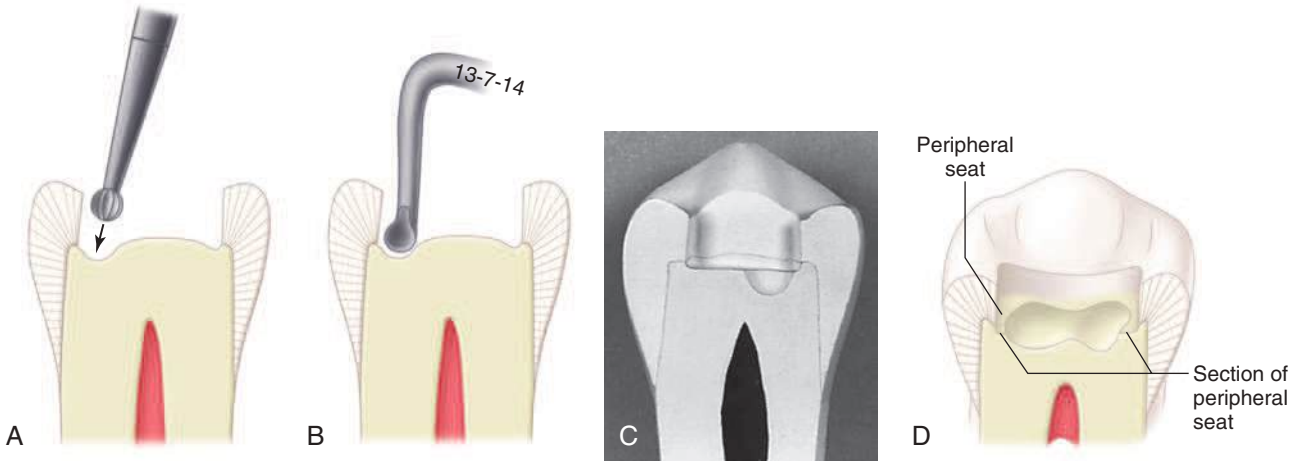
• **Fig. 10.36** Examples of more extensive Class I amalgam tooth preparation outline forms. A, Occlusal outline form in the mandibular second premolar. B, Occlusolingual outline form in the maxillary first molar. C, Occlusofacial outline form in the mandibular first molar.



• **Fig. 10.37** Removal of enamel fissure extending over most of the pulpal floor. A, Full-length occlusal fissure remnant remaining on the pulpal floor after the initial tooth preparation. B and C, The pulpal floor is deepened to a maximum depth of 2 mm to eliminate the fissure or uncover dentin caries lesion.



• **Fig. 10.38** Removal of enamel pit and fissure and dentin caries lesion that is limited to a few small pit-and-fissure remnants. A, Two pit remnants remain on the pulpal floor after the initial tooth preparation. B, Carious enamel and dentin caries lesion have been removed.



• **Fig. 10.39** A and B, Removal of dentin caries lesion is accomplished with round burs *(A)* or spoon excavators *(B)*. C and D, The resistance form may be improved with a flat floor peripheral to the excavated area(s).

using a discoid-type spoon excavator or a slowly revolving round carbide bur of appropriate size (Fig. 10.39A and B).

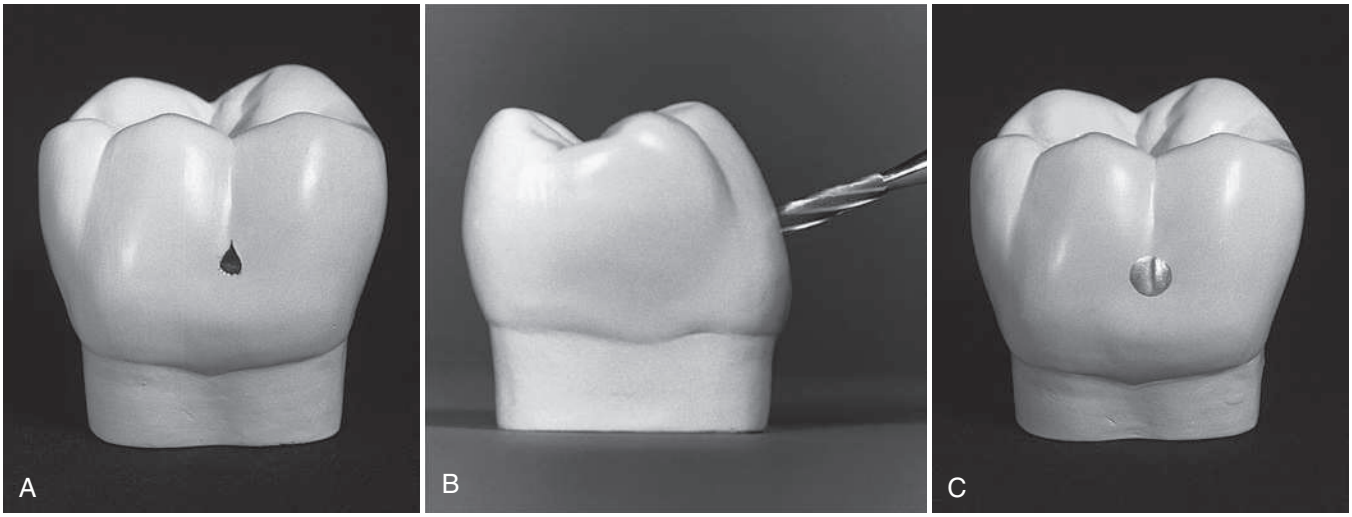
The removal of carious dentin will not affect the resistance form of the tooth because the periphery would not need further extension. In addition, it will not affect the resistance form of the restoration as it will rest on the pulpal wall peripheral to the excavated area or areas. The peripheral pulpal floor should be at the previously described initial pulpal floor depth just inside the DEJ (see Fig. 10.39C and D). Usually, no secondary resistance or retention form features are necessary Class I amalgam preparations.

External walls may already have been finished during initial preparation steps; however, operator assessment of the external walls of the preparation is accomplished at this point, and any additional finish is accomplished as described previously and in Chapter 4. An occlusal cavosurface bevel is contraindicated in the tooth preparation for an amalgam restoration.[92] It is important to provide an approximate 90- to 100-degree cavosurface angle, which should result in 80- to 90-degree amalgam at the margins.[92] This butt-joint marginal area (which approximates 90-degree cavosurface marginal angles for both enamel and amalgam) allows strength for both. Amalgam is a brittle material with low edge strength and tends to chip under occlusal stress if its angle at the margins is less than 80 degrees. The transition between the external and internal walls should be gently rounded.[93] The preparation should then be cleaned, carefully inspected as indicated previously in General Concepts Guiding Preparation for Amalgam Restorations, and any final modifications completed.

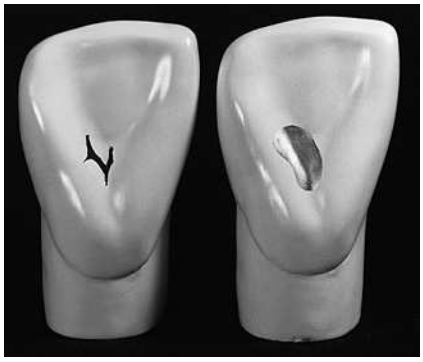
Other Class I Amalgam Preparations

Class I preparations, especially those in esthetically important areas, may be restored with composite because of their small size and the maximal thickness of enamel available for bonding. However, the following preparations may also be restored with amalgam:

1. Facial pit of the mandibular molar.
2. Lingual pit of the maxillary lateral incisor.
3. Occlusal pits of the mandibular first premolar.
4. Occlusal pits and fissures of the maxillary first molar.
5. Occlusal pits and fissures of the mandibular second premolar.



• **Fig. 10.40** Mandibular molar. A, Facial pit with a caries lesion. B, The bur positioned perpendicular to the tooth surface for entry. C, Outline of restoration.



• **Fig. 10.41** Carious lingual pit and fissure and restoration on the maxillary lateral incisor.

Examples of some of these types of preparations and restorations are provided in Figs. 10.40, 10.41, 10.42, 10.43, 10.44, and 10.45.

Restorative Technique for Class I Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed in the preparation according to manufacturer recommendations before amalgam condensation (Fig. 10.46).[72]

Matrix Placement

Generally matrices are unnecessary for a conservative Class I amalgam restoration except as specified in later sections.

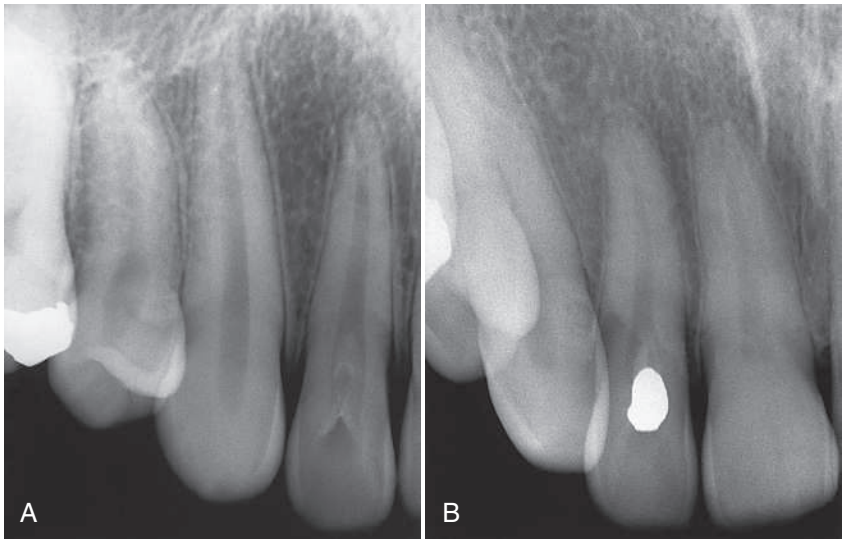
Insertion and Carving of the Amalgam

The triturated amalgam (Fig. 10.47A) is placed into an amalgam well. The outline of the tooth preparation should be reviewed before inserting amalgam to establish a mental image that will later aid in carving amalgam to the cavosurface margin. Preoperative occlusal contact locations should be recalled (see Fig. 10.47B). Amalgam should be carefully condensed into the pulpal line angles (see Fig. 10.47C). The preparation should be overpacked 1 mm or more using heavy pressure (see Fig. 10.47D). This ensures that the cavosurface margins are completely covered with well-condensed amalgam. Final condensation over cavosurface margins should be done perpendicular to the external enamel surface adjacent to the margins. The overpacked amalgam should then be precarve burnished.

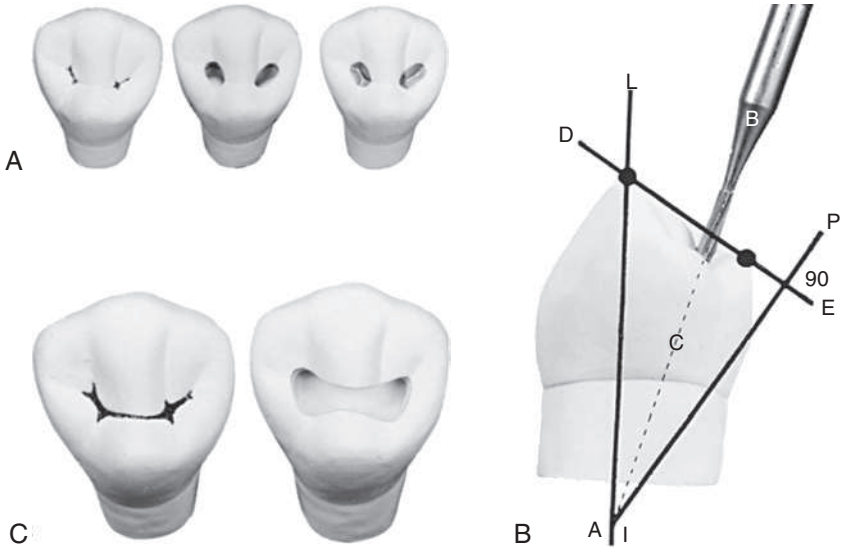
Finishing and Polishing of the Amalgam

With care, carving may begin immediately after condensation. Sharp discoid-cleoid carvers of suitable sizes are useful for this. The larger end of the discoid-cleoid instrument (No. 3/6) is used first, followed by the smaller instrument (No. 4/5) in regions not accessible to the larger instrument. Alternatively, the Hollenback carver may be used, with care not to overcarve the groove or fossa areas. All carving should be done with the edge of the blade perpendicular to the margins as the instrument is moved parallel to the margins. Part of the edge of the carving blade should rest on the unprepared tooth surface adjacent to the preparation margin (see Fig. 10.47F). Using this surface as a guide helps prevent overcarving amalgam at the margins and produces a continuity of surface contour across the preparation margins. During carving, thin amalgam should be removed from areas of enameloplasty. Thin amalgam left in these areas may fracture because of its low edge strength. Deep occlusal anatomy should not be carved into the restoration because these may thin the amalgam at the margins and weaken the restoration (see Fig. 10.47G). Undercarving leaves thin portions of amalgam (subject to fracture) on the unprepared tooth surface. The thin portion of amalgam extending beyond the margin is referred to as *flash*. The mesial and distal fossae should be carved slightly deeper than the proximal marginal ridges (see Fig. 10.47H).

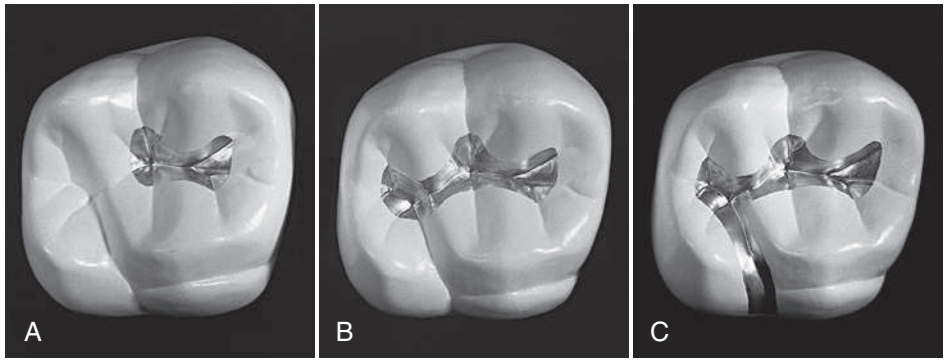
After carving is completed, the outline of the amalgam margin should reflect the contour and location of the prepared cavosurface margin, revealing a regular (i.e., not ragged) outline with gentle curves. An amalgam outline that is larger or irregular is undercarved and requires further carving or finishing (Fig. 10.48). An amalgam restoration that is more than minimally overcarved (i.e., a submarginal defect >0.2 mm) should be replaced.[94] If the total carving time is short enough, the smoothness of the carved surface may be improved by wiping with a small, damp ball of cotton held with the cotton pliers. Alternatively, postcarve burnishing may be



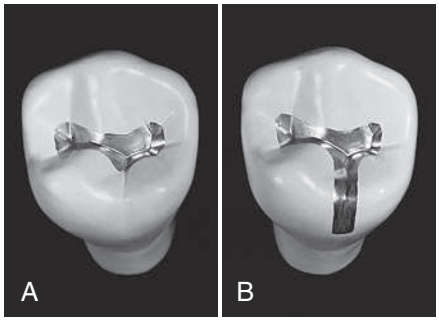
• **Fig. 10.42** Maxillary lateral incisor. A, Preoperative radiograph of dens in dente. B, Radiograph of restoration after 13 years. (Courtesy Dr. Ludwig Scott.)



• **Fig. 10.43** A, Preparation design and restoration of carious occlusal pits on the mandibular first premolar. B, Bur tilt for entry. The cutting instrument is held such that its long axis *(broken line, CI)* is parallel with the bisector *(B)* of the angle formed by the long axis of the tooth *(LA)* and the line *(P)* that is perpendicular to the plane *(DE)* drawn through the facial and lingual cusp points. This dotted line *(CI)* is the bur position for entry. C, Conventional outline, including occlusal pits and central fissure.



• **Fig. 10.44** Maxillary first molar. A, Outline necessary to include the carious mesial and central pits connected by the fissure. B, Preparation outline extended from outline in *A* to include distal pit and connecting deep fissure in oblique ridge. C, Preparation outline extended from outline in *B* to include distal oblique and lingual fissures.



• **Fig. 10.45** Mandibular second premolar. A, Typical occlusal outline. B, Extension through the lingual ridge enamel is necessary when enameloplasty does not eliminate the lingual fissure.



• **Fig. 10.46** Use of microbrush to apply the dentin desensitizer in the tooth preparation. (Courtesy Aldridge D. Wilder, DDS.)

utilized to refine the anatomy and leave a smooth, satin finish. All shavings from the carving procedure should be removed from the mouth with the aid of the oral evacuator.

The occlusion on the restoration is then evaluated and adjusted so that all markings on the restoration and adjacent teeth are uniform (see Fig. 10.23).

Extensive caries lesions require a more extensive restoration (which is a clear indication for amalgam as compared with composite resin). Use of amalgam in large Class I restorations provides good wear resistance and stable occlusal contact relationships. For very large Class I restorations, a bonding system may be used, although this book does not promote such use. The perceived benefits of bonded amalgams have not been substantiated.[85,95-99] Bonded amalgams have no advantage as compared with the conventional technique described above. Carving the extensive Class I restoration is often more complex because more cuspal inclines are included in the preparation. Appropriate contours, occlusal contacts, and groove/fossa anatomy must be provided. Finishing and polishing indications and techniques are as described previously.

Tooth Preparation for Class I Occlusolingual and Occlusofacial Amalgam Restorations

Occlusolingual amalgam restorations may be used on maxillary molars when a lingual fissure connects with the distal oblique fissure and distal pit on the occlusal surface (Fig. 10.49). Composite may also be used as the restorative material in smaller lesions.

After local anesthesia and evaluation of the occlusal contacts, the use of a rubber dam is generally recommended for isolation of the operating field. Alternatively, typical Class I preparations may be adequately isolated with cotton rolls.

Tooth Preparation

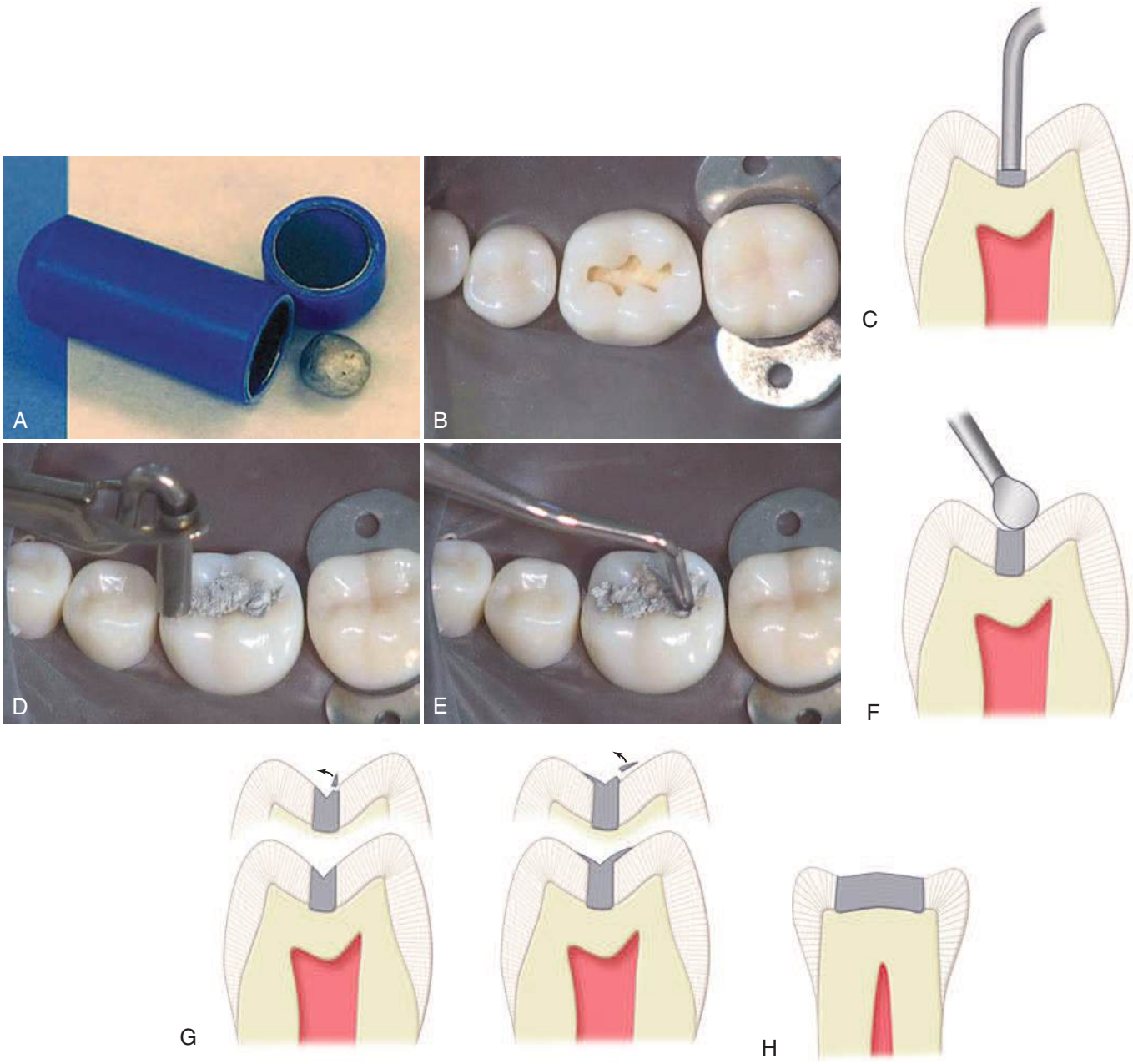
The initial tooth preparation involves the establishment of the outline, primary resistance, and primary retention forms, as well as initial preparation depth. The accepted principles of the outline form (previously presented) should be observed with special attention to the following:

1. The tooth preparation should be no wider than necessary; ideally, the mesiodistal width of the lingual extension should not exceed 1 mm except for extension necessary to remove carious or undermined enamel or to include unusual fissuring.
2. When indicated, the tooth preparation should be more at the expense of the oblique ridge, rather than centering over the fissure (weakening the small distolingual cusp).
3. Especially on smaller teeth, the occlusal portion may have a slight distal tilt to conserve the dentin support of the distal marginal ridge (Fig. 10.50).
4. The margins should extend as little as possible onto the oblique ridge, distolingual cusp, and distal marginal ridge.

These objectives help conserve the dentinal support (i.e., strength of the tooth) and aid in establishing an enamel cavosurface angle as close to 90 degrees as possible (Fig. 10.51). They also help to minimize deterioration of the restoration margins by locating the margins away from enamel eminences where occlusal forces may be concentrated.

The distal pit is identified with indirect vision and entered with the end of the No. 245 bur (Fig. 10.52A). The long axis of the bur usually should be parallel to the long axis of the tooth crown. The dentinal support and strength of the distal marginal ridge and the distolingual cusp should be conserved by positioning the bur such that it cuts more of the tooth structure mesial to the pit rather than distal to the pit (e.g., 70:30 rather than 50:50), if needed. The initial cut is to the level of the DEJ (a depth of 1.5–2 mm) (see Fig. 10.52B). At this depth, the pulpal floor is usually in dentin. When the entry cut is made (see Fig. 10.52C), the bur (maintaining the initial established depth) is moved to include any remaining fissures facial to the point of entry (see Fig. 10.52D). The bur is then moved along the fissure toward the lingual surface (see Fig. 10.52E). As with Class I occlusal preparations, a slight distal inclination of the bur is indicated occasionally (particularly in smaller teeth) to conserve the dentinal support and strength of the marginal ridge and the distolingual cusp. To ensure adequate strength for the marginal ridge, the distopulpal line angle should not approach the distal surface of the tooth closer than 2 mm. On large molars, the bur position should remain parallel to the long axis of the tooth, particularly if the bur is offset slightly mesial to the center of the fissure. Keeping the bur parallel to the long axis of the tooth creates a distal wall with slight occlusal convergence, providing favorable enamel and amalgam angles. The bur is moved lingually along the fissure, maintaining a uniform depth until the preparation is extended onto the lingual surface (see Fig. 10.52F). The pulpal floor should follow the contour of the occlusal surface and the DEJ, which usually rises occlusally as the bur moves lingually.

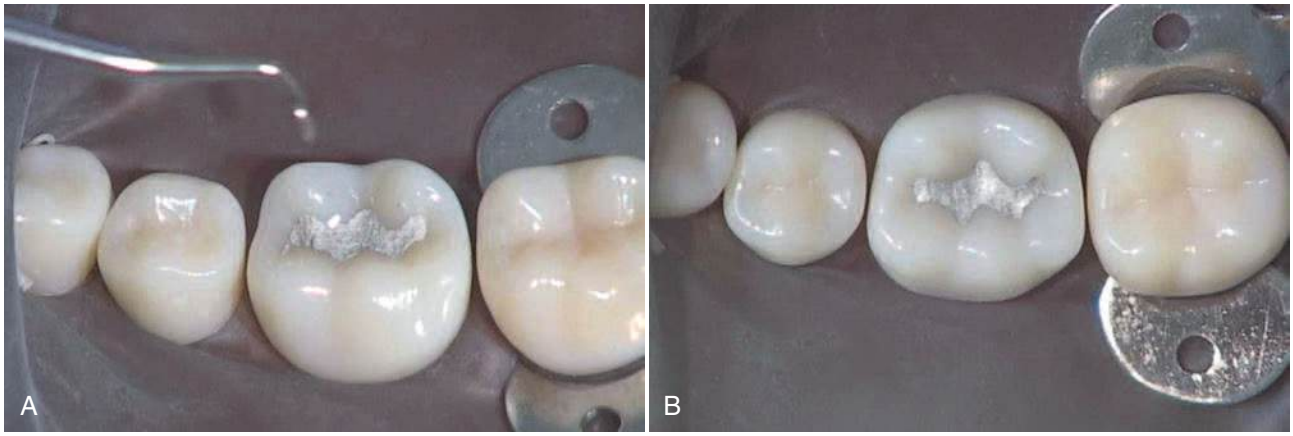
The mesial and distal walls of the occlusal portion of the preparation should converge occlusally because of the shape of the



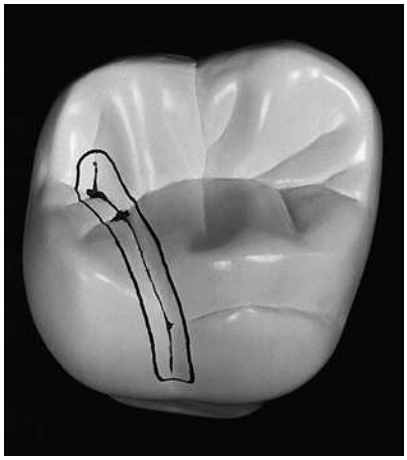
• **Fig. 10.47** Restoration of occlusal tooth preparation. A, Properly triturated amalgam is a homogeneous mass with slightly reflective surface. A properly mixed mass will flatten slightly if dropped on a tabletop. B, The operator should have a mental image of the outline form of the preparation before condensing amalgam to aid in locating the cavosurface margins during the carving procedure. C, Amalgam should be inserted incrementally and condensed with overlapping strokes. D and E, The tooth preparation should be overpacked to ensure well-condensed marginal amalgam that is not mercury rich. F, The carver should rest partially on the external tooth surface adjacent to the margins to prevent overcarving. G, Deep occlusal grooves invite chipping of amalgam at the margins. Thin portions of amalgam left on the external surfaces soon break away, giving the appearance that amalgam has grown out of the preparation. H, Carve fossae slightly deeper than the proximal marginal ridges. (A, From Darby ML, Walsh MM: *Dental hygiene: theory and practice*, ed 3, St. Louis, Saunders, 2010. B, D, and E, Courtesy Aldridge D. Wilder, DDS.)

bur. If the slight distal bur tilt was required, the mesial and distal walls still should converge relative to each other (although the distal wall may be divergent occlusally, relative to the tooth's long axis). Convergence of the mesial and distal walls ensures occlusal retention form is adequate.

The lingual portion is prepared at this point by using one of two techniques. In one technique, the lingual surface is prepared with the bur's long axis parallel with the lingual surface (Fig. 10.53A and B). The tip of the bur should be located at the gingival extent of the lingual fissure. The bur should be controlled so that it does



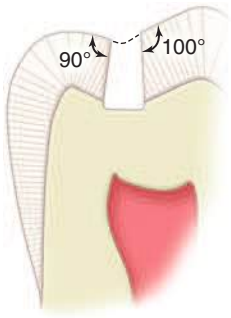
• **Fig. 10.48** A, Undercarved amalgam with flash beyond the margins. The restoration outline is irregular and larger than the preparation outline in Fig. 10.36B. B, Correctly carved amalgam restoration. (Courtesy Aldridge D. Wilder, DDS.)



• **Fig. 10.49** Outline of margins for occlusolingual tooth preparation.



• **Fig. 10.50** Small distal inclination of the bur on smaller teeth may be indicated to conserve the dentinal support and the strength of the marginal ridge.



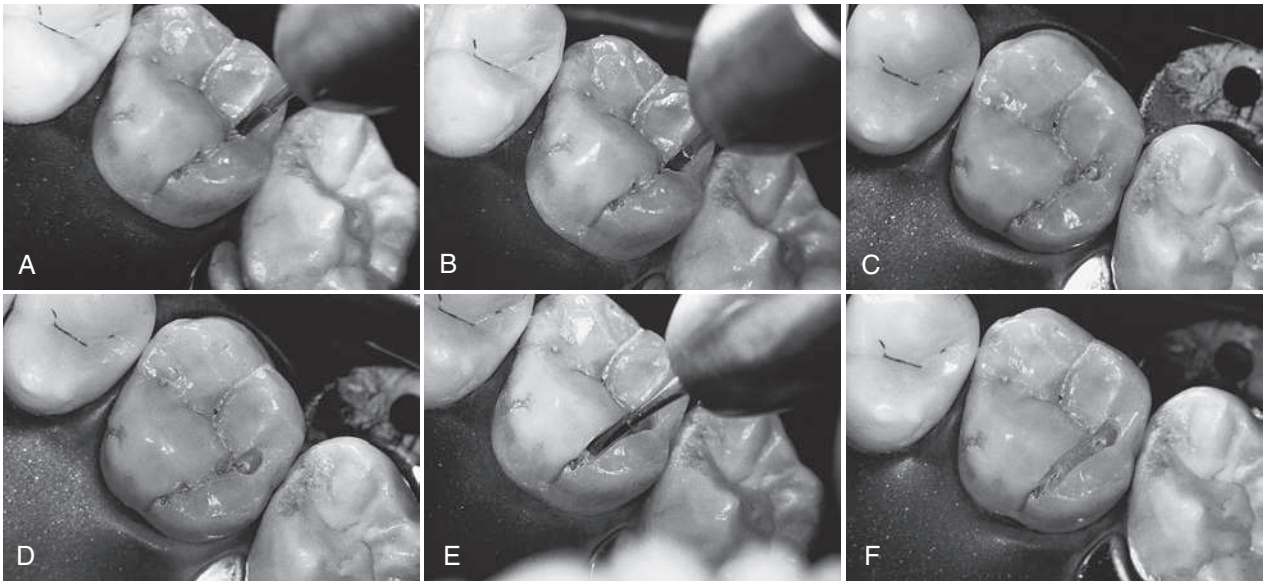
• **Fig. 10.51** Enamel cavosurface angles of 90 to 100 degrees are ideal.

not "roll out" onto the lingual surface, which may "round over" or damage the cavosurface margin. The facial inclination of the bur must be altered as the cutting progresses to establish the axial wall of the lingual portion at a uniform depth of 0.5 mm inside the DEJ (see Fig. 10.53C). The axial wall should follow the contour of the lingual surface of the tooth with an axial depth of 0.2 mm inside the DEJ at the preparation periphery. When the caries lesion affecting the lingual surface is wide mesiodistally, the resultant mesial and distal external walls may diverge to the lingual. In this case the amalgam will require retention grooves to prevent lingual displacement. An axial depth of 0.5 mm inside the DEJ is indicated if retentive grooves are required. A second, alternate approach to preparing the lingual aspect of the preparation may also be considered. In this case, the No. 245 bur is held perpendicular to the cusp ridge and then the lingual surface, as it extends the preparation from the occlusal surface gingivally (to include the entire defect). This technique also results in opposing preparation walls that converge lingually.

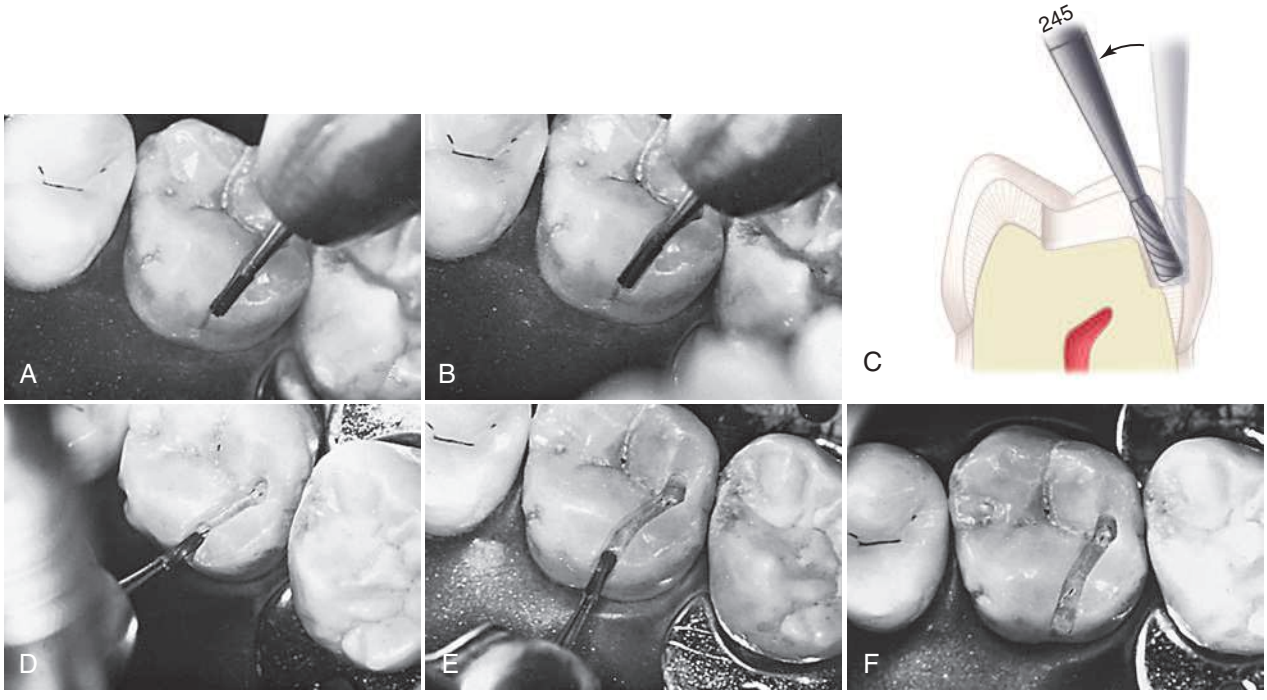
The No. 245 bur may be used with its long axis perpendicular to the axial wall to accentuate (i.e., refine) the mesioaxial and distoaxial line angles; this also results in the mesial and distal walls converging lingually because of the shape of the bur (see Fig. 10.53D and E). During this step, the axial wall depth is not altered (see Fig. 10.53F). The occlusal and lingual convergences usually provide a sufficient preparation retention form, and retention grooves are not needed.

The axiopulpal line angle must be rounded to limit the areas of stress concentration and ensure adequate preparation depth and amalgam thickness (Fig. 10.54). Initial tooth preparation of the occlusolingual preparation is now complete. As mentioned previously, enameloplasty may be performed to conserve the tooth structure and limit extension.

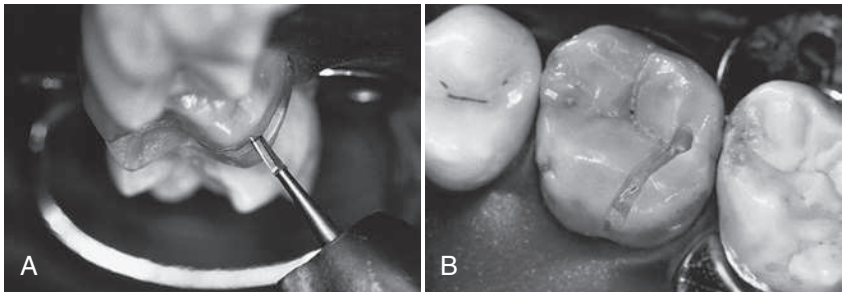
Additional retention in the lingual extension may be required if the extension is wide mesiodistally or if it was prepared without a lingual convergence. If additional retention is required, the No.



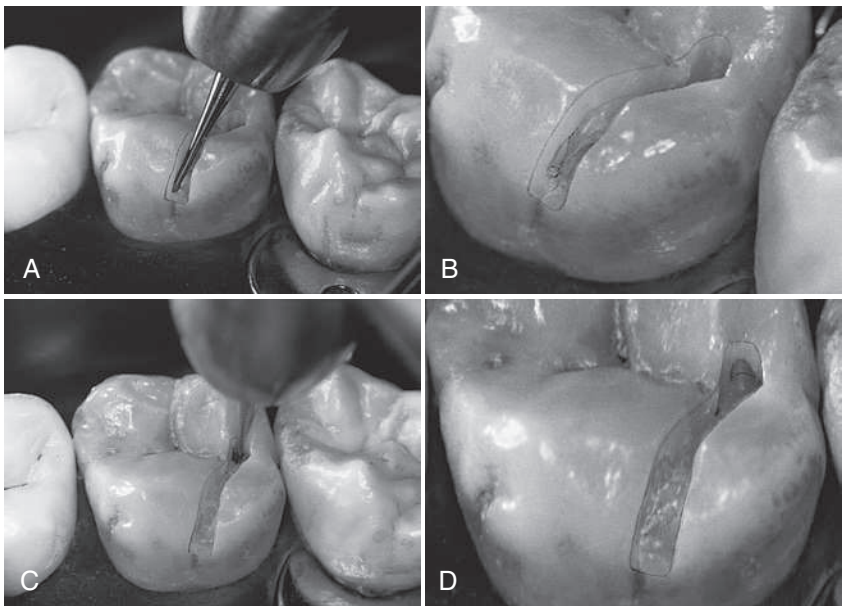
• **Fig. 10.52** Occlusolingual tooth preparation. A, No. 245 bur positioned for entry. B, Penetration to a minimal depth of 1.5 to 2 mm. C, Entry cut. D, The remaining fissures facial to the point of entry are removed with the same bur. E and F, A cut lingually along the fissure until the bur has extended the preparation onto the lingual surface.



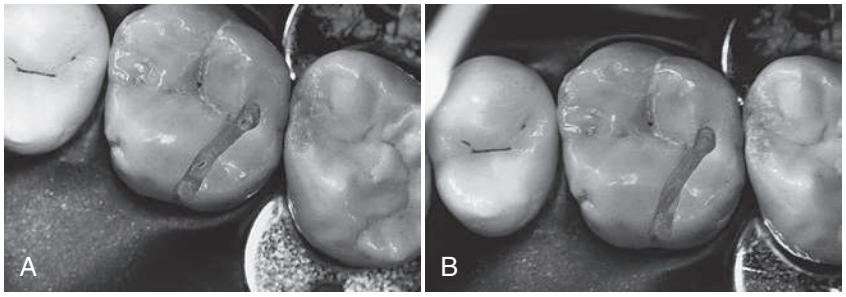
• **Fig. 10.53** Occlusolingual tooth preparation. A, Position of bur to cut the lingual portion. B, Initial entry of the bur for cutting the lingual portion. C, The inclination of the bur is altered to establish the correct axial wall depth. D and E, The bur is directed perpendicular to the axial wall to accentuate the mesioaxial and distoaxial line angles. F, The axial wall depth should be 0.2 to 0.5 mm inside the DEJ.



• **Fig. 10.54** A, Bur position for rounding the axiopulpal line angle. B, Axiopulpal line angle rounded.



• **Fig. 10.55** Secondary retention form. A, Bur position for preparing groove in mesioaxial line angle. B, Completed groove is internal to the DEJ. C, Bur position for the retention cove in the faciopulpal line angle. D, Completed cove is internal to the DEJ.



• **Fig. 10.56** A, Any remaining pit and fissure in enamel and soft dentin on established pulpal and axial walls are removed. B, Completed tooth preparation.

¼ or No. 169 bur may be used to prepare grooves into the mesioaxial and distoaxial line angles (Fig. 10.55A). If these angles are in enamel, the axial wall must be deepened to 0.5 mm axially of the DEJ to allow the grooves to be prepared in dentin and to not undermine enamel. The depth of the grooves at the gingival floor is one half the diameter of the No. ¼ bur. The cutting direction for each groove is the bisector of the respective line angle. The grooves are prepared in the mesioaxial and distoaxial line angles of the ideally positioned axial wall and 0.2 mm axial to the DEJ. The grooves should diminish in depth toward the occlusal surface, terminating midway along the axial wall (see Fig. 10.55B). The adequacy of the groove should be tested by inserting the tine of an explorer into the groove and detecting resistance to lingual movement. The depth of the groove should prevent the explorer from being withdrawn lingually. (See Secondary Resistance and Retention Forms for a description of placing retentive grooves in the proximal boxes of Class II amalgam preparations; the techniques are similar.)

Extension of a facial occlusal fissure may have required a slight divergence occlusally to the facial wall to conserve support of the facial ridge. If so, and if deemed necessary, the No. ¼ round bur may be used to prepare a retention cove in the faciopulpal line angle (see Fig. 10.55C and D). The tip of the No. 245 bur held parallel to the long axis of the tooth crown also might be used to prepare this cove. Care should be taken so as not to undermine the occlusal enamel. This retentive cove is recommended only if occlusal convergence of the mesial and distal walls of the occlusal portion is absent or inadequate.

The final tooth preparation is accomplished by removal of the remaining caries lesion on the pulpal and axial walls (Fig. 10.56) with an appropriate round bur, a discoid-type spoon excavator, or both. A RMGI is placed in areas of deep caries removal for pulpal protection. The external walls are finished. Any irregularities at the margins may indicate weak enamel that may be removed by the side of the No. 245 bur rotating at low speed. The preparation should then be cleaned, carefully inspected as indicated in General Concepts Guiding Preparation for Amalgam Restorations, and any final modifications completed.

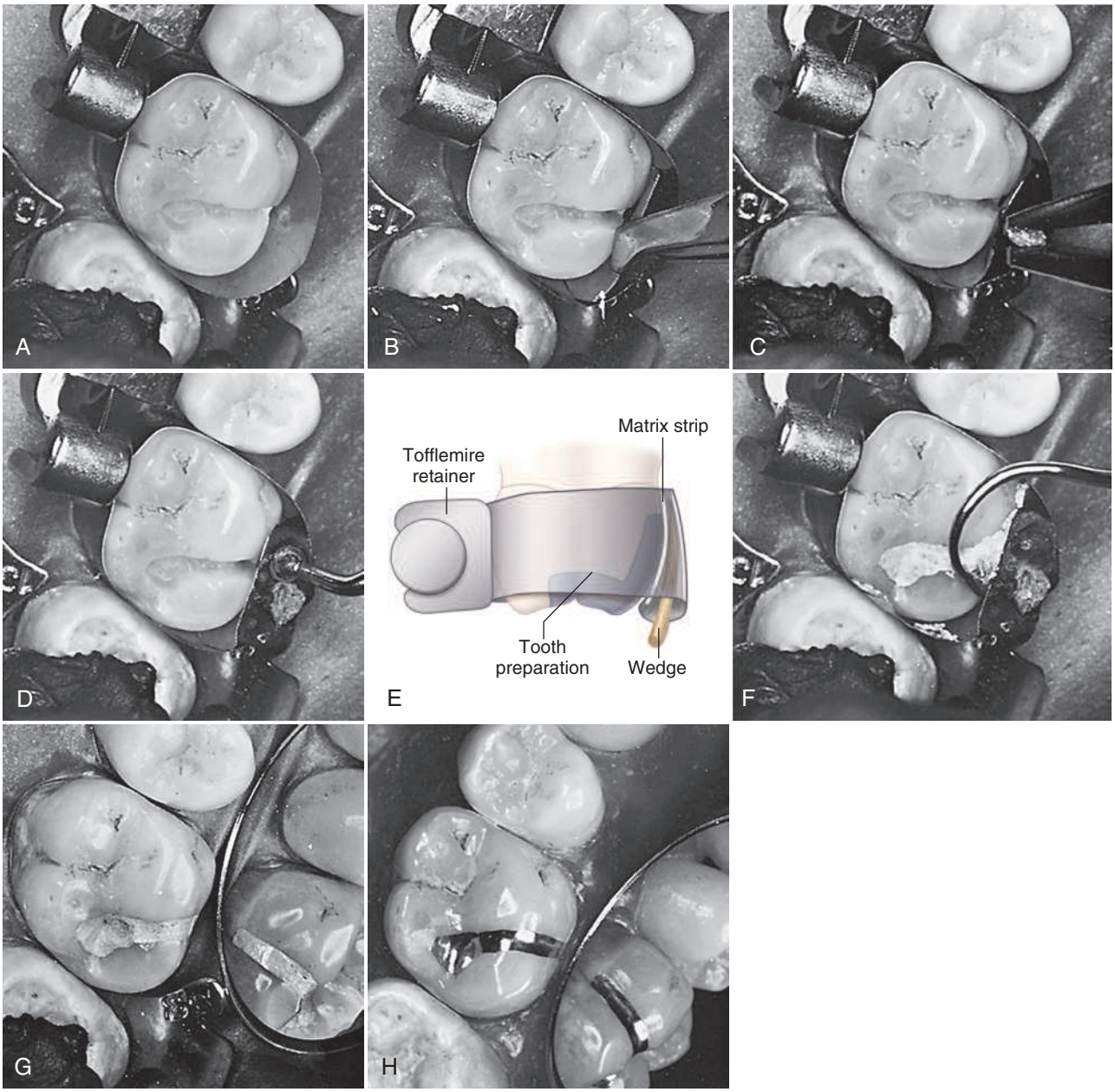
Restorative Technique for Class I Occlusolingual or Occlusofacial Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed over the prepared tooth structure and rinsed per manufacturer instructions. Excess moisture is removed using air stream without desiccating the dentin.

Matrix Placement (If Necessary)

Using a matrix to support the lingual portion of the restoration during condensation is occasionally necessary. A matrix is helpful to prevent "land sliding" during condensation and to ensure marginal adaptation and strength of the restoration. The Tofflemire matrix retainer is used to secure a matrix band to the tooth (as described later). Because this type of matrix band does not intimately adapt to the lingual groove area of the tooth (Fig. 10.57A), an additional step may be necessary to provide a matrix that is rigid on the lingual portion of the tooth preparation. If so, a piece of stainless steel matrix material (0.05 mm thick and 8 mm wide) is cut to fit between the lingual surface of the tooth and the band already in place (see Fig. 10.57B). The gingival edge of this segment of matrix material is placed slightly gingival to the gingival edge of the band to help secure the band segment. A quick-setting, rigid polyvinyl siloxane (PVS)–based material may be used between the sectional matrix and the Tofflemire matrix band, to prevent lingual displacement of the sectional matrix during condensation of the amalgam. Alternatively, green stick compound may be used. In



• **Fig. 10.57** Matrix for occlusolingual tooth preparation. A, Matrix band secured to the tooth with Tofflemire retainer. B, Positioning a small strip of stainless steel matrix material between the tooth and the band already in place. C, Inserting the wedge and the rigid supporting material. D, Compressing the rigid supporting material gingivally (or holding the steel strip in correct position while the material sets), which adapts the steel strip to the lingual surface. E, Cross section of the tooth preparation and the matrix construction. F, Using the explorer to remove excess amalgam adjacent to the lingual matrix. G, Carving completed. H, Polished restoration.

this case, the end of a toothpick wedge is covered with softened (heated) compound. The compound-coated wedge is then immediately inserted between the Tofflemire band and the cut piece of matrix material (see Fig. 10.57C). While the compound is still soft, a suitable burnisher is used to press the compound gingivally, securing the matrix tightly against the gingival cavosurface margin and the lingual surface of the tooth to provide a rigid, lingual matrix (see Fig. 10.57D and E). This matrix for the occlusolingual amalgam restoration is referred to as the *Barton matrix*. Occasionally the piece of strip matrix may be positioned appropriately by using only the wedge (without the rigid PVS or compound support).

Insertion of the Amalgam

Insertion of amalgam is accomplished as previously described in General Concepts Guiding Restoration with Amalgam and for the Class I occlusal tooth preparation. Condensation is begun at the gingival wall at the lingual component of the preparation. If a matrix is not used, care must be taken to ensure that "land sliding" of the amalgam does not occur because two adjoining surfaces of the tooth are being restored. For this technique, the last increments of amalgam may be condensed on the lingual surface with the side of a large condenser. Its long, broadly rounded contour approximates the rectangular shape of the lingual groove preparation. Appropriate care should be taken (when condensing the occlusal surface) to prevent fracturing the lingual amalgam. Another technique is to have the assistant secure the condensed lingual surface with a broad condenser nib, while the operator completes the condensation of the occlusal surface. Regardless of the technique used, the amalgam must be well condensed.

Contouring and Finishing of the Amalgam Restoration

When the preparation is sufficiently overfilled, carving of the occlusal surface may begin immediately with a sharp discoid-cleoid instrument or a Hollenback carver. All carving should be done with the edge of the blade perpendicular to the margin and the blade moving parallel to the margin. To prevent overcarving, the blade edge should be guided by the unprepared tooth surface adjacent to the margin. An explorer is used to remove excess amalgam adjacent to the lingual matrix before matrix removal (see Fig. 10.57F). After the occlusal carving is complete, the Tofflemire retainer is loosened from the band, and the band is removed with No. 110 pliers. The free ends of the band are pushed one at a time, lingually and occlusally, through the proximal contacts. After the lingual carving is complete (see Fig. 10.57G), the rubber dam is removed and the restoration is adjusted to ensure proper occlusion. Most amalgams do not require finishing and polishing. Fig. 10.57H illustrates a polished occlusolingual restoration.

Class I Occlusofacial Amalgam Restorations

Occasionally, mandibular molars exhibit fissures that extend from the occlusal surface through the facial cusp ridge and onto the facial surface. The preparation and restoration of these defects are very similar to those described for Class I occlusolingual amalgam restorations. Although these may be restored with composite, an illustration of preparation and restoration with amalgam is provided in Fig. 10.58. The amalgam restoration may be polished after it is completely set. The shape of abrasive points may need to be modified to allow optimal polishing (see Fig. 10.58I and J).

Clinical Technique for Class II Amalgam Restorations

Class II restorations restore defects that affect one or both of the proximal surfaces of posterior teeth (see Fig. 10.26B). Amalgam restorations that restore one or both of the proximal surfaces of the tooth may provide years of service to the patient when (1) the operating field is isolated, (2) the tooth preparation is correct, (3) the matrix is suitable, and (4) the restorative material is manipulated properly. Application of the principles discussed here will result in high-quality, durable Class II amalgam restorations.

Initial Clinical Procedures

Occlusal contacts should be marked with articulating paper before tooth preparation. A mental image of these contacts will serve as a guide in tooth preparation and restoration. Any opposing "plunging cusp" (that is not aligned well with the occlusal plane) or other sharply pointed cusp may need to be recontoured to reduce the risk of fracture of the new restoration. Before tooth preparation for amalgam, the placement of a rubber dam is generally recommended. It is especially beneficial when the caries lesion is extensive. If an existing defective restoration has rough proximal contacts, the restoration may be removed before rubber dam application. Soft dentin should be removed with the rubber dam in place, especially if pulpal exposure is a possibility. Insertion of an interproximal wedge or wedges is useful to depress and protect the rubber dam and underlying soft tissue, separate teeth slightly, and may serve as a guide to prevent gingival overextension of the proximal boxes.

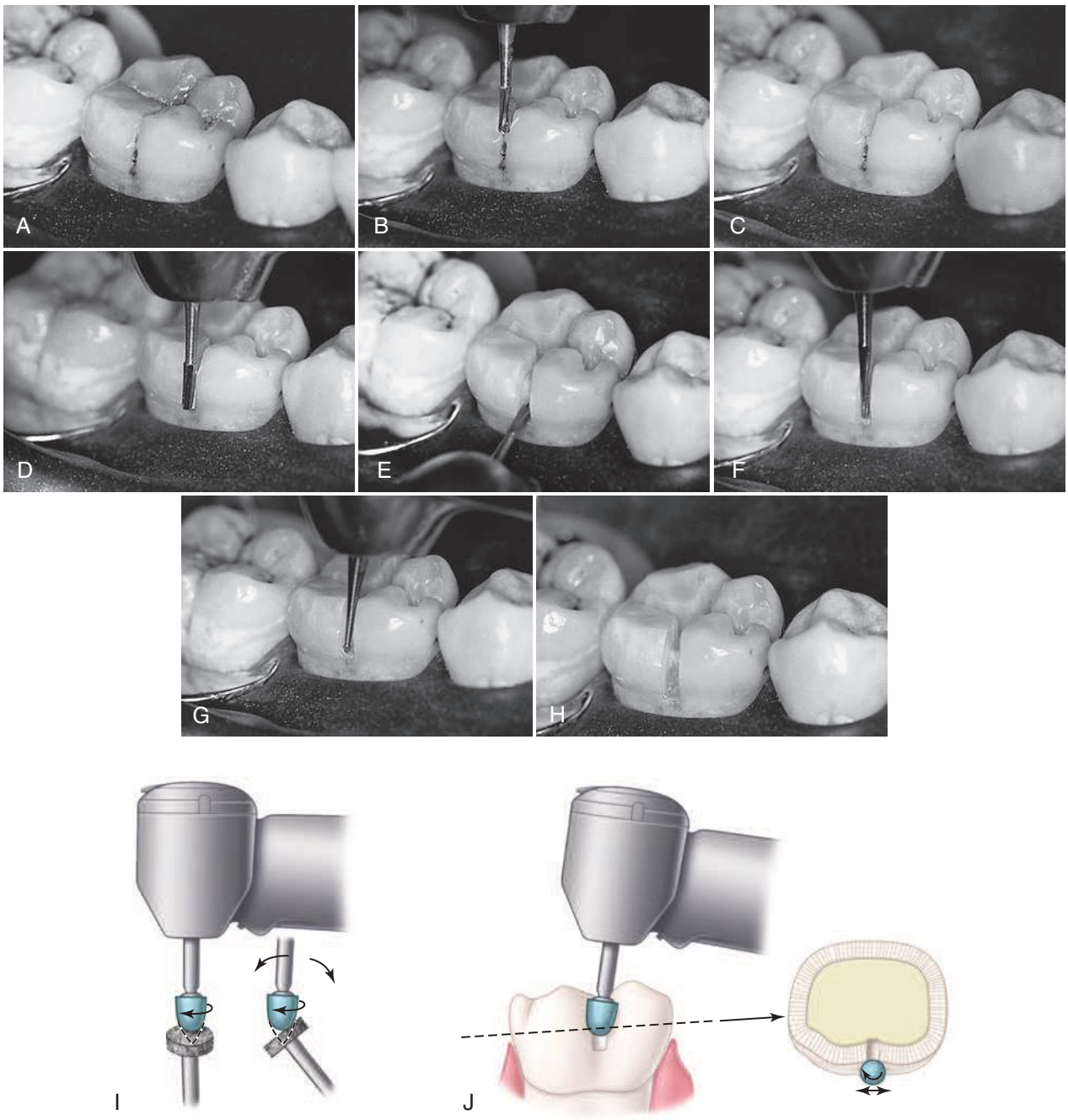
Tooth Preparation for Class II Amalgam Restorations That Involve Only One Proximal Surface

This section introduces the principles and techniques of a Class II tooth preparation for an amalgam restoration involving a caries lesion on one proximal surface. A small, conservative mesiocclusal tooth preparation on the mandibular second premolar is presented to illustrate basic concepts. Composite may also be an acceptable restorative material for this preparation. Preoperative placement of a wedge ("pre-wedging") in the space adjacent to the proximal surface with the caries lesion will create space between the teeth. This lowers the risk of iatrogenic damage to the adjacent tooth during preparation.

Initial Tooth Preparation

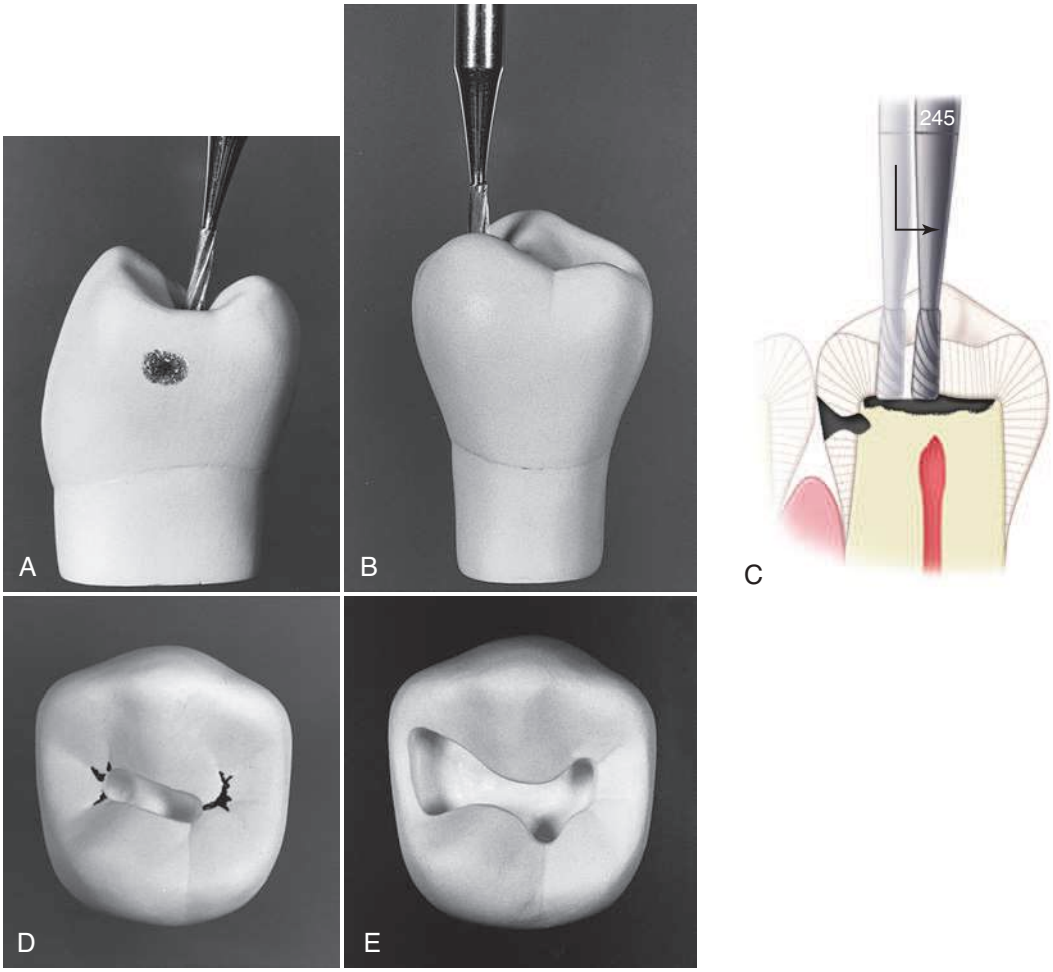
Occlusal Outline Form (Occlusal Step)

The occlusal outline form of a Class II tooth preparation for amalgam is similar to that for a Class I tooth preparation. Using high speed with air-water spray, the operator enters the pit nearest the involved proximal surface with a punch cut using a No. 245 bur oriented as illustrated in Fig. 10.59A and B. The bur should be rotating when it is applied to the tooth and should not stop rotating until removed. Viewed from the proximal and lingual (facial) aspects, the long axis of the bur and the long axis of the tooth crown should remain parallel during the cutting procedures. The dentin caries lesion initially spreads at the DEJ; therefore the goal of the initial cut is to reach the DEJ. The DEJ location in posterior teeth is approximately 1.5 to 2.0 mm from the occlusal



• **Fig. 10.58** Fissure extension. A, Facial occlusal fissure continuous with the fissure on the facial surface. B, Extension through the facial ridge onto the facial surface. C, Appearance of the tooth preparation after extension through the ridge. D, The facial surface portion of the extension is cut with the side of the bur. E, The line angles are sharpened by directing the bur from the facial aspect. F, Sharpening the line angles from the occlusal direction with a No. 169 L bur. G, Ensuring the retention form by preparing retention grooves with a No. ¼ round bur. H, Completed tooth preparation. I, The rubber polishing point may be reshaped (blunted as indicated) on a coarse diamond wheel. J, Proper orientation of the rubber point when polishing the facial surface groove area.

surface. As the bur enters the pit, a target depth of 0.1 to 0.2 mm into dentin (i.e., just into dentin) should be established. This depth is one half to two thirds the length of the cutting portion of a No. 245 bur, or approximately 1.5 mm as measured from the central fissure and 2 mm from the preparation external wall. While maintaining the same depth and bur orientation, the bur is moved to extend the outline to include the carious central fissure and the opposite carious pit (see Fig. 10.59C and D). The isthmus width should be as narrow as possible, preferably no wider than one quarter the intercuspal distance.[54,55,100,101] Ideally the preparation should be the width of the No. 245 bur. Narrow restorations provide a greater length of clinical service.[57,60] Generally the amount of remaining tooth structure is more important to restoration longevity than is the restorative material used.[102] Ultimately, the

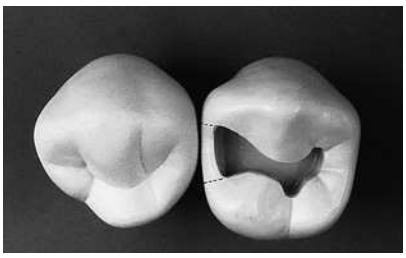


• **Fig. 10.59** Entry and occlusal step. A, Bur position for entry, as viewed proximally. Note the slight lingual tilt of the bur. B, Bur position as viewed lingually. C, The tooth is entered with a punch cut, and extension is done distally along central fissure at a uniform depth of 1.5 to 2 mm (1.5 mm at fissure; because of the inclination of the unprepared tooth surface, the corresponding measurement on the prepared wall is greater). D, Occlusal view of *C*. E, Completed occlusal step.

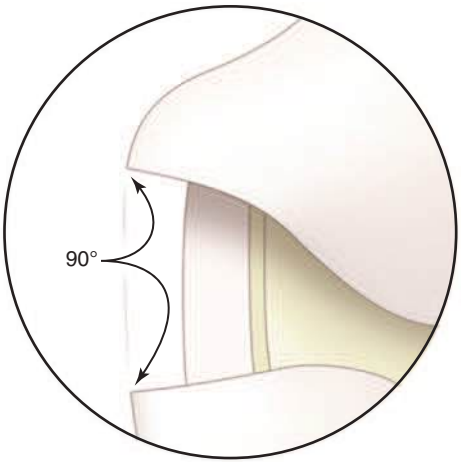
extension of the caries lesion at the DEJ will determine the amount of preparation extension and resultant width. The pulpal floor of the preparation should follow the slight rise and fall of the DEJ along the central fissure in teeth with prominent triangular ridges.

Maintaining the bur parallel to the long axis of the tooth crown creates facial, lingual, and distal walls with a slight occlusal convergence as well as external wall orientation favorable for amalgam use. The facial, lingual, and distal walls should be extended until a sound DEJ is reached. Proper extension will result in the formation of the peripheral seat, which aids in the primary resistance form. It may be necessary to tilt the bur to diverge occlusally at the distal wall if further distal extension would undermine the marginal ridge of its dentinal support. During development of the distal pit area of the preparation, extension to include any distofacial and distolingual developmental fissures radiating from the pit may be indicated. The distal pit area (in this example) provides a dovetail retention form, which will prevent mesial displacement of the completed Class II restoration. However, the dovetail feature is not required in the occlusal step of a single proximal surface preparation unless a fissure emanating from an occlusal pit indicates it. This type of retention form (i.e., form that provides resistance to mesial displacement of the restoration) also is provided by any extension of the central fissure preparation that is not in a straight direction from pit to pit (see Fig. 10.59E). A dovetail outline form in the distal pit is not required if radiating fissures are not present.[103,104] Enameloplasty should be performed, where indicated, to conserve the tooth structure.

Before extending into the involved proximal marginal ridge (the mesial ridge, in this example), the final locations of the facial and lingual walls of the proximal box are estimated visually. Visual assessment prevents overextension of the occlusal outline form (i.e., occlusal step) where it joins the proximal outline form (i.e., proximal box). Fig. 10.60 illustrates visualization of the final location of the proximoocclusal margins before preparing the proximal box. Showing the view from the occlusal aspect, Fig. 10.61 illustrates a reverse curve in the occlusal outline of a Class II preparation, which often results when developing the mesiofacial wall perpendicular to the enamel rod direction while, at the same time, conserving as much of the facial cusp structure as possible.[101] The extension into the mesiofacial cusp is limited to that amount required to permit a 90-degree mesiofacial margin. Lingually, the reverse curve usually is minimal (if necessary at all) because of the relationship of the central fissure relative to the faciolingual position of the proximal contact.



• **Fig. 10.60** Visualize final location of proximoocclusal margins *(dotted lines)* before preparing the proximal box.



• **Fig. 10.61** The reverse curve in the occlusal outline usually is created when the mesiofacial enamel wall is parallel to the enamel rod direction. Lingually, the reverse curve is very slight, often unnecessary.

While maintaining the established pulpal depth and with the bur parallel to the long axis of the tooth crown, the preparation is extended mesially, stopping approximately 0.8 mm short of cutting through the marginal ridge into the contact area. The occlusal step in this region is made slightly wider faciolingually than in the Class I preparation because additional width is necessary for the proximal box. The proper depth of the occlusal portion of the preparation increases the strength of the restoration however, more than does faciolingual width (see Fig. 10.59C and E). Although this extension includes part of the mesial marginal ridge, it also exposes the marginal ridge DEJ. The location of the DEJ is an important guide in the development of the proximal preparation.

Proximal Outline Form (the Proximal "Box")

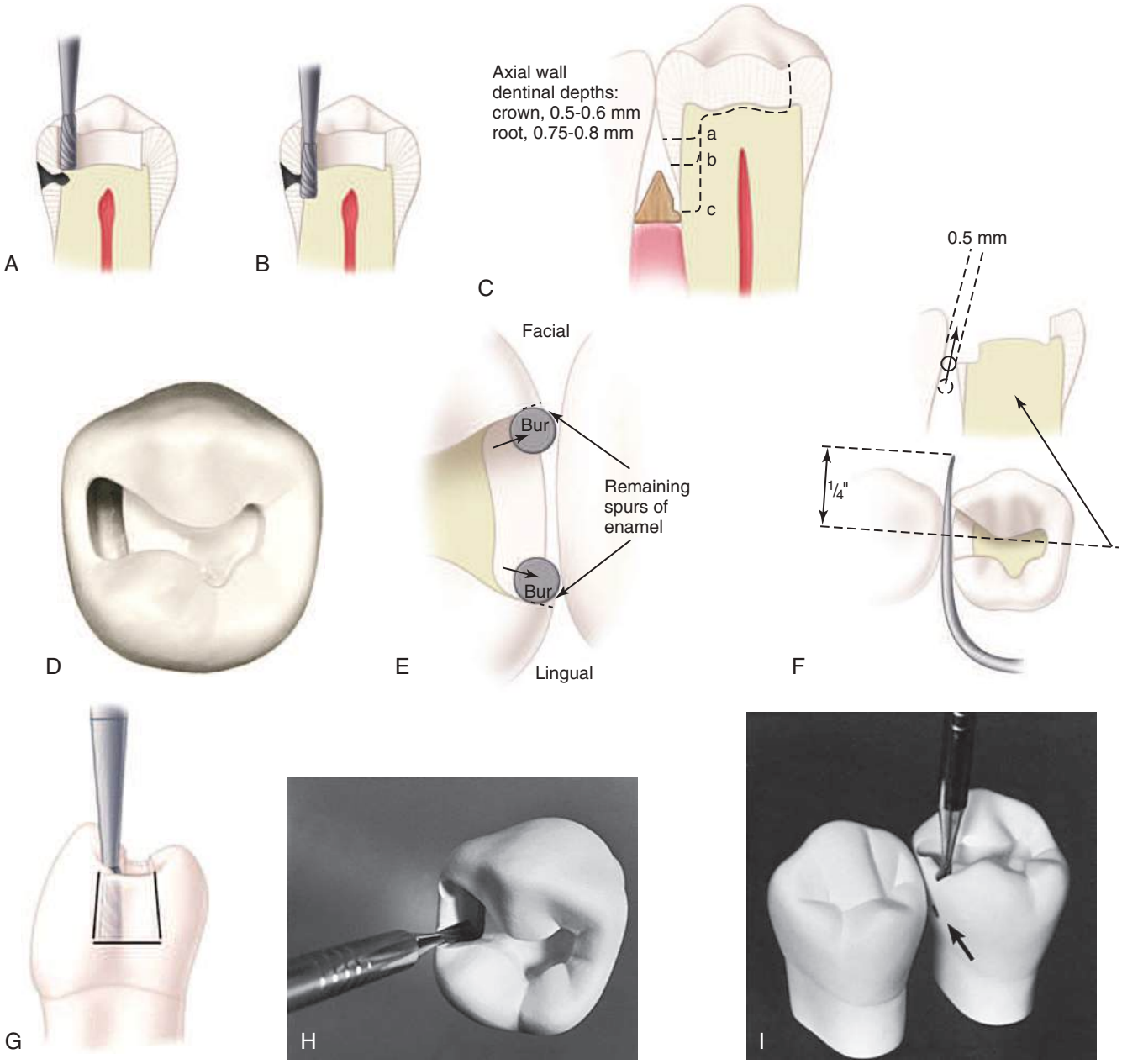
Before initiating the proximal outline form (i.e., the proximal "box"), the operator should visualize the desired final location of the facial and lingual walls relative to the contact. The objectives for the extension of the proximal margins are to (1) include all caries lesion, defects, or existing restorative material; (2) create approximately 90-degree cavosurface margins (i.e., butt-joint margins); and (3) establish (ideally) not more than 0.5-mm clearance with the adjacent proximal surface facially, lingually, and gingivally.

The initial procedure in preparing the outline form of the proximal box is the isolation of the proximal (i.e., in this case, mesial) enamel by the proximal ditch cut. While maintaining the same orientation of the bur, it is positioned over the DEJ in the pulpal floor next to the remaining mesial marginal ridge (Fig. 10.62A). The end of the bur is allowed to cut a ditch gingivally along the exposed proximal DEJ, two thirds at the expense of enamel and one third at the expense of dentin. The 0.8-mm-diameter bur end cuts approximately 0.5 to 0.6 mm into enamel and 0.2 to 0.3 mm into dentin. Pressure is directed gingivally and lightly toward the mesial surface to keep the bur against the proximal enamel, while the bur is moved facially and lingually along the DEJ. The ditch is extended gingivally just beyond the caries lesion or the proximal contact, whichever is greater (see Fig. 10.62B). Axial wall dentinal depths will vary based on the gingival extension of the preparation (see Fig. 10.62C). The axial wall follows the faciolingual contour of the proximal surface and the DEJ (see Fig. 10.62D).

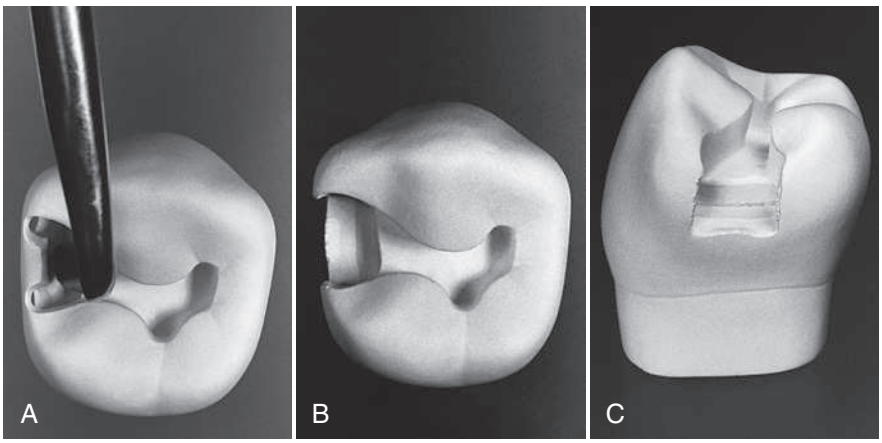
It is necessary to visualize the completed mesiofacial and mesiolingual margins as right-angle projections of the facial and lingual limits of the ditch to establish the proper faciolingual ditch extension (see Fig. 10.62E). When preparing a tooth with a small lesion, these margins may clear the adjacent tooth by only 0.2 to 0.3 mm.[101] A guide for the gingival extension is the visualization that the finished gingival margin will be only slightly gingival to the gingival limit of the ditch. This gingival margin will likely clear the adjacent tooth by only 0.5 mm when treating an early cavitated proximal lesion (see Fig. 10.62F).[103] Recall that the caries lesion develops immediately gingival to the proximal contact and therefore the preparation will always result in gingival clearance, and that the amount of clearance will depend on the size of the caries lesion. Clearance of the proximal margins (i.e., mesiofacial, mesiolingual, gingival) greater than 0.5 mm is excessive, unless indicated to include the caries lesion, undermined enamel, or existing restorative material. The location of the final proximal margins (i.e., facial, lingual, gingival) should be established with hand instruments (i.e., chisels, hatchets, trimmers) and not the No. 245 bur, so as to prevent unnecessary overextension (see Fig. 10.62E). Extending gingival margins into the gingival sulcus should be avoided, where possible, because subgingival margins are more difficult to restore and may be a contributing factor to periodontal disease.[105-107]

The depth of the axial dentinal wall should be adjusted to approximately 0.5 mm if retention grooves are deemed necessary (i.e., if caries lesion removal requires placement/orientation of facial and lingual walls that are very divergent). This will allow the grooves to be prepared into the axiolingual and axiofacial line angles without undermining the proximal wall enamel. If the proximal ditch cut is entirely in dentin, the initial axial wall is too deep. Because the proximal enamel becomes thinner from the occlusal to the gingival aspect, the end of the bur comes closer to the external tooth surface as the cutting progresses gingivally (see Fig. 10.62B, C, and I). Premolars may have proximal boxes that are shallower pulpally than are molars because premolars typically have thinner enamel. In the tooth crown, the ideal dentinal depth of the axial wall of the proximal boxes of premolars and molars should be the same (two thirds to three fourths the diameter of the No. 245 bur [or 0.5–0.6 mm]).[101] When the extension places the gingival margin in cementum, the initial pulpal depth of the axiogingival line angle should be 0.7 to 0.8 mm (the diameter of the tip end of the No. 245 bur is 0.8 mm). The bur may shave the side of the wedge that is protecting the rubber dam and the underlying gingiva (see Fig. 10.62C).

The gingival extension of the proximal ditch may be measured by first noting the depth of the nonrotating bur in the ditch. The dentist removes the bur from the preparation and holds it in the facial embrasure at the same level to observe the relationship of



• **Fig. 10.62** Isolation of proximal enamel. A, Bur position to begin the proximal ditch cut. B, The proximal ditch is extended gingivally to the desired level of the gingival wall (i.e., floor). C, Variance in the pulpal depth of the axiogingival line angle as the extension of the gingival wall varies: *(a)* at minimal gingival extension; *(b)* at moderate extension; *(c)* at extension that places gingival margin in cementum, whereupon the pulpal depth is 0.75 to 0.8 mm and the bur may shave the side of wedge. D, The proximal ditch cut results in the axial wall that follows the outside contour of the proximal surface. E, The position of the proximal walls (i.e., facial, lingual, gingival) should not be overextended with the No. 245 bur, considering additional extension will occur when the remaining spurs of enamel are removed. F, When a small lesion is prepared, the gingival margin should clear the adjacent tooth by only 0.5 mm. This clearance may be measured with the side of the explorer. The diameter of the tine of a No. 23 explorer is ~0.5 mm at 6 mm from its tip. G, The faciolingual dimension of the proximal ditch is greater at the gingival level than at the occlusal level to provide occlusal convergence of the facial and lingual proximal box walls. H, To isolate and weaken the proximal enamel further, the bur is moved toward and perpendicular to the proximal surface. I, The side of the bur may emerge slightly through the proximal surface at the level of the gingival floor *(arrow).*



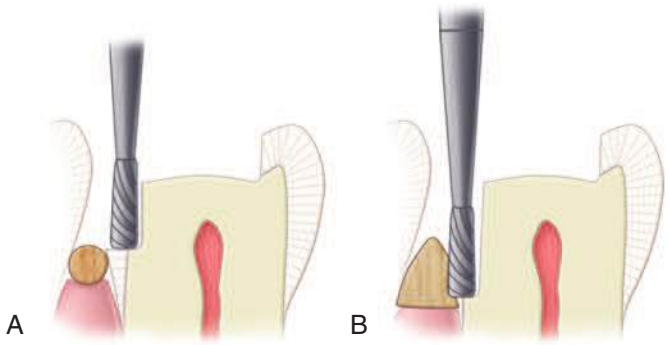
• **Fig. 10.63** Removing isolated enamel. A, Using a spoon excavator to fracture the weakened proximal enamel. B, Occlusal view with the proximal enamel removed. C, Proximal view with the proximal enamel removed.

the end of the bur to the proximal contact. A periodontal probe also may be used.

The proximal ditch cut may diverge gingivally (i.e., converge occlusally) to ensure that the faciolingual dimension at the gingival aspect is greater than at the occlusal aspect (see Fig. 10.62G). The shape of the No. 245 bur will provide this divergence. The gingival divergence contributes to the retention form and provides for the desirable extension of the facial and lingual proximal margins to include defective tooth structure or old restorative material at the gingival level, while conserving the marginal ridge and providing for 90-degree amalgam at the margins on this ridge.[55]

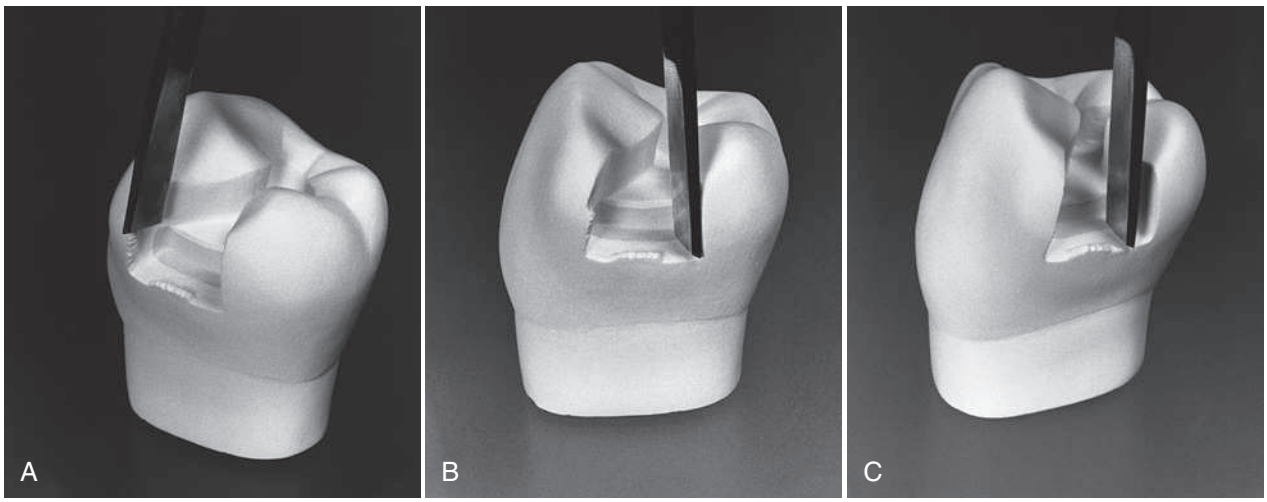
Occasionally it is permissible not to extend the outline of the proximal box facially or lingually beyond the proximal contact to conserve the tooth structure.[103] An example of this modification is a narrow proximal lesion where broad proximal contact is present in a patient with low risk for caries. If it is necessary to extend 1 mm or more just to arbitrarily create clearance (i.e., to "break the contact"), the proximal margin is left in contact with the adjacent proximal surface. It is usually the facial margins that remain in contact as proximal caries lesions tend to develop slightly more toward the lingual.

The proximal extensions are completed when two cuts, one starting at the facial limit of the proximal ditch and the other starting at the lingual limit, extending toward and perpendicular to the proximal surface (until the bur is nearly through enamel at the contact level), are made (see Fig. 10.62H). The side of the bur may emerge slightly through the surface at the level of the gingival floor (see Fig. 10.62I); this weakens the remaining enamel by which the isolated portion is held. If this level is judged to be insufficiently gingival, additional gingival extension should be accomplished using the isolated proximal enamel, that is still in place, as a mental barrier to help the operator guide the bur. Ensuring the presence of the proximal enamel surface helps to limit the likelihood of iatrogenically damaging the proximal surface of the adjacent tooth. At this stage, however, the remaining wall of enamel often breaks away during cutting, especially when high speed is used. At such times, if additional use of the bur is indicated, a matrix band may be used around the adjacent tooth to limit potential marring of its proximal surface; however, the relative amount of protection afforded by the metal matrix material is very limited and serves more as a visual guide than an actual physical barrier. A No. 245 bur, rotating between 200,000 and 400,000 rpm, will rapidly cut through any matrix material and damage the adjacent proximal surface. The isolated enamel, if still in place, may be fractured with a spoon excavator (Fig. 10.63) or by mesial movement with the side of the nonrotating bur.

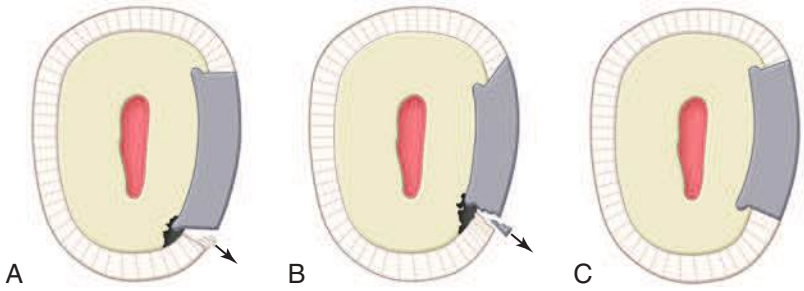


• **Fig. 10.64** Wedging. A, A round toothpick wedge placed in the gingival embrasure protects the gingiva and the rubber dam during preparation of the proximal box. B, A triangular wedge is indicated when a deep gingival extension of the proximal box is anticipated because the wedge's greatest cross-sectional dimension is at its base. Consequently it more readily engages the remaining clinical tooth surface. Wedging increases the interproximal space and limits the potential for iatrogenic damage of adjacent tooth (or restoration) surfaces.

To protect the gingiva and the rubber dam when extending the gingival wall apically, a wooden wedge should already be in place in the gingival embrasure to depress soft tissue and the rubber dam.[55] A round toothpick wedge is preferred unless a deep gingival extension is anticipated (Fig. 10.64A). A triangular (i.e., anatomic) wedge is more appropriate for deep gingival extensions because the greatest cross-sectional dimension of the wedge is at its base; as the gingival wall is cut, the bur's end corner may shave the wedge slightly (see Fig. 10.64B). With a sharp enamel hatchet (10-7-14), bin-angle chisel (12-7-8), or both, the dentist cleaves away any remaining undermined proximal enamel (Fig. 10.65), establishing the proper orientation of the mesiolingual and mesiofacial walls. Proximal walls that result in cavosurface angles of 90 degrees are desired.[55] Cavosurface angles of 90 degrees ensure that no undermined enamel rods remain on the proximal margins and



• **Fig. 10.65** Removing the remaining undermined proximal enamel with an enamel hatchet on the facial proximal wall (A), the lingual proximal wall (B), and the gingival wall (C).



• **Fig. 10.66** Direction of mesiofacial and mesiolingual walls. A, Failure caused by a weak enamel margin. B, Failure caused by a weak amalgam margin. C, Proper direction to the proximal walls results in full-length enamel rods and 90-degree amalgam at the preparation margin. Retention grooves have been cut 0.2 mm inside the DEJ, and their direction of depth is parallel to the DEJ.

that the maximal edge strength of amalgam is maintained. The cutting edge of the instrument should not be aggressively forced against the gingival wall because this may cause a craze line (i.e., fracture) that extends gingivally in enamel, perhaps to the cervical line. Fig. 10.66 shows the importance of the correct direction of the mesiofacial and mesiolingual walls, dictated by enamel rod direction and the physical properties of amalgam. If hand instruments were not used to remove the remaining spurs of enamel, the proximal margins would have undermined enamel. To create 90-degree facial and lingual proximal margins with the No. 245 bur, the proximal margins would have to be significantly overextended for an otherwise conservative preparation. The weakened enamel along the gingival wall is removed by using the enamel hatchet in a scraping motion (see Fig. 10.65C).

When the isolation of the proximal enamel has been executed properly, the proximal box may be completed easily with hand-cutting instruments. Otherwise more cutting with rotary instruments is indicated. When a rotary instrument is used in a proximal box after the proximal enamel is removed, there is increased risk that the instrument may either mar the adjacent proximal surface or "crawl out" of the box into the gingiva or across the proximal margins. The latter mishap produces a rounded cavosurface angle, which results in a weak amalgam margin of less than 90 degrees if not corrected. The risk of this occurring is markedly reduced when high-speed operation is used. When finishing enamel margins by using a rotary instrument, intermittent application of the bur along with air coolant alone (i.e., no water spray) is used to improve visualization and precision.

The primary resistance form is provided by (1) the pulpal and gingival walls being relatively level (i.e., perpendicular to forces directed along the long axis of the tooth); (2) restricting the extension of the walls to allow sufficient dentin support to remain (and therefore strong cusps and ridge areas) while at the same time establishing the peripheral seat; (3) restricting the occlusal outline form (where possible) to areas receiving minimal occlusal contact;[57] (4) use of a reverse curve to optimize the strength of the amalgam and tooth structure at the junction of the occlusal step and proximal box; (5) slight rounding of the internal line angles to reduce stress concentration in the tooth structure (automatically created by bur design except for the axiopulpal line angle); and (6) providing enough thickness of the restorative material to prevent its flexure and resultant fracture from the forces of mastication. The primary retention form is provided by the occlusal convergence of the facial and lingual walls and by the dovetail design of the occlusal step, if present.

After completing the initial tooth preparation, the adjacent proximal surface should be evaluated. An adjacent proximal restoration may require recontouring to reestablish the normal anatomic convex shape; this may be done with abrasive finishing strips, disks, finishing burs, or a combination of these. Any iatrogenic

damage that compromises the convex adjacent proximal surface should be corrected by recontouring or restoration.

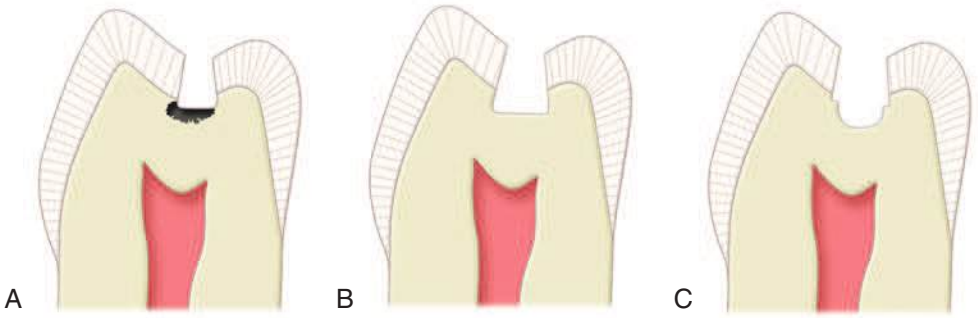
Final Tooth Preparation

Removing enamel pit-and-fissure remnants and soft dentin on the pulpal wall in Class II preparations is accomplished in the same manner as in the Class I preparation. Soft dentin is removed with a slowly revolving round bur of appropriate size, a discoid-type spoon excavator, or both. In general, the caries lesion removal should stop when a hard or firm feel, with an explorer or small spoon excavator, is achieved; this often occurs before all of the stained or discolored dentin is removed. The exception to this is when using the selective caries removal protocol (see Chapter 2). Removing enamel pit-and-fissure remnants and carious dentin should not affect the resistance form. To achieve an enhanced resistance form, the occlusal step should have pulpal seats at the initial preparation depth, perpendicular to the long axis of the tooth in sound tooth structure and peripheral to the excavated area (Fig. 10.67). Carious dentin in the axial wall is removed with appropriate round burs, spoon excavators, or both so as to conserve as much tooth structure as possible (Fig. 10.68).

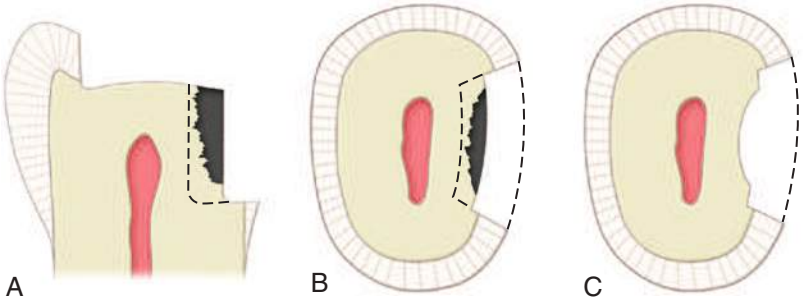
Any remaining old restorative material (including base and liner) may be left if no evidence of a recurrent caries lesion exists, if its periphery is intact, and if the tooth has been asymptomatic (assuming the pulp is vital). This concept is particularly important if removal of all remaining restorative material may increase the risk of pulpal exposure. Appropriate steps to protect the pulp should be taken as indicated in General Concepts Guiding Preparation for Amalgam Restorations.

After completion of the minimal gingival extension (i.e., the gingivoaxial line angle is in sound dentin), a remnant of the enamel portion of a caries lesion may remain on the gingival floor (wall), seen in the form of a decalcified (i.e., white, chalky) or faulty area bordering the margin (Fig. 10.69). This situation dictates further extension of a part or all of the gingival floor to place it in sound tooth structure. Extension of the entire gingival wall to include a large caries lesion may place the gingival margin so deep that proper matrix application and wedging become extremely difficult. Fig. 10.70A illustrates an outline form that extends gingivally in the central portion of the gingival wall to include a caries lesion that is deep gingivally, although leaving the facial and lingual gingival corners at a more occlusal position. This partial extension of the gingival wall permits wedging of the matrix band where otherwise it may be difficult and damaging to soft tissue. In this instance, the gingival wedge may not tightly support a small portion of the band. Special care must be exercised by placing small amounts of amalgam in this area first and thoroughly condensing with light pressure. In addition, care is exercised in carving the restoration in this area to remove any excess that may have extruded gingivally during condensation.

Fig. 10.70B illustrates removal of a caries lesion facially and gingivally beyond the conventional margin position. Such minor variations from the ideal preparation form permit conservation of healthy tooth structure. A partial extension of a facial or lingual wall is permissible if (1) the entire wall is not weakened, (2) the



• **Fig. 10.67** Management of small- to moderate-sized caries lesion on the pulpal wall. A, Soft dentin extending beyond the ideal pulpal wall position. B, Incorrect lowering of the pulpal wall to include soft dentin. C, Correct extension facially and lingually beyond the soft dentin. Note the caries removal below the ideal pulpal wall level and the facial and lingual seats at the ideal pulpal wall level.

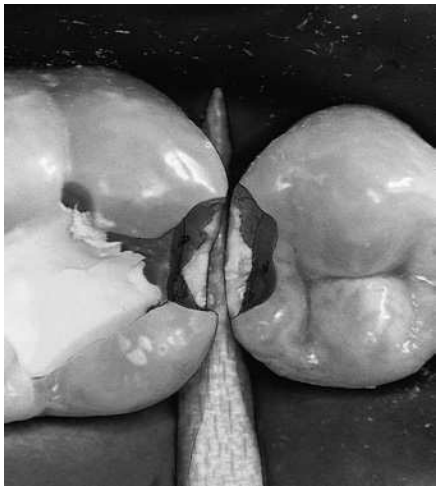


• **Fig. 10.68** Management of a moderate to extensive caries lesion. A and B, Soft dentin on the axial wall does not call for the preparation of the whole axial wall toward the pulp *(dotted lines).* C, Soft dentin extending pulpally from the ideal axial wall position is conservatively removed with a round bur.

extension remains accessible and visible, (3) sufficient gingival seats remain to support the restoration, and (4) a butt-joint fit at the amalgam–enamel margin (90-degree amalgam angle and 90-degree cavosurface angle) is possible.

Conserving as much tooth structure as possible by limiting the extensions of external walls helps to provide secondary resistance by limiting the risk of future tooth fracture from oblique forces. Ensuring that all internal line angles are rounded is necessary to limit areas of stress concentration in the remaining tooth structure and in the new restoration and improves secondary resistance of both. Use of the gingival margin trimmer or a bur to round the axiopulpal line angle (Fig. 10.71) helps to increase the bulk of restorative material and decrease the stress concentration within the restorative material.

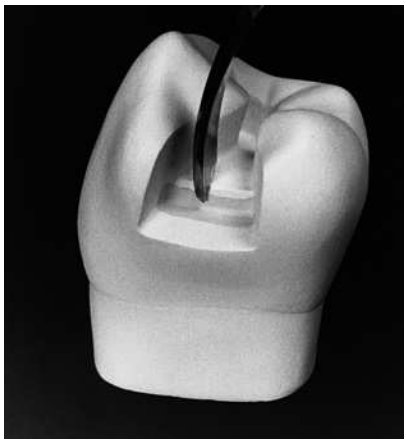
The secondary retention forms for the occlusal and proximal portions of the preparation should be independent of each other. The occlusal convergence of the facial and lingual walls provide a sufficient retention form to the occlusal portion of the tooth preparation.



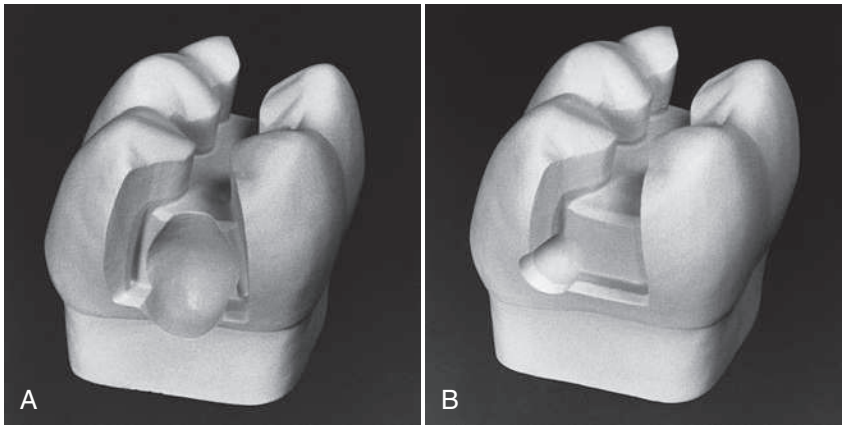
• **Fig. 10.69** Remnant of caries lesion bordering the enamel margin after insufficient gingival extension. Such a lesion indicates extending part or all of the gingival floor to place it in sound tooth structure. (Courtesy Dr. C. L. Sockwell.)

Proximal Retention Grooves

To enhance the retention form of the proximal portion, proximal grooves may be indicated to counter proximal displacement of the amalgam restoration.[55,108-110] The use of retention grooves in proximal boxes is controversial. It has been reported that proximal retention grooves in the axiofacial and axiolingual line angles may increase the fracture resistance and significantly strengthen the isthmus of a Class II amalgam restoration and that these grooves are significantly superior to the axiogingival grooves in increasing the restoration's fracture strength.[111-114] Other investigators have suggested, however, that retention grooves located occlusal to the axiopulpal line angle provide more resistance than do conventional grooves.[108] It also has been reported that proximal retention grooves are unnecessary in preparations that include dovetails, as part of the occlusal preparation, when high-copper amalgam is used.[109,115] Evidence suggests that retentive grooves may not be needed in conservative, narrow proximal boxes.[110] The use of retention grooves is recommended in tooth preparations with extensive proximal boxes. Generally the depth of the grooves should be approximately half of the diameter of the tip of the No. 169 or No. ¼ round bur (i.e., ~0.25–0.5 mm). The depth of retention grooves in extensively wide proximal boxes may need to be 0.5 mm or greater at the gingival aspect.



• **Fig. 10.71** Rounding the axiopulpal line angle.



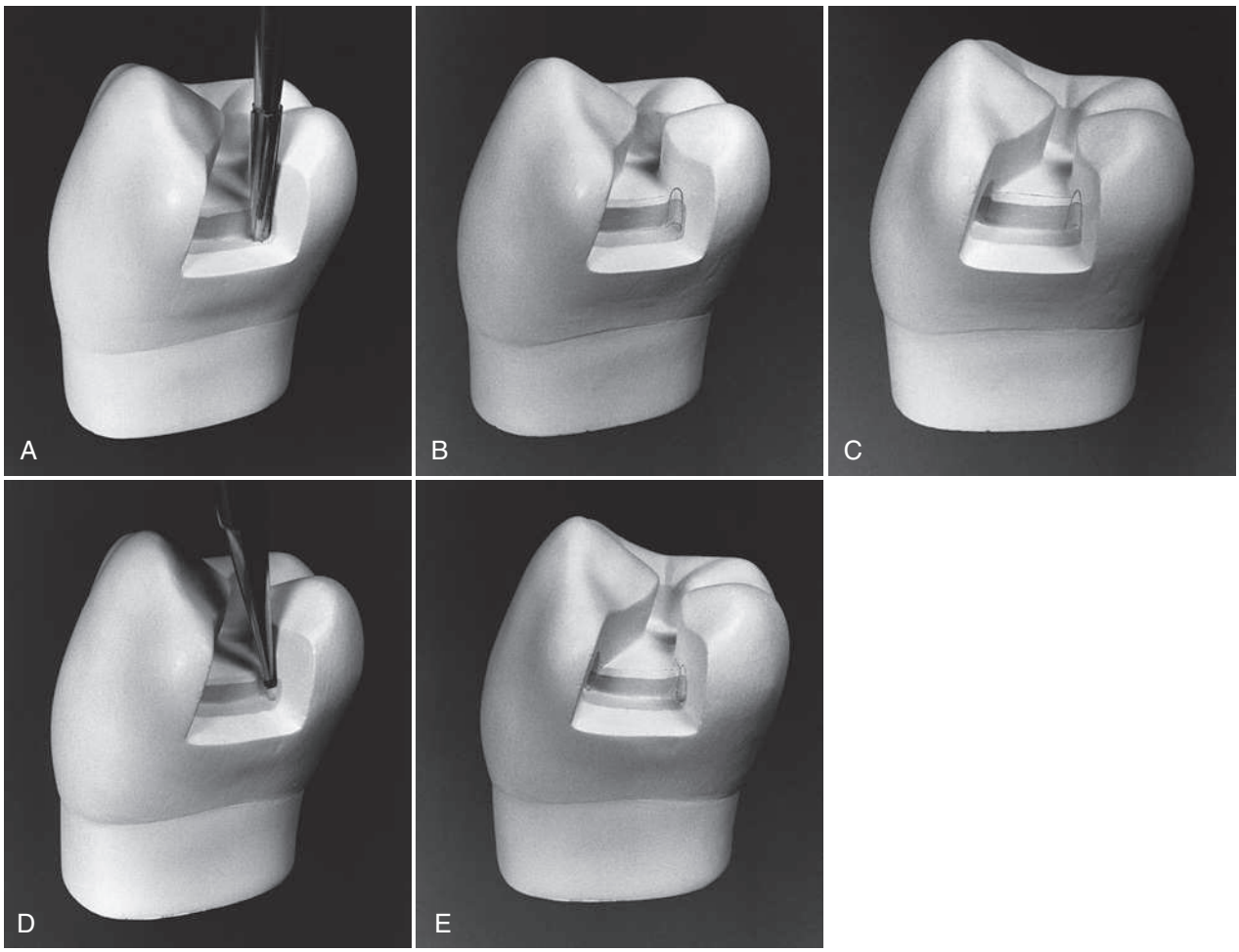
• **Fig. 10.70** A, Outline form that permits extension of the center portion of the gingival wall to facilitate proper matrix construction and wedging in situations where the caries lesion extends deep gingivally. B, Outline form that permits partial wall extension facially and gingivally to conserve the tooth structure.

Retention grooves are prepared with a No. 169 L or No. ¼ round bur with air coolant only (to improve visualization) and reduced speed (to improve tactile "feel" and control). The bur is placed in the properly positioned axiolingual line angle and directed (i.e., translated) to bisect the angle (Fig. 10.72) approximately parallel to the DEJ (Fig. 10.73). This positions the retention groove 0.2 mm inside the DEJ, maintaining the enamel support. The bur is tilted to allow cutting to the depth of the diameter of the end of the bur (which is initially positioned at the axiolinguogingival point angle) and to permit the groove to diminish in depth occlusally. The facial groove in the axiofacial line angle is prepared in a similar manner. When the axiofacial and axiolingual line angles are less than 2 mm in length, the tilt of the bur is reduced slightly so that the proximal grooves are extended occlusally to disappear midway between the DEJ and the enamel margin (see Fig. 10.72B and C).

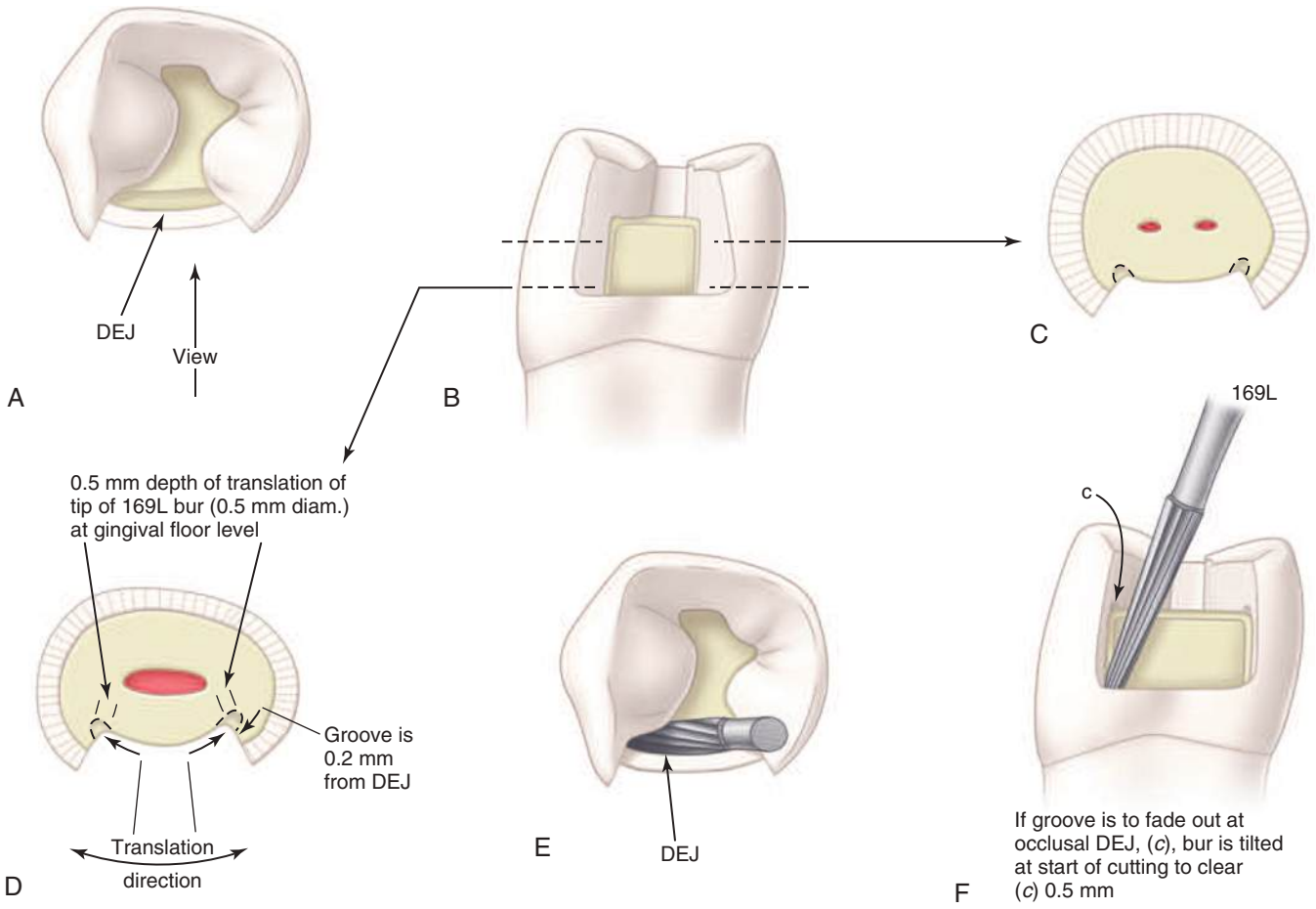
The four characteristics or determinants of proximal grooves are (1) position, (2) translation, (3) depth, and (4) occlusogingival orientation (see Fig. 10.73). *Position* refers to the axiofacial and axiolingual line angles of initial tooth preparation (0.5 mm axial to the DEJ). *Translation* refers to the direction of movement of the axis of the bur. *Depth* refers to the extent of translation (i.e., depth of the retention groove). *Occlusogingival orientation* is considered when using the No. 169 L bur and refers to the tilt of the bur. The tilt dictates the occlusal height of the groove, given a constant depth. When using the No. ¼ bur to prepare the proximal groove, the rotating bur is carried into the axiolinguogingival (or axiofaciogingival) point angle, then moved parallel to the DEJ to the depth of ~0.25 to 0.5 mm. It is then drawn occlusally along the axiolingual (or axiofacial) line angle, allowing the groove to become shallower and to terminate at the axiolinguopulpal (or axiofaciopulpal) point angle (or more occlusally if the line angles are <2 mm in length) (see Fig. 10.72D and E).

Regardless of the method used in placing the grooves, extreme care is necessary to prevent the removal of dentin that immediately supports the proximal enamel. In addition, it is essential that the grooves are not prepared entirely in the axial wall (i.e., incorrect translation [moving the bur only in an axial direction]) because groove placement in this area will not affect any resistance to proximal displacement of the amalgam (i.e., will not increase retention) and will increase the risk of iatrogenic pulpal involvement.

An improperly positioned axiofacial or axiolingual line angle must not be used as a positional guide for the proximal groove. If the axial line angle is too shallow, the groove may remove the necessary dentinal support of enamel. If the line angle is too deep, preparation of the groove may result in exposure of the pulp. *Retention grooves, when used, always should be placed 0.2 mm inside*



• **Fig. 10.72** Proximal retention grooves. A, Position of the No. 169 L bur to prepare the retention groove as the bur is moved lingually and pulpally. B, Lingual groove. Note the dentin support of the proximal enamel. C, Completed grooves. D, Grooves prepared with a No. ¼ round bur. E, Completed grooves.



• **Fig. 10.73** Four characteristics of retentive grooves. A, Occlusal view of the mesiooccl usal preparation before placement of the retention grooves. B, Proximal view of the mesiooccl usal preparation. C and D, Position, translation, and depth. E and F, Occlusogingival orientation.

the DEJ of the facial and lingual proximal walls regardless of the depth of the axial wall and associated line angles.

Proximal Walls

The preparation walls and margins should not have unsupported enamel and marginal irregularities.[116] No occlusal cavosurface bevel is indicated in the tooth preparation for amalgam. Ideally, a 90-degree cavosurface angle (maximum of 100 degrees) should be present at the proximal margin. The occlusal line angle may be 90 to 100 degrees. This angle aids in obtaining a marginal amalgam angle of 90 degrees (≥80 degrees). Clinical experience has established that this "butt-joint" relationship of enamel and amalgam creates the strongest margin.[55] Amalgam is a brittle material and may fracture under occlusal load if its angle at the margin is less than 80 degrees.

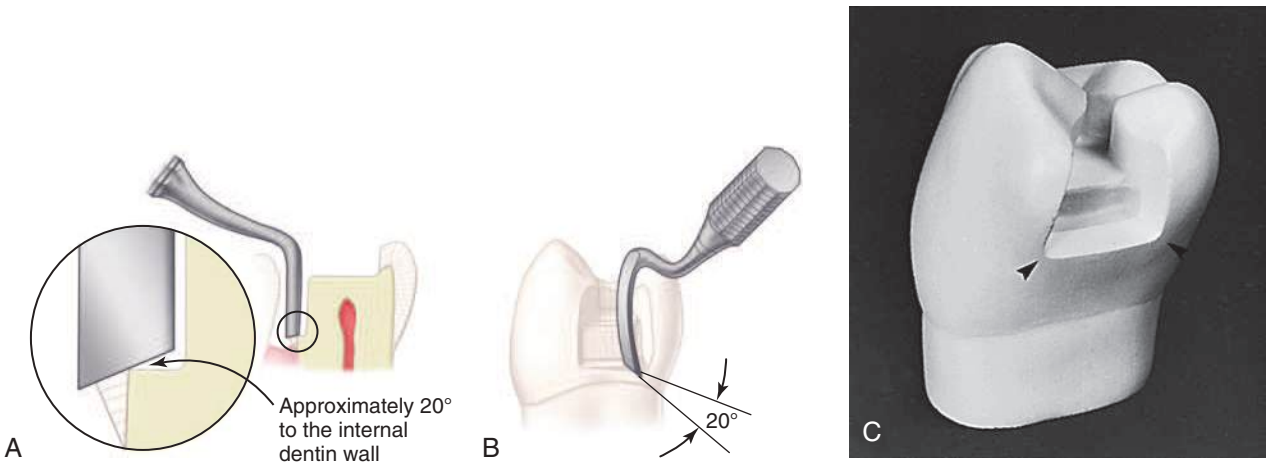
A sharp mesial gingival margin trimmer (13-85-10-14, R and L) is used to remove unsupported gingival enamel by the establishment of a 90-degree gingival cavosurface angle at the gingival margin (Fig. 10.74). The resulting enamel gingival margin has a 6-centigrade (20-degree) gingival declination. The marginal configuration is angled no more than necessary to ensure that full-length enamel rods form the gingival margin and is no wider than the enamel (see Fig. 10.74). When the gingival margin is positioned gingival to the cementoenamel junction (CEJ) (i.e., on the tooth root), the gingival cavosurface configuration is 90 degrees to the external surface of the root.[55] A sharp distal gingival margin trimmer (13-95-10-14, R and L) is similarly used for distal proximal preparations. Alternatively, the side of an explorer tine may be used to remove any friable enamel at the gingival margin. The tine is placed in the gingival embrasure apical to the gingival margin. With some pressure against the prepared tooth, the tine is moved occlusally across the gingival margin to "trim" the margin of friable, unsupported enamel.

Variations of Proximal Surface Tooth Preparations

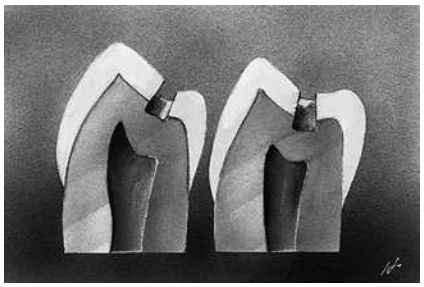
This section describes variations in tooth preparation for some small Class II amalgam restorations. In most clinical situations, the restoration of these preparations would likely be with composite unless the patient has high caries risk. If amalgam is used, the features presented should be considered in the tooth preparation portion of the procedure.

Mandibular First Premolar

The technique of the Class II tooth preparation for amalgam for the mandibular first premolar must be modified because the



• **Fig. 10.74** A, The marginal configuration of the enamel portion of the gingival wall is established with a gingival margin trimmer to ensure full-length enamel rods forming the gingival margin. B and C, The sharp angles at the linguogingival and faciogingival corners are rounded by rotational sweeping with a gingival margin trimmer.



• **Fig. 10.75** The mandibular first and second premolars are compared. Note differences in the sizes of the pulp chambers, lingual cusps, and direction of pulpal walls.

morphologic structure of this tooth is different from other posterior teeth. The relationship of the pulp chamber to the DEJ and the relatively small size of the lingual cusp are illustrated in Fig. 10.75. (This figure also illustrates the correct position of the pulpal wall and how it differs in position and orientation compared with the second premolar.) Incorrect preparation of the central groove area may weaken the lingual cusp, and excessive extension in a facial direction may approach or expose the facial pulp horn. When preparing the occlusal portion, the bur is tilted slightly lingually to establish the correct pulpal wall direction (see Fig. 10.43B).

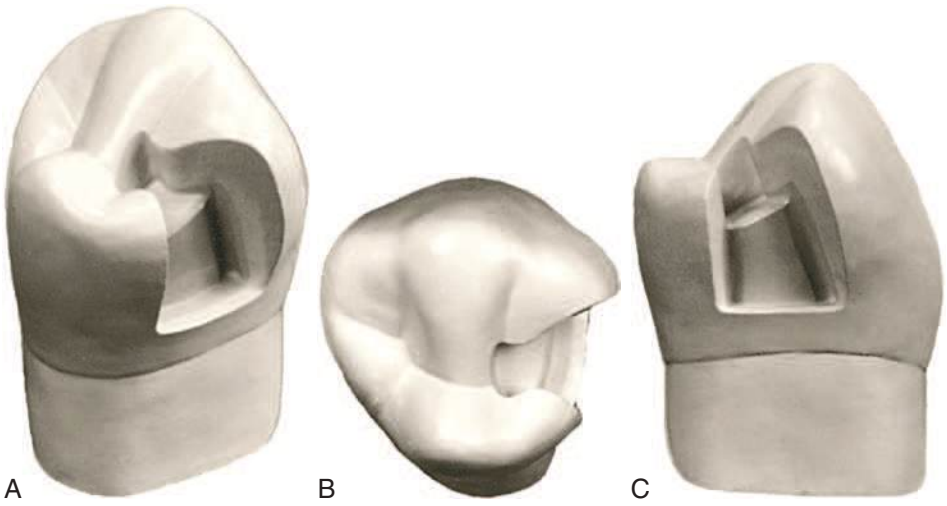
The mandibular first premolar presents with unique occlusal morphology, most notably the presence of a large transverse ridge of enamel. Most often the coalescence of the enamel lobes is complete and no caries-prone central fissure results. Therefore a caries lesion in the mesial pit or distal pit may be included as part of the Class II preparation, but with an outline form that does not extend to or across the transverse ridge (Fig. 10.76A). The level of divergence of the facial and lingual proximal walls (see Fig. 10.76B) is such that secondary retention grooves are indicated (see Fig. 10.76C). If the opposite pit or proximal surface is faulty, it is restored with a separate restoration so as to not compromise the resistance form of the tooth.

For a preparation that does not cross the transverse ridge, the proximal box is prepared before the occlusal portion to prevent removing the tooth structure that will form the isthmus between the occlusal dovetail and the proximal box. The pit adjacent to the involved proximal surface is entered with the No. 245 bur. Immediately after the entry, the bur is directed into the proximal marginal ridge and then gingivally until the proximal DEJ is visible. The bur axis for the proximal ditch cut should be parallel to the tooth crown, which is tilted slightly lingually for mandibular posterior teeth (see Fig. 10.76C). The proximal enamel is isolated and the proximal box completed as previously described for the mandibular second premolar. The bur is then returned to the area of entry, and the occlusal step is prepared with a dovetail, if needed. When preparing the occlusal portion, the bur is tilted slightly lingual to establish the correct pulpal wall direction (which maintains the dentin support for the small lingual cusp and prevents encroachment on the facial pulp horn). The primary difference in tooth preparation on this tooth, compared with the preparation on other posterior teeth, is the facial inclination of the pulpal wall. The isthmus is broadened as necessary, but maintains the dovetail retention form. Fig. 10.76B illustrates the correct occlusal outline form. Removing any remaining caries lesion (if present) and inserting necessary liners, bases, or both precede the placement of proximal grooves and the finishing of the enamel margins (see Fig. 10.76C).

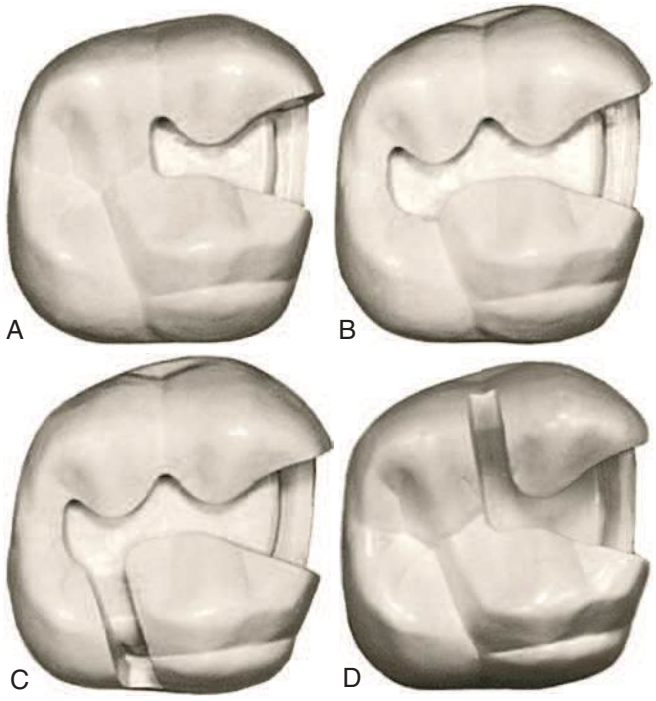
Maxillary First Molar

When mesial and distal proximal surface amalgam restorations are indicated on the maxillary first molar that has an unaffected oblique ridge, separate two-surface tooth preparations are indicated (rather than a mesioocclusodistal preparation). The use of two preparations is preferred because the strength of the tooth crown is significantly greater when the oblique ridge is intact.[55] The mesioocclusal tooth preparation is generally uncomplicated (Fig. 10.77A). Occasionally, extension through the oblique ridge and into the distal pit is necessary because of the extent of the caries lesion. The outline of the occlusolingual pit-and-fissure portion is similar to that of the Class I occlusolingual preparation. Fig. 10.77B and C illustrates a mesioocclusal preparation extended to include the distal pit and the outline form that includes the distal oblique and lingual fissures.

When the occlusal fissure extends into the facial cusp ridge and cannot be removed by enameloplasty, the defect should be eliminated by extension of the tooth preparation. Occasionally this may be accomplished by tilting the bur to create an occlusal divergence



• **Fig. 10.76** The mandibular first premolar with a sound transverse ridge. A, A two-surface tooth preparation that does not include the opposite pit. B, Occlusal outline form. C, Proximal view of the completed preparation.



• **Fig. 10.77** Maxillary first molar. A, Conventional mesiooccluslal preparation. B, Mesiooccluslal preparation extended to include the distal pit. C, Mesiooccluslingual preparation, including the distal pit and the distal oblique and lingual fissures. D, Mesiooccluslal preparation with facial fissure extension.

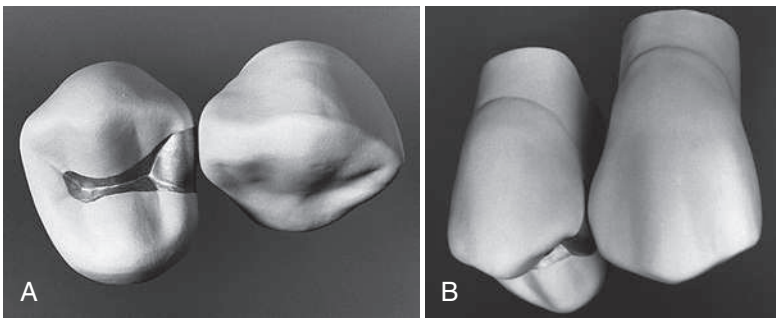
of the facial wall while maintaining the dentin support of the ridge. If it is not possible to eliminate this fault without extending the margin to the height of the cusp ridge or undermining the enamel margin, the preparation should be extended facially through the ridge (see Fig. 10.77D). The pulpal wall of this facial extension may have remaining enamel, but a depth of 1.5 to 2 mm is necessary to provide sufficient bulk of amalgam material for adequate strength (adequate resistance to flexure under load). For the best esthetic results, minimal extension of the proximal mesiofacial margin is indicated.

The distooccluslal tooth preparation may take one of several outlines, depending on the occlusal anatomy. The occlusal outline is determined by the pit-and-fissure pattern and by the amount and extension of the caries lesion. An extension onto the lingual surface to include a lingual fissure should be prepared only after the distolingual proximal margin is established. This approach may allow conservation of more tooth structure between the distolingual wall and the lingual fissure extension, resulting in more strength of the distolingual cusp. It is accomplished by preparing the lingual fissure extension more at the expense of the mesiolingual cusp than the distolingual cusp. Nevertheless, the distolingual cusp on many maxillary molars may be weakened during the distooccluslingual tooth preparation because of the small cuspal portion remaining between the lingual fissure preparation and the distolingual proximal wall. In addition, removal of the caries lesion may reveal a weakened distolingual cusp. Reduction of the distolingual cusp and coverage with amalgam is often necessary to provide the proper resistance form (see the mesiooccluslodistolingual [MODL] example in Clinical Technique for Complex Amalgam Restorations).

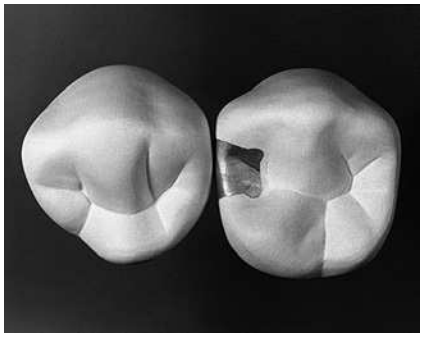
Maxillary First Premolar

A Class II amalgam tooth preparation involving the mesial surface of a maxillary first premolar requires special attention because the mesiofacial embrasure is esthetically prominent. The occlusogingival preparation of the facial wall of the mesial box should be parallel to the long axis of the tooth instead of converging occlusally to minimize an unesthetic display of amalgam in the faciogingival corner of the restoration. In addition, the facial extension of the mesiofacial proximal wall should be minimal so that the mesiofacial proximal margin of the preparation only minimally clears the contact as the margin is finished with an appropriate enamel hatchet or chisel (Fig. 10.78).

If the mesial proximal involvement (1) is limited to a fissure in the marginal ridge that is at risk for caries, (2) is not treatable by enameloplasty, and (3) does not involve the proximal contact, then the proximal portion of the tooth preparation is prepared by extending through the fault with the No. 245 bur so that the margins are lingual to the contact. Often this means that the proximal box is the faciolingual width of the bur, and the gingival



• **Fig. 10.78** To produce an inconspicuous margin on the maxillary first premolar, the mesiofacial wall does not diverge gingivally, and facial extension with a No. 245 bur should be minimal so that the mesiofacial proximal margin of the preparation minimally clears the contact as the margin is finished. A, Occlusal view. B, Facial view.



• **Fig. 10.79** A simple box restoration without the occlusal step is permissible when restoring a small proximal lesion in the tooth without either occlusal fissures or previously inserted occlusal restoration and when the involved marginal ridge does not support occlusal contact. The proximal grooves extend to the occlusal surface.

floor may be at the same depth as the pulpal floor. The retention form for this extension is provided by the slight occlusal convergence of the facial and lingual walls (see Fig. 10.34).

If the proximal caries lesion is limited to the mesiolingual embrasure, the mesial proximal contact should not be included in the tooth preparation. If only the lingual aspect of the mesial proximal contact is carious, the mesiofacial wall may be left in contact with the adjacent tooth (reducing the display of amalgam). A Class II tooth preparation involving the distal surface of the maxillary first premolar is similar to the preparation of the mandibular second premolar described earlier.

Box-Only Preparation

When restoring a small, cavitated proximal lesion in a tooth with neither occlusal fissures nor a previously inserted occlusal restoration, a proximal box preparation without an occlusal step has been recommended.[54,55] This preparation should be limited to a proximal surface with a narrow proximal contact (allowing minimal facial and lingual extensions) and a marginal ridge with no occlusal contact. To maximize retention, preparations with facial and lingual walls that almost oppose each other are recommended. As in the typical preparation, the facial and lingual proximal walls converge occlusally. Retention grooves are mandatory in box-only preparations.[117] The proximal retention grooves should have a 0.5-mm depth at the gingival point angle and extend occlusally to be visible in the occlusal outline form (Fig. 10.79).

Tooth Preparation for Class II Amalgam Restorations Involving Both Proximal Surfaces

Perhaps the best indications for the use of amalgam restorations are moderate and large Class II defects that include both proximal surfaces and much of the occlusal surface. This section describes factors to consider when amalgam is used for moderate and large Class II restorations. The tooth preparation techniques presented for a two-surface Class II restoration apply to larger Class II restorations as well. When the defect is large, however, certain modifications in tooth preparation may be necessary.

Occlusal Extensions

Larger Class II defects may require greater extension of the occlusal surface outline form. This may require extending into grooves that are fissured, extending the outline form up the cuspal inclines, or reducing and covering the cusps that have inadequate dentin support. These alterations may be accomplished easily by following the principles presented previously. The depth and extension of the initial preparation of fissures occur at the same initial pulpal floor depth (i.e., at the level of the DEJ) and follow the DEJ as the preparation is extended in a facial or lingual direction.

If an occlusal outline form extends up a cuspal incline, the extension also should maintain the pulpal floor at the level of the DEJ. This extension usually requires some alteration in the orientation of the No. 245 bur—a slight lingual tilt when extending in a facial direction and a slight facial tilt when extending in a lingual direction (see Fig. 10.35). Maintaining the correct pulpal floor depth preserves tooth structure and reduces the potential for pulpal encroachment. The prepared facial (or lingual) cavosurface margin still should result in a 90-degree amalgam margin. If it is necessary to extend through the cusp ridge onto the facial or lingual surface, the preparation is accomplished as described for the occlusolingual Class I restoration.

When the occlusal outline form extends from a primary fissure to within two thirds of the distance to a cusp tip, that cusp is usually sufficiently weakened so as to require reduction and coverage with the restoration. Leaving the cusp in a weakened state may be acceptable if the cusp is very large. Routine preparations in some teeth may predispose some cusps for reduction and coverage with amalgam. The small distal cusp of the mandibular first molars, the distolingual cusp of maxillary molars, and the lingual cusp of some mandibular premolars (especially first premolars) may be weakened when normal preparations of surrounding areas of the tooth are included. See the section Clinical Technique for Complex

Amalgam Restorations for information on when and how to reduce and cover cusps with amalgam.

Proximal Extensions

Larger Class II caries lesions often require larger proximal box preparations. These may include not only increased faciolingual or gingival extensions but also extension around a facial or lingual line angle. Large proximal box preparations may also need secondary retention features (i.e., retention grooves) for adequate retention form because the proximal external walls are excessively divergent. Extensive proximal boxes are usually prepared in the same fashion as more conservative proximal boxes but may require modifications. For increased faciolingual extensions, it may be necessary to tilt the No. 245 bur to include extensive proximal faults that are gingival to the contact area. Tilting the bur lingually when extending a facial proximal wall, or facially when extending a lingual proximal wall, conserves more of the marginal ridge and cuspal tooth structure. Although this action enhances preservation of some tooth structure strength, it results in a more occlusally convergent wall, which increases the difficulty of amalgam condensation in the gingival corners of the preparation.

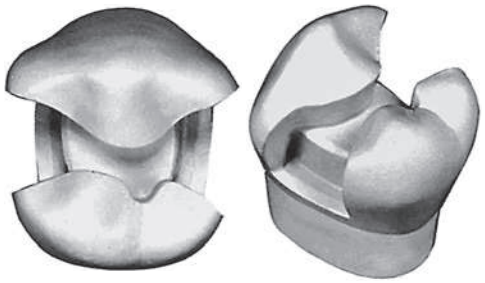
When proximal extension around a line angle is necessary, it usually is associated with a reduction of the involved cusp (see Clinical Technique for Complex Amalgam Restorations). Such proximal extension is necessary because of a severely defective (or fractured) cusp or a cervical lesion that extends from the facial (or lingual) surface into the proximal area. Often these areas are included in the preparation by extending the gingival floor of the proximal box around the line angle, using the same criteria for preparation as the typical proximal box: (1) Facial (or lingual) extension results in an occlusogingival wall that has a 90-degree cavosurface margin, and (2) the axial depth is 0.5 mm inside the DEJ.

When the proximal defect is extensive gingivally, isolation of the area, tooth preparation, matrix placement, and condensation and carving of amalgam are more difficult. If the proximal box is extended onto the root surface, the axial wall depth is no longer dictated by the DEJ. Any root-surface preparation for amalgam should result in an initial axial wall depth of approximately 0.8 mm. This axial depth provides appropriate strength for amalgam, limits any potential pulp compromise, and creates enough dimension for the placement of ~0.25 to 0.5 mm deep retention grooves while preserving the strength of the adjacent, remaining marginal dentin and cementum. The extent of the preparation onto the root surface, the contour of the tooth, or both may require that the bur be tilted toward the adjacent tooth when preparing the gingival floor of the proximal box. This tilting may result in an axial wall that has two planes, the more gingival plane angled slightly internally. It also may cause more difficulty in retention groove placement. The more occlusal part of the axial wall may be overreduced if the bur is not tilted.

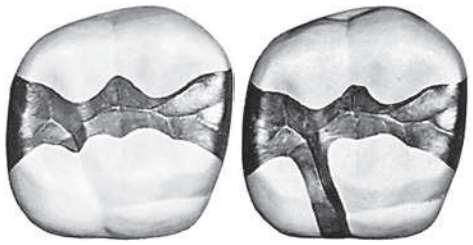
Examples of Moderate Class II Amalgam Tooth Preparations That Involve Both Proximal Surfaces

Mandibular Second Premolar

A moderate mesioocclusodistal tooth preparation in a mandibular second premolar is illustrated in Fig. 10.80. Note the similarity with the two-surface mesioocclusal preparation.



• **Fig. 10.80** Mesioocclusodistal (MOD) preparation on the mandibular second premolar that had a moderately sized MOD caries lesion.



• **Fig. 10.81** Typical three- and four-surface restorations for the maxillary first molar. (See Fig. 10.167 for preparation of the distolingual cusp for coverage with amalgam.)

Mandibular First Premolar

When a mesioocclusodistal amalgam tooth preparation is needed for the mandibular first premolar, the support of the small lingual cusp may be conserved by preparing the occlusal step more at the expense of tooth structure facial to the central groove than lingual. In addition, the bur is tilted slightly lingually to establish the correct pulpal wall direction. Despite these precautions, the lingual cusp may need to be reduced for coverage with amalgam if the lingual margin of the occlusal step extends more than two thirds the distance from the central fissure to the cuspal eminence (see Clinical Technique for Complex Amalgam Restorations).

Maxillary First Molar

The mesioocclusodistal tooth preparation of the maxillary first molar may require extending through the oblique ridge to unite the proximal preparations with the occlusal step. Extending the preparation through the oblique ridge is indicated only if (1) the ridge is undermined by a caries lesion, (2) it is crossed by a deep fissure, or (3) occlusal portions of the separate mesioocclusal and distoocclusal outline forms leave less than 0.5 mm of the tooth structure between them. The remainder of the outline form is similar to the two-surface outline forms described previously in this chapter. Fig. 10.81 illustrates typical three-surface and four-surface restorations for this tooth.

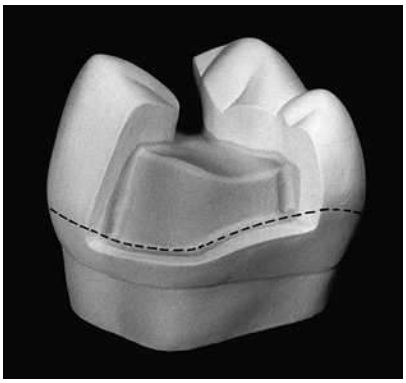
Maxillary Second Molar With a Caries Lesion on the Distal Portion of the Facial Surface

Close examination of the distal portion of the facial surface of the maxillary second molar may reveal decalcification, cavitation, or both. When enamel is only slightly demineralized (i.e., softened and rough), polishing with sandpaper disks may eliminate the superficial fault. Careful brushing technique, daily use of fluoride (i.e., rinses, toothpaste), and periodic applications of a fluoride varnish may prevent further mineral loss. When decalcification is as deep as the DEJ and a distal proximal caries lesion is also present,

however, the entire distofacial cusp may need to be reduced and included in a mesioocclusodistofacial tooth preparation. The facial lesion may be restored separately if it is judged that the distofacial cusp would not be significantly weakened if left unrestored by amalgam. In that case the mesioocclusodistal preparation would be restored first, followed by preparation and restoration of the facial lesion. When such sequential preparations are contraindicated, the preparation outline is extended gingivally to include the distofacial cusp (just beyond the caries lesion) and mesially to include the facial groove (Fig. 10.82). The No. 245 bur should be used to create a gingival floor (i.e., shoulder) perpendicular to the occlusal force when extending the distal gingival floor to include the affected facial surface. Inclusion of distofacial caries often indicates a gingival margin that follows the gingival tissue level. The width of the shoulder should be approximately 1 mm (or 0.5 mm inside the DEJ), whichever is greater. Some resistance form is provided by the shoulder. A retention groove should be placed in the axiofacial line angle of this distofacial extension, similar to the grooves placed in the proximal boxes. For additional retention a slot may be placed, as discussed in the section on Clinical Technique for Complex Amalgam Restorations.

Mandibular First Molar

The distal cusp on the mandibular first molar may be weakened when positioning the distofacial wall and margin. Facial extension of the distofacial margin to clear the distal contact often places the occlusal outline in the center of the cusp; this dictates relocation of the margin to provide a sound enamel wall and 90-degree amalgam that is not on a cuspal eminence. When the distal cusp is small or weakened or both, extension of the distal gingival floor and distofacial wall to include the distal cusp places the margin just mesial to the distofacial groove. Fig. 10.83 illustrates the ideal distofacial extension and a preparation design that includes the distal cusp.



• **Fig. 10.82** Mesioocclusodistofacial preparation of the maxillary second molar showing extension to include moderate to extensive caries lesion in the distal half of the facial surface. The outline includes the distofacial cusp and the facial groove. The dotted line represents the soft tissue level.

Modifications in Tooth Preparation for Proximal Surfaces

Slot Preparation for Root Caries

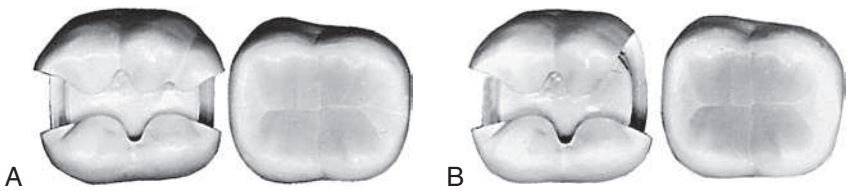
Older patients who have gingival recession that exposes cementum (or dentin) may experience caries lesion development on the proximal root surface that is appreciably gingival to the proximal contact (Fig. 10.84A). Assuming that the contact does not need restoring, the tooth preparation usually is approached from the facial direction and has the form of a slot (see Fig. 10.84B and E). A lingual approach is used when the caries is limited to the linguoproximal surface. Amalgam is particularly indicated for slot preparations if isolation is difficult.[5]

The initial outline form is prepared from a facial approach with a No. 2 or No. 4 bur using high speed and air-water spray. Outline form extension to sound tooth structure is at a limited depth axially (i.e., 0.75–1 mm at the gingival aspect [if no enamel is present], increasing to 1–1.25 mm at the occlusal wall [if the margin is in enamel]) (see Fig. 10.84B). If the occlusal margin is in enamel, the axial depth should be 0.5 mm inside the DEJ. During this extension, the bur should not remove any soft dentin from the axial wall deeper than the outline form initial depth. The remaining soft dentin (if any) is removed during final tooth preparation (see Fig. 10.84C). The external walls should form a 90-degree cavosurface angle. With a facial approach, the lingual wall should face facially as much as possible; this aids condensation of amalgam during insertion. The facial wall must be extended to provide access and visibility (convenience form) (see Fig. 10.84D). In the final tooth preparation, the No. 2 or No. 4 bur should be used to remove any remaining soft dentin on the axial wall. A liner or base (or both) is applied, as indicated.

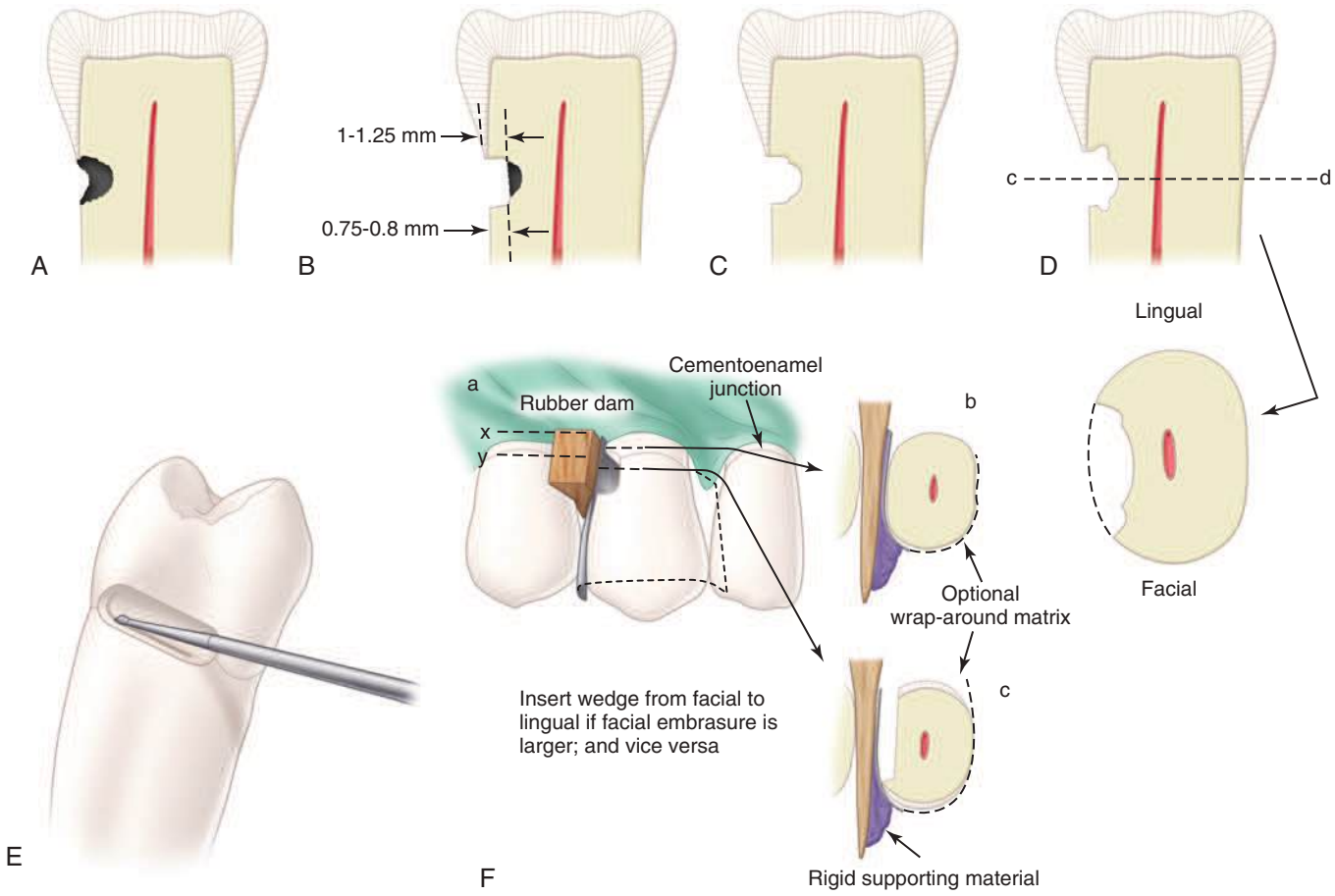
A No. ¼ bur is used to create retention grooves in the occlusoaxial and gingivoaxial line angles, 0.2 mm inside the DEJ or 0.3 to 0.5 mm inside the cemental cavosurface margin (see Fig. 10.84D and E). The depth of these grooves is ~0.25 to 0.5 mm and the bur is directed to bisect the angle formed by the junction of the occlusal (or gingival) and axial walls. Ideally, the direction of the occlusal groove is slightly more occlusal than axial, and the direction of the gingival groove would be slightly more gingival than axial. A dentin desensitizer may be placed before or after application of the matrix. The matrix for the slot preparation is illustrated in Fig. 10.84F.

Rotated Teeth

Tooth preparation for rotated teeth follows the same principles as for normally aligned teeth. The outline form for a mesioocclusal tooth preparation on the rotated mandibular second premolar



• **Fig. 10.83** Mandibular first molar. A, Ideal distofacial extension. B, Entire distal cusp included in preparation outline form.



• **Fig. 10.84** Slot preparation. A, Mesiodistal longitudinal section illustrating a caries lesion. The proximal contact is not involved. B, Initial tooth preparation. C, Tooth preparation with soft carious dentin removed. D, The retention grooves are shown in longitudinal section, and the transverse section through plane *cd* illustrates the contour of the axial wall and the direction of the facial and lingual walls. E, Preparing the retention form to complete the tooth preparation. F, Matrix for slot preparation: *(a)* facial view of wedged matrix; *(b)* wedged matrix as viewed in transverse cross section *(x),* gingival to gingival floor; *(c)* wedged matrix as viewed in transverse cross section *(y),* occlusal to gingival floor.

(Fig. 10.85A) differs from normal in that its proximal box is displaced facially because the proximal caries lesion involves the mesiofacial line angle of the tooth crown. When the tooth is rotated 90 degrees and the "proximal" lesion is on the facial or lingual surface, and orthodontic correction is declined or ruled out, the preparation may require an isthmus that includes the cuspal eminence (see Fig. 10.85B). If the lesion is small, consideration should be given to slot preparation. In this instance, the occlusal margin may be in the contact area or slightly occlusal to it (see Fig. 10.85C).

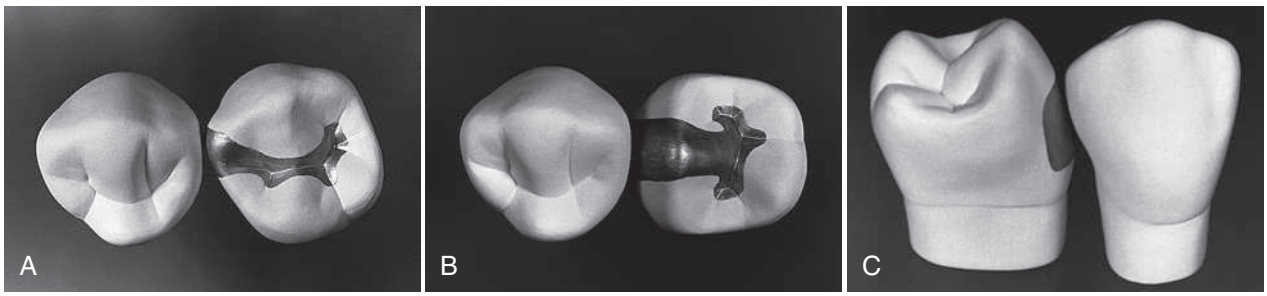
Unusual Outline Forms

Outline forms should conform to the restoration requirements of the tooth and not to the classic example of a Class II tooth preparation. As mentioned earlier, a dovetail feature is not required in the occlusal step of a single proximal surface preparation unless a fissure emanating from the occlusal step is involved in the preparation. Another example is an occlusal fissure that is segmented by coalesced enamel (as illustrated previously for mandibular premolars and the maxillary first molars). This condition should be treated with individual amalgam restorations if the preparations are separated by approximately 0.5 mm or more of sound tooth structure (Fig. 10.86).[54,104]

Adjoining Restorations

It is permissible to repair or replace a defective portion of an existing amalgam restoration if the remaining portion of the original restoration has adequate resistance and retention forms. Adjoining restorations on the occlusal surface occur more often in molars because the dovetail of the new restoration usually may be prepared without eliminating the dovetail of the existing restoration. Where the two restorations adjoin, care should be taken to ensure that the outline of the second restoration does not weaken the amalgam margin of the first (Fig. 10.87A). The intersecting margins of the two restorations should be at a 90-degree angle as much as possible. The decision to join two restorations is based on the assumption that the first restoration, or a part of it, does not need to be replaced and that the procedure for the single proximal restoration (compared with a mesioocclusodistal restoration) is less complicated and conserves tooth structure.

Occasionally, preparing an amalgam restoration in two or more phases is indicated, such as for a Class II lesion that is contiguous



• **Fig. 10.85** Restoration outlines for rotated teeth. A, Mesioocclusal outline for the mandibular premolar with 45-degree rotation. B, Mesioocclusal outline for the mandibular premolar with 90-degree rotation. C, Slot preparation outline for the restoration of a small mesial lesion involving the proximal contact of the mandibular premolar with 90-degree rotation.



• **Fig. 10.86** Restoration of the mesioocclusal tooth preparation, with the central fissure segmented by coalesced enamel.

with a Class V lesion. Preparing both lesions before placing amalgam introduces condensation problems that may be eliminated by preparing and restoring the Class II lesion before preparing and restoring the Class V lesion (see Fig. 10.87B). It is better to condense amalgam against a carious wall of the first preparation than to attempt condensation where no wall exists.

Abutment Teeth for a Removable Partial Denture

When the tooth is an abutment for a removable partial denture (RPD), the occlusoproximal outline form adjacent to the edentulous region may need additional extension if a rest seat is planned. Additional extension must be sufficient facially, lingually, and axially to allow for the rest seat to be in the proximoocclusal restoration without jeopardizing its strength. The facial and lingual proximal walls and the respective occlusal margins must be extended so that the entire rest seat may be prepared in the amalgam without encroaching on the restoration margins. If the rest seat is to be within the amalgam, it is recommended that a minimum of 0.5 mm of amalgam be present between the rest seat and the margins (Fig. 10.88A). The portion of the pulpal wall apical to the planned rest seat is deepened 0.5 mm so that the total depth of the axiopulpal line angle measured on the faciolingual wall is 2.5 mm (see Fig. 10.88B). A rest seat used for a tissue-borne (i.e., distal extension) partial denture may involve amalgam and enamel (see Fig. 10.88C). In this case, no modification of the outline form of the tooth preparation is indicated (see Fig. 10.88C). Note Fig. 10.88D illustrates the relationship of the tissue-borne RPD with the abutment tooth (see Fig. 10.88C).

All aspects of the Class II preparation, regardless of various levels of complexity, should be accomplished with precision. The preparation should then be cleaned, carefully inspected as indicated in General Concepts Guiding Preparation for Amalgam Restorations, and any final modifications completed.

Restorative Technique for Class II Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed on the prepared tooth structure per manufacturer instructions. Excess moisture is removed without desiccating the dentin.

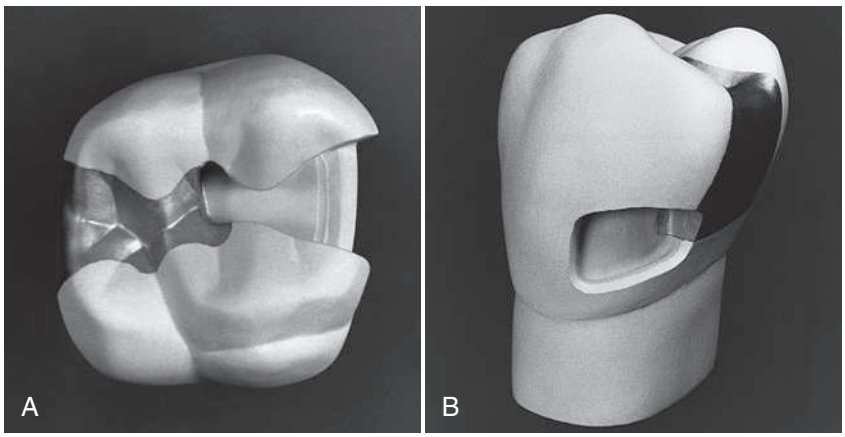
Matrix Placement

The primary function of the matrix is to enable proper restoration of anatomic contours and contact areas. The qualities of a good matrix include (1) rigidity, (2) establishment of proper anatomic contour, (3) restoration of correct proximal contact relationships, (4) prevention of gingival excess, (5) convenient application, and (6) ease of removal. The following information presents the technique of placement for the universal (Tofflemire) and precontoured matrix systems.

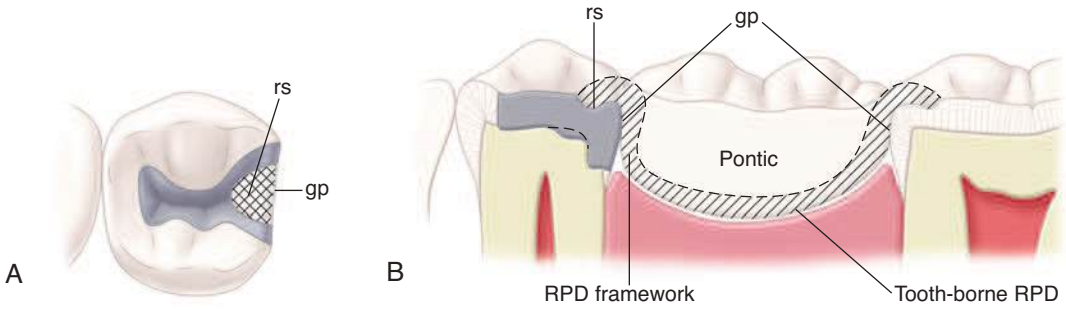
Universal Matrix

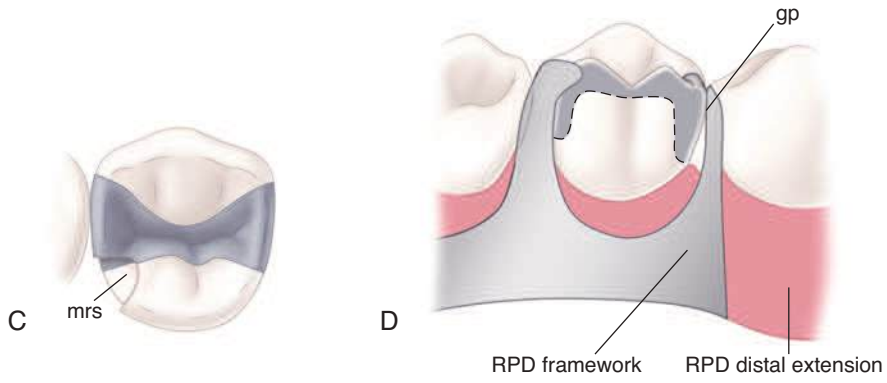
The universal matrix system (designed by B.R. Tofflemire) is ideally indicated when three surfaces (i.e., mesial, occlusal, distal) of a posterior tooth have been prepared (Fig. 10.89). This system is also commonly used for the two-surface Class II restoration. A definite advantage of the Tofflemire matrix retainer is that it may be positioned on the facial or lingual aspect of the tooth. However, lingual positioning requires the contra-angled design of the retainer (which may be used on the facial aspect as well) (Fig. 10.90; also see Fig. 10.89). The retainer and the band are generally stable when in place. The retainer is separated easily from the band to expedite removal of the matrix system from the tooth. Matrix bands of various occlusogingival widths are available (see Fig. 10.89). A small Tofflemire retainer is available for use with the primary dentition.

The universal matrix band itself does not meet all the requirements of an ideal matrix band. The conventional, flat universal matrix band must be shaped (i.e., burnished) to reproduce natural anatomic contour and resultant proximal contact. Uncontoured bands are available in two thicknesses, 0.05 mm and 0.038 mm. Burnishing the thinner band to create contour is more difficult, and the band is less likely to retain its contour when tightened around the tooth. The uncontoured band must be burnished before assembling the matrix band and the retainer. Burnishing must



• **Fig. 10.87** Adjoining restorations. A, Adjoining mesioocclusal tooth preparation with distoocclusal restoration so that the new preparation does not weaken the amalgam margin of the existing restoration. B, Preparing and restoring a Class II lesion before preparing and restoring a Class V lesion contiguous with it eliminates condensation problems that occur when both lesions are prepared before either is restored.

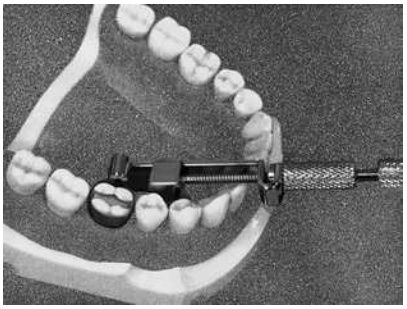




• **Fig. 10.88** Abutment teeth with Class II restorations designed for a removable partial denture (RPD). A, Occlusal view showing the location of the rest seat *(rs)* and the guide plane *(gp)* for a tooth-borne RPD. B, Cross-sectional view illustrating deepened pulpal wall in the area of the rest seat *(rs)* to provide adequate thickness of amalgam. Note the relationship of the guide planes *(gp)* to the tooth-borne RPD. C, Occlusal view showing the mesial rest seat *(mrs)* for the tissue-borne (i.e., distal extension) RPD prepared partly in a previous MOD amalgam restoration and partly in natural tooth structure. D, Lingual view of a distal extension RPD showing the relationship of the RPD to the Class II restoration and the guide plan (*gp*). Note that the figure shows a M-D cross section through the MOD amalgam and the occlusal portion of the facial aspect of the distal marginal ridge is visible.



• **Fig. 10.89** Straight (A) and contra-angled (B) universal (Tofflemire) retainers. Bands with varying occlusogingival widths are available.



• **Fig. 10.90** Lingual positioning requires a contra-angled universal retainer.

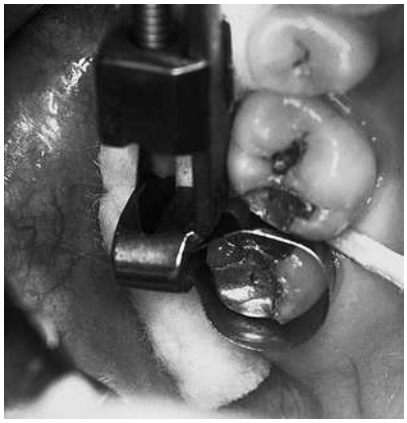
occur in the areas corresponding to the proximal surface to be restored after the band is positioned around the tooth.

Precontoured bands for the universal retainer are commercially available and need little or no adjustment before being placed in the retainer or after being positioned around the tooth (Fig. 10.91). Although precontoured bands are more expensive, they generally are preferred over the uncontoured bands. The difference in cost may be justified as less chair time is required during the restorative phase. When cotton roll isolation is used, the Tofflemire retainer helps hold the cotton roll in place (Fig. 10.92).

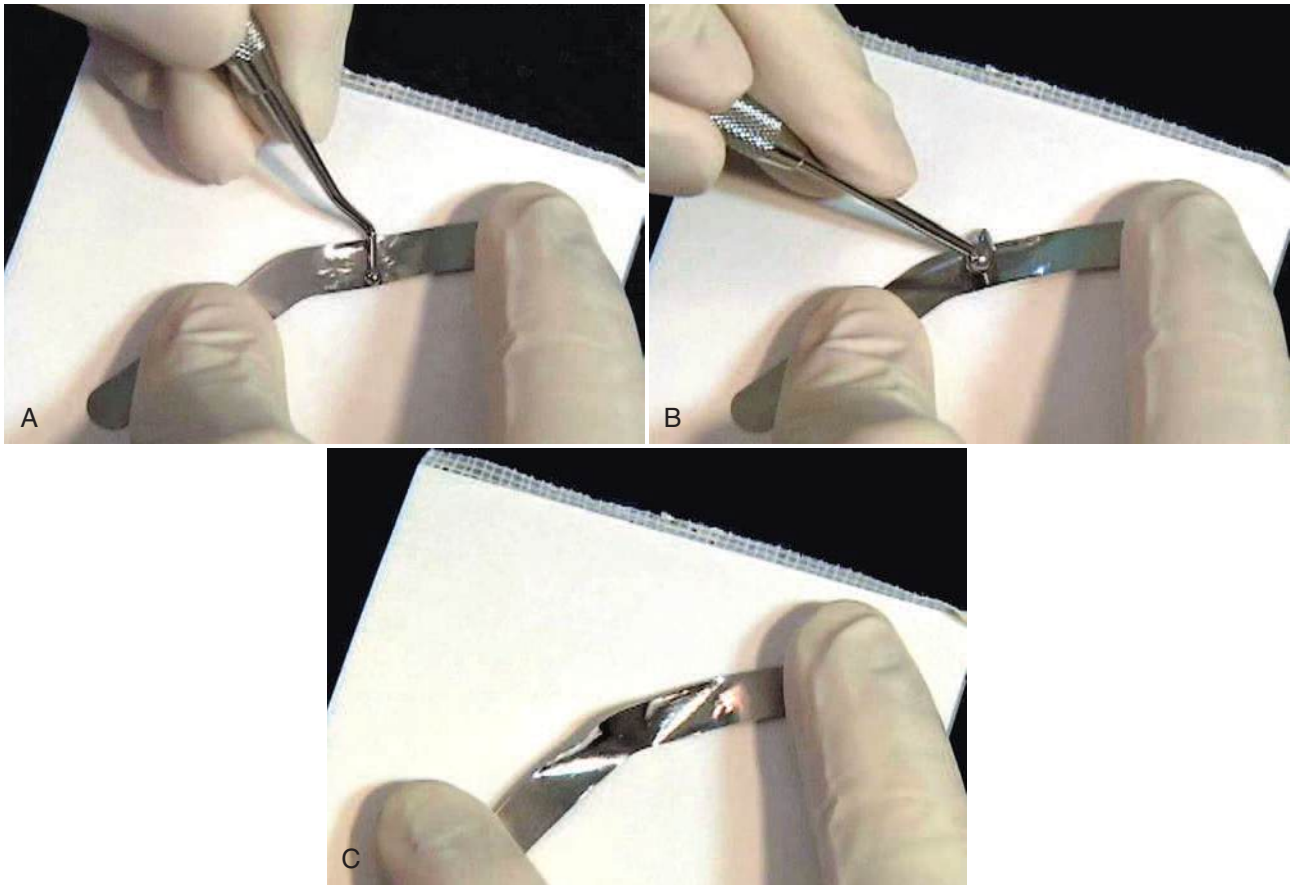
"Burnishing the matrix band" means that the metal band is intentionally deformed occlusogingivally with a suitable hand instrument to produce a rounded or convex surface that (when in place around the tooth) produces a restoration that is symmetric in contour with the adjacent proximal surface (Fig. 10.93). The No. 26-28 burnisher is generally recommended for burnishing the band. The band should be placed on a resilient paper pad because distortion of the metal matrix will not occur on a hard, nonresilient surface. The smaller round burnisher tip should first be used with firm pressure in back-and-forth, overlapping strokes along the length of the band until the band is deformed occlusogingivally in the appropriate areas. When the band is deformed, the larger egg-shaped end may be used to smooth the distorted band. If a convex surface is not obvious in the burnished areas when the band is removed from the pad, it has not been adequately burnished. The band may be burnished with the larger egg-shaped burnisher only, but more work is required to do so. It is not necessary to burnish the entire length of the band; focus effort only on distorting



• **Fig. 10.91** Precontoured bands for a universal retainer. Pictured: Waterpik Getz Contour Matrix Bands. (Courtesy Water Pik Inc., Fort Collins, CO.)



• **Fig. 10.92** Tofflemire retainer maintaining a cotton roll in the maxillary vestibule during condensation.



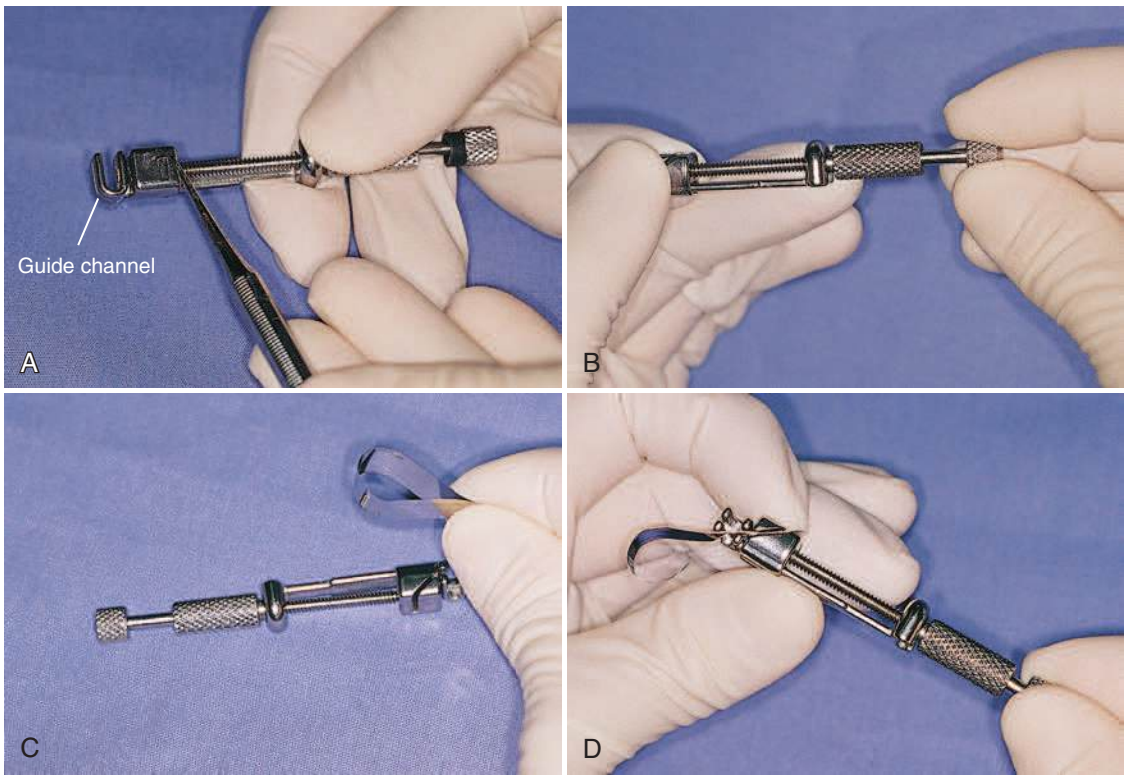
• **Fig. 10.93** Burnishing the matrix band. A, With the band on the pad, a small burnisher is used to deform the band. B, A large burnisher used to smooth the band contour. C, Burnished matrix band for mesiooccl usodistal tooth preparation. (Courtesy Aldridge D. Wilder, DDS.)

the matrix where the proximal contact areas are planned to occur.

To prepare the retainer to receive the band, the larger of the knurled nuts is turned counterclockwise until the locking vise is positioned adjacent to the guide channel on the end of the retainer (Fig. 10.94A). Next, while holding the large nut, the dentist turns the smaller knurled nut counterclockwise until the pointed spindle is free of the slot in the locking vise (see Fig. 10.94B). The matrix band is folded end to end, forming a loop (see Fig. 10.94C). When the band is folded, the gingival edge has a smaller circumference as compared with the occlusal edge so as to accommodate the difference in tooth circumferences at the gingival and, the more occlusal, contact levels. The band is positioned in the retainer so that the slotted side of the retainer is directed gingivally to permit easy subsequent separation (in an occlusal direction) of the retainer from the band after the restoration has been inserted. This is accomplished by placing the occlusal edge of the band in the correct guide channel (i.e., right, left, or parallel to the long axis of the retainer), depending on the location of the tooth. The two ends of the band are placed in the slot of the locking vise so that the ends are aligned with the edge of the vice, and the smaller of the knurled nuts is turned clockwise to tighten the pointed spindle against the band within the locking vice (see Fig. 10.94D). If proximal wedges were used during tooth preparation, the wedges are removed at this point and the matrix band is fitted around the tooth (allowing the gingival edge of the band to be positioned at least 1 mm apical to the gingival margin). Damaging the gingival attachment with the sharp edge of the matrix band should be avoided. Care should be taken not to trap the rubber dam between the band and the gingival margin. If the dam material is trapped between the band and the tooth, the septum of the dam should be stretched and depressed gingivally to reposition the dam material. Once the matrix band is in place, the larger knurled nut is rotated clockwise to tighten the band slightly. The explorer should be used along the gingival margin to determine that the gingival edge of the band extends beyond the preparation margins. When the band is correctly positioned, the band is securely tightened around the tooth.

When one of the proximal margins is deeper gingivally, the Tofflemire mesioocclusodistal band may be modified to prevent damage to the gingival tissue or attachment on the more shallow side. The band may be trimmed for the shallow gingival margin, permitting the matrix to extend farther gingivally for the deeper gingival margin (Fig. 10.95).

The mouth mirror is positioned lingually to observe the proximal contours of the matrix through the interproximal space (Fig. 10.96). The occlusogingival contour should be convex, with the height of contour at proper contact level such that the metal band is touching the adjacent proximal surface in the precise location where the contact should occur. The matrix is also observed from an occlusal aspect allowing evaluation of the position of the contact area in a faciolingual direction. If proper proximal contour and contact is not observed, it may be necessary to remove the retainer and reburnish the band. Minor alterations in contour and contact may be accomplished without removal from the tooth. The backside of the blade of the 15-8-14 spoon excavator (i.e., Black spoon) is



• **Fig. 10.94** Positioning the band in a universal retainer. A, Explorer pointing to the locking vise located immediately adjacent to the guide channel. B, The pointed spindle is released from the locking vise when the small knurled nut is turned counterclockwise. C, The band is folded to form a loop and to be positioned in the retainer (occlusal edge of band first). The gingival aspect of the vise is shown in this view. D, The spindle is tightened against the band in the locking vise. (From Daniel SJ, Harfst SA, Wilder RS: *Mosby's dental hygiene: Concepts, cases, and competencies*, ed 2, St. Louis, Mosby, 2008.)

excellent for improving contour in the area where the proximal contact is to occur. If a smaller burnishing instrument is used, the dentist should use care not to create a grooved or bumpy surface that would result in a restoration with an irregular proximal surface. Ideally the band should be positioned 1 mm apical to the gingival margin or deep enough to be engaged by the wedge (whichever is less) and 1 to 2 mm above the adjacent marginal ridge(s) to allow for adequate condensation of the amalgam in the marginal ridge areas. A minor modification of the matrix may be indicated for restoring the proximal surface that is planned for a guide plane for a RPD.



• **Fig. 10.95** The band may be trimmed for the shallower gingival margin, permitting the matrix to extend farther gingivally for the deeper gingival margin on the other proximal surface.

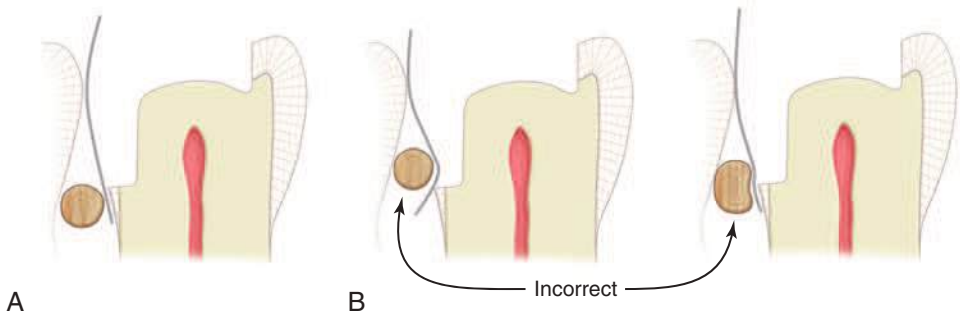


• **Fig. 10.96** Using a mirror from the facial or lingual position to evaluate the proximal contour of the matrix band.

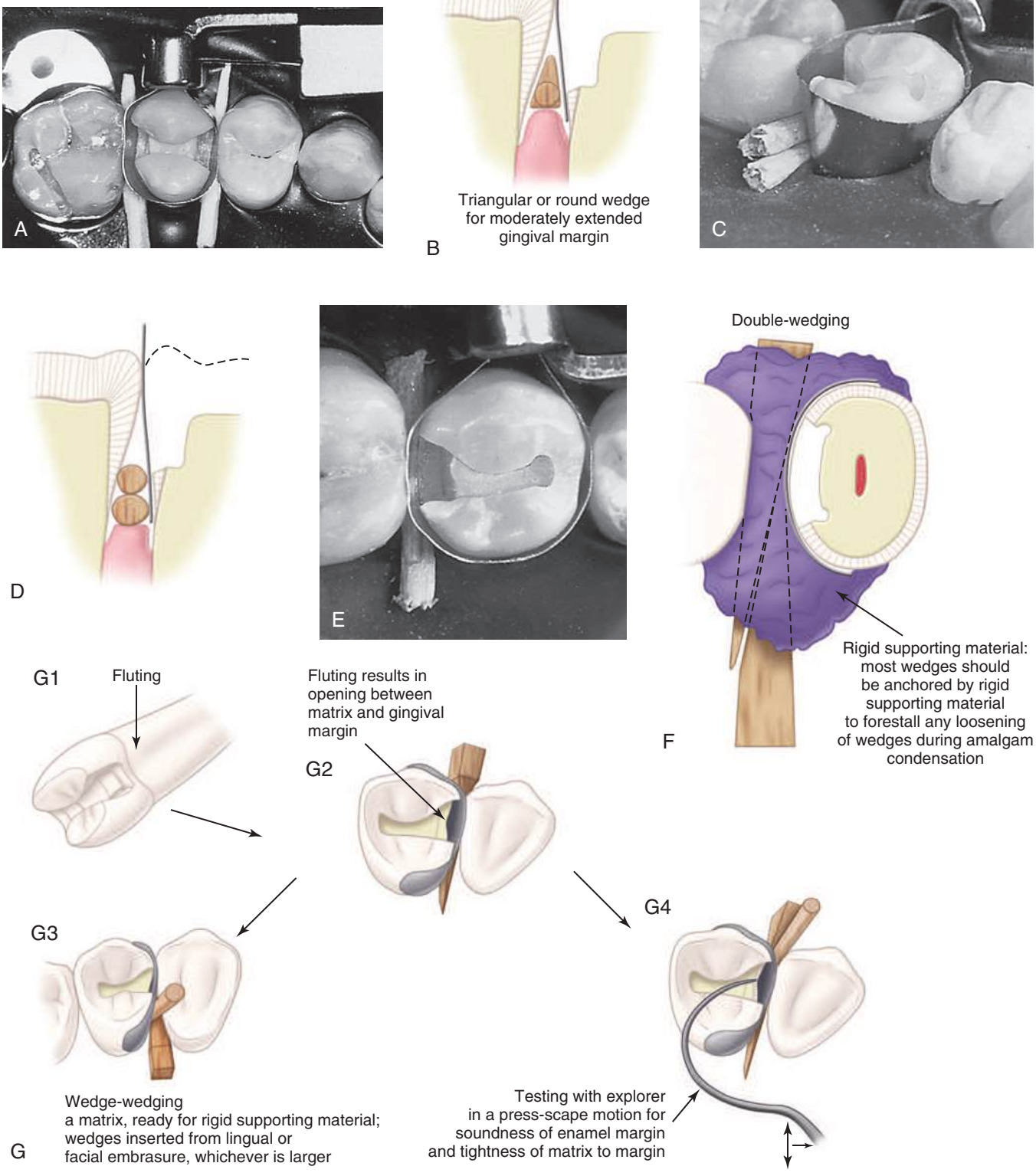
After the matrix contour and extension are evaluated, a wedge is placed in the gingival embrasure(s) using the following technique: (1) Break off approximately 1.2 cm of a round toothpick; (2) grasp the broken end of the wedge with the No. 110 pliers; (3) insert the pointed tip from the lingual or facial embrasure (whichever is larger), slightly gingival to the gingival margin; and (4) wedge the band tightly against the tooth and margin (Fig. 10.97A). The wedge may be lightly moistened with water or a water-soluble lubricant to facilitate its placement. If the wedge is placed occlusal to the gingival margin, the band is pressed into the preparation, which will result in an abnormal concavity in the proximal surface of the restoration (see Fig. 10.97B). The wedge should not be so far apical to the gingival margin that the band will not be held tightly against the gingival margin. This improper wedge placement results in gingival excess of amalgam (i.e., amalgam "overhang") caused by the band moving slightly away from the margin during condensation of the amalgam. Such an overhang often goes undetected and may result in irritation of the gingiva or an area of plaque accumulation, which may increase the risk of secondary caries. If the wedge is significantly apical of the gingival margin, a second (usually smaller) wedge may be placed on top of the first to wedge adequately the matrix against the margin (Fig. 10.98). This type of wedging is particularly useful for patients whose interproximal tissue level has receded.

The gingival wedge should be tight enough to prevent any possibility of an overhang of amalgam in at least the middle two thirds of the gingival margin (see Fig. 10.98A and B). Occasionally, double wedging is permitted (if access allows), securing the matrix when the proximal box is wide faciolingually. *Double wedging* refers to using two wedges: one from the lingual embrasure and one from the facial embrasure (see Fig. 10.98E and F). Two wedges help ensure that the gingival corners of a wide proximal box may be properly condensed; they also help minimize gingival excess. Double wedging should be used only if the middle two thirds of the proximal margins are not able to be adequately wedged. Any amalgam excess that forms in the facial and lingual corners is accessible to carving, and therefore proper wedging to prevent gingival excess of amalgam in the middle two thirds of the proximal box (see Fig. 10.98B) is the primary objective.

Occasionally a concavity may be present on the proximal surface that is adjacent to the gingival margin. This may occur on a surface with a fluted root, such as the mesial surface of the maxillary first premolar (see Fig. 10.98G1). A gingival margin located in this area may be concave (see Fig. 10.98G2). To wedge a matrix band tightly against such a margin, a second pointed wedge may be inserted between the first wedge and the band (see Fig. 10.98G3 and G4).

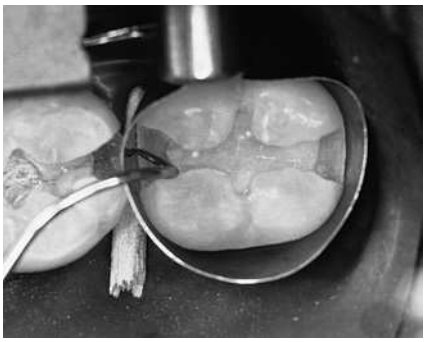


• **Fig. 10.97** A, Correct wedge position. B, Incorrect wedge positions.



• **Fig. 10.98** Various double-wedging techniques. A and B, Proper wedging for the matrix for a typical mesioocclusodistal preparation. C and D, Technique to allow wedging near the gingival margin of the preparation when the proximal box is shallow gingivally, the interproximal tissue level has receded, or both. E and F, Double wedging may be used with faciolingually wide proximal boxes to provide maximal closure of the band along the gingival margin. G, Another technique may be used on the mesial aspect of the maxillary first premolars to adapt the matrix to the fluted (i.e., concave) area of the gingival margin *(G1, G2)*; a second wedge inserted from the lingual embrasure *(G3)*; testing the adaptation of the band after insertion of the wedges from the facial aspect *(G4)*.

The wedging action between teeth should provide enough separation of adjacent teeth to compensate for the thickness of the matrix band. This ensures a positive contact relationship after the matrix is removed (after the condensation and initial carving of amalgam). If a Tofflemire retainer is used to restore a two-surface Class II preparation, the single wedge must provide enough separation to compensate for two thicknesses of band material. The tightness of the wedge is tested by pressing the tip of an explorer firmly at several points along the middle two thirds of the gingival margin (against the matrix band) to verify that it cannot be moved away from the gingival margin (Fig. 10.99). As an additional test, the dentist attempts to remove the wedge (using the explorer with moderate pressure) after first having set the explorer tip into the wood near the broken end. Moderate pulling should not cause dislodgment. Often the rubber dam has a tendency to loosen the wedge through rebounding of the dam stretched during wedge placement. Lubrication of the wedge and stretching the interproximal dam septa, in the direction opposite to the direction of wedge placement, before and during wedge placement may prevent rebound dislodgment. The stretched dam is released after the wedge is inserted.

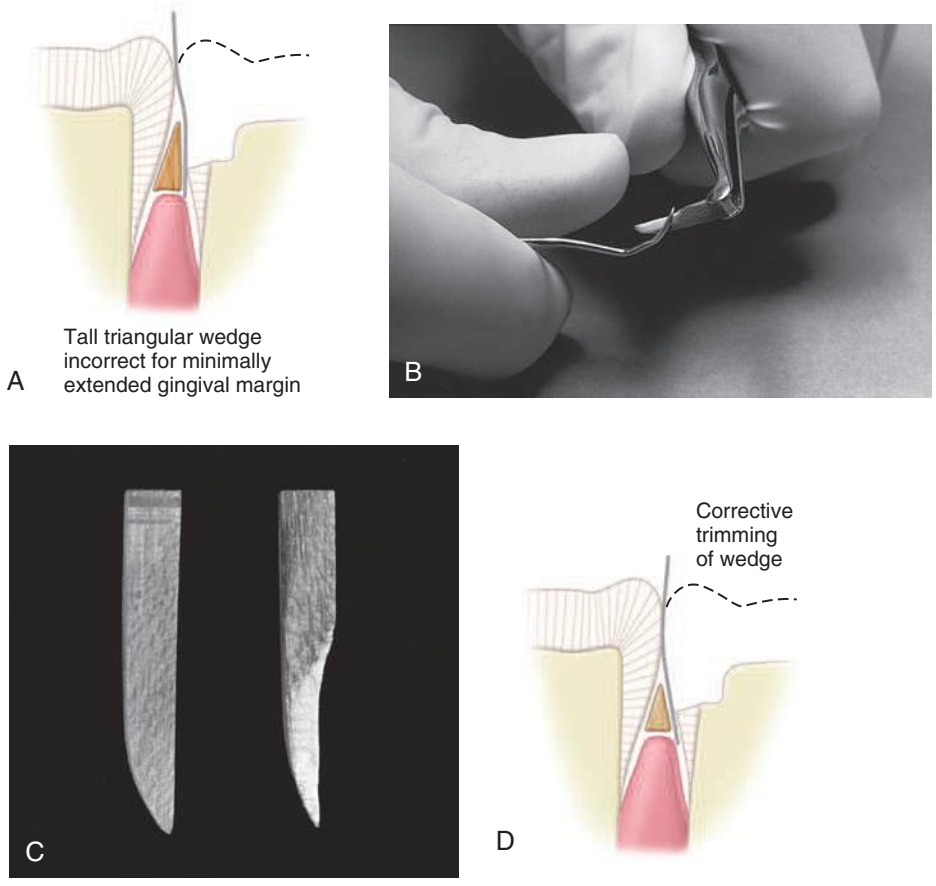


• **Fig. 10.99** Use of the explorer tip (with pressure) to ensure proper adaptation of the band to the gingival margin.

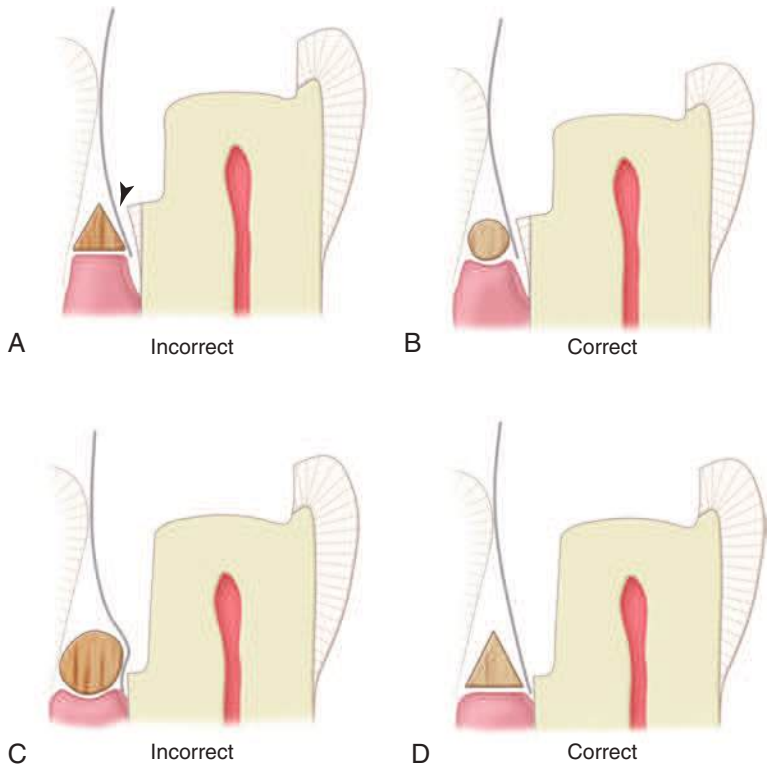
Some situations may require a triangular wedge that is modified (by knife or scalpel blade) to conform to the approximating tooth contours (Fig. 10.100). The round toothpick wedge is usually the wedge of choice with conservative proximal boxes, however, because its wedging action is more occlusal (i.e., nearer the gingival margin) than with the triangular wedge (Fig. 10.101A and B).

The triangular (i.e., anatomic) wedge is recommended for a preparation with a deep gingival margin. The triangular wedge is positioned similarly to the round wedge, and the goal is the same. When the gingival margin is deep (cervically), the base of the triangular wedge more readily engages the tooth gingival to the margin without causing excessive soft tissue displacement. The anatomic wedge is preferred for deeply extended gingival margins because its greatest cross-sectional dimension is at its base (see Fig. 10.101C and D).

To maintain gingival isolation attained by an anatomic wedge placed before the preparation of a deeply extended gingival margin,



• **Fig. 10.100** Modified triangular (i.e., anatomic) wedge. A, Depending on the proximal convexity, a triangular wedge may distort the matrix contour. B, A sharp-bladed instrument may be used to modify the triangular steepness of the wedge. C, Modified and unmodified wedges are compared. D, Properly modified triangular wedge prevents distortion of the matrix contour.

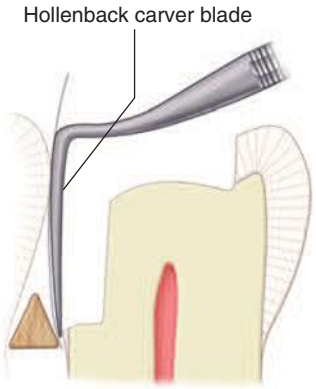


• **Fig. 10.101** Indications for the use of a round toothpick wedge versus a triangular (i.e., anatomic) wedge. A, As a rule, the triangular wedge does not firmly support the matrix band against the gingival margin in conservative Class II preparations *(arrowhead)*. B, The round toothpick wedge is preferred for these preparations because its wedging action is nearer the gingival margin. C, In Class II preparations with deep gingival margins, the round toothpick wedge crimps the matrix band contour if it is placed occlusal to the gingival margin. D, The triangular wedge is preferred with these preparations because its greatest width is at its base.

it may be appropriate to withdraw the wedge a small distance to allow passage of the band between the loosened wedge and the gingival margin. Tilting (i.e., canting) the matrix into place helps the gingival edge of the band slide between the loosened wedge and the gingival margin. The band is tightened, and the same wedge is firmly reinserted.

Supporting the matrix material with the blade of a Hollenback carver during the insertion of the wedge for the difficult deep gingival restoration may be helpful.[55] The tip of the blade is placed between the matrix and the gingival margin, and then the "heel" of the blade is leaned against the matrix and adjacent tooth (Fig. 10.102). In this position, the blade supports the matrix to help in positioning the wedge sufficiently gingivally and preventing the wedge from pushing the matrix into the preparation. After the wedge is properly inserted, the blade is gently removed.

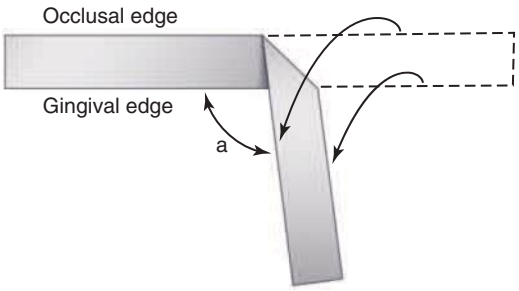
All aspects of the band and wedge are assessed, and any desired final corrections are made. The matrix band and wedge must provide appropriate proximal contact and contour (see Fig. 10.96). After wedging, relaxing the tension of the band by turning the larger knurled nut of the Tofflemire retainer a quarter turn counterclockwise will help to ensure proximal contact. If loosening the Tofflemire band still does not allow for contact with an adjacent tooth, a custom-made band with a smaller angle may be used. Reducing the angle of the band increases the difference in length (i.e., circumferences) of the gingival and occlusal edges. To reduce the angle of the band, the operator folds it as shown in Fig. 10.103,



• **Fig. 10.102** Supporting the matrix with the blade of a Hollenback carver during wedge insertion.

burnishes for appropriate occlusogingival contour (in the contact areas), and inserts the band into the Tofflemire retainer.

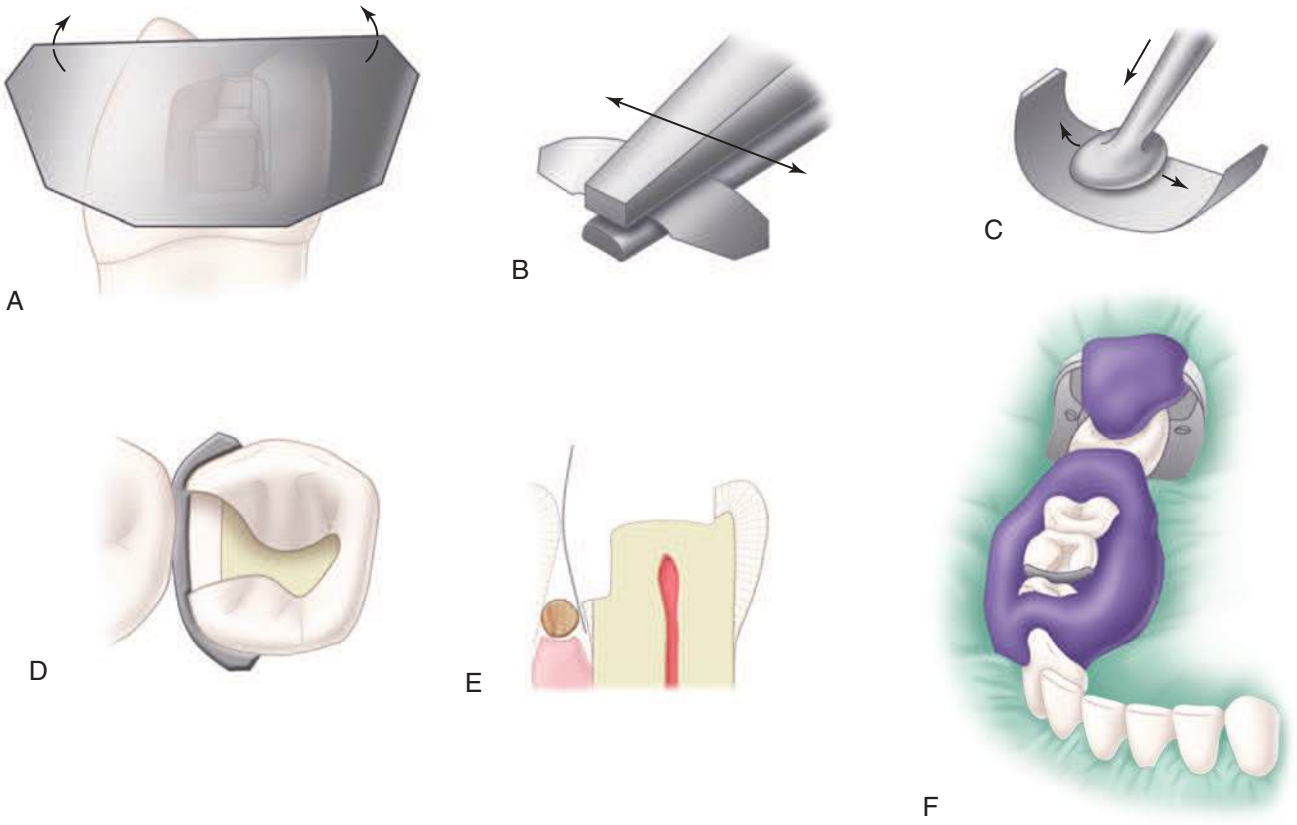
Occasionally the interdental spacing is so large that a suitably trimmed tongue blade is required to wedge the matrix (Fig. 10.104). Occasionally it is impossible to use a wedge to secure the matrix band. In this case the band must be sufficiently tight to minimize the gingival amalgam excess. Because the band is not wedged, special care must be exercised to place small amounts of amalgam



• **Fig. 10.103** The custom-made matrix strip is folded, as indicated by arrows. The smaller angle *(a)* compared with the angle of the commercial strip increases the difference between the lengths of the gingival and occlusal edges.



• **Fig. 10.104** A custom-made tongue blade wedge may be used when excessive space exists between adjacent teeth.



• **Fig. 10.105** Rigid material–supported sectional matrix. A, The shape of the stainless steel strip after trimming. B, The strip contoured to the circumferential contour of the tooth (fingers can be used). C, Burnishing the strip to produce occlusogingival contact contour (left and right arrows indicate the short, back-and-forth motion of the burnisher). D, Contoured strip in position. E, Matrix strip properly wedged. F, Completed rigid material–supported sectional matrix.

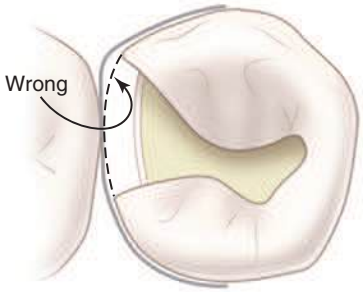
in the gingival floor and condense the first 1 mm of amalgam lightly but thoroughly in a gingival and lateral direction. The condensation of additional increments is then carefully continued in a gingival direction using a larger condenser with firm pressure. Condensation against an unwedged matrix may cause a gross extrusion of amalgam beyond the gingival margin. The excess amalgam must be removed with a suitable carver immediately after matrix removal.

Rigid-Material Supported Sectional Matrix

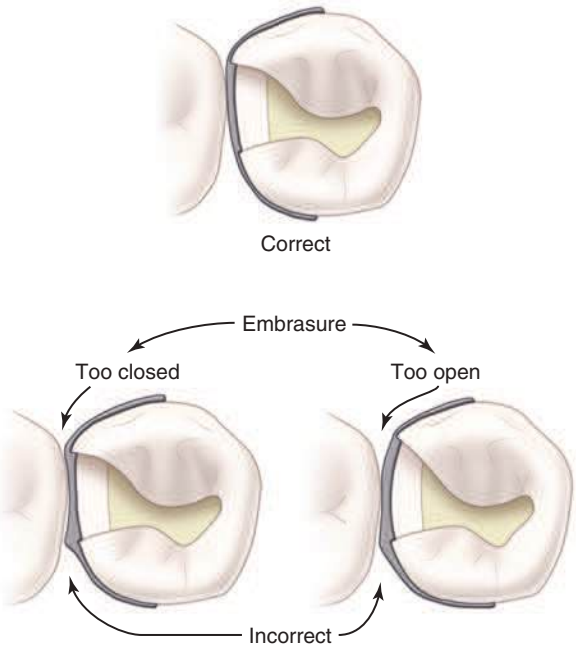
An alternative to the universal matrix band is the use of a properly contoured sectional matrix that is wedged and supported by a material that is rigid enough to resist condensation pressure. The supporting material selected must be easy to place and to remove. Examples include light-cured, thermoplastic, and quick-setting rigid polyvinyl siloxane (PVS) materials (Fig. 10.105). The gingival wedge is positioned interproximally to secure the band tightly at

the gingival margin to prevent any excess of amalgam (i.e., overhang). The wedge also separates teeth slightly to compensate for the thickness of the section of matrix material. The wedge is placed before the application of rigid supporting material.

The proximal surface contour of the matrix should allow the normal slight convexity between the occlusal and middle thirds of the proximal surface when viewed from the lingual (and facial) aspect. Proximal surface restorations often display an occlusogingival proximal contour that is too straight, thereby causing the contact relationship to be located too far occlusally (with little or no occlusal embrasure). This condition allows food impaction between teeth, with resultant injury to the interproximal gingiva and supporting tissues, and invites conditions leading to demineralization of the adjacent proximal surfaces. The proximal surface contour of the matrix should also provide the correct form to the proximal line angles, both facially and lingually. If these contours are not present, the facial and lingual embrasures of the restoration are too open, inviting food impaction and injury to underlying supporting tissues. Correct and incorrect contours and matrix correction steps are illustrated in Figs. 10.106 and 10.107.



• **Fig. 10.106** Alteration of the sectional matrix contour to provide the correct form to the proximofacial line angle region.



• **Fig. 10.107** Correct or incorrect facial and lingual embrasure form is determined by the shape of the sectional matrix strip.

The matrix should be tight against the facial and lingual margins on the proximal surface so that the amalgam may be well condensed at the preparation margins. In addition, when the matrix is tight against the tooth, minimal carving is necessary on the proximal margins after the matrix is removed.

Precontoured Sectional Matrix Strips

Commercially available sectional metal strips (e.g., Palodent System; DENTSPLY Caulk, Milford, DE) are precontoured and ready for application to the tooth (Fig. 10.108). These strips have limited application when used for amalgam because of their rounded contour. They usually are most suitable for mandibular first premolars and the distal surface of maxillary canines. The contact area of the adjacent tooth occasionally is too close to allow placement of the contoured Palodent strip without causing a dent in the strip's contact area, making it unusable.

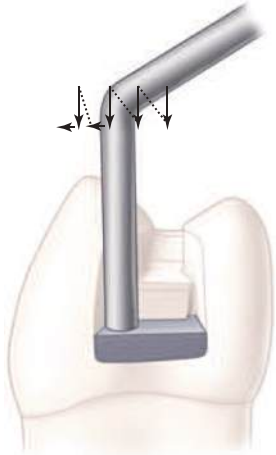
Condensation and Carving of the Amalgam

Proper condensation of amalgam results in close material adaptation to the preparation walls (limiting marginal voids) and minimal inclusion of internal voids. In addition, careful overstepping condensation technique eliminates excess mercury and results in a stronger restoration that is less prone to corrosion. Care should be taken to choose condensers that are best suited for use in each part of the tooth preparation and that may be used without binding in the preparation or between the preparation and the matrix band.

Condensation should occur in the proximal box(es) first by transfer of enough amalgam with the amalgam carrier to fill the gingival portion by about 1 mm. This first increment is condensed in a gingival and lateral direction with sufficient force to adapt the amalgam to the gingival floor. Additional amalgam is carefully condensed against the proximal margins of the preparation and into the proximal retention grooves (if present). Firm, facially and lingually directed condensation (i.e., lateral condensation) with the side of the condenser should be accomplished along with the application of gingivally directed force (Fig. 10.109).[118] Condensation strokes in a gingival and lateral direction help to ensure that no voids occur internally or at the faciogingival or linguogingival line angles. Mesial (or distal) condensation of the amalgam in the proximal box is accomplished to ensure proper contour and proximal contact with the adjacent tooth.



• **Fig. 10.108** Precontoured metal strips. (The Palodent System; courtesy of DENTSPLY Caulk, Milford, DE.)

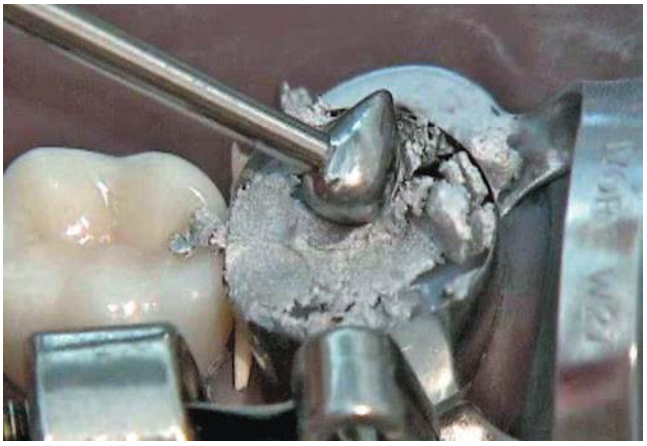


• **Fig. 10.109** Lateral and occlusogingival force is necessary to condense amalgam properly into the proximal grooves and into the angles at the junction of the matrix band and the margins of the preparation.

The procedure of adding and condensing continues until amalgam reaches the level of the pulpal wall. The size of the condenser is changed (usually to a larger one), if indicated, and amalgam is condensed in the remaining proximal portion of the preparation concurrently with the occlusal portion. It may be necessary to return to a smaller condenser when condensing in a narrow extension of the preparation or near the proximal margins. A smaller condenser face is more effective at condensing, provided it does not significantly penetrate the amalgam already in place. The occlusal margins are covered and overpacked by at least 1 mm using a large condenser, ensuring that the margins are well condensed, especially in the area of the marginal ridge.[119] Proper condensation of the amalgam in the marginal ridge area(s) reduces the risk of marginal ridge fracture during matrix removal. The plasticity and wetness of the amalgam mass should be monitored during condensation. Proper condensation requires that the mix should be neither wet (i.e., mercury rich) nor dry and crumbly (i.e., mercury poor). Amalgam that is beginning to set should be discarded and a new mix obtained to complete the condensation. Before carving procedures are initiated, precarve burnishing of the occlusal portion with a large egg-shaped or ball burnisher should be accomplished (Fig. 10.110).[79] As mentioned previously, precarve burnishing is a form of condensation.

Carving the Occlusal Portion

Carving of the occlusal portion (including the occlusal embrasure or embrasures) may be accomplished with the matrix assembly in place while the amalgam is reaching its initial set. Early removal of the band increases the risk of marginal ridge fracture. Careful carving of the occlusal portion should begin immediately after condensation and burnishing. Sharp discoid instruments of suitable size as well as the Hollenback carver may be effective at reestablishing the occlusal anatomy. The marginal ridge is carved confluent with the tooth's anatomy such that it duplicates the height and shape of the adjacent marginal ridge (see Fig. 10.18). An explorer or small Hollenback carver may be used to carefully define the occlusal embrasure. Remembering the pattern of occlusal contacts (that were evaluated before tooth preparation), observing the height of the adjacent marginal ridge, and knowing where the preparation cavosurface margins are located all aid in completing the carving of the occlusal surface, including the marginal ridge and occlusal



• **Fig. 10.110** Precarve burnishing with a large burnisher. (Courtesy Aldridge D. Wilder, DDS.)



• **Fig. 10.111** Using No. 110 pliers, the matrix band should be removed in a linguoocclusal *(arrow)* or facioocclusal direction (not just in an occlusal direction).

embrasure. If the restoration has extensive proximal involvement of the tooth, the occlusal carving should be accomplished quickly.

Removal of the Matrix Band and Completion of Carving

At the appropriate time, the matrix retainer first is removed from the band by turning the small knurled nut counterclockwise to retract the pointed spindle from the vice. The end of an index finger may be placed on the occlusal surface of the tooth and matrix band to provide stabilization while the retainer is removed. Next, any rigid supporting material, if used, is removed. The matrix band is removed next. The No. 110 pliers are used to tease the band free, from one contact area at a time, by pushing or pulling the band in a linguoocclusal (or facioocclusal) direction and, if possible, in the direction of wedge insertion (Fig. 10.111). The wedge may be left in place to provide separation of teeth while the matrix band is removed, and then subsequently removed. By maintaining slight interdental separation, the wedge reduces the risk of fracturing the newly condensed marginal ridge(s). Band removal in a straight occlusal direction should be avoided so as to limit the risk of marginal ridge fracture.

Matrix band removal allows access to carve the proximal portions of the restoration. Efficient use of time is required so as to gain access to the proximal areas while the amalgam remains carvable.

Proper matrix form should result in a proximal surface that is nearly completed, with proper contact evident and minimal carving required except to remove a possible small amount of excess amalgam at the proximal facial and lingual margins, at the faciogingival and linguogingival corners, and along the gingival margin. Amalgam knives (scalers, No. 34 and No. 35) are ideal for removing gingival amalgam excess (see Figs. 10.21 and 10.22). They also are ideal for refining the embrasure form around the proximal contacts (see Fig. 10.19). The secondary (or "back") edges on the blades of amalgam knives are occasionally helpful, using a push stroke. The Hollenback carver No. 3 and (occasionally) the side of the explorer may also be suitable instruments for carving these areas. However, the explorer may not refine the margins and contour as effectively as the amalgam knife if the set of the amalgam is close to completion. When the proximal carving is completed, the occlusal surface contouring is completed. Occasionally, occlusal contouring may require the use of an abrasive stone or finishing bur if the setting of the amalgam is nearing completion.

When carving the margins of the restoration, the cutting surface of the carving instrument is held in a perpendicular orientation. Carving should be parallel to the margins, however, with the adjacent tooth surface being used to guide the carver. The existence of the proximal contact is verified visually by using the mouth mirror. When carving is completed, the rubber dam is removed and the occlusion is assessed and adjusted, as needed.

Abutment teeth for a tooth-supported RPD must provide amalgam contour to allow defining (by carving or [later] disking) a guide plane extending from the marginal ridge 2.5 mm gingivally. Normal proximal contour, rather than leaving an overcontour, is usually sufficient, however, and best for the development of a guide plane. Guide plane development results in a gingival embrasure between the natural tooth and RPD framework that is less open and less likely to trap food (see Fig. 10.88B and D).

An amalgam restoration of a proximal abutment tooth for a tissue-supported (i.e., distal extension) RPD is carved to provide a guide plan and adequate gingival embrasure space (see Fig. 10.88D). This allows for protection of the distogingival tissue, associated with the distal abutment, when occlusal loading of the RPD distal extension causes compression of the alveolar ridge soft tissue.

Before the patient is dismissed, thin unwaxed dental floss may be passed through the proximal contact(s) once to remove any amalgam shavings on the proximal surface of the restoration and to assess the gingival margin. Passing the floss multiple times through the contact area of an amalgam that is still early in its set may result in removal of some surface amalgam and loss of contact with the adjacent tooth. When positioned in the gingival embrasure, the floss is wrapped around the restored tooth and positioned apical to the gingival margin of the restoration. The floss is moved in an occlusogingival direction. The floss not only removes amalgam shavings but also smooths the proximal amalgam and detects any gingival overhang of amalgam. If an overhang is detected, further use of an amalgam knife is necessary.[120] Alternatively, a finishing strip may be used to remove the amalgam excess and finish the proximal surface gingival to the contact area. Floss also may be used to verify that the intensity of the contact is similar to that of neighboring teeth. Final rinsing and evacuation of the oral cavity is then accomplished. The patient is advised to avoid chewing with the restored tooth for 24 hours.

Finishing and Polishing of the Amalgam

Finishing and polishing of the occlusal surface of the Class II amalgam should follow the approach presented in General Concepts Guiding Restoration with Amalgam. Finishing and polishing of the proximal surface may be indicated where the proximal amalgam is accessible. This area usually includes the facial and lingual margins and areas occlusal to the contact. The remainder of the proximal surface is often inaccessible; however, the matrix band should have imparted sufficient smoothness to it.

If amalgam along the facial and lingual proximal margins was slightly overcarved, the enamel margin may be felt as the explorer tip passes from amalgam across the margin onto the external enamel surface. Finishing burs or sandpaper disks, rotating at slow speed, may be used to smooth the enamel–amalgam margin. Sandpaper disks also may be used to smooth and contour the marginal ridge. Inappropriate use of sandpaper disks may "ledge" the restoration around the contact, however, resulting in inappropriate proximal contours.

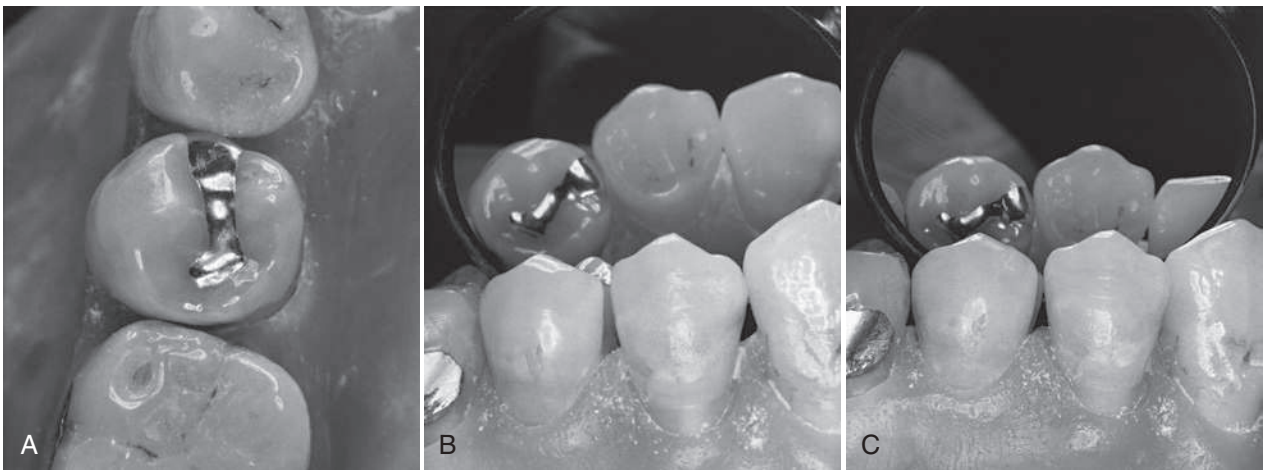
In small preparations, the facial and lingual proximal margins are generally inaccessible for finishing and polishing. Fine abrasive disks or the tip of a sharpened rubber polishing point or edge of a sharped abrasive rubber polishing cup may be used to polish the proximal portion that is accessible. When proximal margins are inaccessible to finishing and polishing with disks or rubber polishing points, and some excess amalgam remains (e.g., at the gingival corners and margins), amalgam knives occasionally may be used to trim amalgam back to the margin and to improve the contour, though a flame-shaped finishing bur may ultimately be required. Fig. 10.112 provides an example of a finished and polished Class II amalgam restoration.

Quadrant Dentistry

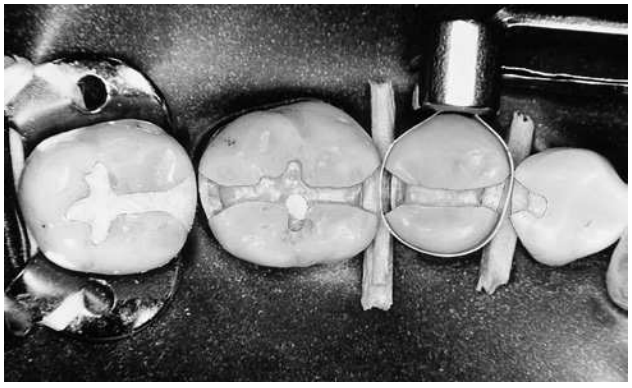
When several teeth in a quadrant have caries lesions, the experienced dentist usually treats them all at the same appointment rather than individually. Quadrant dentistry increases efficiency and reduces chair time for the patient. The use of the rubber dam is particularly important in quadrant dentistry. For maximal efficiency, when a quadrant of amalgam tooth preparations is planned, each rotary or hand instrument should be used on every tooth where it is needed before being exchanged.

When restoring a quadrant of Class II amalgam tooth preparations, it is permissible to apply matrix bands on alternate preparations in the quadrant and restore teeth two at a time. Banding adjacent preparations requires excessive wedging to compensate for a double thickness of band material and makes the control of proximal contours and interproximal contacts difficult. Extensive tooth preparations may need to be restored one at a time. If proximal boxes differ in size, teeth with smaller boxes should be restored first because often the proximal margins are inaccessible to carving if the larger adjacent box is restored first. In addition, smaller boxes may be restored more quickly and accurately because more tooth structure remains to guide the carver. If the larger proximal box is restored first, the gingival contour of the restoration may be damaged when the wedge is inserted to secure the matrix band for the second, smaller restoration. If the adjacent proximal boxes are similar in size, the banding of alternate preparations should be started with the most posterior preparation because this allows the patient to close slightly as subsequent restorations are inserted (Fig. 10.113).

Before restoring the second of two adjacent teeth, the proximal contour of the first restoration should be carefully established. Its anatomy serves as the guide to establish proper proximal contours leading to correct embrasure forms and contact position/size. If necessary, a finishing strip may be used to refine the contour of the first proximal amalgam (Fig. 10.114). The finishing strip is



• **Fig. 10.112** Polished mesiooccclusal amalgam restoration. Note the conservative extension. A, Occlusal view. B, Mesiofacial and occlusal views of the mesiofacial margin. C, Facial and occlusal views of the proximal surface contour and the location of the proximal contact.

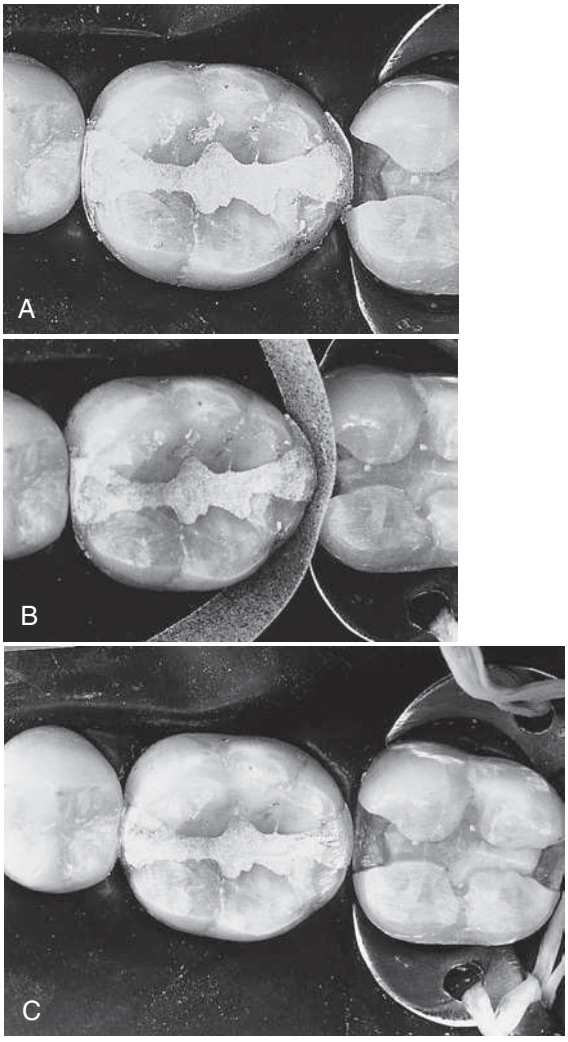


• **Fig. 10.113** Quadrant dentistry. Unless otherwise indicated, a quadrant of Class II preparations with similarly sized proximal boxes may be restored using two bands simultaneously if they are placed on every other prepared tooth. It is recommended that the most distal tooth be restored first.

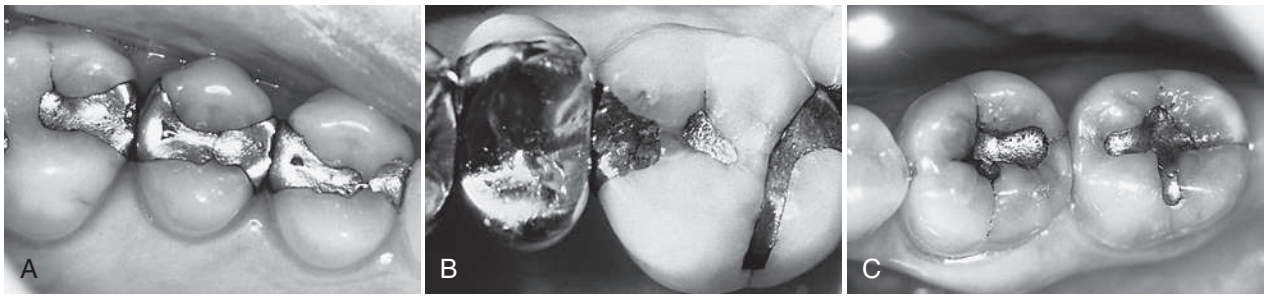
indicated, however, only where the proximal clearance is present. Using a finishing strip between contacting amalgam restorations may compromise or eliminate the proximal contact. Properly placed amalgam restorations have the potential to last for decades (Fig. 10.115).

Clinical Technique for Class III Amalgam Restorations

Class III restorations are indicated for defects located on the proximal surface of anterior teeth that do not affect the incisal edge. Part of the facial or lingual surfaces also may be involved. The Class III amalgam restoration is rarely used and has been supplanted by tooth-colored restorations (primarily composite), which have become increasingly wear resistant and color stable. For esthetic reasons, use of amalgam is best suited for caries lesions that may be accessed from the lingual rather than from the facial. The Class III amalgam restoration is generally reserved for the distal surface of maxillary and mandibular canines if (1) the preparation is extensive with only minimal facial involvement, (2)



• **Fig. 10.114** When the first of two adjacent Class II preparations is restored, proper contour may be established by using a finishing strip before the second is restored. A, Before using the strip. B, Removing the over-contour. C, Verification of proper contour may be accomplished by viewing the restoration with a mirror from the occlusal, facial, and lingual positions. The proximal contour also may be evaluated after matrix placement by noting the symmetry between the restored surface and the burnished matrix band.



• **Fig. 10.115** Clinical examples of long-term amalgam restorations: A, 44-year-old amalgams; B, 58-year-old amalgams in the first molar; C, 65-year-old amalgams in molars. (A and B, Courtesy Drs. John Osborne and James Summitt.)



• **Fig. 10.116** Restoration for Class III tooth preparation using facial approach on mandibular canine. The restoration is 5 years old. (Courtesy Dr. C.L. Sockwell.)

the gingival margin primarily involves cementum, and/or (3) moisture control is difficult.

Initial Procedures

Appropriate review of the patient record (including medical history), treatment plan, and radiographs should be accomplished. Anesthesia is usually necessary when a vital tooth is to be restored. The gingival extension of the preparation should be anticipated. Prewedging in the gingival embrasure of the proximal site to be restored provides (1) better protection of soft tissue and the rubber dam, (2) better access because of the slight separation of teeth, and (3) better reestablishment of the proximal contact. The use of a rubber dam is generally recommended; however, cotton roll isolation is also acceptable.

Tooth Preparation for Class III Amalgam Restorations

A lingual access preparation on the distal surface of the maxillary canine is described here because the use of amalgam in that location is more likely. A facial approach for a mandibular canine may be indicated, however, if the lesion is more facial than lingual. The mandibular restoration is often not visible at conversational distance (Fig. 10.116). The outline form of the Class III amalgam preparation may include only the proximal surface. A lingual dovetail may be indicated if one existed previously or if additional retention is needed for a larger restoration.

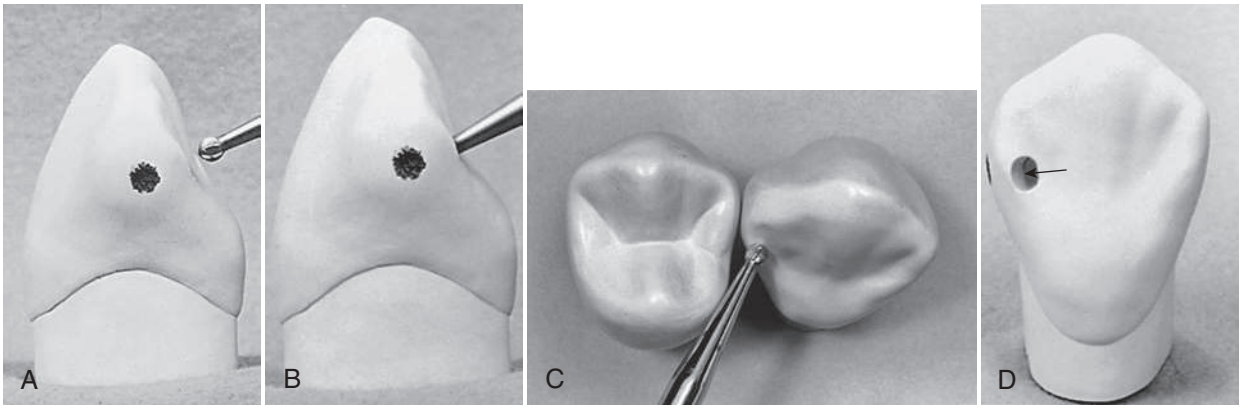
Initial Tooth Preparation

Bur size selection depends on the size of the lesion. Bur options may include a No. 2 (or smaller) round bur or a No. 330 bur. The bur is positioned so that the entry cut penetrates into the caries lesion, which is usually slightly gingival to the contact area. Ideally the bur is positioned so that its long axis is perpendicular to the lingual surface of the tooth, but directed at a mesial angle as close to the adjacent tooth as possible. (The bur position may be described as perpendicular to the distolingual line angle of the tooth.) This position conserves the marginal ridge enamel (Fig. 10.117A–C). Penetration through enamel positions the bur so that additional cutting isolates the proximal enamel affected by the caries lesion and removes some or all of the soft dentin. In addition, penetration should be at a limited initial axial depth (i.e., 0.5–0.6 mm) inside the DEJ (see Fig. 10.117C and D) or at a 0.75- to 0.8-mm axial depth when the gingival margin is on the root surface (in cementum/dentin) (Fig. 10.118). This 0.75-mm axial depth on the root surface allows a 0.25-mm distance (the diameter of the No. ¼ bur is 0.5 mm) between the retention groove (which is placed later) and the gingival cavosurface margin. Soft dentin that is deeper than this limited initial axial depth is removed later during final tooth preparation.

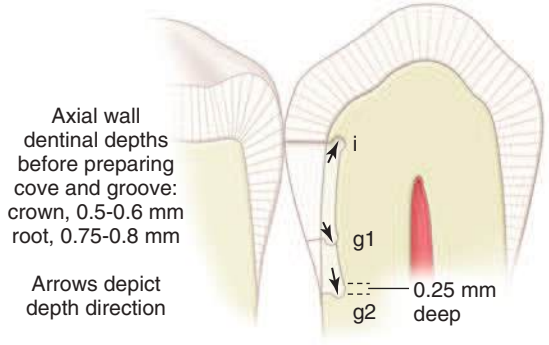
For a small lesion, the facial margin is extended 0.2 to 0.3 mm into the facial embrasure (if necessary), with a curved outline from the incisal to the gingival margin (resulting in a less visible margin). The lingual outline blends with the incisal and gingival margins in a smooth curve, creating a preparation with little or no lingual wall. The cavosurface angle should be 90 degrees at all margins. The facial, incisal, and gingival walls should meet the axial wall at approximately right angles (although the lingual wall meets the axial wall at an obtuse angle or may be continuous with the axial wall) (Figs. 10.119 and 10.120). If a large round bur is used, the internal angles are more rounded. The axial wall should be uniformly deep into dentin and follow the faciolingual contour of the external tooth surface (see Fig. 10.120). The initial axial wall depth may be in firm dentin (i.e., shallow lesion), in soft dentin (i.e., moderate to deep lesion), or in existing restorative material, if a restoration is being replaced.

Incisal extension to remove carious tooth structure may eliminate the proximal contact (Fig. 10.121). It is important to conserve as much of the distoincisal tooth structure (i.e., sometimes referred to as the "incisal canopy") as possible to reduce the risk for subsequent fracture. When possible (i.e., when the incisal enamel has dentin support), it is best to leave the incisal margin in contact with the adjacent tooth.

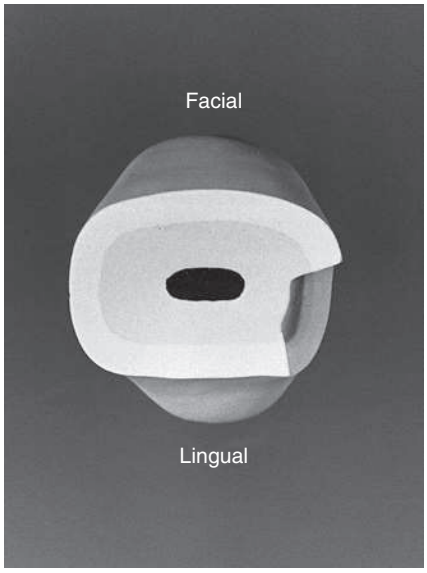
When preparing a gingival wall that is near the level of the rubber dam or apical to it, it is beneficial to initially place a wedge



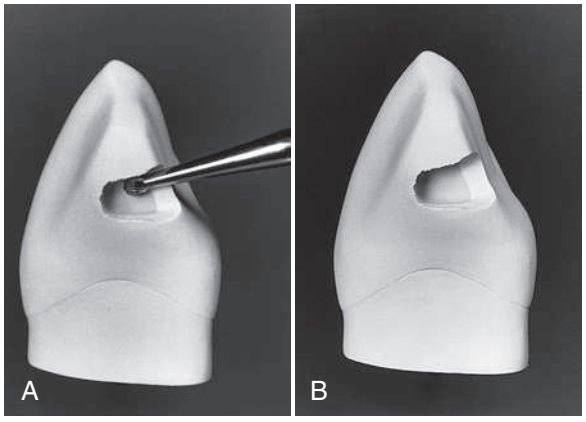
• **Fig. 10.117** Entry for Class III tooth preparation on maxillary canine. A, Bur position is perpendicular to the enamel surface at the point of entry. B, Initial penetration through enamel is directed toward cavitated, caries lesion. C, Initial entry should isolate the proximal enamel, while preserving as much of the marginal ridge as possible. D, Initial cutting reveals the DEJ *(arrow).*



• **Fig. 10.118** Mesiodistal vertical section showing location, depth direction *(arrows),* and direction depth of the retention form in Class III tooth preparations of different gingival depths. Incisal cove *(i)*; gingival groove *(g1),* enamel margin; gingival groove *(g2),* root-surface margin. Distance from outer aspect of *g2* groove to margin is approximately 0.3 mm; bur head diameter is 0.5 mm; direction depth of groove is half this diameter (or approximately 0.3 mm [0.25 mm]).



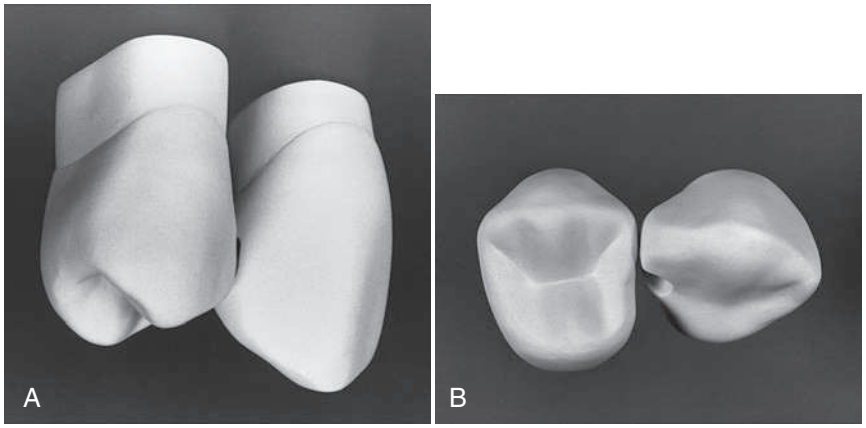
• **Fig. 10.120** Transverse section of mandibular lateral incisor illustrating that the lingual wall of a Class III tooth preparation may meet the axial wall at an obtuse angle and that the axial wall is a uniform depth into dentin and follows the faciolingual contour of the external tooth surface.



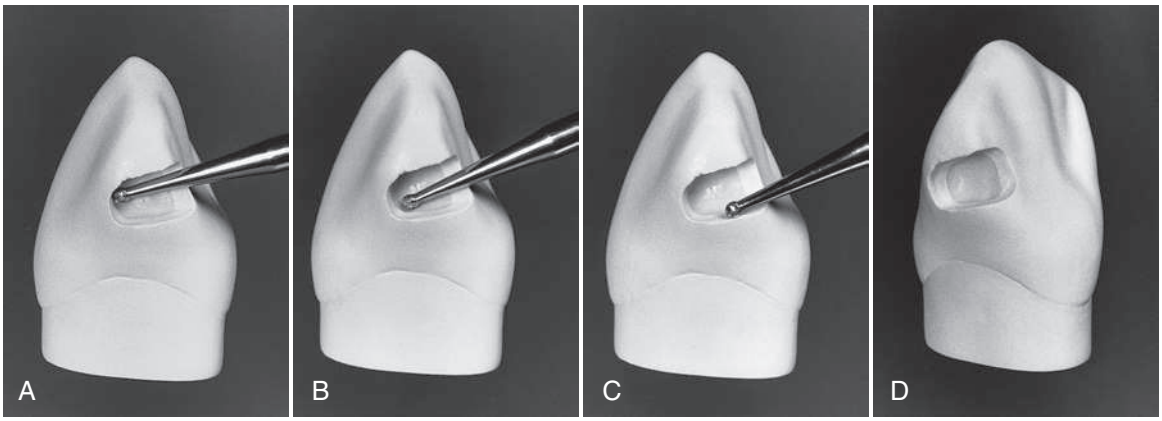
• **Fig. 10.119** Class III tooth preparation on maxillary canine. A, Round bur shaping the incisal area. The incisal angle remains. B, Initial shape of the preparation accomplished with a round bur.

in the gingival embrasure to depress and protect soft tissue and the rubber dam. As the bur is preparing the gingival wall, it may lightly shave the wedge. A triangular (i.e., anatomic) wedge, rather than a round wedge, is used for a deep gingival margin.

The initial tooth preparation is completed by using a No. ½ round bur to accentuate the axial line angles (Fig. 10.122A and B), particularly the axiogingival angle. This facilitates the subsequent placement of retention grooves and leaves the internal line angles slightly rounded. Rounded internal preparation angles permit more complete condensation of the amalgam. The No. ½ round bur also may be used to smooth any roughened, undermined enamel produced at the gingival and facial cavosurface margins (see Fig. 10.122C). The incisal margin of the minimally extended preparation is often not accessible to the larger round bur without damaging



• **Fig. 10.121** Distofacial (A) and incisal (B) views of the canine to show the curved proximal outline necessary to preserve the distoincisal corner of the tooth. The incisal margin of this preparation example is located slightly incisally of the proximal contact (but whenever possible, the margin may be in the contact area).



• **Fig. 10.122** Refining proximal portion. A–C, A small, round bur is used to shape the preparation walls, define line angles, and initiate removal of any undermined enamel along the gingival and facial margins. D, Tooth preparation completed, except for the final finishing of the enamel margins and placement of secondary retention.

("marring") the adjacent tooth (see Fig. 10.122D). Further finishing of the incisal margin is presented later. At this point the initial tooth preparation is completed.

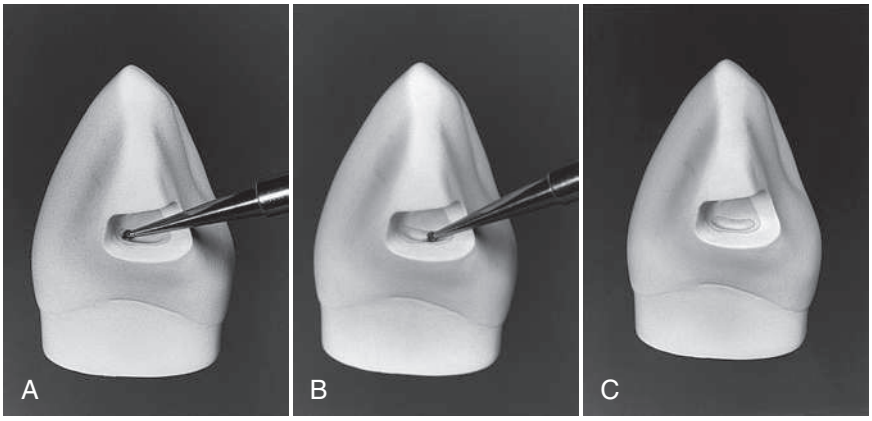
Final Tooth Preparation

Final tooth preparation involves removing any remaining soft dentin, protecting the pulp, developing secondary resistance and retention forms, finishing external walls, and debridement (cleaning)/inspecting. Any remaining soft dentin on the axial wall is removed by using a slowly revolving round bur (No. 2 or No. 4), appropriate spoon excavators, or both. (See Chapter 2 for the management of soft dentin.)

For the Class III amalgam restoration, resistance form against postrestorative fracture is provided by (1) cavosurface and amalgam margins of 90 degrees, (2) enamel walls supported by sound dentin, (3) sufficient bulk of amalgam, and (4) rounded preparation internal angles. The boxlike preparation form provides primary retention form. Secondary retention form is provided by a gingival groove, an incisal cove, and sometimes a lingual dovetail.

The gingival retention groove is prepared by placing a No. ¼ round bur (rotating at low speed) in the axiofaciogingival point angle. It is positioned in the dentin to maintain 0.2 mm of dentin between the groove and the DEJ. The rotating bur is moved lingually along the axiogingival line angle, with the angle of cutting generally bisecting the angle between the gingival and axial walls. Ideally the direction of the gingival groove is slightly more gingival than axial (and the direction of an incisal [i.e., occlusal] groove would be slightly more incisal [i.e., occlusal] than axial) (Fig. 10.123; see also Fig. 10.118).

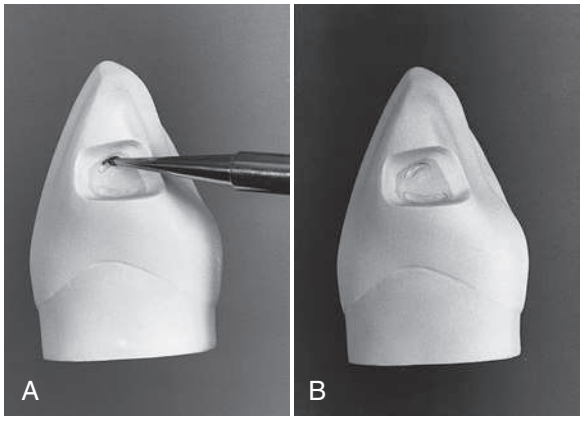
Alternatively if less retention form is needed, two gingival coves may be used as opposed to a continuous groove. One each may be placed in the axiofaciogingival and axiogingivolingual point angles. The diameter of the ¼ round bur is 0.5 mm, and the depth of the groove should be ~0.25 to 0.5 mm. Note the location and depthwise direction of the groove, where the gingival wall remains in enamel (Fig. 10.124). When preparing a retention groove on the root surface (gingival wall in cementum or dentin), the angle of cutting is more gingival, resulting in the distance from the gingival cavosurface margin to the groove being approximately 0.3 mm (see Fig. 10.118). Careful technique is necessary in preparing the gingival retention groove. If the dentin that supports gingival enamel is removed, enamel is subject to fracture. In addition, if the groove is placed only in the axial wall, no effective retention form is developed and the risk of pulpal involvement is increased.



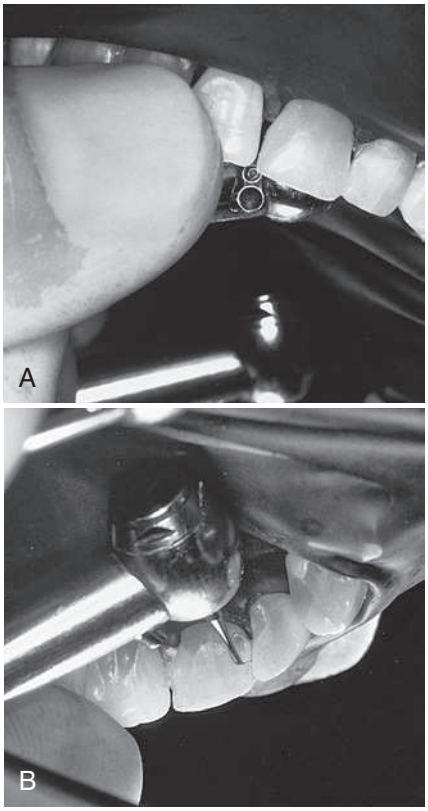
• **Fig. 10.123** Preparing the gingival retention form. A, Position of No. ¼ round bur in axiofaciogingival point angle. B, Advancing the bur lingually to prepare the groove along the axiogingival line angle. (See Fig. 10.118 regarding location, depth direction, and direction depth of groove.) C, Completed gingival retention groove.



• **Fig. 10.124** Completed Class III tooth preparation for amalgam restoration.



• **Fig. 10.125** Preparing the incisal retention cove. A, Position of No. ¼ round bur in the axioincisal point angle. B, Completed incisal cove.

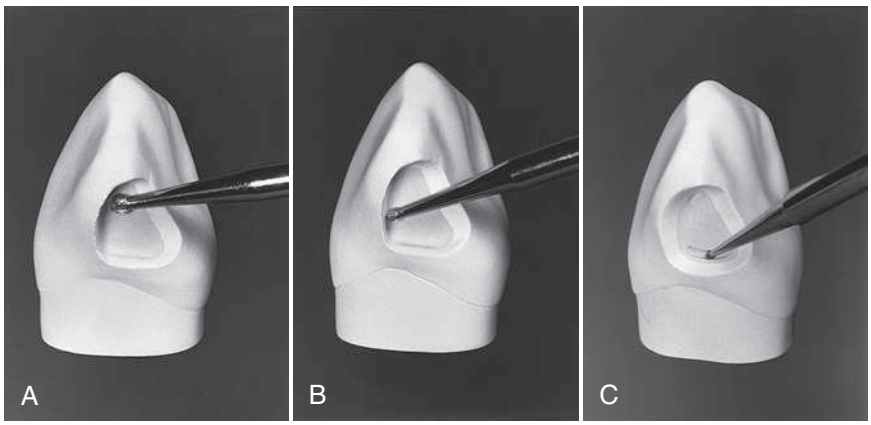


• **Fig. 10.126** Use of the palm-and-thumb grasp to place the incisal retention cove. A, Hand position showing thumb rest. B, Handpiece position for preparing the incisal retention.

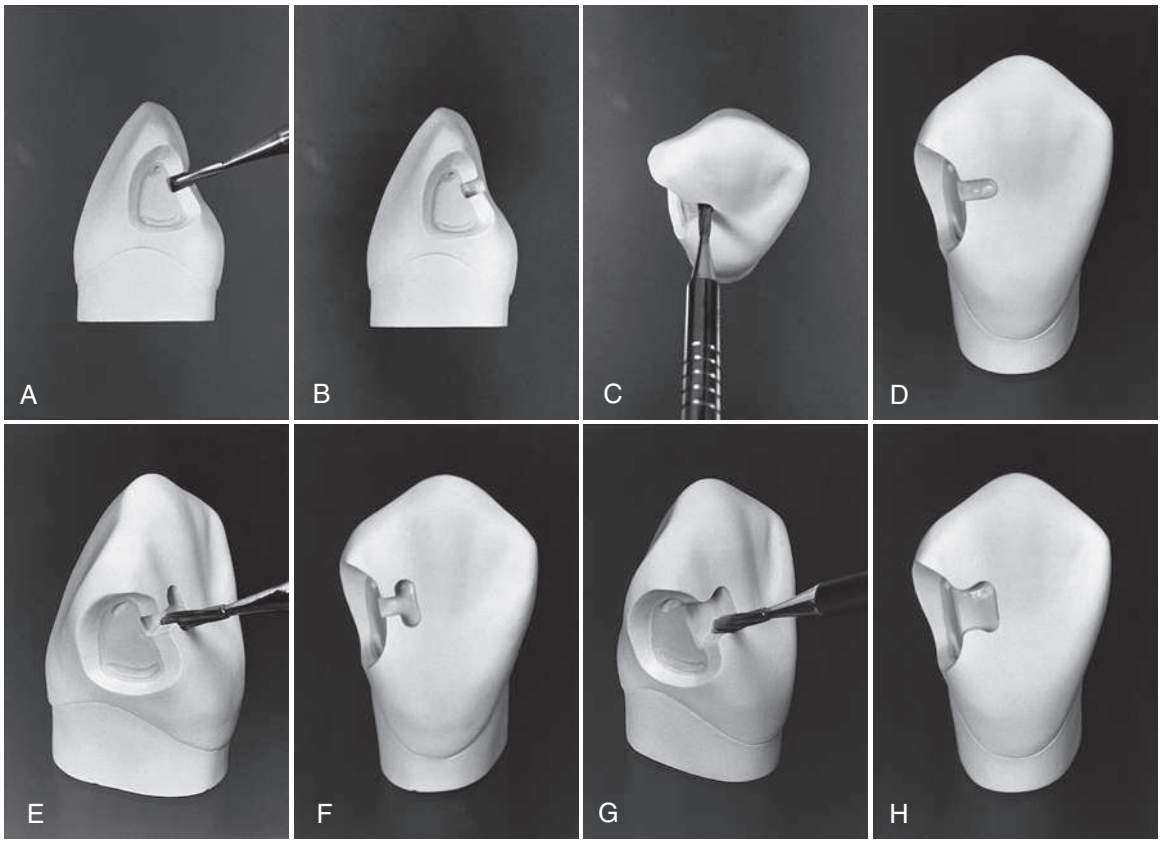
An incisal retention cove is prepared at the axiofacioincisal point angle with a No. ¼ round bur in dentin, being careful not to undermine enamel. It is directed similarly into the incisal point angle and prepared to half the diameter of the bur (Fig. 10.125). Undermining the incisal enamel should be avoided. For the maxillary canine, the palm-and-thumb grasp may be used to direct the bur incisally (Fig. 10.126). This completes the typical Class III amalgam tooth preparation (see Fig. 10.124).

A lingual dovetail is not required in small or moderately sized Class III amalgam restorations. It may be used in large preparations, especially preparations with excessive incisal extension in which additional retention form is needed. The dovetail may not be necessary (even in large preparations), however, if an incisal secondary retention form is possible (Fig. 10.127).

If a lingual dovetail is needed, it is prepared only after initial preparation of the proximal portion has been completed. Otherwise

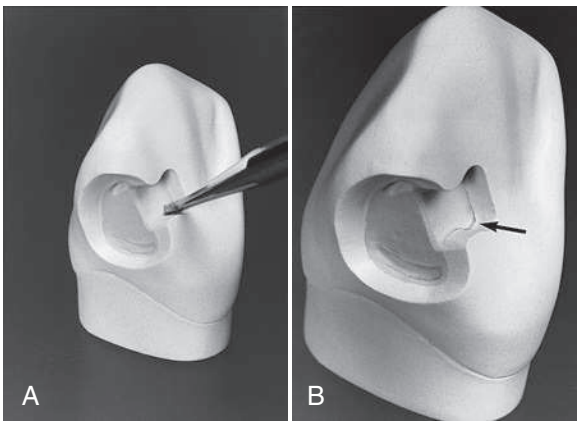


• **Fig. 10.127** Extensive Class III tooth preparation. A, Initial tooth preparation with No. 2 round bur. B, Defining line angles and removing undermined enamel with No. ½ round bur. C, Placing the retention groove using No. ¼ round bur. Note the completed incisal cove.



• **Fig. 10.128** Lingual dovetail providing additional retention for extensive amalgam restoration. A, Bur position at correct depth and angulation to begin cutting. B, Initial cut in beginning dovetail. C, Bur moved to most mesial extent of dovetail. D, If possible, cutting should not extend beyond the midlingual position. E, Bur cutting gingival extension of the dovetail. F, Incisal and gingival extensions of the dovetail. G, Completing the isthmus. The proximal and lingual portions are connected by the incisal and gingival walls in smooth curves. H, Completed lingual dovetail.

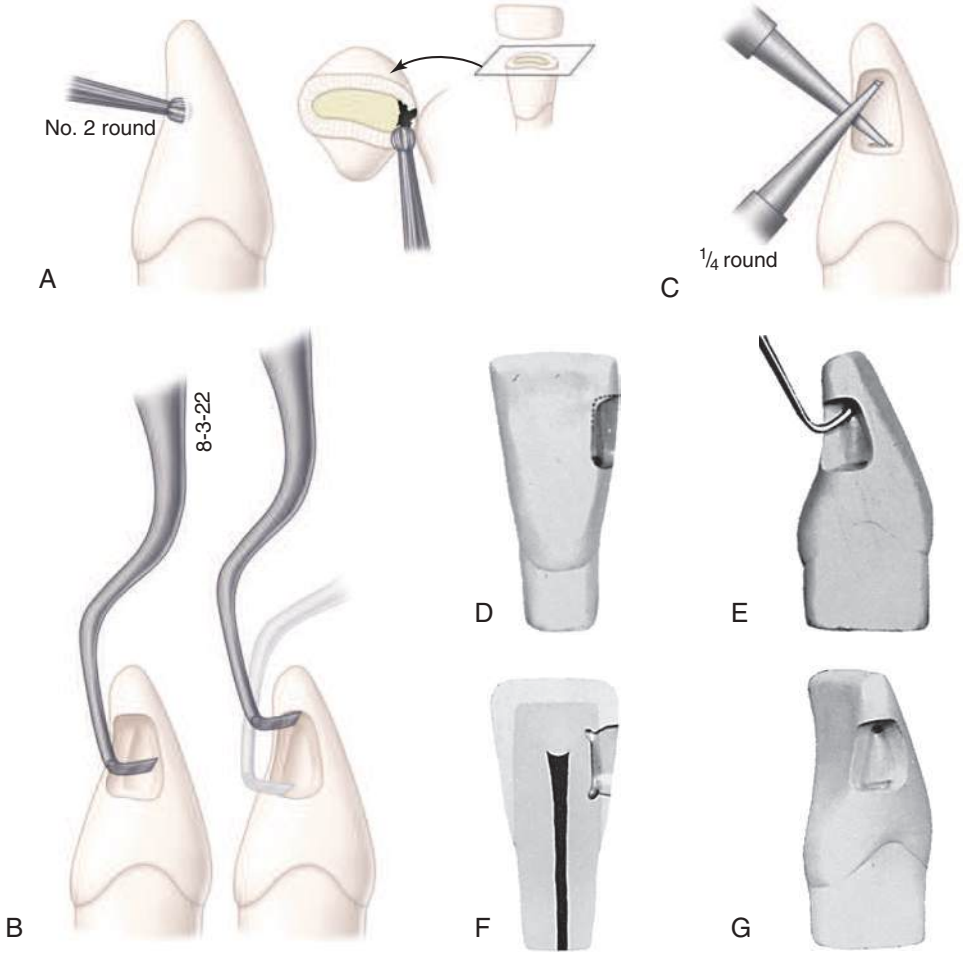
the tooth structure needed for the isthmus between the proximal portion and the dovetail may be removed when the proximal outline form is prepared. The lingual dovetail should be conservative, generally not extending beyond the mesiodistal midpoint of the lingual surface; this varies according to the extent of the proximal caries lesion. The depth of the axial wall of the dovetail should be just inside the DEJ and should be parallel to the lingual surface of the tooth. The No. 245 bur is positioned in the proximal portion at the correct depth and angulation and moved in a mesial direction (Fig. 10.128A and B). The correct angulation places the long axis of the bur perpendicular to the lingual surface. The bur is moved to the point that corresponds to the most mesial extent of the dovetail (see Fig. 10.128C and D). The bur is then moved incisally and gingivally to create sufficient incisogingival dimension to the



• **Fig. 10.129** Ensuring retention in lingual dovetail (often optional). A, Position of No. 33½ bur for cutting the retention cove. B, Preparation of the cove should not remove the dentinal support of the lingual enamel *(arrow)*.

dovetail (approximately 2.5 mm) (see Fig. 10.128E and F). The incisal and gingival walls of the isthmus are prepared in smooth curves connecting the dovetail to the proximal outline form (see Fig. 10.128G and H).

The gingival margin trimmer may be used to round the axiopulpal line angle (i.e., the junction of the proximal and dovetail preparation). This increases the strength of the restoration at the junction of the proximal and lingual portions by providing bulk and limiting stress concentration. The slight lingual convergence of the dovetail's external walls (prepared with the No. 245 bur) usually provides a sufficient retention form. Retention coves, one in the incisal corner and one in the gingival corner (Fig. 10.129), may be placed in the dovetail to enhance retention. The coves are prepared with the No. ¼ round bur or the No. 33½ bur without compromising the dentinal support of the lingual enamel. This preparation may require deepening of the axial wall. Unsupported enamel is removed, the walls or margins are smoothed, and the cavosurface angles are refined, where indicated. The 8-3-22 hoe is recommended for finishing minimally extended margins (Fig. 10.130). If the gingival margin is in enamel, correct marginal configuration (approximately 20 degrees gingival declination) is



• **Fig. 10.130** Class III tooth preparation for amalgam restoration on the mandibular incisor. A, Entering the tooth from the lingual approach. B, Finishing the facial, incisal, and lingual enamel margins with an 8-3-22 triple angle hoe. Note how the secondary cutting edge of the blade is used on the lingual enamel. C, Placing incisal and gingival retention forms with No. ¼ round bur. D, Dotted line indicates the outline of the additional extension that is sometimes necessary for access in placing the incisal retention cove. E, Position of a bi-beveled hatchet to place the incisal retention cove. F, The axial wall forms a convex surface over the pulp. G, Completed tooth preparation. Note the gingival retention groove.



• **Fig. 10.131** Completed distolingual Class III tooth preparation for amalgam.

necessary to ensure full-length enamel rods forming the cavosurface margin. All the walls of the preparation should meet the external tooth surface to form a right angle (i.e., butt joint) (Fig. 10.131; see also Fig. 10.124). The various steps involved in the preparation of the dovetail are shown in Fig. 10.132.

The completed tooth preparation should be cleaned of any residual debris and inspected. Careful assessment should be made to see that all of the caries lesion has been removed (as indicated), that the depth and retention are appropriate, and that cavosurface margins provide for sufficient amalgam bulk.

Restorative Technique for Class III Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed on the prepared tooth structure per manufacturer instructions. Excess moisture is removed without desiccating the dentin.

Matrix Placement

The wedged, rigid material–supported sectional matrix may be used for the Class III amalgam restoration. It is essential to trim the lingual portion of the sectional matrix material correctly to avoid covering the preparation and blocking access for insertion of the amalgam. Stainless steel matrix material is obtained, 8 mm wide and 0.05 mm thick, which covers one third of the facial surface and extends through the proximal to the lingual surface. The lingual portion is trimmed at an angle that corresponds approximately to the slope of the lingual surface of the tooth (Fig. 10.133). The section of matrix is burnished on a resilient paper pad to create the desired proximal surface form. The sectional matrix is placed in position and wedged from the facial or lingual embrasure, whichever is greater. The facial and lingual portions of the matrix are stabilized with a rigid material (see Fig. 10.132G). Precontoured metallic matrices may be used (instead of custom-made matrices) if the contour of the precontoured matrix coincides with that of the proximal surface being restored. If the preparation is small and the matrix is sufficiently rigid, additional rigid material to support the matrix may not be required.

Condensation and Carving

Insertion of the amalgam, initial carving, matrix removal, wedge removal, and final carving are similar to the techniques for Class II preparations. When properly placed, conservative restorations in incisors and canines are relatively inconspicuous (Fig. 10.134). Fig. 10.135 illustrates a Class III amalgam restoration in a mandibular incisor.

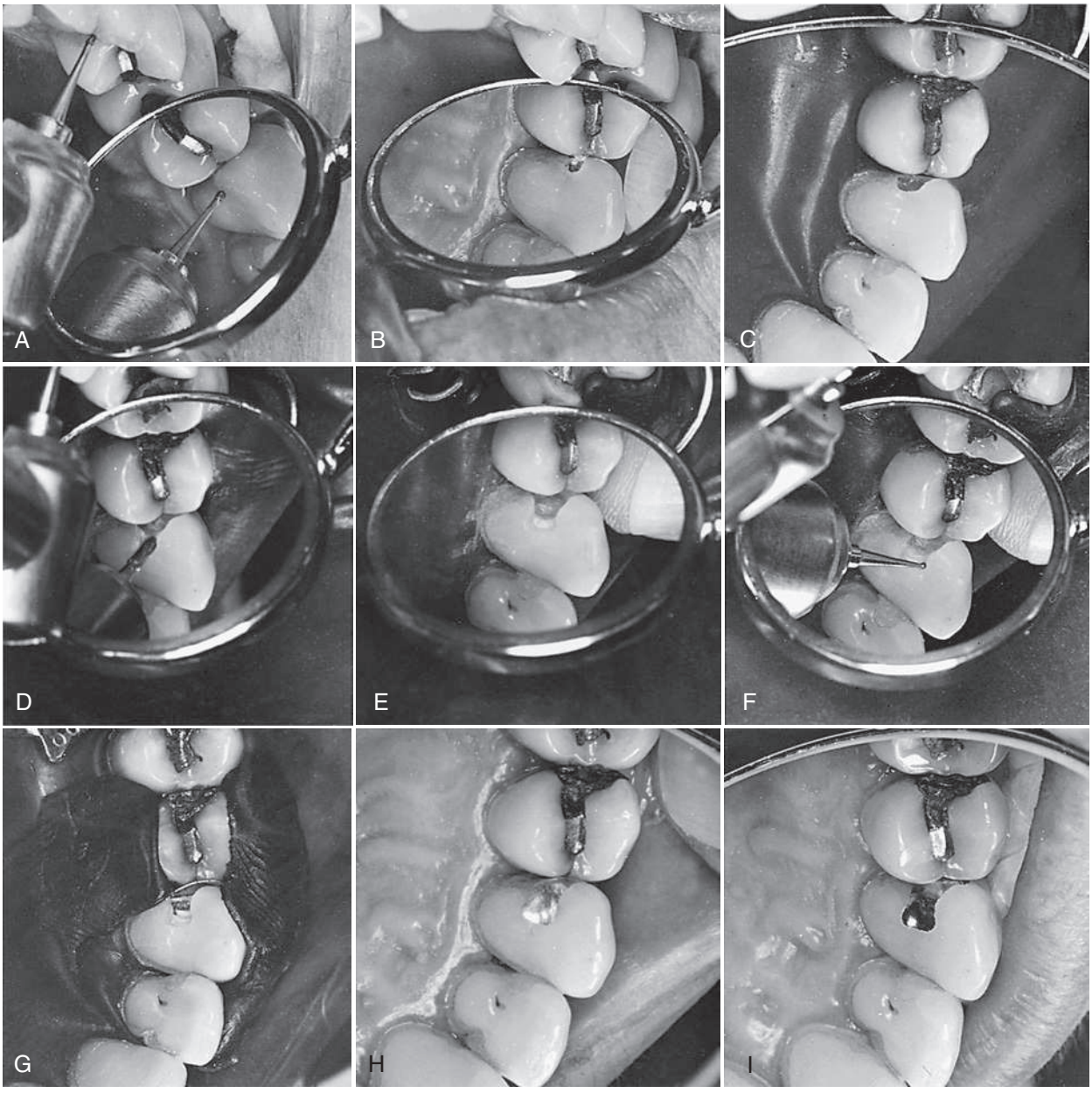
Finishing and Polishing

Finishing and polishing of the Class III amalgam should follow the approach presented in General Concepts Guiding Restoration with Amalgam.

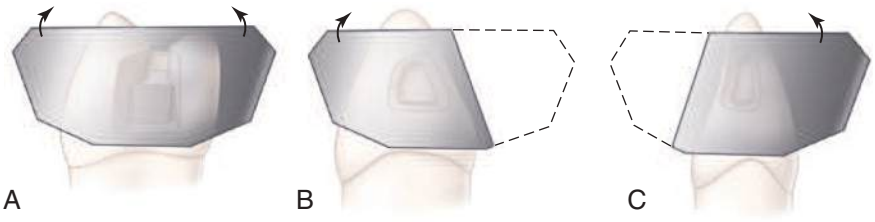
Clinical Technique for Class V Amalgam Restorations

Class V restorations are indicated to restore defects on the facial or lingual cervical one third of any tooth. Class V amalgam restorations may be especially technique sensitive because of location, extent of the caries lesion, and limited access and visibility. Cervical caries lesions usually develop because of the chronic presence of an acidogenic biofilm located in the non–self-cleansing area just gingival to the height of contour. Patients with gingival recession are predisposed to Class V *root caries lesions* because dentin is more susceptible to demineralization than enamel. Patients with a reduced salivary flow caused by certain medical conditions (e.g., Sjögren's syndrome), medications, or head and neck radiation therapy are also predisposed to cervical and root caries lesions. These patients usually have less saliva to buffer the acids produced by oral bacteria. Class V restorations may be used to treat both cervical and root caries lesions. Incipient, smooth-surface enamel demineralization appears as a chalky white line just occlusal or incisal to the crest of the marginal gingiva (usually on the facial surface) (Fig. 10.136). These areas often are overlooked in the oral examination unless teeth are isolated with cotton rolls, free of debris, and dried gently with the air syringe. When the incipient cervical caries has not decalcified the enamel sufficiently to result in cavitation (i.e., a break in the continuity of the surface), the lesion may be remineralized by appropriate techniques, including patient motivation toward proper diet and hygiene. Occasionally an enamel surface that is only slightly cavitated may be treated successfully by smoothing with sandpaper disks, polishing, and treating with a fluoride varnish or infiltrated with a low viscosity resin in an attempt to prevent further demineralization, cavitation, and surgical management. Prophylactic, preventive treatment alone will be inadequate if the caries lesion has progressed to decalcify and soften enamel to an appreciable depth. In this instance, a Class V tooth preparation and restoration is indicated, particularly if the caries lesion has penetrated to the DEJ (Fig. 10.137A). When numerous cervical lesions are present (see Fig. 10.137B), high caries activity is obvious. In addition to the restorative treatment, the patient should be instructed and encouraged to implement an aggressive prevention program to avoid subsequent development of additional caries lesions. Class V amalgam restorations may be used anywhere in the mouth. As with Class III amalgam restorations, they generally are reserved for nonesthetic areas, for areas where access and visibility are limited and where moisture control is difficult, such as areas that are significantly deep gingivally. Because of limited access and visibility, many Class V restorations are difficult and present special problems during the preparation and restorative procedures.

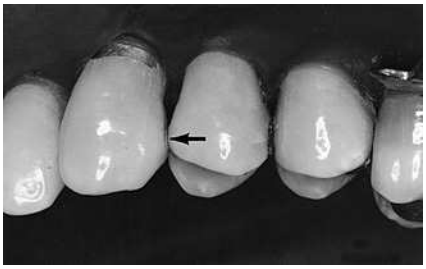
Occasionally amalgam is preferred when the caries lesion extends gingivally enough that a mucoperiosteal flap must be reflected for adequate access and visibility (Fig. 10.138). Proper surgical technique must be followed, with special attention to complete debridement



• **Fig. 10.132** Distolingual tooth preparation and restoration. A, Bur position for entry. B, Penetration made through the lingual enamel to the caries lesion. C, Proximal portion completed, except for the retention form. D, Preparing the dovetail. E, Completed preparation, except for the retention groove and the coves. F, Bur position for the incisal cove in the dovetail. G, Rigid material–supported sectional matrix ready for the insertion of amalgam. H, Carving completed and rubber dam removed. I, Polished restoration.



• **Fig. 10.133** Matrix strip design. A, Design required for rigid material–supported matrix for Class II tooth preparations. B, Alteration necessary for Class III preparation on the maxillary canine. C, Alteration necessary for the mandibular incisor. The strip material is cut to approximate the slope of the lingual surface.



• **Fig. 10.134** Inconspicuous facial margin *(arrow)* of Class III amalgam restoration on the maxillary canine.



• **Fig. 10.135** Class III amalgam restoration on the mandibular incisor *(arrowhead).*



• **Fig. 10.136** Incipient caries lesions of enamel appear as white spots. The affected surface may be smooth (i.e., noncavitated). White spots are more visible when dried. (From Cobourne MT, DiBiase AT: *Handbook of orthodontics*, Edinburgh, 2010, Mosby.)

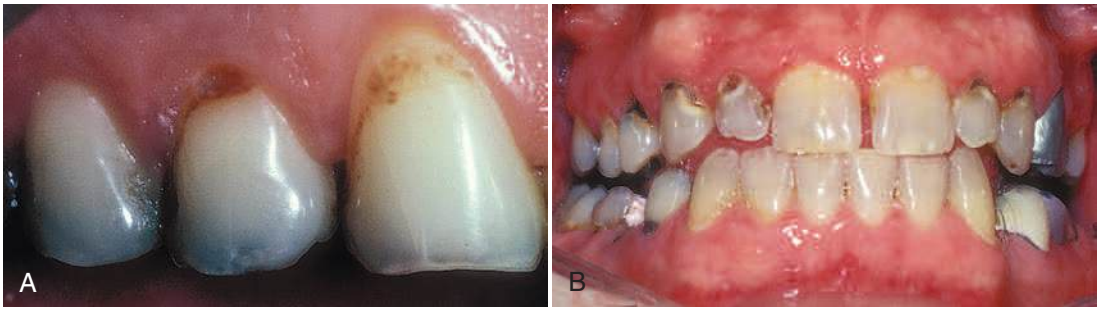
of amalgam particles before closure of the surgical site. Class V amalgam restorations usually are contraindicated in esthetically important areas because many patients object to metal restorations that are visible (Fig. 10.139). Generally Class V amalgams placed on the facial surface of mandibular canines, premolars, and molars are not readily visible. Amalgams placed on maxillary premolars and first molars may be visible.

Amalgam may be used in the cervical areas of partial denture abutment teeth because amalgam resists wear as clasps move over the restoration. Contours prepared in the restoration for clasp retentive areas may be achieved relatively easily and maintained when an amalgam restoration is used.

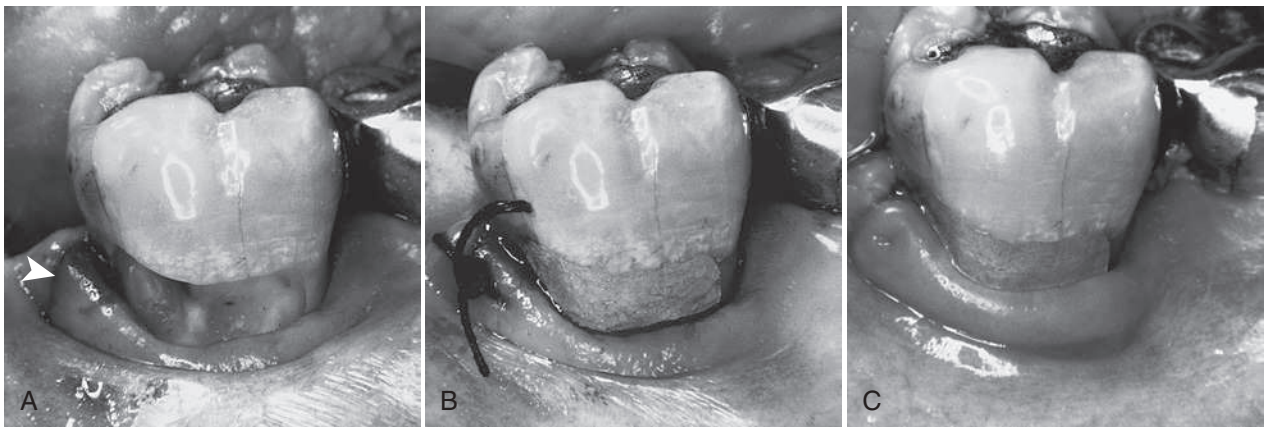
Initial Procedures

Proper isolation prevents moisture contamination of the operating site, enhances asepsis, and facilitates access and visibility. Moisture in the form of saliva, gingival sulcular fluid, or gingival hemorrhage must be excluded during caries lesion removal, liner application, and insertion and carving of amalgam. Moisture impairs visual assessment, may contaminate the pulp during caries lesion removal (especially with a pulpal exposure), and may negatively affect the physical properties of the amalgam. The gingival margin of Class V tooth preparations is often apical to the gingival crest. Such a gingival margin necessitates use of appropriate rubber dam and retainer, or displacement of the free gingiva with a retraction cord, to protect it and to provide access. Isolation helps eliminate seepage of sulcular fluid into the tooth preparation or restorative materials.

Isolation objectives are met through local anesthesia and use of (1) a cotton roll and a retraction cord or (2) a rubber dam and a suitable cervical retainer (Fig. 10.140). Isolation with a cotton roll and a retraction cord is satisfactory when properly performed (Fig. 10.141). This type of isolation is practical and probably the approach most often used. The retraction cord should be placed in the sulcus before initial tooth preparation to reduce the possibility of cutting instruments damaging the free gingiva. The cord should produce a temporary, adequate, atraumatic lateral displacement of the free gingiva. The cord may be treated with hemostatic preparations containing aluminum chloride or ferrous sulfate. Alternatively the cord may be treated with epinephrine; however, caution must be used as epinephrine on abraded gingiva is absorbed rapidly into the circulatory system and may cause an increase in blood pressure, elevated heart rate, and possible dysrhythmia. The retraction cord may be braided, twisted, or woven. The diameter of the cord should be easily accommodated in the gingival sulcus.



• **Fig. 10.137** Cervical caries lesions. A, Cavitation involving enamel and dentin in several teeth. B, Relatively high caries index is obvious when numerous cervical lesions are present. (From Perry DA, Beemsterboer PL: *Periodontology for the dental hygienist*, ed 3, St. Louis, 2007, Saunders.)



• **Fig. 10.138** Surgical access. A, Class V preparation requiring mucoperiosteal flap reflection with a releasing incision *(arrowhead).* B, Completed restoration with suture in place. C, Suture removed 1 week after the procedure.



• **Fig. 10.139** Patients may object to metal restorations that are visible during conversation.



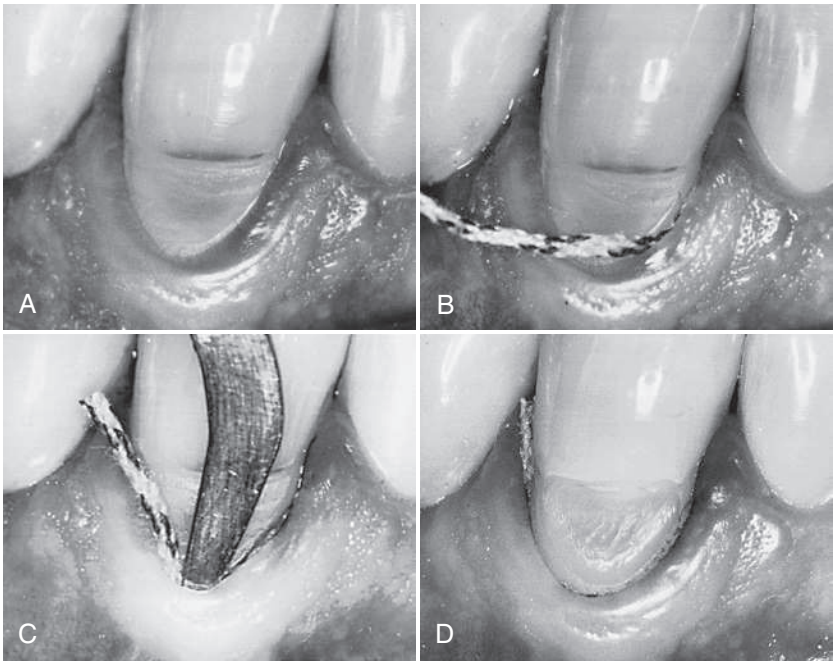
• **Fig. 10.140** A rubber dam and a No. 212 retainer may be required to isolate the caries lesion properly.

If significant blanching of the free gingiva is observed (or if too much pressure has to be applied to place the cord), an oversized cord may have been selected. If so, it should be exchanged for a cord of smaller diameter. An appropriate amount is cut to a length 6 mm longer than the gingival margin. Some operators prefer to place the cord in a dappen dish, wet it with a drop of hemostatic solution, and blot it with gauze (5 × 5 cm) to remove excess liquid. A braided or woven cord is usually easier to use because it does not unravel during placement. A larger cord may be inserted over the first cord if the sulcus is large enough to accommodate two cords. A thin, blunt-edged instrument blade or the side of an explorer tine may be used to gently insert the cord progressively into place. A slight backward direction of the instrument as it steps along the cord helps prevent dislodgment of previously inserted cord (see Fig. 10.141). In addition, using a second instrument stepping along behind the first instrument may help prevent dislodgment of cord. Additionally using the air syringe or cotton pellets to reduce or absorb the sulcular fluid in the cord already placed is helpful during cord placement. The cord results in adequate retraction (displacement) in a short time. The cord may be moistened with the hemostatic solution before, during, or after placement if slight hemorrhage is anticipated or observed. Alternatively the cord may be used dry.

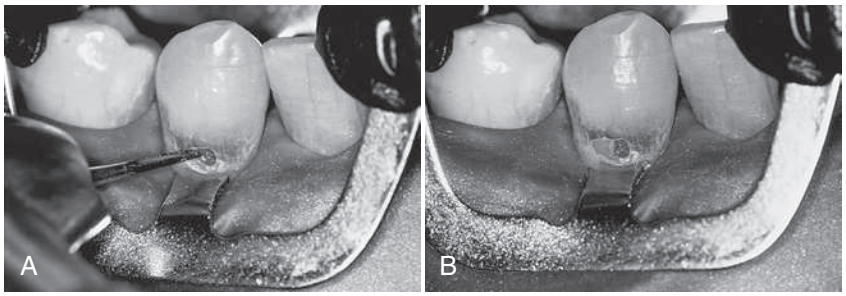
The cord usually remains in place during tooth preparation as well as insertion and initial carving of amalgam. While carving amalgam at the gingival margin, the presence of the cord may cause difficulty in detection of the unprepared tooth surface apical to the margin. This may limit proper carving, result in an overcontour and marginal excess. In this instance, after carving gross excess, the cord may be teased from the sulcus before the carving is completed.

Tooth Preparation for Class V Amalgam Restorations

Proper outline form for Class V amalgam tooth preparations results in extending the cavosurface margins to sound tooth structure while maintaining a limited axial depth of 0.5 mm inside the DEJ and 0.75 mm inside cementum (when on the root surface). The outline form for the Class V amalgam tooth preparation is determined primarily by the location and size of the caries lesion or old restorative material. Clinical judgment determines final preparation outline, especially when the cavosurface margins approach or extend into areas of enamel decalcification. The operator must observe the prepared enamel wall to evaluate the depth of the decalcified enamel and to determine if cavitation exists peripheral to the wall. When no cavitation has occurred and when the decalcification does not extend appreciably into the enamel, extension of the outline form often should cease. In some cases, if all decalcification were included in the outline form, the preparation would extend into the proximal cervical areas (if not circumferentially around the tooth). Such a preparation would be difficult and perhaps unrestorable. A full-coverage indirect restoration should be considered for teeth with extensive circumferential cervical decalcification (but only after caries risk has been reduced).



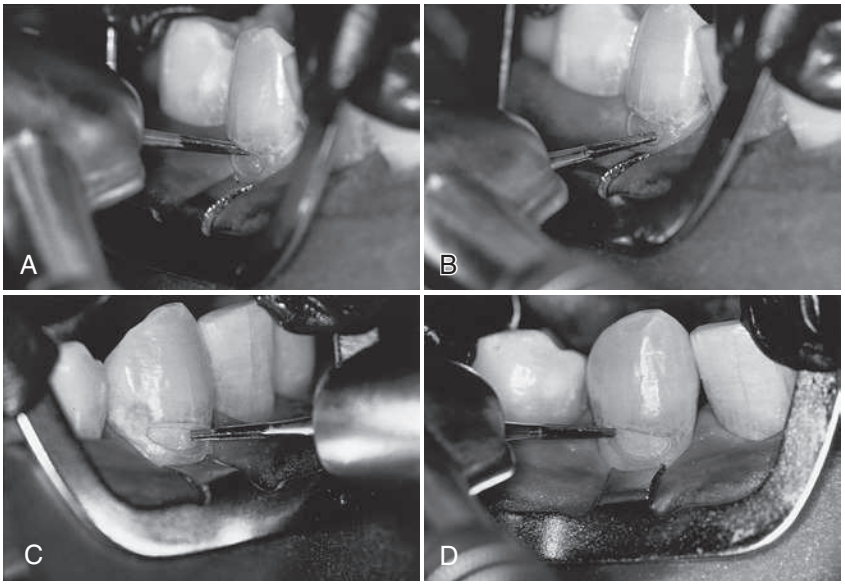
• **Fig. 10.141** Use of the retraction cord for isolation of a Class V lesion. A, Preoperative view. B, Cord placement initiated. C, Cord placement using a thin, flat-bladed instrument. D, Cord placement completed.



• **Fig. 10.142** Starting the Class V tooth preparation. A, Bur positioned for entry into caries lesion. B, The entry cut is the beginning of the outline form having a limited axial depth.

Initial Tooth Preparation

A Class V amalgam restoration is not used often in a mandibular canine, but it is presented here for illustration. The same general principles for tooth preparation apply for all other tooth locations. A tapered fissure bur of suitable size (e.g., No. 271) is used to enter the caries lesion (or existing restoration) to a limited initial axial depth of 0.5 mm inside the DEJ (Fig. 10.142). This depth is usually 1 to 1.25 mm total axial depth, depending on the incisogingival (i.e., occlusogingival) location. The enamel of the incisal (occlusal) external wall of the preparation is considerably thicker than the cervical. If the preparation is on the root surface, however, the initial axial depth is approximately 0.75 mm. The end of the bur at the initial depth may be in sound dentin, in soft dentin, or in old restorative material. The edge of the bur end may be used to penetrate the area; this is more efficient than using the flat end of the bur, reducing the possibility of the bur's tendency to "crawl" (i.e., move laterally instead of cut axially) on the external surface. When the entry is made, the bur orientation is adjusted to ensure that all external walls are perpendicular to the external tooth surface (i.e., parallel to the enamel rods) (Fig. 10.143). This requires frequent changing of the bur orientation so as to accommodate the cervical mesiodistal and incisogingival (or occlusogingival) convexity of the tooth. The preparation is extended incisally, gingivally, mesially, and distally until the cavosurface margins are positioned in sound tooth structure such that an initial axial depth of 0.5 mm inside the DEJ (or 0.75 mm if on the root surface) is established. When extending mesially and distally, it may be necessary to protect the rubber dam from the bur by placing a flat-bladed instrument over the dam (Fig. 10.144). Because the axial wall follows the mesiodistal and incisogingival (or occlusogingival) contours of the facial surface of the tooth, it usually is convex in both directions (Fig. 10.145). In addition, the axial wall usually is slightly deeper at the incisal wall, where more enamel (i.e., approximately 1–1.25 mm in depth) is present than at the gingival wall, where little or no enamel (i.e., approximately 0.75–1 mm in depth) may be present. A depth of 0.5 mm inside the DEJ permits placement of necessary retention grooves without undermining enamel. This subtle difference in depth serves also to increase the thickness of the remaining dentin (between the axial wall and the pulp) in the gingival aspect of the preparation to aid in protecting the pulp.



• **Fig. 10.143** When extending incisally (A), gingivally (B), mesially (C), and distally (D), the bur is positioned to prepare these walls perpendicular to the external tooth surface.



• **Fig. 10.144** The flat-bladed instrument protects the rubber dam from the bur.

Alternatively, an appropriate carbide bur (usually No. 2 or No. 4) may be used for the initial tooth preparation. Round burs are indicated in areas inaccessible to a fissure bur, which must be oriented perpendicular to the tooth surface. If needed, smaller round burs may also be used to define the internal angles in these preparations, enhancing proper placement of the retention grooves.

Final Tooth Preparation

Final tooth preparation involves the removal of any remaining soft dentin, pulp protection, retention form, the finishing of external walls, and cleaning/inspecting. Specifically any remaining soft axial wall dentin is removed (as indicated) with a No. 2 or No. 4 round bur. Any old restorative material remaining (including base and liner) may be left if (1) no evidence of recurrent caries exists, (2) the periphery of the base and liner is intact, and (3) the tooth is asymptomatic. With proper outline form, the axial line angles are already in sound dentin. If needed, an appropriate liner or base is applied.

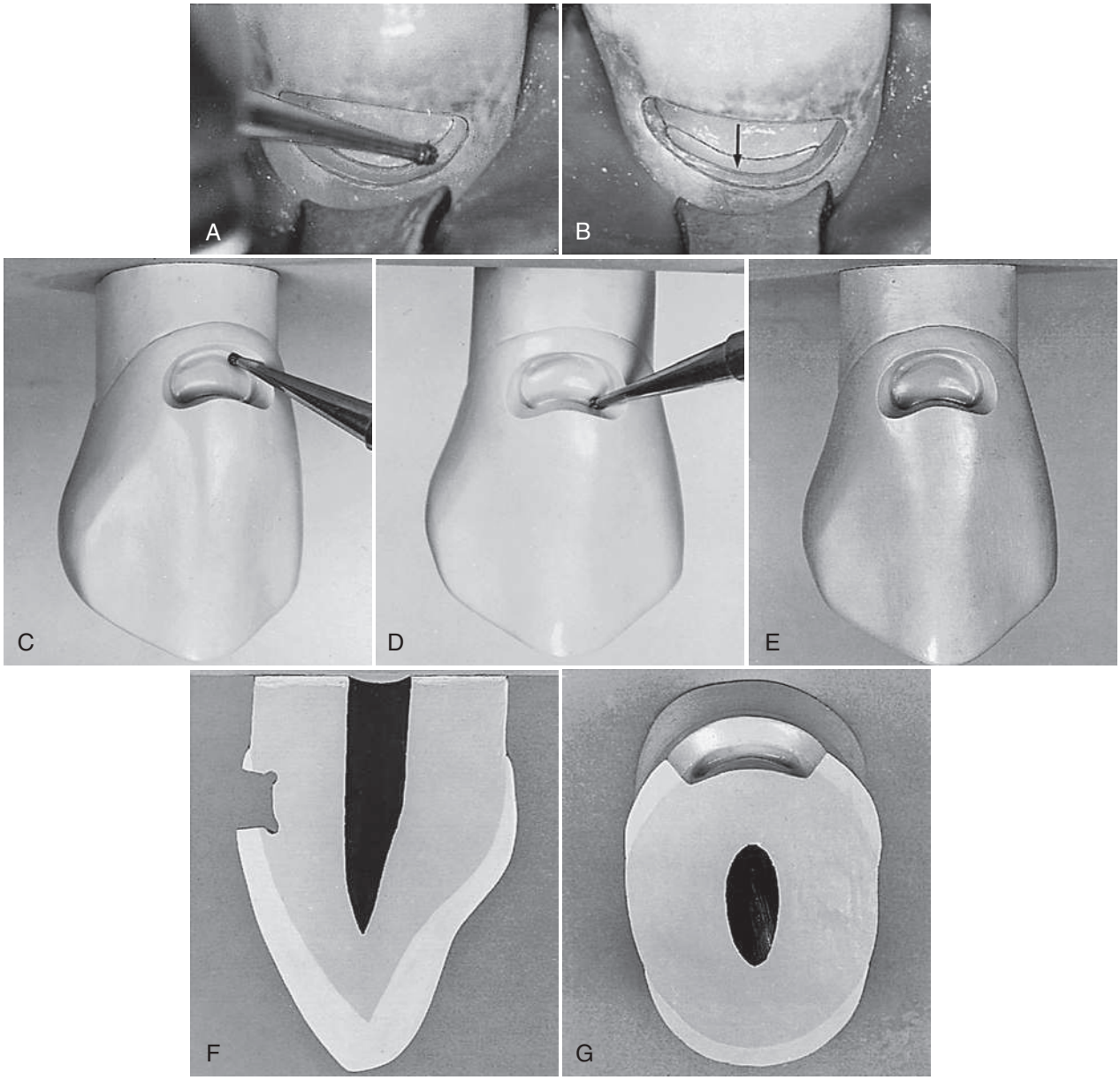
Because the mesial, distal, gingival, and incisal walls of the tooth preparation are perpendicular to the external tooth surface, they usually diverge. Consequently this form provides no inherent retention, and retention form must be provided to retain the amalgam. A No. ¼ round bur is used to prepare two retention grooves, one along the incisoaxial line angle and the other along the gingivoaxial line angle (see Fig. 10.145). The handpiece is positioned so that the No. ¼ round bur is directed generally to bisect the angle formed at the junction of the axial wall and the incisal (i.e., occlusal) wall. Ideally the direction of the incisal (i.e., occlusal) groove is slightly more incisal (i.e., occlusal) than axial, and the direction of the gingival groove is slightly more gingival than axial. Alternatively, four retention coves may be prepared, one in each of the four axial point angles of the preparation (Fig. 10.146).

Using four coves instead of two full-length grooves conserves the dentin near the pulp, reducing the possibility of a mechanical pulp exposure. The depth of the grooves (or coves) should be approximately 0.25 to 0.5 mm. It is important that the retention grooves be adequate because they provide the only retention form to the preparation. Regardless, the grooves should not remove dentin immediately supporting enamel. In a large Class V amalgam preparation, extending the retention groove circumferentially around all the internal line angles of the tooth preparation may enhance the retention form.

If access is inadequate for use of the No. ¼ round bur, an angle-former chisel may be used to prepare the retention form. Additionally, a No. 33½ bur may be used. Both methods result in retention grooves that are angular, but positioned in the same location and approximately to the same depth as when the No. ¼ round bur is used. The rounded retention form created with the No. ¼ round bur is generally preferred, however, because amalgam is able to be condensed into rounded areas more readily than into sharp areas, resulting in better adaptation of amalgam into the retention grooves. If necessary, suitable hand instruments (e.g., chisels, margin trimmers) are used to plane the enamel margins, verifying soundness and 90-degree cavosurface angles. Finally the preparation is cleaned of debris and inspected for completeness.

Large Preparations That Include Line Angles

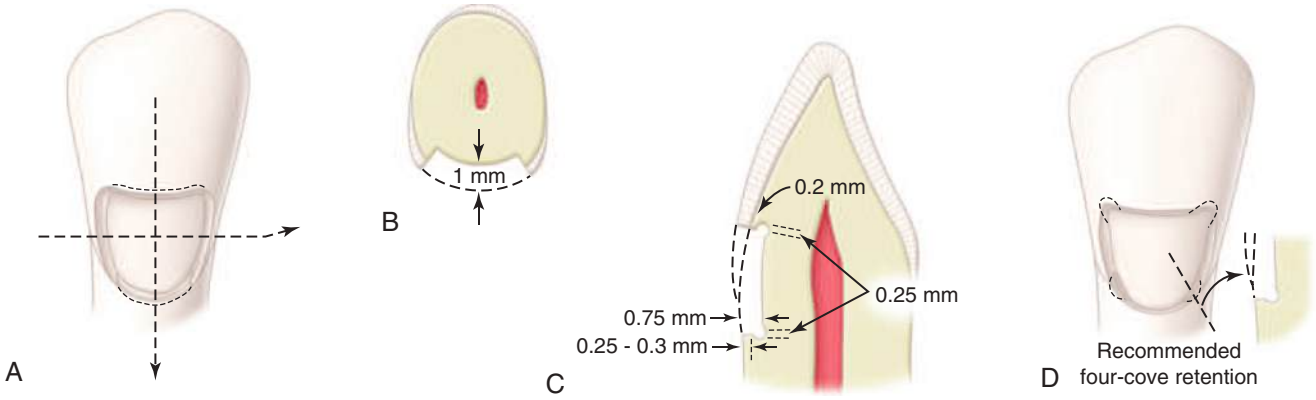
A caries lesion on the facial (or lingual) surface may extend beyond the line angles of the tooth. Maxillary molars, particularly the second molars, are most commonly affected by these extensive defects (Fig. 10.147A). In this example, if the remainder of the



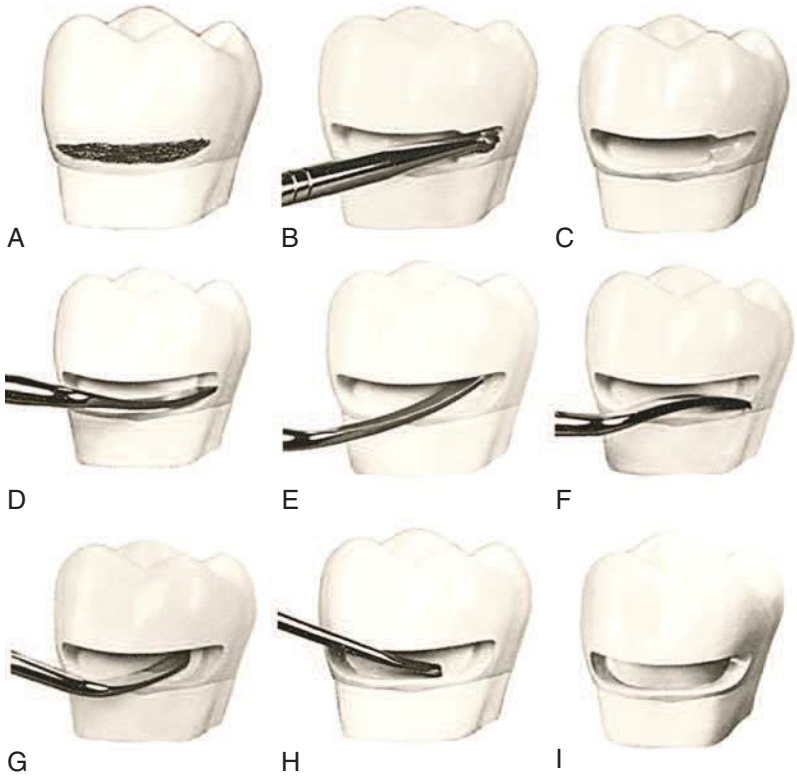
• **Fig. 10.145** Retention form. A, A No. ¼ round bur positioned to prepare the gingival retention groove. B, Gingival retention groove *(arrow)* prepared along the gingivoaxial line angle generally to bisect the angle formed by the gingival and axial walls. Ideally the direction of preparation is slightly more gingival than pulpal. An incisal retention groove is prepared along the incisoaxial line angle and directed similarly. C and D, A groove is placed with a No. ¼ round bur along the gingivoaxial and incisoaxial line angles 0.2 mm inside the DEJ and ~0.25 – 0.5 mm deep. Note the slight pulpal inclination of the shank of the No. ¼ round bur. E, Facial view. F, Incisogingival section. Grooves depthwise are directed mostly incisally (gingivally) and slightly pulpally. G, Mesiodistal section.

distal surface is sound and the distal caries lesion is accessible facially, the facial restoration should extend around the line angle. This prevents the need for a Class II proximal restoration to restore the distal surface. However, the operator should strive not to overextend the occlusal wall of the distal surface preserving as much sound tooth structure as possible to support the distal marginal ridge. As much of the preparation as possible should be completed with a fissure bur. A round bur approximately the same diameter as the fissure bur is then used to initiate the distal portion of the preparation (see Fig. 10.147B and C). Smaller round burs should be used to accentuate the internal line angles of the distal portion. Preparing the facial portion first provides better access and visibility to the distal portion. Occasionally hand instruments may be useful for completing the distal half of the preparation when space for the handpiece is limited (see Fig. 10.147D–F).

Grooves placed along the entire length of the occlusoaxial and gingivoaxial line angles help ensure retention of the restoration. The No. ¼ round bur is used as previously described to prepare the retention grooves. A gingival margin trimmer or a 7-85-2½-6 angle-former chisel may be used in the distal half of the preparation to provide retention form when access for the handpiece is limited (see Fig. 10.147G and H).



• **Fig. 10.146** A–C, Extended Class V tooth preparation (*A*) with the axial wall contoured parallel to the DEJ mesiodistally (*B*) and incisogingivally (*C*). The axial wall pulpal depth is 1 mm in the crown and 0.75 mm in the root. In addition, note location and direction depth (~0.25 – 0.5 mm) of the retention grooves and the dimension of the gingival wall (0.25 mm) from the root surface to the retention groove. D, Large Class V preparation with retention coves prepared in the four axial point angles.



• **Fig. 10.147** Tooth preparation on a maxillary molar. A, Caries lesion extending around distofacial corner of the tooth. B and C, Distal extension is accomplished with round bur. D–F, A gingival margin trimmer may be useful in completing the distal half of the preparation when handpiece access is limited. G, A gingival margin trimmer may be used to provide the retention grooves. H, An angle-former chisel may be used to prepare the retention grooves in the distal portion of the preparation. I, Completed tooth preparation.

Because of the proximity of the coronoid process, access to the facial surfaces of maxillary molars, particularly the second molars, is often limited. Having the patient partially close and shift the mandible toward the tooth being restored improves access and visibility (Fig. 10.148).

If the Class V outline form approaches an existing proximal restoration, it is better to extend slightly into the bulk of the proximal restoration rather than to leave a thin section of the tooth structure between the two restorations (Fig. 10.149). In this illustration the previously placed amalgam serves as the distal wall of the preparation. When proper treatment requires Class II and V amalgam restorations on the same tooth, the Class II preparation and restoration is completed before initiating the Class V restoration. If the Class V restoration was accomplished first, it might be damaged by the matrix band and wedge needed for the Class II restoration.



• **Fig. 10.148** The mandible shifted laterally for improved access and visibility.



• **Fig. 10.149** When a Class V outline form closely approaches an existing restoration, the preparation should be extended to remove the remaining thin enamel wall to achieve adjoining restorations.

The preparation should then be cleaned, carefully inspected as indicated in General Concepts Guiding Preparation for Amalgam Restorations, and any final modifications completed.

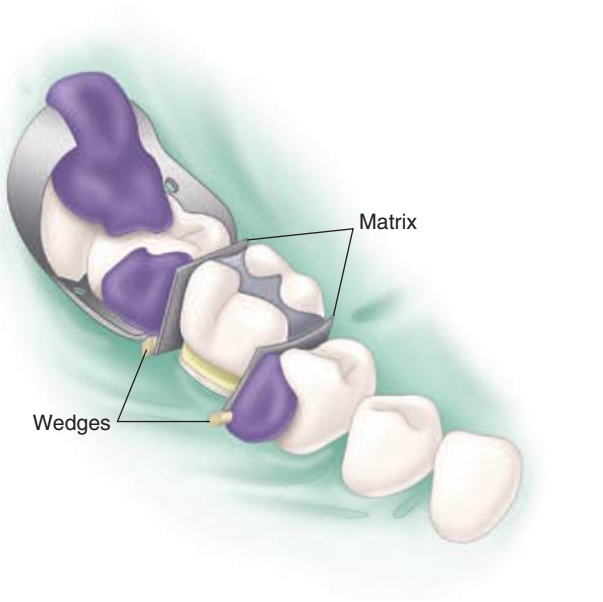
Restorative Technique for Class V Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed over the prepared tooth structure per manufacturer instructions. Excess moisture is removed without desiccating the dentin.

Matrix Placement

Most Class V amalgam restorations are accomplished without the use of any type of matrix. The most difficult condensation occurs in a tooth preparation with an axial wall that is convex mesiodistally. Two alternative methods for insertion may be used. The preferred method is the application of a matrix that confines amalgam in the mesial and distal portions of the preparation (Fig. 10.150). Short lengths of stainless steel matrix material, one each for the mesial and distal surfaces, are passed through the proximal contacts, carefully guided into the gingival sulcus, and wedged. The strips must be wide enough to extend occlusally through the respective proximal contacts and long enough to extend slightly past the facial line angles. The strip usually requires rigid material support for stability. The strips offer resistance against condensing the mesial and distal portions, which provides support for condensing the center of the restoration. The gingival edge of the steel strip often



• **Fig. 10.150** Application of the matrix to confine amalgam in the mesial and distal extensions of the preparation.

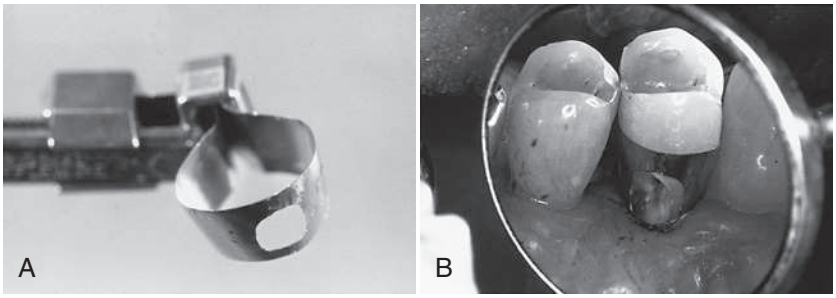
must be trimmed to conform to the circumferential contour (level) of the base of the gingival sulcus to prevent soft tissue damage. Rather than using two short pieces, the operator may use a longer length that is passed through one proximal contact, extended around the lingual surface, and passed through the other contact, forming a U-shaped matrix. Trimming the gingival edge to conform to the interproximal soft tissue anatomy usually is more difficult with one matrix strip than when two strips are used.

A conventional Tofflemire band and retainer may be used with a window cut into the band, with a high-speed handpiece bur, to allow access to the preparation for condensation (Fig. 10.151). Alternatively, the tooth may be prepared and restored in sections without using a matrix. Each successive section of the preparation should be extended slightly into the previously condensed portion to ensure caries lesion removal.

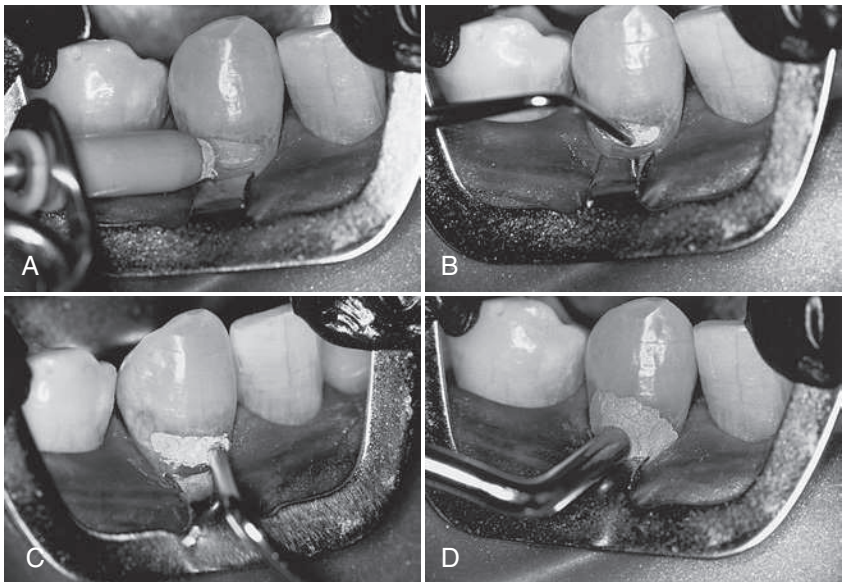
Insertion and Carving of the Amalgam

The amalgam carrier is used to insert the mixed amalgam into the preparation in small increments (Fig. 10.152A). Amalgam is condensed into the retention areas first by using an appropriate condenser (see Fig. 10.152B). Amalgam is then condensed against the mesial and distal walls of the preparation (see Fig. 10.152C). Finally a sufficient bulk of amalgam is placed in the central portion to allow for carving the correct contour (see Fig. 10.152D). As the surface of the restoration becomes more convex, condensation becomes increasingly difficult. It is important to guard against the "land sliding" of amalgam during overpacking. A large condenser or flat-bladed instrument held against amalgam may help resist pressure applied elsewhere on the restoration (Fig. 10.153).

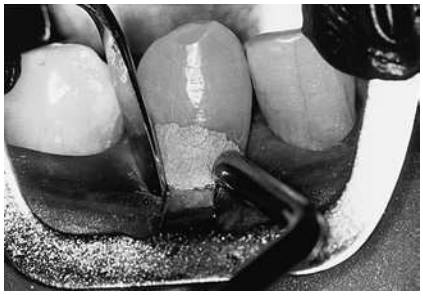
Carving may begin immediately after the insertion of amalgam (Fig. 10.154). All carving should be done using the side of the explorer tine or a Hollenback No. 3 carver held parallel to the margins. The side of the carving instrument always should rest on the unprepared tooth surface adjacent to the prepared cavosurface margin; this prevents overcarving or submargination of the restoration. The carving procedure is begun by removing excess amalgam to expose the incisal (or occlusal) margin. Removal of excess material



• **Fig. 10.151** Customized matrix band used to restore an area of proximal root caries. A, Conventional Tofflemire matrix with window cut into the band using a high-speed bur to allow access for condensation. B, Matrix in place around the tooth, allowing lingual access to preparation.



• **Fig. 10.152** Inserting amalgam. A, Place amalgam into the preparation in small increments. B, Condense first into the retention grooves with a small condenser. C, Condense against the mesial and distal walls. D, Overfill and provide sufficient bulk to allow for carving.



• **Fig. 10.153** Use a large condenser or a flat-bladed instrument to offer resistance to condensation pressure applied elsewhere on the restoration.

continues until the mesial and distal margins are exposed. Finally, material excess at the gingival margin is carved away. Carving the marginal areas should result in developing the desired convex contours in the completed restoration. Improper use of the carving instruments results in a poorly contoured restoration. Note in Fig. 10.20 how carving instruments are positioned to provide the desired contours. No amalgam excess should remain at the margins because amalgam may break away, creating a defect at the margin, or cause gingival inflammation and increased risk of secondary caries from biofilm accumulation.

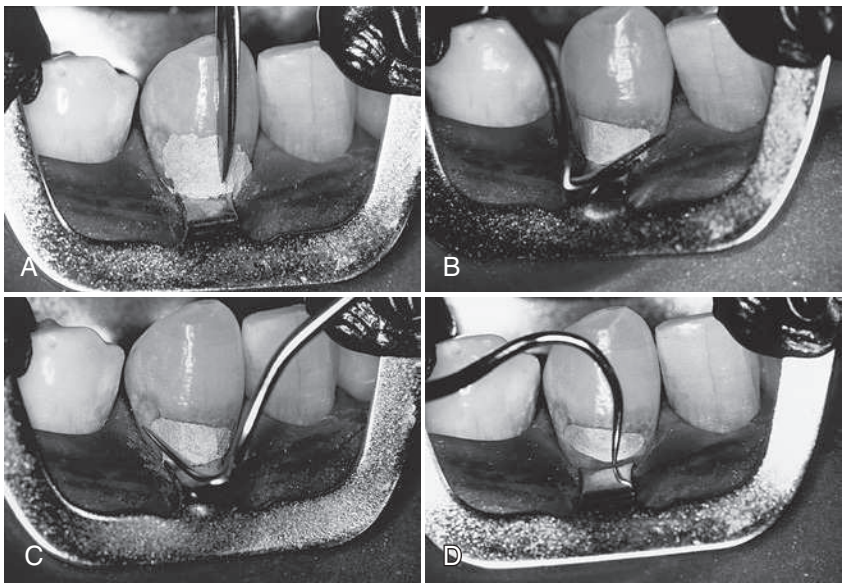
In some instances it is appropriate to change facial contours because of altered soft tissue levels (e.g., cervical lesions in periodontally treated patients). Facial contours may be increased (or relocated) only enough to prevent food impaction into the gingival sulcus and to provide access for the patient to clean the area. Overcontouring must be avoided as it results in reduced stimulation and cleansing of the gingiva during mastication.

When a rubber dam and the No. 212 cervical retainer have been used for isolation, the retainer is removed using care to open the jaws of the retainer wide enough to prevent damage to the surface of the restoration. The rubber dam is removed and the area is examined carefully to ensure that no amalgam particles remain in the sulcus.

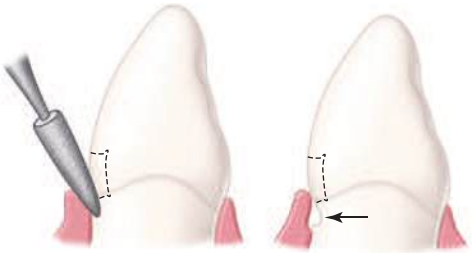
When a retraction cord is used for isolation, it may interfere with carving any excess amalgam at the gingival margin. If so, removal of the gross excess of amalgam is followed by careful removal of the cord, followed by completion of the carving along the margin.

Finishing and Polishing of the Amalgam

If carving procedures were performed correctly, no finishing of the restoration should be required. A slightly moistened cotton



• **Fig. 10.154** Carving and contouring the restoration. A, Begin the carving procedure by removing any excess and locating the incisal margin. B and C, An explorer may be used to remove the excess and locate the mesial and distal margins. D, Remove the excess and locate the gingival margin.



• **Fig. 10.155** Incorrect use of a pointed stone at the gingival margin results in the removal of cementum, notching of the tooth structure gingival to the margin, or both.

pellet held in cotton pliers may be used to further smooth the carved restoration. Additional finishing and polishing of amalgam restorations may be necessary, however, to correct a marginal discrepancy or to improve contour. Care is required when using stones or any rotating cutting instruments on margins positioned below the CEJ because of the possibility of removing cementum, notching the tooth structure gingival to the margin, or both (Fig. 10.155). Fig. 10.156 illustrates reshaping a rubber abrasive point to allow optimal access to the gingival portion of a Class V amalgam restoration. A rubber abrasive cup may be preferred for finishing convex surfaces, like Class V restorations, if used carefully.

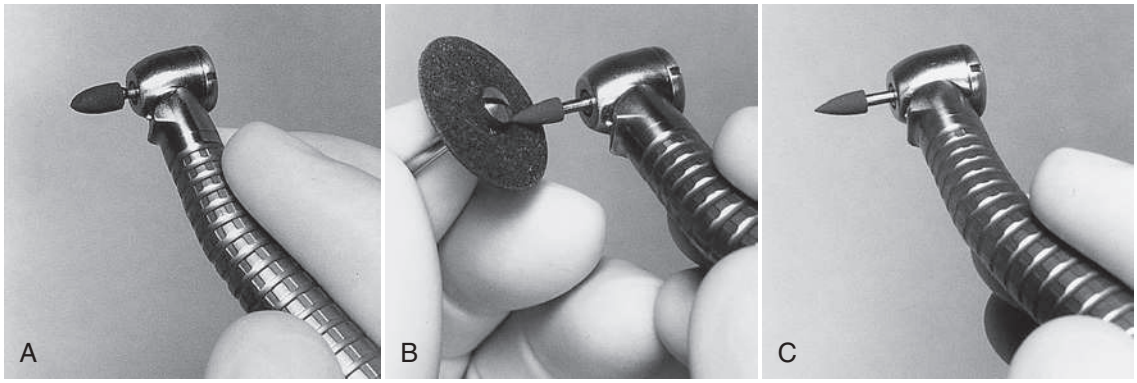
One measure of clinical success of cervical amalgam restorations is the length of time the restoration serves without failing (Fig. 10.157). Properly placed Class V amalgams have the potential to be clinically acceptable for many years. Some cervical amalgam restorations show evidence of failure, however, even after a short period. Inattention to tooth preparation principles, improper manipulation of the restorative material, and moisture contamination contribute to early failure. Extended service life of the restoration depends on the operator's care in following accepted treatment techniques and proper care by the patient. Any additional finishing and polishing of the Class V amalgam should follow the approach presented in General Concepts Guiding Restoration with Amalgam.

Clinical Technique for Class VI Amalgam Restorations

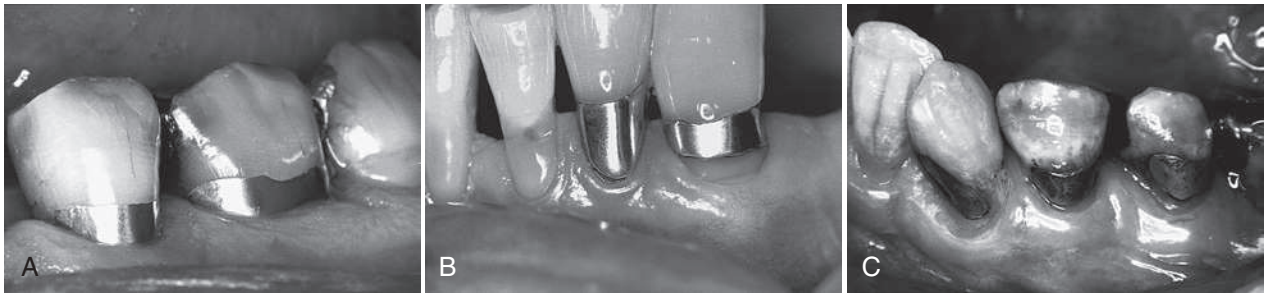
The Class VI tooth preparation is used to restore the incisal edge of anterior teeth or the cusp tip regions of posterior teeth (see Fig. 10.26C). Such tooth preparations are frequently indicated where attrition (loss of tooth substance from the occluding of food, abrasives, and opposing teeth) has removed enamel to expose the underlying dentin on those areas (Fig. 10.158). When the softer dentin is exposed, it wears faster than the surrounding enamel, resulting in "cupped-out" areas of the cusp tips. Intrinsic or extrinsic erosion also may be contributing to the excessive loss of tooth structure. Diagnostic steps, to identify the etiology of the tooth structure loss, are indicated and may involve referral for evaluation of potential gastroesophageal reflux disease. As the dentin support is lost, enamel begins to fracture away and more dentin is exposed, which often causes sensitivity. Sensitivity to hot and cold is a frequent complaint with Class VI lesions, and some patients are bothered by food impaction in deeper depressions. Enamel edges may become jagged and sharp to the tongue, lips, or cheek. Lip, tongue, or cheek biting is occasionally a complaint. Rounding and smoothing such incisoaxial (or occlusoaxial) edges is an excellent service to the patient. Early recognition and restoration of these lesions is recommended to limit the loss of dentin and the subsequent loss of enamel supported by this dentin.

The Class VI tooth preparation may be indicated to restore the hypoplastic pit occasionally found on cusp tips (Fig. 10.159). Such developmental faults are vulnerable to caries lesion formation, especially in high-risk patients, and should be restored as soon as they are detected.

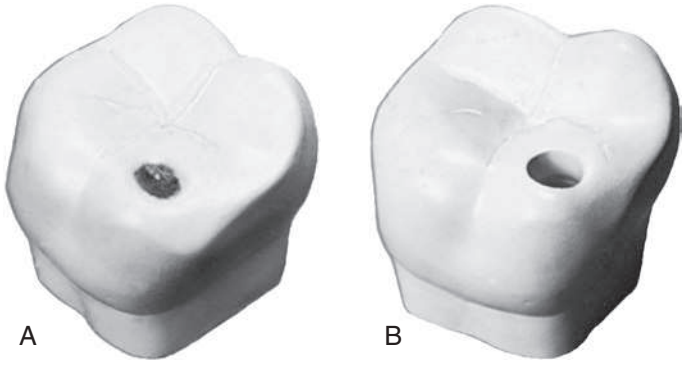
Amalgam may be selected for posterior Class VI preparations because of its wear resistance and longevity. Moisture control for Class VI restorations is usually achieved with cotton roll isolation. Class VI amalgam preparations may be accomplished with a small tapered fissure bur (e.g., No. 169 L or 271) and involves extension to place the cavosurface margin on enamel that has sound dentin support (see Fig. 10.158B). The preparation walls may need to



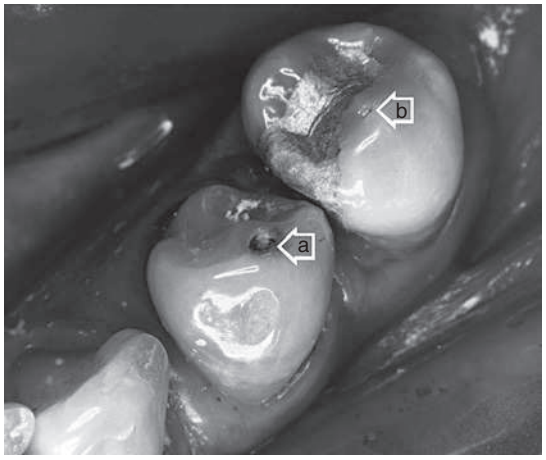
• **Fig. 10.156** Reshaping a rubber abrasive point against a mounted carborundum disk.



• **Fig. 10.157** A, Six-year-old cervical amalgam restorations. B, After 16 years, some abrasion and erosion are evident at the gingival margin of the lateral incisor and canine restorations. C, Twenty-year-old cervical amalgam restorations.



• **Fig. 10.158** Class VI preparation. A, Exposed dentin on the mesiofacial cusp. B, Tooth preparation necessary to restore the involved area.



• **Fig. 10.159** Class VI lesions. Carious cusp tip fault on the first premolar *(a).* Noncarious fault on the second premolar *(b).*

diverge occlusally to ensure a 90-degree cavosurface margin. A depth of 1.5 mm is sufficient to provide bulk of material for strength. Retention of the restoration is ensured by the creation of small undercuts along the internal line angles similar to other areas where secondary retention is indicated. Conservative tooth preparation is particularly important with Class VI preparations because it is easy to undermine the enamel on incisal edges and cusp tips. The preparation should be cleaned, carefully inspected as indicated in General Concepts Guiding Preparation for Amalgam Restorations, and any final modifications completed. Insertion, carving, finishing, and polishing are similar to procedures described in General Concepts Guiding Restoration with Amalgam.

Some older patients have excessive occlusal wear of most of their teeth in the form of large concave areas with much exposed dentin. Teeth with excessive wear may require indirect restorations or, at least, efforts to limit further loss of tooth structure.

Clinical Technique for Complex Amalgam Restorations

Indications

Complex posterior restorations are indicated when tooth structure is missing due to cusp fracture, severe caries lesion development,

or when replacement of existing restorative material is necessary. Complex amalgam preparations should be considered when large amounts of tooth structure are missing, when one or more cusps need to be covered, and when increased resistance and retention forms are needed (Fig. 10.160).[121] Complex amalgams may be used as (1) definitive (final) restorations, (2) foundations, (3) control restorations in teeth that have a questionable pulpal or periodontal prognosis, or (4) control restorations in teeth with acute or severe caries lesions. When determining the appropriateness of a complex amalgam restoration, the factors discussed in the following sections must be considered.

Resistance and Retention Forms

In a tooth with a severe caries lesion or existing restorative material, any undermined enamel or weak tooth structure subject to fracture must be removed and replaced. Usually a weakened tooth is best restored with a properly designed indirect (usually cast) restoration that prevents tooth fracture caused by mastication forces (see Online Chapter 18). In selected cases, amalgam preparations that improve the resistance form of a tooth may be designed (Fig. 10.161).

When conventional retention features are not adequate because of insufficient remaining tooth structure, the retention form may be enhanced by using auxiliary features such as slots and pins. The type of retention features needed depend on the amount of tooth structure remaining and the tooth being restored. As more tooth structure is lost, more auxiliary retention is required. Slots and pins provide additional resistance and retention form to the tooth when remaining vertical walls are inadequate.

Status and Prognosis of the Tooth

A tooth with a severe caries lesion that may require endodontic therapy or crown lengthening or that has an uncertain periodontal prognosis often is treated initially with a control restoration. A control restoration helps (1) protect the pulp from the oral cavity (i.e., fluids, thermal stresses, pH changes, bacteria), (2) provide an anatomic contour that is consistent with gingival health, (3) facilitate control of acidogenic biofilm and resultant caries risk, and (4) provide some resistance against tooth fracture (or propagation of an existing fracture). (See Chapter 2 for caries-control rationale and techniques.)

The status and prognosis of the tooth determine the size, number, and placement of retention features. Larger restorations generally require more retention. The size, number, and location of retention features demand greater care in smaller teeth. Carelessness may increase the risk of pulpal irritation or exposure.

Role of the Tooth in Overall Treatment Plan

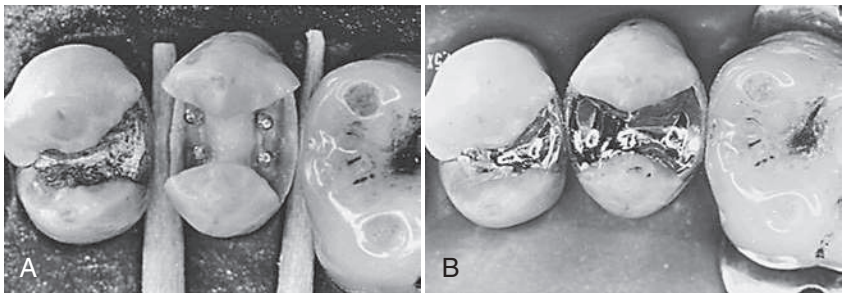
The restorative treatment choice for a tooth is influenced by its role in the overall treatment plan. Although complex amalgam restorations are used occasionally as an alternative to indirect restorations, particularly due to cost savings, they often are used as foundations for full-coverage restorations. Abutment teeth for fixed prostheses may use a complex restoration as a foundation (Fig. 10.162). Extensive caries or previous restorations on abutment



• **Fig. 10.160** Mesioocclusodistolingual (MODL) complex amalgam tooth No. 3.



• **Fig. 10.162** Mesioocclusodistofacial (MODF) amalgam foundation for abutment tooth No. 15 in preparation for a 3 unit fixed partial denture.



• **Fig. 10.161** Maxillary second premolar weakened by an extensive caries lesion and by the small fracture line extending mesiodistally on the center of the excavated dentinal wall. A, Minikin pins placed in the gingival floor improve resistance form after amalgam has been placed. B, Restorations polished.



• **Fig. 10.163** Mesioocclusodistolingual (MODL) complex amalgam tooth #19.

teeth for removable prostheses generally indicate an indirect restoration for the resistance and retention forms and for development of external surface contours for retention of the prosthesis. A tooth may be treated with a complex direct restoration if adequate resistance and retention forms are provided. For patients with periodontal and orthodontic problems, the complex restoration may be the restoration of choice until the final phase of treatment, when indirect restorations may be preferred.

Occlusion and Economics

Complex amalgam restorations are sometimes indicated as interim restorations for teeth that require elaborate occlusal alterations, ranging from vertical dimension changes to correcting occlusal plane discrepancies. When cost of indirect restorations is a major factor for the patient, the complex direct amalgam restoration may be an appropriate treatment option, provided that adequate resistance and retention forms are included (Fig. 10.163).[122,123]

Age and Health of Patient

For some older patients and/or those who are debilitated, complex amalgam restoration may be the treatment preferred over the more expensive and time-consuming indirect restoration.

Contraindications

The complex amalgam restoration may be contraindicated if the tooth cannot be restored properly with direct restoration because of anatomic or functional considerations (or both). The complex amalgam restoration also may be contraindicated if the area to be restored has esthetic importance for the patient.

Advantages

Conservation of Tooth Structure

The preparation for a complex amalgam restoration is usually more conservative than the preparation for an indirect restoration.

Appointment Time

The complex restoration may be completed in one appointment. An indirect restoration generally requires at least two appointments unless it is done using a chairside computer-assisted design/computer-assisted machining (CAD/CAM) system.

Resistance and Retention Forms

Amalgam restorations with cusp coverage significantly increase the fracture resistance of weakened teeth compared with amalgam restorations without cusp coverage.[124] Resistance and retention forms may be significantly increased by the use of slots and pins (discussed in subsequent sections).

Reduced Cost

The complex amalgam restoration may be utilized to reinforce and stabilize compromised posterior teeth at a much reduced cost to the patient. It may serve as a definitive final restoration or an intermediate-term restoration with a long-term goal of indirect, more costly restoration and protection of the tooth.

Disadvantages

Tooth Anatomy

Proper contours and occlusal contacts and anatomy are sometimes difficult to achieve with large, complex restorations.

Resistance Form

Resistance form is more difficult to develop with a complex amalgam as compared to the preparation of a tooth for a cusp-covering onlay (skirting axial line angles of the tooth [see Online Chapter 18]) or a full crown. The complex amalgam restoration does not protect the tooth from fracture as effectively as a full-coverage indirect restoration.

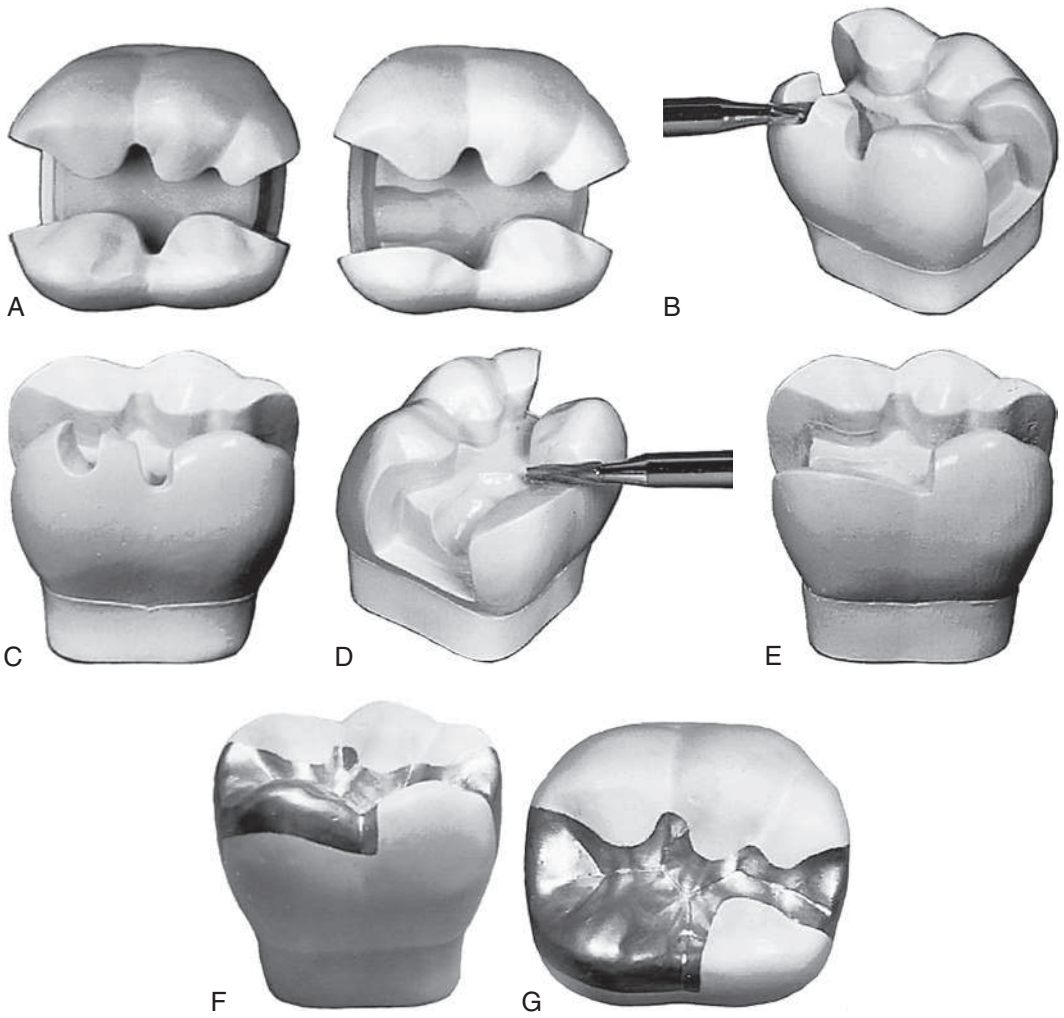
Tooth Preparation for Complex Amalgam Restorations

In this section various types of Complex Amalgam preparation procedures are discussed before the restoration technique is presented. Also the word "vertical" is used to describe tooth preparation walls and other preparation aspects that are approximately parallel to the long axis of the tooth. The word "horizontal" is used to describe the walls and other aspects that are approximately perpendicular to the long axis of the tooth. Every effort should be made to protect the pulp as early as possible when accomplishing the procedures associated with complex amalgam restorations.

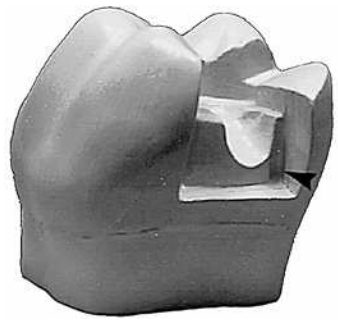
Preparation for Cusp Coverage Complex Amalgams

General concepts of initial tooth preparation presented in earlier sections apply to the complex amalgam preparations described here. When a caries lesion is extensive with resultant loss of dentin necessary to support occlusally loaded areas of the tooth, reduction of one or more of the cusps followed by coverage with amalgam may be indicated. When the facial or lingual extension exceeds two thirds the distance from a primary fissure toward the cusp tip (or when the faciolingual extension of the occlusal preparation exceeds two thirds the distance between the facial and lingual cusp tips), reduction of the cusp for coverage with amalgam usually is required for the development of adequate resistance form (Fig. 10.164A; see also Fig. 4.13). For cusps prone to fracture, coverage reduces the risk of fracture and extends the life of the restoration.[125,126] Complex amalgam restorations that cover one or more cusps have documented longevity of 72% after 15 years.[123,127]

The operator uses the bur to reduce the cusp, following the mesiodistal inclines of the cusp; this results in a uniform reduction (see Fig. 10.164B–E). If only one of two facial (or lingual) cusps is to be covered, the cusp reduction should extend slightly beyond the facial (or lingual) groove area, provide the correct amount of tooth structure removal, and meet the adjacent, unreduced cusp to create a 90-degree cavosurface margin. This approach results in adequate thickness and edge strength of the amalgam (see Fig. 10.164F and G). Cusp reduction and coverage reduce the amount of vertical preparation wall height and increase the need for the use of secondary retention features (Fig. 10.165). An increased retention form may be provided by proximal box retention grooves



• **Fig. 10.164** Covering the cusp with amalgam. A, Comparison of the mesial aspects of normally extended *(left)* and extensive *(right)* mesiooccclusodistal tooth preparation. The resistance form of the mesiolingual cusp with extensive preparation is compromised and cusp coverage with amalgam is indicated. B, Preparing depth cuts. C, Depth cuts prepared. D, Reducing the cusp. E, Cusp reduced. F and G, Final restoration.



• **Fig. 10.165** A retention groove is associated with the remaining vertical wall of a cusp requiring reduction.

but may require the use of slots and/or pins (as described in subsequent sections). If indicated, cusp reduction and coverage increase the resistance form of the tooth.[123,128]

Reduction of the unsupported cusp should be accomplished during the initial tooth preparation because it improves access and visibility for subsequent steps. If the cusp to be reduced is located at the correct occlusal height before preparation, depth cuts should be made on the remaining occlusal surface of each cusp to be covered, using the side of a carbide fissure bur or a suitable diamond instrument (see Fig. 10.164B). The depth cuts should be a minimum of 2 mm for functional cusps and 1.5 mm for nonfunctional cusps.[126] These dimensions provide adequate strength for amalgam by preventing flexure of the restoration during loading. To reduce the cusp, the dentist orients the bur parallel to the cuspal incline (lingual incline for facial cusp reductions and facial inclines for lingual cusp reduction) and makes several depth cuts in the cusp (to a depth of 1.5 or 2 mm as indicated). The depth cuts provide guides for the correct amount of cusp reduction. Without depth cuts, after the beginning reduction of the cusp, the operator may no longer know how much more reduction is necessary. When using the restoration to correct an occlusal relationship, if the unreduced cusp height is located at less than the correct occlusal height, the depth cuts may be less. Likewise, if the unreduced cusp height is located higher than the correct occlusal height, the depth cuts may be deeper. The goal is to ensure that the final restoration has restored cusps with

a minimal thickness of 2 mm of amalgam for functional cusps and 1.5 mm of amalgam for nonfunctional cusps (see Fig. 10.164C), while developing an appropriate occlusal relationship. Using the depth cuts as a guide, the reduction is completed to provide for a uniform reduction of tooth structure (see Fig. 10.164D). The occlusal contour of the reduced cusp should be similar to the normal contour of the unreduced cusp. Any sharp internal corners of the tooth preparation formed at the junction of prepared surfaces should be rounded to reduce stress concentration in the amalgam and improve its resistance to fracture from occlusal forces (see Fig. 10.164E). When reducing only one of two facial or lingual cusps, the cusp reduction should be extended just past the facial or lingual groove, creating a vertical wall against the adjacent unreduced cusp. Fig. 10.164F and G illustrate a final restoration.

Mandibular First Premolar

The lingual cusp of the mandibular first premolar may need to be reduced for coverage with amalgam if the lingual margin of the occlusal step extends more than two thirds the distance from the central fissure to the cuspal eminence (Fig. 10.166). Special attention is given to such cusp reduction because retention is severely diminished when the cusp is overreduced resulting in elimination of the lingual wall of the occlusal portion. Depth cuts of 1.5 mm aid the operator in establishing the correct amount of cusp reduction and in conserving a small portion of the lingual wall in the occlusal step. It is acceptable when restoring diminutive nonfunctional cusps, such as the lingual cusp of a mandibular first premolar, to reduce the cusp only 0.5 to 1 mm and restore the cusp to achieve an amalgam thickness of 1.5 mm. This procedure conserves more of the lingual wall of the isthmus for added retention form.

Maxillary First Molar

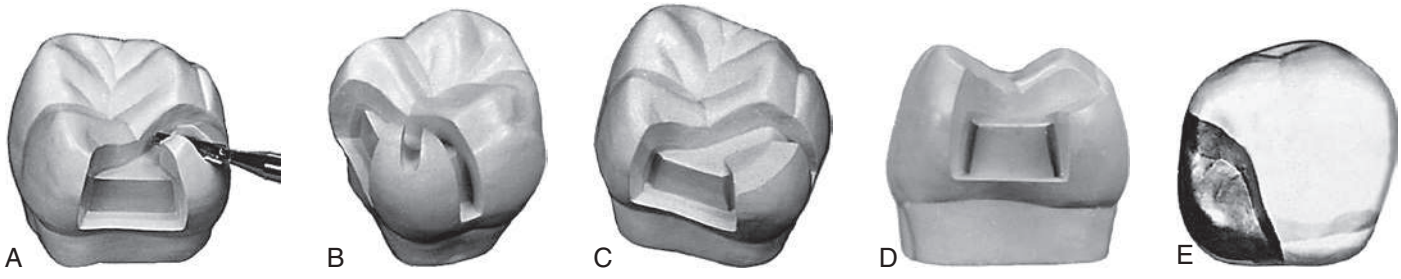
The procedure for reducing the distolingual cusp of a maxillary first molar for coverage is illustrated in Fig. 10.167. Extending the facial or lingual wall of a proximal box to include the entire cusp is accomplished (if necessary) to include weak or carious tooth structure or existing restorative material. The lingual groove may be extended arbitrarily to increase the retention form (see Fig. 10.167B and C). When possible, opposing vertical walls should be formed to converge occlusally, to enhance the primary retention form (see Fig. 10.167C and D). The pulpal and gingival walls should be relatively flat and perpendicular to the long axis of the tooth. Secondary retention features may be indicated when the divergence of proximal walls results in limited primary retention form (see Fig. 10.167D). Mesial extension of the preparation into the mesiolingual cusp should be enough to allow placement of the distooblique and lingual groove within the amalgam restoration (see Fig. 10.167E).

Mandibular First Molar

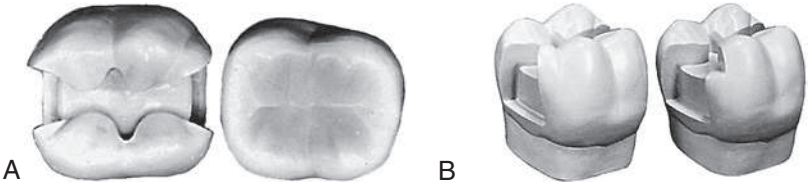
Reducing the distal cusp is an alternative to extending the entire distofacial wall when the occlusal margin crosses the cuspal eminence (Fig. 10.168A; compare with Fig. 10.83). A minimal reduction



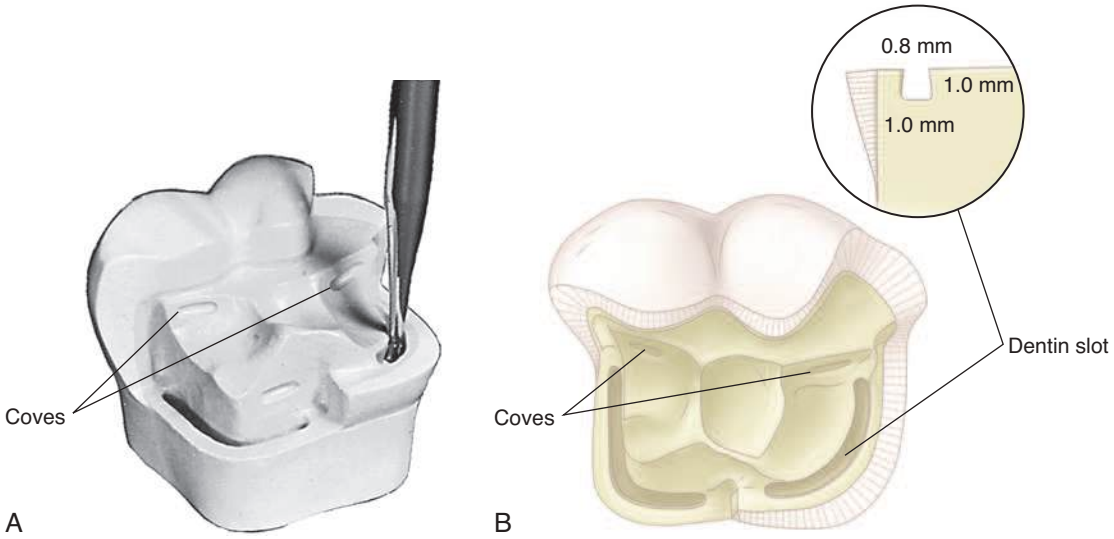
• **Fig. 10.166** Mandibular first premolar with the lingual cusp reduced for coverage with amalgam.



• **Fig. 10.167** Reduction of the distolingual cusp of the maxillary molar. A, Cutting a depth gauge groove with the side of the bur. B, Completed depth gauge groove. C and D, Completed cusp reduction. E, Distoocclusolingual complex amalgam.



• **Fig. 10.168** Mandibular first molar. A, Reduction of the distal cusp is indicated when the occlusal margin crosses the cuspal eminence. B, Distofacial view of the distal cusp shown in *A* before reduction for coverage *(left)*; the distal cusp after reduction *(right).* A reduction of 2 mm is necessary to provide for minimal 2-mm thickness of amalgam.



• **Fig. 10.169** Slots. A and B, With a No. 330 bur, dentinal slots are prepared approximately 1 mm deep and 0.5 to 1 mm inside the DEJ.

of 2 mm will allow a 2-mm thickness of amalgam over the reduced cusp (see Fig. 10.168B). The cusp reduction should result in a butt joint between the tooth structure and amalgam. When possible, reducing and covering the distal cusp is more desirable than extending the distofacial proximal margin because this conserves tooth structure of the functional cusp, and the remaining portion of the cusp helps in applying the matrix for the development of proper distofacial embrasure form. The plane of the reduced cusp should parallel the facial (or lingual) outline of the unreduced cusp mesiodistally and the cuspal incline emanating from the central groove faciolingually.

Tooth Preparation for Slot-Retained Amalgam Restorations

When loss of vertical coronal height is approximately 2 to 4 mm, then use of slots, in horizontal areas where no vertical walls remain, may be indicated to gain secondary retention form for the complex amalgam restoration (Figs. 10.169, 10.170, and 10.171). A slot is a horizontal retention groove in dentin (see Fig. 10.169). Slots are placed in the gingival floor of a preparation with a No. 330 bur (see Fig. 10.169B, inset). In general, slots should be about 1 mm wide and 1 mm deep, should be placed in the line-angle areas of the tooth, should be 2 to 4 mm in length (depending on the distance between remaining vertical walls), and be positioned 0.5 to 1 mm inside the DEJ. In general, one slot per missing axial line angle should be used. Increasing the width and depth of the slot does not increase the retentive strength of the amalgam restoration.[129] Slot length depends on the extent of the tooth preparation. A slot may be continuous or segmented, depending on the amount of missing tooth structure and whether pin retention is being planned (see subsequent sections). Shorter slots provide as much resistance to horizontal force as do longer slots.[130] Preparations with shorter slots fail less frequently than do preparations with longer slots.[130] Slot design should be such that it is independently retentive as a singular feature (the internal walls of the slot should be convergent) (see Fig. 10.169B, inset). In addition, the walls of the slot should work together with other vertical walls to create



• **Fig. 10.170** Coves prepared in dentin with No. ¼ bur, where appropriate.



• **Fig. 10.171** Vertical grooves prepared in dentin with a No. ¼ round or No. 169L bur, where appropriate.

an overall convergence of the preparation so as to enhance retention of the restoration.

In addition to any slot placement, retention grooves and coves may be utilized in association with remaining vertical walls (Figs. 10.172 and 10.173). Coves and retention grooves should be prepared when possible. Coves are prepared in a horizontal plane, and grooves are prepared in a vertical plane.

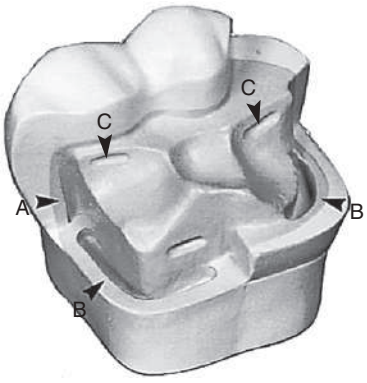
Coves also may be used in preparations using slots (see Fig. 10.169). Proximal retention grooves, as described for wide Class II preparations, also are placed in the proximal box and in other locations where sufficient vertical tooth preparation permits (see Figs. 10.165 and 10.171).

Tooth Preparation for Pin-Retained Amalgam Restorations

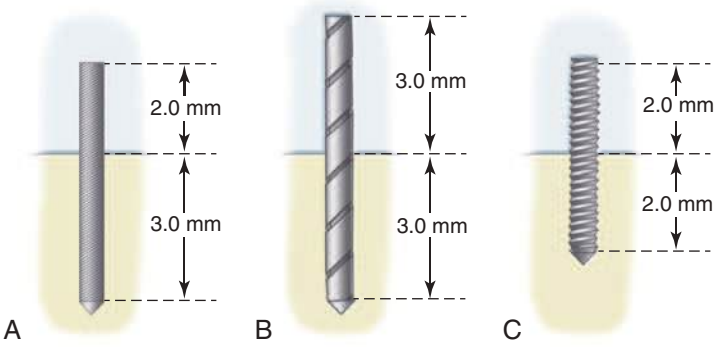
When severe carious destruction or cusp fracture has resulted in the loss of 4 mm or more of vertical coronal height, consideration should be given to the use of a parapulpal retention pin to regain some of the lost vertical height. A *pin-retained amalgam* (also referred to as a "pin-amalgam") is defined as any restoration requiring the placement of one or more pins in dentin to provide adequate resistance and retention forms. Pins are used whenever adequate resistance and retention forms are not able to be established with slots, grooves, or undercuts only.[131] Pins placed into prepared pinholes (also referred to as *pin channels*) are used to provide auxiliary resistance and retention forms. The pin-retained amalgam is an important adjunct in the restoration of teeth with extensive caries lesions or fractures.[132] Amalgam restorations including pins have significantly greater retention compared with restorations using boxes only or restorations relying solely on bonding systems.[133] However, caution is indicated when using pins. Preparing pinholes and placing pins may create craze lines or fractures and internal stresses in dentin.[134-136] Such craze lines and internal stress areas may have little or no clinical significance, but they may be important when minimal dentin is present. Preparation steps for pin retention have increased risk of penetrating into the pulp or perforating the external tooth surface. The use of pins decreases the tensile and horizontal strength of associated amalgam restorations.[137,138]

Types of Parapulpal Retention Pins

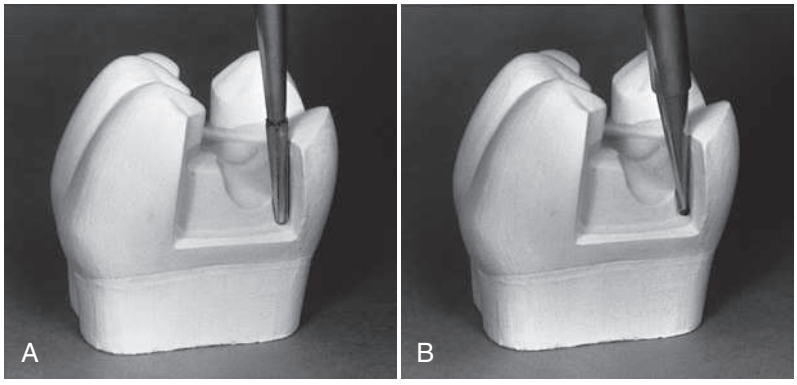
The most frequently used type of parapulpal pin is the self-threading pin. The pin-retained amalgam restoration using self-threading pins originally was described by Going in 1966.[139] Friction-locked and cemented pins, although still available, are rarely used (Fig. 10.174). The diameter of the prepared pinhole is 0.0381 to 0.1016 mm smaller than the diameter of the pin (Table 10.1). The threads engage dentin as the pin is inserted, thus retaining the pin. The elasticity (resiliency) of dentin permits insertion of a threaded pin into a hole of smaller diameter.[140] Although the threads of self-threading pins do not engage dentin for their entire width, self-threading pins are the most retentive of the three types of pins (Fig. 10.175), being three to six times more retentive than cemented pins.[141-143]



• **Fig. 10.172** Groove (A), slots (B), and coves (C).



• **Fig. 10.174** Three types of pins. A, Cemented. B, Friction locked. C, Self-threading.



• **Fig. 10.173** Placement of retention grooves. A, Position of No. 169L bur to prepare the retention groove. B, Groove prepared with No. ¼ bur.

Vertical and horizontal stresses, potentially resulting in craze lines (cracks), are generated in dentin when a self-threading pin is inserted. Craze lines in dentin are related to the size of the pin. The insertion of 0.787-mm self-threading pins produces more dentinal craze lines than does the insertion of 0.533-mm self-threading pins.[144] Some evidence suggests, however, that self-threading pins may not cause dentinal crazing.[140] Pulpal stress is maximal when the self-threading pin is inserted perpendicular to the pulp.[145] The depth of the pinhole varies from 1.3 to 2 mm, depending on the diameter of the pin used.[146] Recommended pinhole depth is generally accepted to be 2 mm.

Several styles of self-threading pins are available. The Thread Mate System (TMS) (Coltène/Whaledent Inc., Cuyahoga, Ohio) is the most widely used self-threading pin system because of its (1) versatility, (2) wide range of pin sizes, (3) color-coding system, and (4) greater retentiveness.[147,148] TMS pins are available in gold-plated stainless steel or in titanium. Titanium alloy pins (Max Restorative Pins, Coltène/Whaledent Inc.) are also available.

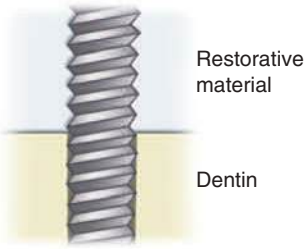
Factors Affecting Retention of the Pin in Dentin and Amalgam

Type

With regard to the retentiveness of the pin in dentin, the self-threading pin is the most retentive, the friction-locked pin is intermediate, and the cemented pin is the least retentive.[141]

Surface Characteristics

The number and depth of the elevations (serrations or threads) on the pin influence the retention of the pin in the amalgam restoration. The shape of the self-threading pin results in the greatest retention of amalgam.



• **Fig. 10.175** The complete width of the threads of self-threading pins does not engage dentin.

Orientation, Number, and Diameter

Placing pins in a nonparallel manner increases their ability to retain the restoration; however, bending pins to improve their retention in amalgam is not advised as the bends may interfere with adequate condensation of amalgam around the pin and decrease amalgam retention. In addition, bending also may weaken the pin and risk fracture of the adjacent dentin. Pins should be bent only to provide for an adequate amount of amalgam (approximately 1 mm) between the pin and the external surface of the finished restoration (on its lateral surface). Only the manufacturer's specific bending tool should be used to bend a pin, not other hand instruments (see subsequent section, Pin Insertion).

In general, increasing the number of pins increases their retention in dentin and amalgam. However, the potential benefit of increasing the number of pins must be compared with the potential problems. As the number of pins increases, (1) the crazing of dentin and the potential for fracture increase, (2) the amount of available dentin between the pins decreases, and (3) the strength of the amalgam restoration decreases.[149,150] Also, as the diameter of the pin increases, retention in dentin and amalgam generally increases. As the number, depth, and diameter of pins increase, the danger of perforating into the pulp or the external tooth surface increases. Numerous long pins also may severely compromise the condensation of amalgam and its adaptation to the pins. A pin technique (i.e., a balance between pin orientation, size, and number) that permits optimal retention with minimal danger to the remaining tooth structure should be used.[151]

Extension Into Dentin and Amalgam

Self-threading pin extension into dentin and amalgam should be approximately 1.5 to 2 mm to preserve the strength of dentin and amalgam.[141] Extension greater than this is unnecessary for pin retention and is contraindicated.

Pin Placement Factors and Techniques

Pin Size

Four sizes of TMS pins are available (Fig. 10.176), each with a corresponding color-coded drill (see Table 10.1). Familiarity with drill sizes and their corresponding colors is necessary to ensure that a proper-sized pinhole is prepared for the desired pin. It is

TABLE 10.1 The Thread Mate System (TMS) Pins

Name	Illustration (Not to Scale)	Color Code	Pin Diameter (inches/mm)*	Drill Diameter (inches/mm)*	Total Pin Length (mm)	Pin Length Extending From Dentin (mm)
Regular (standard)		Gold	0.031/0.78	0.027/0.68	7.1	5.1
Regular (self-shearing)		Gold	0.031/0.78	0.027/0.68	8.2	3.2
Regular (two-in-one)		Gold	0.031/0.78	0.027/0.68	9.5	2.8
Minim (standard)		Silver	0.024/0.61	0.021/0.53	6.7	4.7
Minim (two-in-one)		Silver	0.024/0.61	0.021/0.53	9.5	2.8
Minikin (self-shearing)		Red	0.019/0.48	0.017/0.43	7.1	1.5
Minuta (self-shearing)		Pink	0.015/0.38	0.0135/0.34	6.2	1

*1 mm = 0.03937 inch.

difficult to specify a particular size of pin that is always appropriate for a particular tooth. Two determining factors for selecting the appropriate-sized pin are 1) the amount of dentin available to safely receive the pin and 2) the amount of retention desired. In the TMS system, the pins of choice for severely involved posterior teeth are the Minikin (0.48 mm) and, occasionally, the Minim (0.61 mm). The Minikin pins usually are selected to reduce the risk of dentin crazing, pulpal penetration, and root-surface perforation. The Minim pins usually are used as a backup in case the pinhole for the Minikin is overprepared or the pin threads strip dentin during placement resulting in a lack of Minikin pin retention in the dentin.

Larger diameter pins have the greatest retention.[152] For example, the Minuta (0.38 mm) pin is approximately half as retentive as the Minikin and one third as retentive as the Minim pin.[147,148] The Minuta is usually too small to provide adequate retention in posterior teeth. The Regular (0.78 mm) or largest diameter pin is rarely used because a significant amount of crazing in the tooth (dentin and enamel) may be created during its insertion.[144,153] Of the four types of pins, the Regular pin is associated with the highest incidence of dentinal crack communication with the pulp chamber.[136]

Number of Pins

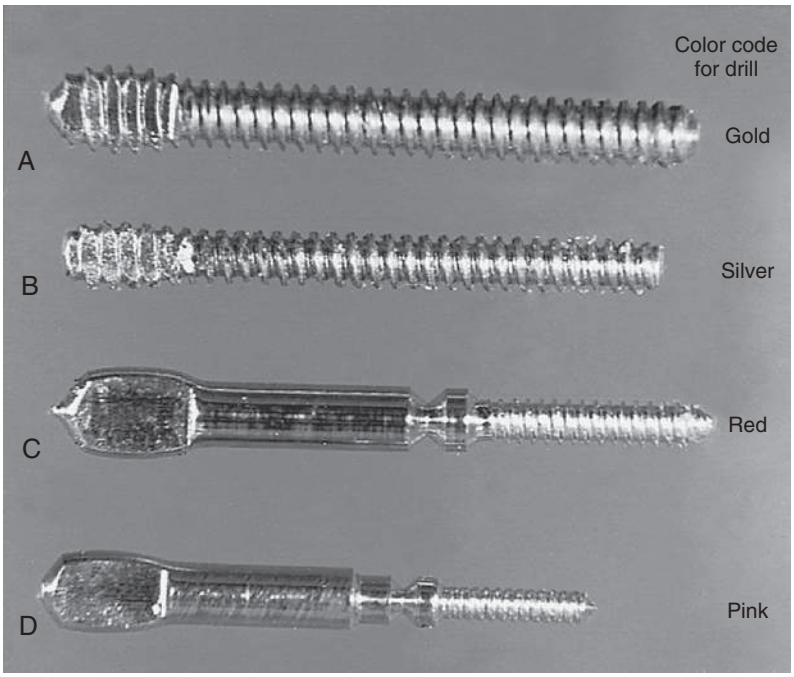
Factors to be considered when deciding how many pins are required include (1) the amount of missing tooth structure, (2) the amount of dentin available to safely receive the pins, (3) the amount of retention required, and (4) the size of the pins. As a general rule, one pin per missing axial line angle should be used. The fewest pins possible should be used to achieve the desired retention for a given restoration. Although the retention of the restoration increases as the number of pins increases, an excessive number of pins may fracture the tooth and significantly weaken the amalgam restoration.

Pinhole Location

Factors that aid in determining the pinhole locations include (1) knowledge of normal pulp anatomy and external tooth contours, (2) a current radiograph of the tooth, (3) a periodontal probe, and (4) the patient's age. Although the radiograph is only a two-dimensional image of the tooth, it may give an indication of the position of the pulp chamber and the contour of the mesial and distal surfaces of the tooth. Consideration also must be given to the placement of pins in areas where the greatest bulk of amalgam will occur so as to compensate (through the reinforcing tendency of the amalgam restoration) for the weakening effect of the pins on the tooth structure.[154] Areas of occlusal contacts on the restoration must be anticipated because a pin oriented vertically and positioned directly below an occlusal contact point weakens amalgam significantly.[155] Occlusal clearance should be sufficient to provide 2 mm of amalgam over the pin.[156,157]

Several attempts have been made to identify the ideal location of the pinhole.[135,144,158,159] The following principles of pin placement are recommended. In the cervical third of molars and premolars (where most pins are located), pinholes should be located near the line angles of the tooth except as described later.[151,160] The pinhole should be positioned no closer than 0.5 to 1 mm to the DEJ or no closer than 1 to 1.5 mm to the external surface of the tooth, whichever distance is greater (Fig. 10.177). Before the final decision is made about the location of the pinhole, the operator should probe the gingival crevice carefully to determine if any abnormal contours exist that would predispose the tooth to an external perforation. The pinhole should be parallel to the adjacent external surface of the tooth.

The position of a pinhole must not result in the pin being so close to a vertical wall of tooth structure that condensation of amalgam against the pin or wall is jeopardized (Fig. 10.178A). It may be necessary to first prepare a recess in the vertical wall with the No. 245 bur to permit proper pinhole preparation and to

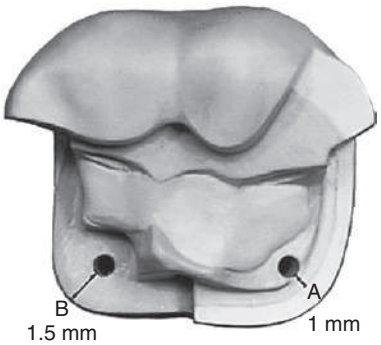


• **Fig. 10.176** Four sizes of the Thread Mate System (TMS) pins. A, Regular (0.78 mm). B, Minim (0.61 mm). C, Minikin (0.48 mm). D, Minuta (0.38 mm).

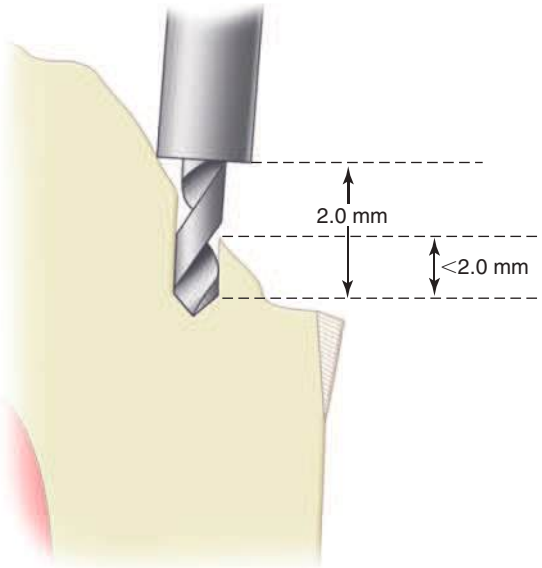
provide a minimum 0.5-mm clearance around the circumference of the pin for adequate condensation of amalgam (see Fig. 10.178B and C).[161] If necessary, after a pin is inappropriately placed, the operator should provide clearance around the pin to provide sufficient space for the smallest condenser nib to ensure that amalgam is condensed adequately around the pin. A No. 169L bur may be used, taking care not to damage or weaken the pin. Pinholes should be prepared on a flat surface that is perpendicular to the proposed direction of the pinhole. This will help prevent the drill tip from slipping or "crawling" and will allow a depth-limiting drill (discussed later) to prepare the pinhole as deeply as intended (Fig. 10.179).

Whenever three or more pinholes are placed, they should be located at different vertical levels on the tooth, if possible; this reduces stresses resulting from pin placement in the same horizontal plane of the tooth. Spacing between pins, or the interpin distance, must be considered when two or more pinholes are prepared. The optimal interpin distance depends on the size of pin to be used. The minimal interpin distance is 3 mm for the Minikin pin and 5 mm for the Minim pin.[160] Maximal interpin distance results in lower levels of stress in dentin.[162]

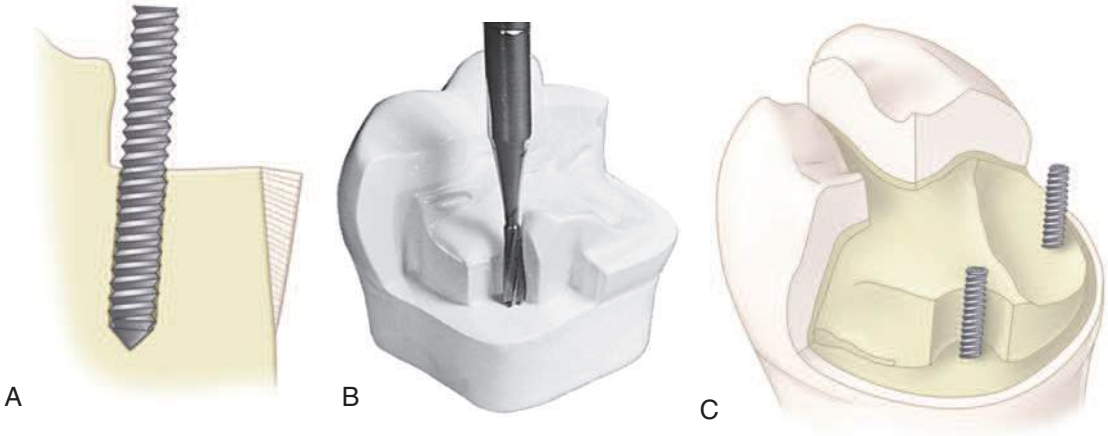
Several posterior teeth have anatomic features that may preclude safe pinhole placement. Fluted and furcal areas should be avoided.[160] External perforation may result from pinhole placement (1) over the prominent mesial concavity of the maxillary first premolar; (2) at the midlingual and midfacial bifurcations of the mandibular first and second molars; and (3) at the midfacial, midmesial, and middistal furcations of the maxillary first and second molars. Pulpal penetration may result from pin placement at the mesiofacial corner of the maxillary first molar and the mandibular first molar. Mandibular posterior teeth (with their lingual crown tilt), teeth that are rotated in the arch, and teeth that are abnormally tilted in the arch warrant careful attention before and during pinhole placement. When possible, the location of pinholes on the distal surface of mandibular molars and lingual surface of maxillary molars should be avoided. Obtaining the proper direction for preparing a pinhole in these locations is difficult because of the abrupt flaring of the roots just apical to the CEJ (Fig. 10.180). If the pinhole is placed parallel to the external surface of the tooth crown in these



• **Fig. 10.177** Pinhole position. A, Position relative to the DEJ. B, Position relative to external tooth surface.



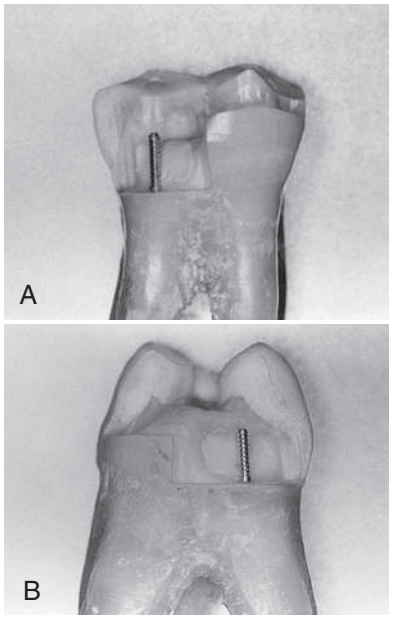
• **Fig. 10.179** Use of a depth-limiting drill to prepare a pinhole in the surface that is not perpendicular to the direction of the pinhole results in a pinhole of inadequate depth.



• **Fig. 10.178** A, Pin placed too close to the vertical wall such that adequate condensation of amalgam is jeopardized. B and C, Recessed area prepared in the vertical wall of the mandibular molar with a No. 245 bur to provide adequate space for amalgam condensation around the pin.

areas, penetration into the pulp is likely.[160] For mandibular second molars that are severely tilted mesially, care must be exercised to orient the drill properly to prevent external perforation on the mesial surface and pulpal penetration on the distal surface (Fig. 10.181). Because of limited interarch space, it is sometimes difficult to orient the twist drill (see subsequent section, Pinhole Preparation) correctly when placing pinholes at the distofacial or distolingual line angles of the mandibular second and third molars (Fig. 10.182).

When the pinhole locations have been determined, a No. ¼ round bur is first used to prepare a pilot hole (dimple) approximately one half the diameter of the bur at each location (Fig. 10.183). The purpose of this pilot hole is to permit more accurate placement of the twist drill and to prevent the twist drill from moving laterally (i.e., "crawling") when it has begun to rotate.



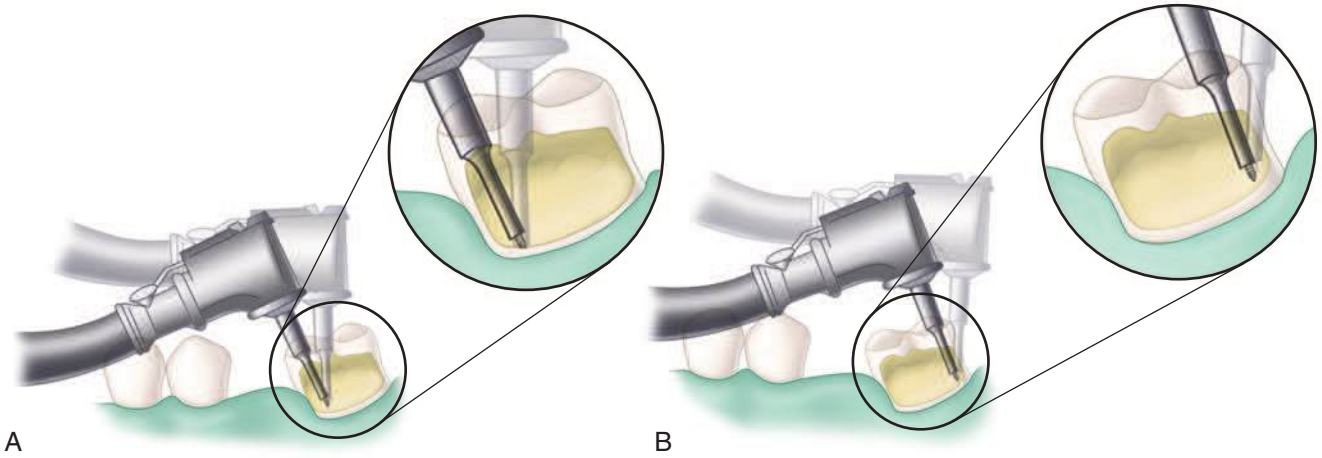
• **Fig. 10.180** Distal flaring of the mandibular molar (A) and palatal root flaring of the maxillary molar (B). Root angulation should be considered before pinhole placement.

Pinhole Preparation

The KODEX drill (a twist drill provided in the TMS system) should be used for preparing pinholes (Fig. 10.184). The aluminum shank of this drill, which acts as a heat absorber, is color coded to allow ready matching with the appropriate pin size (Table 10.2; see also Table 10.1). The drill shanks for the Minuta and Minikin pins are tapered to provide a built-in "wobble" when placed in a latch-type contra-angle handpiece. This wobble allows the drill to be "free floating" and to align itself as the pinhole is prepared. This feature minimizes the potential for dentinal crazing or the breakage of small drills.

Because the optimal depth of the pinhole into the dentin is 2 mm (only 1.5 mm for the Minikin pin), a depth-limiting drill should be used to prepare the hole (see Fig. 10.184). This type of drill is able to prepare the pinhole to the correct depth only when used on a flat surface that is perpendicular to the drill (see Fig. 10.178C; contrast with Fig. 10.179). When the location for starting a pinhole is not perpendicular to the desired pinhole direction, the location area should be flattened or the standard twist drill should be used (see Fig. 10.184). The standard twist drill has blades that are 4 to 5 mm in length, which would allow preparation of a pinhole with an effective depth. Creation of a flat area and use of the depth-limiting drill provide for a more predictable outcome and are recommended.

The technique of creating a properly aligned pinhole is as follows: (1) A pilot hole (dimple) is placed in the horizontal dentin surface to receive the pin; (2) the correct size twist drill (based on the pin size) is placed in a latch-type contra-angle handpiece; (3) the twist drill is placed in the gingival crevice beside the location for the pinhole and positioned such that it lies flat against the external surface of the tooth; (4) without changing the angulation obtained from the gingival crevice position, the handpiece is moved occlusally and the drill placed in the previously prepared pilot hole (Fig. 10.185A); (5) the drill is then viewed from a 90-degree angle to the previous viewing position to ascertain that the drill also is angled correctly in this plane (see Fig. 10.185B); (6) with the drill tip in its proper position and with the handpiece rotating at very low speed (300–500 rpm), pressure is applied to the drill until the depth-limiting portion of the drill is reached (see Fig. 10.185C and D); (7) the pinhole is prepared in one or two movements *without allowing the drill to stop rotating.* While still rotating, the drill is immediately removed from the pinhole.

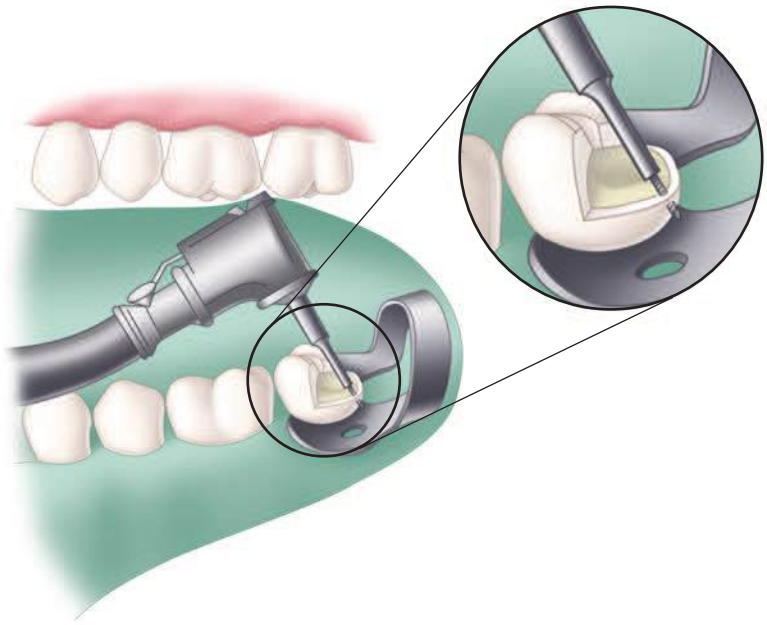


• **Fig. 10.181** Care must be exercised when preparing pinholes in mesially tilted molars to prevent external perforation on mesial surface (A) and pulpal penetration on the distal surface (B).

TABLE 10.2 The Thread Mate System (TMS) Link Series and Link Plus Pins

Name	Illustration (Not to Scale)	Color Code	Pin Diameter (Inches/mm)*	Drill Diameter (Inches/mm)*	Pin Length Extending from Sleeve (mm)	Pin Length Extending From Dentin (mm)
Link Series						
Regular		Gold	0.031/0.78	0.027/0.68	5.5	3.2 (single shear)
Regular		Gold	0.031/0.78	0.027/0.68	7.8	2.6 (double shear)
Minim		Silver	0.024/0.61	0.021/0.53	5.4	3.2 (single shear)
Minim		Silver	0.024/0.61	0.021/0.53	7.6	2.6 (double shear)
Minikin		Red	0.019/0.48	0.017/0.43	6.9	1.5 (single shear)
Minuta		Pink	0.015/0.38	0.0135/0.34	6.3	1 (single shear)
Link Plus						
Minim		Silver	0.024/0.61	0.021/0.53	10.8	2.7 (double shear)

*1 mm = 0.03937 inch.



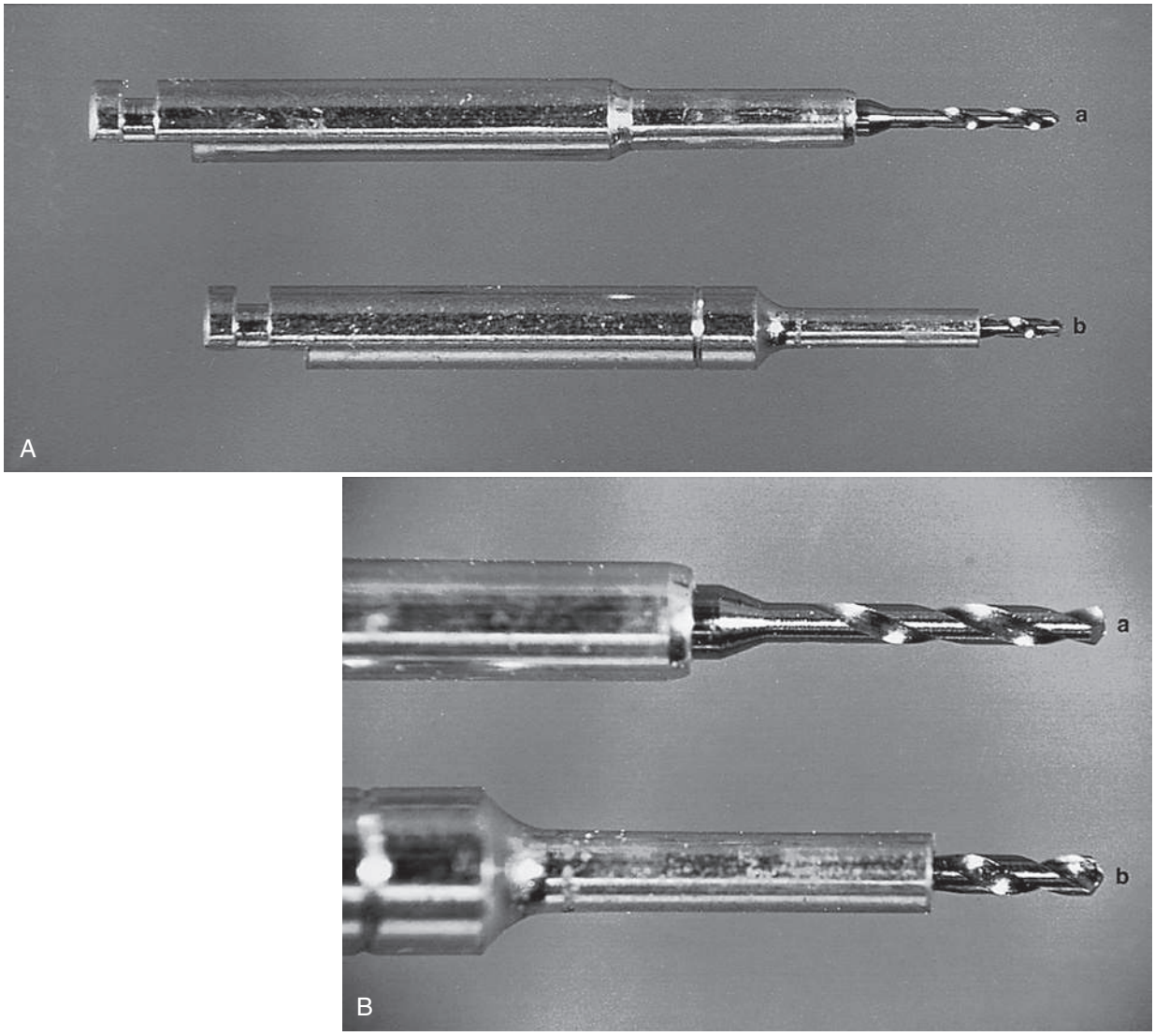
• **Fig. 10.182** When placing pinholes in molars and interarch space is limited, care must be exercised to prevent external perforation on distal surface.



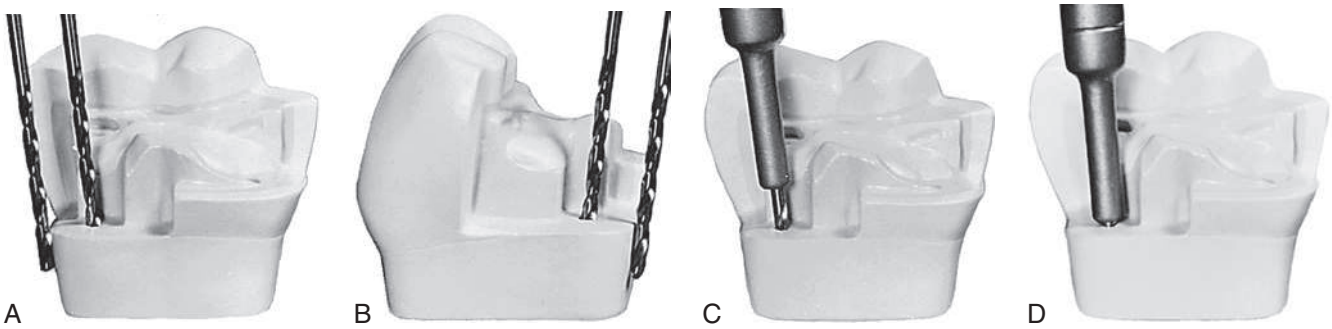
• **Fig. 10.183** Pilot hole (dimple) prepared with a No. ¼ bur.

Incorrect angulation of the drill may result in pulpal exposure or external perforation. If the proximity of an adjacent tooth interferes with placement of the drill into the gingival crevice, a flat, thin-bladed hand instrument is placed into the crevice and against the external surface of the tooth to indicate the proper angulation for the drill.[163] Using more than one or two movements, tilting the handpiece during the drilling procedure, or allowing the drill to rotate more than briefly at the bottom of the pinhole will result in a pinhole that is too large. The twist drill should never stop rotating (from insertion to removal from the pinhole) so as to limit the potential for binding and/or breakage while creating the pinhole.

Dull twist drills used to prepare pinholes may cause increased frictional heat and cracks in the dentin. Standlee et al. showed that a twist drill becomes too dull for use after cutting 20 pinholes



• **Fig. 10.184** A, Two types of KODEX twist drills: standard *(a)* and depth limiting *(b)*. B, Drills enlarged: standard *(a)* and depth limiting *(b)*.



• **Fig. 10.185** Determining the angulation for the twist drill. A, Drill placed in the gingival crevice, positioned flat against the tooth, and moved occlusally into position without changing the angulation obtained. B, *A* repeated while viewing the drill from position 90 degrees left or right of that viewed in *A*. C and D, With twist drill at correct angulation, the pinhole is prepared in one or two thrusts until the depth-limiting portion of drill is reached and then removed while still rotating.

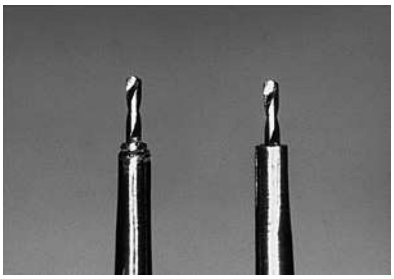
or less, and the signal for discarding the drill is the need for increased pressure on the handpiece.[164] Using a drill when its self-limiting shank shoulder has become rounded is contraindicated (Fig. 10.186). A worn and rounded shoulder may not properly limit pinhole depth and may permit pins to be placed too deeply.

Pin Design

Several designs are available for each of the four sizes of TMS self-threading pins: standard, self-shearing, two-in-one, Link Series, and Link Plus (Fig. 10.187). The Link Series pin is free floating in a plastic sleeve, which allows self-alignment as it is threaded into the pinhole (Fig. 10.188). When the pin reaches the bottom of the hole, the pin shears off at a machined narrow area, leaving a length of pin extending from dentin. The plastic sleeve is then discarded. The Minuta, Minikin, Minim, and Regular pins are available in the Link Series. Link Series pins are recommended because of their versatility, self-aligning ability, and retentiveness.[147]

Link Plus self-threading pins are also self-shearing and are available as single and two-in-one pins contained in color-coded plastic sleeves (Fig. 10.189). This design has a sharper thread, a shoulder stop at 2 mm, and a tapered tip to fit the bottom of the pinhole more readily as prepared by the twist drill. It also provides a 2.7-mm length of pin to extend out of dentin, which usually needs to be shortened with the use of a handpiece bur. Theoretically, and as suggested by Standlee et al., these innovations should reduce the stress created in the surrounding dentin as the pin is inserted and reduce the apical stress at the bottom of the pinhole.[164] Kelsey et al. reported that the two-in-one Link Plus first and second pins seat completely into the pinhole before shearing.[165]

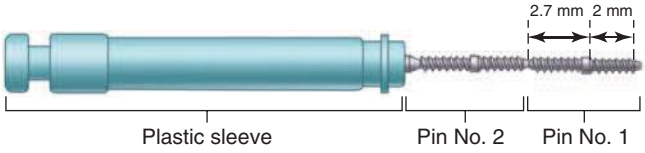
The self-shearing pin has a total length that varies according to the diameter of the pin (see Table 10.1). It also consists of a flattened head to engage the hand wrench or the appropriate handpiece chuck for threading into the pinhole. When the pin approaches the bottom of the pinhole and begins to bind, the head of the pin shears off leaving a length of pin extending from dentin.



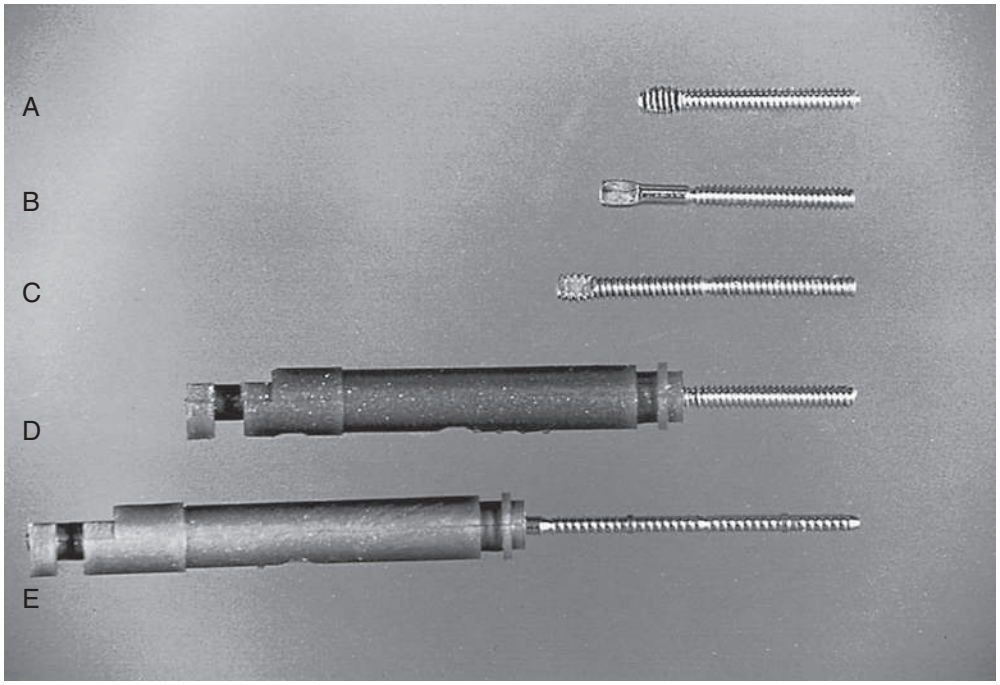
• **Fig. 10.186** Minikin self-limiting drill with worn shank shoulder *(left)* compared with a new drill with an unworn shoulder *(right)*.



• **Fig. 10.188** Cross-sectional view of Link Series pin.



• **Fig. 10.189** Link Plus pin.



• **Fig. 10.187** Five designs of the Thread Mate System (TMS) pins. A, Standard. B, Self-shearing. C, Two-in-one. D, Link Series. E, Link Plus.

The two-in-one pin is actually two pins in one, with each one being shorter than the standard pin. The two-in-one pin is approximately 9.5 mm in length and has a flattened head to aid in its insertion. When the pin reaches the bottom of the pinhole, it shears approximately in half, leaving a length of pin extending from dentin with the other half remaining in the hand wrench or the handpiece chuck. This second pin may be positioned in another pinhole and threaded to place in the same manner as the standard pin. The designs available with each size of pin are shown in Tables 10.1 and 10.2.

Selection of a particular pin design is influenced by the size of the pin being used, the amount of interarch space available, and operator preference. With minimal interarch space, the two-in-one design may be undesirable because of its length. It has been reported that 93% of Link Series and Link Plus two-in-one pins extended to the optimal depth of 2 mm.[166-169] Even so, both the two-in-one pin and the self-shearing pin may occasionally fail to reach the bottom of the pinhole.

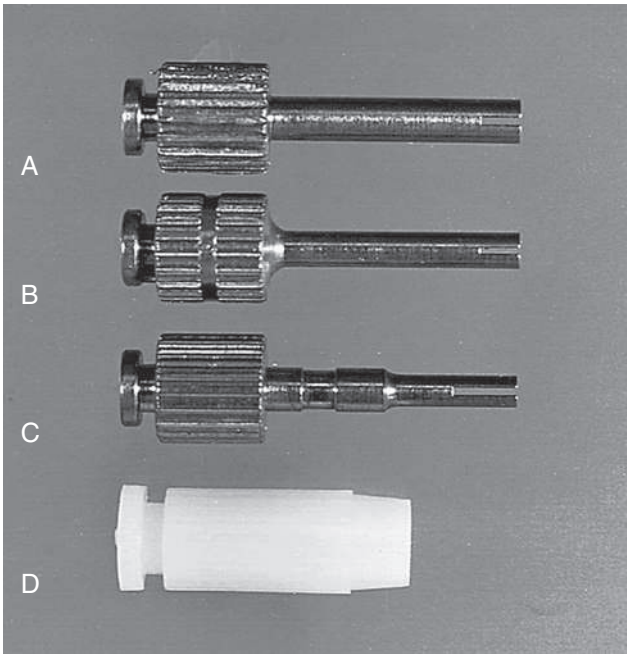
Pin Insertion

The TMS utilizes hand wrenches to place the self-threading pins (Figs. 10.190A–C and 10.191A). Link Series and Link Plus pins are contained in color-coded plastic sleeves that fit a latch-type contra-angle handpiece or a specially designed plastic hand wrench (see Fig. 10.190D). In addition, handpiece chucks are available (Fig. 10.192). The results of studies are conflicting as to which method of pin insertion produces the best results. The latch-type handpiece is recommended for the insertion of the Link Series and the Link Plus pins. The hand wrench is recommended for the insertion of standard pins.

When using the latch-type handpiece, a Link Series or a Link Plus pin is inserted into the handpiece and positioned over the pinhole. The handpiece is activated at low speed until the plastic sleeve shears from the pin. The pin sleeve is then discarded. For low-speed handpieces with a low gear, the low gear should be used. Using the low gear increases the torque and increases the tactile sense of the operator. It also reduces the risk of stripping the threads in dentin during pin insertion.

A standard pin is placed in the appropriate hand wrench (see Fig. 10.191A) and slowly threaded clockwise into the pinhole until a definite resistance (indicating the pin has reached the bottom of the pin hole) is felt (see Fig. 10.191B). The pin should then be rotated one quarter to one half turn counterclockwise to reduce the dentinal stress created by the end of the pin that is pressing on dentin at the bottom of the pinhole.[170] The hand wrench should be removed from the pin carefully. If the hand wrench is used without rubber dam isolation, a gauze throat shield must be in place, and a strand of dental floss, approximately 30 to 38 cm in length, should be tied securely to the end of the wrench (Fig. 10.193) to prevent accidental swallowing or aspiration by the patient.

The lengths of the pins are evaluated after placement (see Fig. 10.191C). Any length of pin greater than 2 mm should be removed. A sharp No. ¼, No. ½, or No. 169L or 245 bur, at high speed and oriented perpendicular to the pin, may be used to remove the excess length (Fig. 10.194A). If oriented otherwise, the rotation



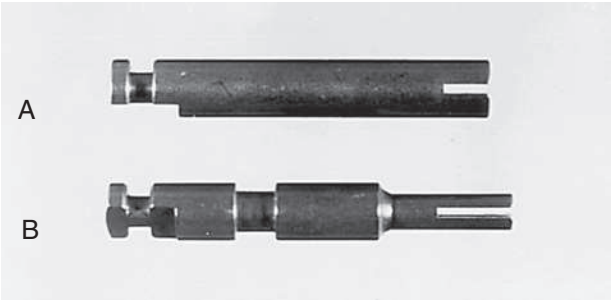
• **Fig. 10.190** Hand wrenches for the Thread Mate System (TMS) pins. A, Regular and Minikin. B, Minim. C, Minuta. D, Link Series and Link Plus.



• **Fig. 10.191** A, Use of a hand wrench to place a pin. B, Threading the pin to the bottom of the pinhole and reversing the wrench one quarter to one half turn. C, Evaluating the length of the pin extending from dentin.

of the bur may loosen the pin by rotating it counterclockwise and unthreading it out of the pinhole. During removal of excess pin length, the assistant may apply a steady stream of air to the pin and have the high-volume evacuator tip positioned to remove the pin segment. Also, during removal of excess pin length, the pin may be stabilized with a small hemostat or cotton pliers. After placement, the pin should be tight, immobile, and not easily withdrawn. Every pin must be assessed with cotton pliers for stability immediately after placement.

The preparation is viewed from all directions with a mirror (particularly from the occlusal direction) to determine if any pins need to be bent so as to be positioned within the anticipated contour of the final restoration such that there will be adequate bulk of amalgam between the pin and the external surface of the final restoration (see Fig. 10.194B and C). When pins require bending, the TMS bending tool (Fig. 10.195A) must be used. The bending tool should be placed on the pin where the pin is to be bent, and with firm controlled pressure, the bending tool should be rotated until the desired amount of bend is achieved (see Fig. 10.195B–D). Use of the bending tool allows placement of the fulcrum at some point along the length of the exposed pin. Other instruments should not be used to bend a pin because the location of the fulcrum would be at the orifice of the pinhole (i.e., the surface of the dentin). Force placed at this area may cause crazing or fracture of dentin, and the abrupt or sharp bend that usually results increases the chances of breaking the pin (Fig. 10.196). Also the operator has less control when pressure is applied with a hand instrument, and the risk of slipping is increased. Pins should only be bent if absolutely necessary, never simply to make them parallel or based on a notion that bending them will increase their relative retentiveness.

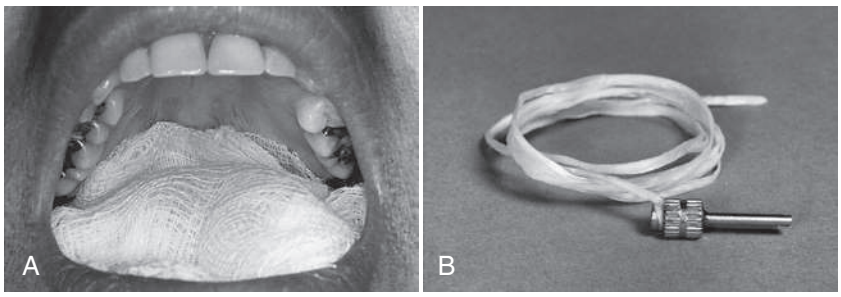


• **Fig. 10.192** Handpiece chucks for the Thread Mate System (TMS) regular self-shearing and Minikin pins (A) and TMS Minuta pins (B).

Possible Problems With Pins

Failure of Pin-Retained Restorations

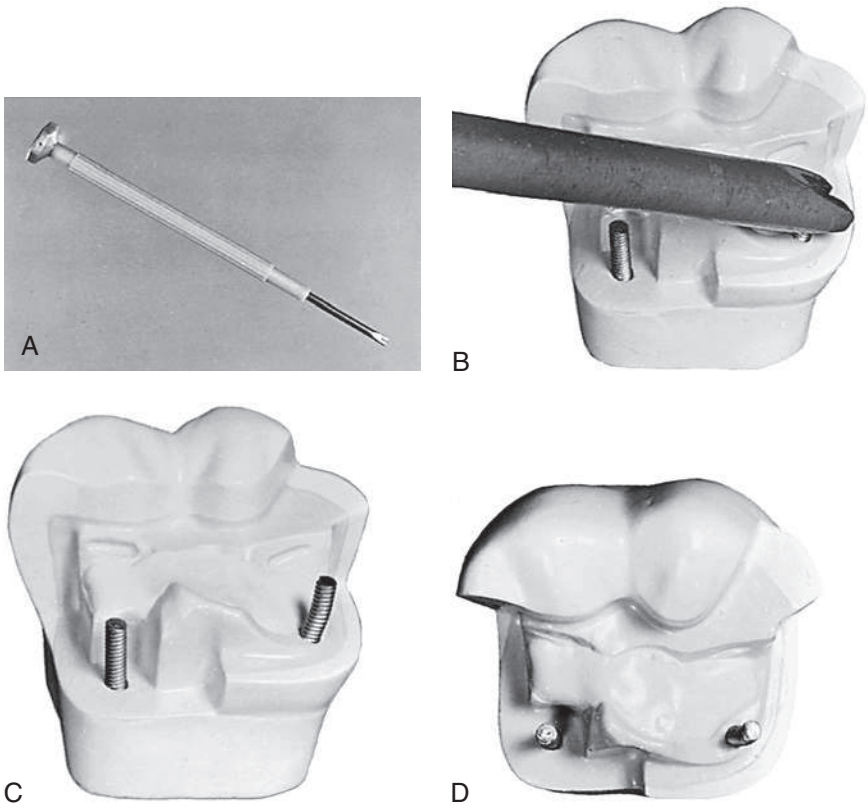
The failure of pin-retained restorations might occur at any of five different locations (Fig. 10.197). Failure may occur (1) within the restoration (restoration fracture), (2) at the interface between the pin and the restorative material (pin–restoration separation), (3) within the pin (pin fracture), (4) at the interface between the pin



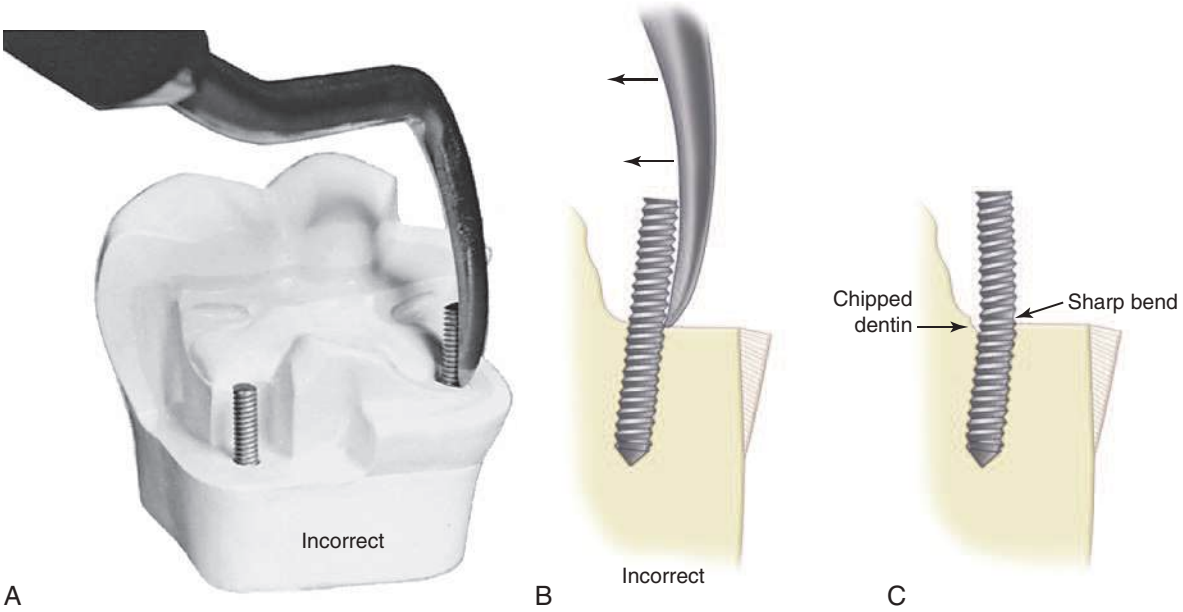
• **Fig. 10.193** Precautions to be taken if a rubber dam is not used. A, Gauze throat shield. B, Hand wrench with 30 to 38 cm of dental tape attached.



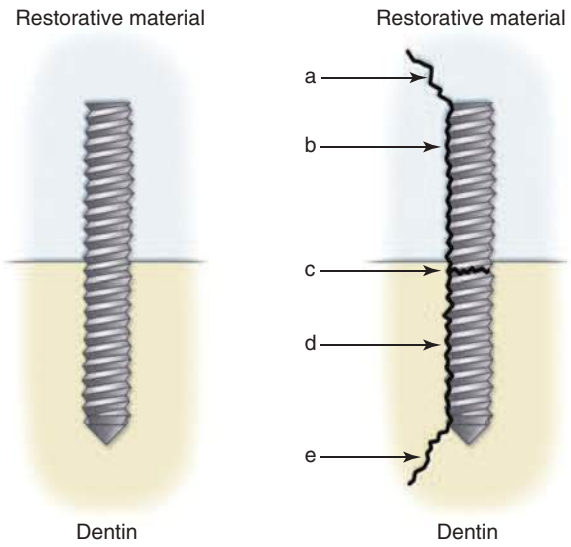
• **Fig. 10.194** A, Use of sharp No. ¼ bur held perpendicular to the pin to shorten the pin. B and C, Evaluating the preparation to determine the need for bending the pins.



• **Fig. 10.195** A, The Thread Mate System (TMS) bending tool. B, Use of the bending tool to bend the pin. C and D, The pin is bent to a position that provides adequate bulk of amalgam between the pin and the external surface of the final restoration.



• **Fig. 10.196** A, A Black spoon excavator or other hand instrument should not be used to bend the pin. B and C, Use of hand instruments may create a sharp bend in the pin and fracture dentin.



• **Fig. 10.197** Five possible locations of failure of pin-retained restorations: fracture of restorative material *(a),* separation of pin from restorative material *(b),* fracture of pin *(c),* separation of pin from dentin *(d),* fracture of dentin *(e)*.

and dentin, and (5) within dentin (dentin fracture). Failure is more likely to occur at the pin–dentin interface than at the pin–restoration interface. The operator must keep these areas of potential failure in mind at all times and apply necessary principles to reduce failure risk.

Broken Drills and Broken Pins

Occasionally, a twist drill breaks if it is stressed laterally or allowed to stop rotating before it is removed from the pinhole. Use of a sharp twist drill helps to limit the possibility of drill breakage. Pins also may break during bending if care is not exercised. The treatment for broken drills and broken pins is to choose an alternative location, at least 1.5 mm remote from the broken item, and prepare another pinhole and place another pin. Removal of a broken pin or drill is difficult, if not impossible, and usually should not be attempted.

Loose Pins

Self-threading pins sometimes do not engage dentin properly because the pinhole was inadvertently prepared too large or a self-shearing pin failed to shear, resulting in stripped-out dentin. The pin should be removed from the tooth and the pinhole re-prepared with the next largest size drill, and the appropriate pin should be inserted. Preparing another pinhole of the same size as the original, 1.5 mm from the original pinhole, also is acceptable.

As described earlier, a properly placed pin may be loosened while being shortened with a bur, if the bur is not held perpendicularly to the pin and the pin stabilized. A loose pin may be removed from the pinhole by holding a rotating bur parallel to the pin and lightly contacting the surface of the pin; this causes the pin to rotate counterclockwise out of the pinhole. A second attempt should be made with the same-size pin. If the second pin fails to firmly engage the dentin, a larger hole is prepared, and the appropriate pin is inserted. Preparing another pinhole of the same size 1.5 mm from the original pinhole also is acceptable.

Penetration Into the Pulp and Perforation of the External Tooth Surface

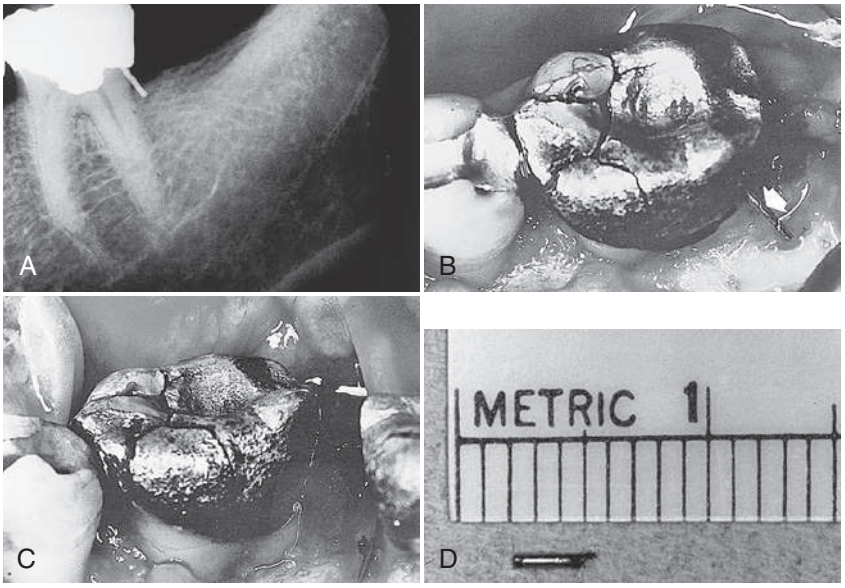
Penetration into the pulp or perforation of the external surface of the tooth is obvious if hemorrhage occurs in the pinhole after removal of the drill. Usually the operator is able to perceive when pulp penetration or root perforation has occurred by an abrupt loss of resistance of the drill to hand pressure. Also if a standard or Link Series pin continues to thread into the tooth beyond the 2-mm depth of the pinhole, this is an indication of a penetration or perforation. A pulpal penetration might be suspected if the patient is anesthetized and has sudden sensitivity when the pinhole is being completed or the pin is being placed. However, if the patient still has profound pulpal anesthesia, pulpal penetration of the pinhole or pin may not be perceived.

Pulpal penetration is treated as any other small mechanical exposure if the tooth has been asymptomatic. When preparation of the pinhole causes a pulp exposure, any hemorrhage is controlled. A calcium hydroxide liner is placed over the opening of the pinhole, and another hole is prepared 1.5 mm away. If the exposure is discovered as the pin is being placed, the pin is removed and the area of pulp penetration treated as described. Although certain studies have shown that the pulp tolerates pin penetration when the pin is placed in a relatively sterile environment, it is not recommended that pins be left in place when a pulpal penetration has occurred.[157,171] If the pin were left in the pulp, (1) the depth of the pin into pulpal tissue would be difficult to determine, (2) considerable postoperative sensitivity might ensue, and (3) the pin location might complicate subsequent endodontic therapy. Regardless of the method of treatment rendered, the patient must be informed of the pulpal penetration or root perforation at the completion of the appointment. The affected tooth should be evaluated periodically using appropriate diagnostic tests including radiographs. The patient should be instructed to inform the dentist if any discomfort develops.

Because most teeth receiving pins have had extensive caries, restorations, or both, the health of the pulp is likely compromised. The ideal treatment of a pulpal penetration for such a compromised tooth generally is endodontic therapy. Furthermore, endodontic treatment should be strongly considered when such a tooth is to receive an indirect restoration.

An external perforation might be suspected if an unanesthetized patient senses pain when a pinhole is being prepared or a pin is being placed in a tooth that has had endodontic therapy. Observation of the angulation of the twist drill or the pin may help in the determination of whether a pulpal penetration or external perforation has occurred. Perforation of the external surface of the tooth may occur occlusal or apical to the gingival attachment. Careful probing and radiographic examination must be used to accurately diagnose the location of a perforation. The method of treatment for a perforation often depends on the experience of the operator and the particular circumstances of the tooth being treated.

Three options are available for perforations that occur occlusal to the gingival attachment: (1) The pin may be cut off flush with the tooth surface and no further treatment rendered; (2) the pin may be cut off flush with the tooth surface and the preparation for an indirect restoration extended gingivally beyond the perforation; or (3) the pin may be removed, if still present, and the external aspect of the pinhole enlarged slightly and restored with amalgam. Surgical reflection of gingival tissue may be necessary to render adequate treatment. The location of perforations occlusal to the attachment often determines the option to be pursued.



• **Fig. 10.198** External perforation of a pin. A, Radiograph showing the external perforation of a pin. B, Surgical access to extruding pin *(arrow)*. C, Pin cut flush with the tooth structure and crown-lengthening procedure performed. D, Length of pin removed.

Two options are available for perforations that occur apical to the gingival attachment: (1) Reflect the tissue surgically, perform any necessary ostectomy, enlarge the pinhole slightly, and restore with amalgam; or (2) perform a crown-lengthening procedure and place the margin of a cast restoration gingival to the perforation (Fig. 10.198). As with perforations located occlusal to the gingival attachment, the location of the perforation and the design of the present or planned restoration help determine which option to pursue. As with pulpal penetration, the patient must be informed of the perforation and the proposed treatment. The prognosis of external perforations is favorable when they are recognized early and treated properly.

Considerations for the Use of Slots or Pins

Slot retention may be used in conjunction with pin retention or as an alternative to it.[172] Some operators use slot retention and pin retention interchangeably. Others more frequently use slot retention in preparations with vertical walls that allow retention grooves to oppose one another. Pin retention is used more frequently in preparations with few or no vertical walls. Slots are particularly indicated in short clinical crowns and when cusp reduction has been limited to 2 or 3 mm.[172] More tooth structure is removed in slot preparation as compared with pin placement; however, slots are less likely to (1) create microfractures in dentin, (2) perforate the tooth surface, or (3) penetrate into the pulp. Slot placement does not elicit an inflammatory response, whereas medium-sized self-threading pins may elicit an inflammatory response if placed within 0.5 mm of the pulp.[158] The retention potential of slots and pins is similar.[173-176]

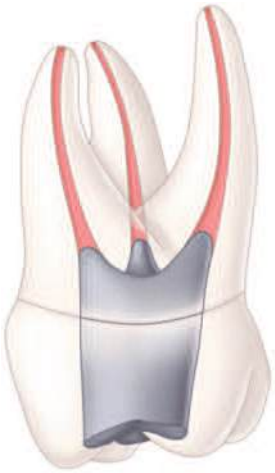
Tooth Preparation for Amalgam Foundations

An amalgam foundation may be used as part of the initial restoration of a severely involved tooth. The tooth is restored so that the restorative material (amalgam, composite, or other) serves to replace lost tooth structure necessary to provide retention and resistance forms for the final indirect restoration (e.g., crown or bridge retainer). The retention of the foundation material should not be compromised by tooth reduction during the final preparation for the indirect restoration. The foundation also should provide the resistance form against forces that otherwise might fracture the remaining tooth structure.

In contrast to a conventional complex amalgam restoration, an amalgam foundation may not depend primarily on the remaining coronal tooth structure for support. Instead it may rely mainly on secondary preparation retention features (proximal retention grooves, coves, slots, pins). A temporary or caries-control restoration may serve as a foundation, but only if the retention and resistance forms of the restoration are appropriate.

A temporary or caries-control restoration is used to restore a tooth when definitive treatment is uncertain or when several teeth require immediate attention for control of caries lesions. It also may be used when a tooth's prognosis is questionable. A temporary or control restoration might depend only on the remaining coronal tooth structure for support, however, using few auxiliary retention features. When preparing a tooth for either a foundation or a temporary restoration, remaining unsupported enamel may be left (except at the gingival aspect) to aid in forming a matrix for amalgam condensation. In this case the remaining unsupported enamel will subsequently be removed when the preparation for the indirect restoration is accomplished. Occasionally, when providing a temporary or control restoration, sufficient retention and resistance forms are included in the preparation to meet the requirements of a foundation.

As a rule, foundations are placed in anticipation of full-coverage indirect restorations. Not all teeth with foundations, however, need to be immediately restored with full-coverage crowns. For example, amalgam may be used as a definitive partial-coverage restoration of endodontically treated teeth if only minimal coronal damage has occurred.[177] The greatest influence on fracture resistance is the amount of remaining tooth structure; therefore restorations that conserve natural tooth structure are preferred.[178]



• **Fig. 10.199** Pulp chamber retention with 2- to 4-mm extension of the foundation into the canal spaces.

Amalgam and composite resin are the direct restorative materials used for foundations. RMGIs may be used to block out minor undercut areas of the preparation for an indirect restoration that do not draw. Amalgam may be preferred by some clinicians because it is easy to use and it is strong. While slots and threaded pins may be used for secondary retention in vital teeth, prefabricated posts and cast post/cores are used to provide additional retention for the foundation material in endodontically treated teeth receiving foundations. Prefabricated posts and cast post/cores are generally used on anterior teeth or single-canal premolars that have little or no remaining coronal tooth structure. Note, however, that there must be adequate remaining natural tooth structure to allow creation of a ferrule so as to achieve long-term clinical success. The pulp chamber and entrance to the root canals of endodontically treated molar teeth typically provide retention for foundations, and it is not necessary to use any form of intraradicular retention (i.e., post material extended further into the root canal space) (Fig. 10.199).

Slot and/or Pin Retention of Foundations

The technique of tooth preparation for a foundation depends on the type of retention that is selected—slot retention, pin retention, or both. The techniques have in common the axial location of the retention. Slots and pins for the retention of amalgam foundations are placed slightly more axial (farther inside the DEJ) than indicated for definitive complex amalgam restorations. The actual pulpal positioning depends on the type of indirect restoration that has been planned. The preparation for an indirect restoration should not eliminate or cut into the foundation's retentive features.

Severely broken teeth with few or no vertical walls, in which an indirect restoration is indicated, may require a pin-retained foundation. The main difference between the use of pins for foundations and the use of pins in definitive complex restorations is the distance of the pinholes from the external surface of the tooth.[179] For foundations, the pinholes must be located farther from the external surface of the tooth (farther internally from the DEJ), and more bending of the pins may be necessary to allow for adequate axial reduction of the foundation without exposing the pins during the cast-metal tooth preparation. Any removal of the restorative material from the circumference of the pin would compromise its retentive effect. If the material is removed from more than half the diameter of the pin, any retentive effect of the pin probably has been eliminated.

The location of the pinhole from the external surface of the tooth for foundations depends on (1) the occlusogingival location of the pin (external morphology of the tooth), (2) the type of restoration to be placed (e.g., a porcelain-fused-to-metal [PFM] requires more reduction than a full gold crown), and (3) the type of margin to be prepared. Preparations with heavily chamfered margins at a normal occlusogingival location require slot and pin placement at a greater axial depth. The length of the pins also must be adjusted to permit adequate occlusal reduction without exposing the pins. Proximal retention grooves still should be used, wherever possible.

Completion of all final preparation steps is accomplished as described previously. An RMGI liner/base may be applied as indicated. A liner or base should not extend closer than 1 mm to a slot or a pin.

Pulp Chamber Retention of Foundations

The pulp chamber may be effectively used to retain an amalgam foundation in multirooted, endodontically treated teeth. This alternative to the use of slots/pins is recommended when (1) dimension of the pulp chamber is adequate to provide retention and bulk of amalgam, and (2) dentin thickness in the region of the pulp chamber is adequate to provide rigidity and strength to the tooth.[180] Additional extension of 2 to 4 mm into the root canal space is recommended when the pulp chamber height is 2 mm or less (see Fig. 10.199).[181] When the pulp chamber height is 4 to 6 mm in depth, no advantage is gained from additional extension into the root canal space. After matrix application, amalgam is thoroughly condensed into the pulp chamber (and radicular space if indicated) and the coronal portion of the tooth. Natural undercuts in the pulp chamber and the divergent root canals provide the necessary retention form. The resistance form against forces that otherwise may cause tooth fracture is improved by gingival extension of the crown preparation approximately 2 mm beyond the foundation onto sound tooth structure to establish the necessary ferrule once the indirect restoration is in place. This extension should have a total taper of opposing walls of less than 10 degrees.[182] If the pulp chamber height is less than 2 mm, the use of a prefabricated post, cast post and core, pins, or slots should be considered for additional retention of the foundation material.

The preparation should then be cleaned, carefully inspected as indicated in General Concepts Guiding Preparation for Amalgam Restorations, and any final modifications completed.

Restorative Technique for Complex Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed over the prepared tooth structure per manufacturer instructions. Excess moisture is removed without desiccating the dentin.

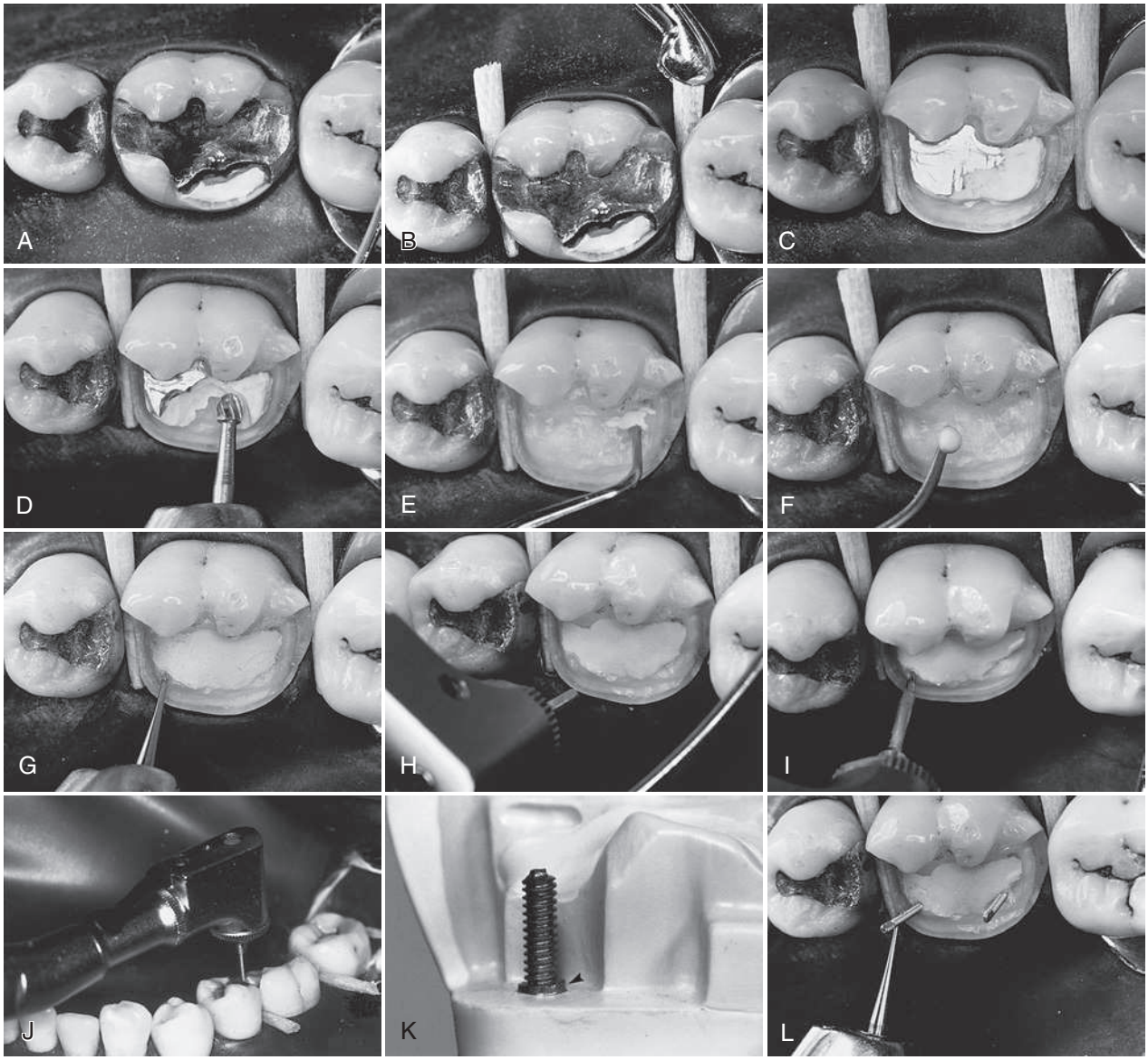
Matrix Placement

One of the most difficult steps in restoring a severely compromised posterior tooth is development of a satisfactory matrix to assist in formation of the anatomic shape of the restoration. Fulfilling the objectives of a matrix is complicated by possible gingival extensions, missing line angles, and restoration of reduced cusps typical of complex tooth preparations.

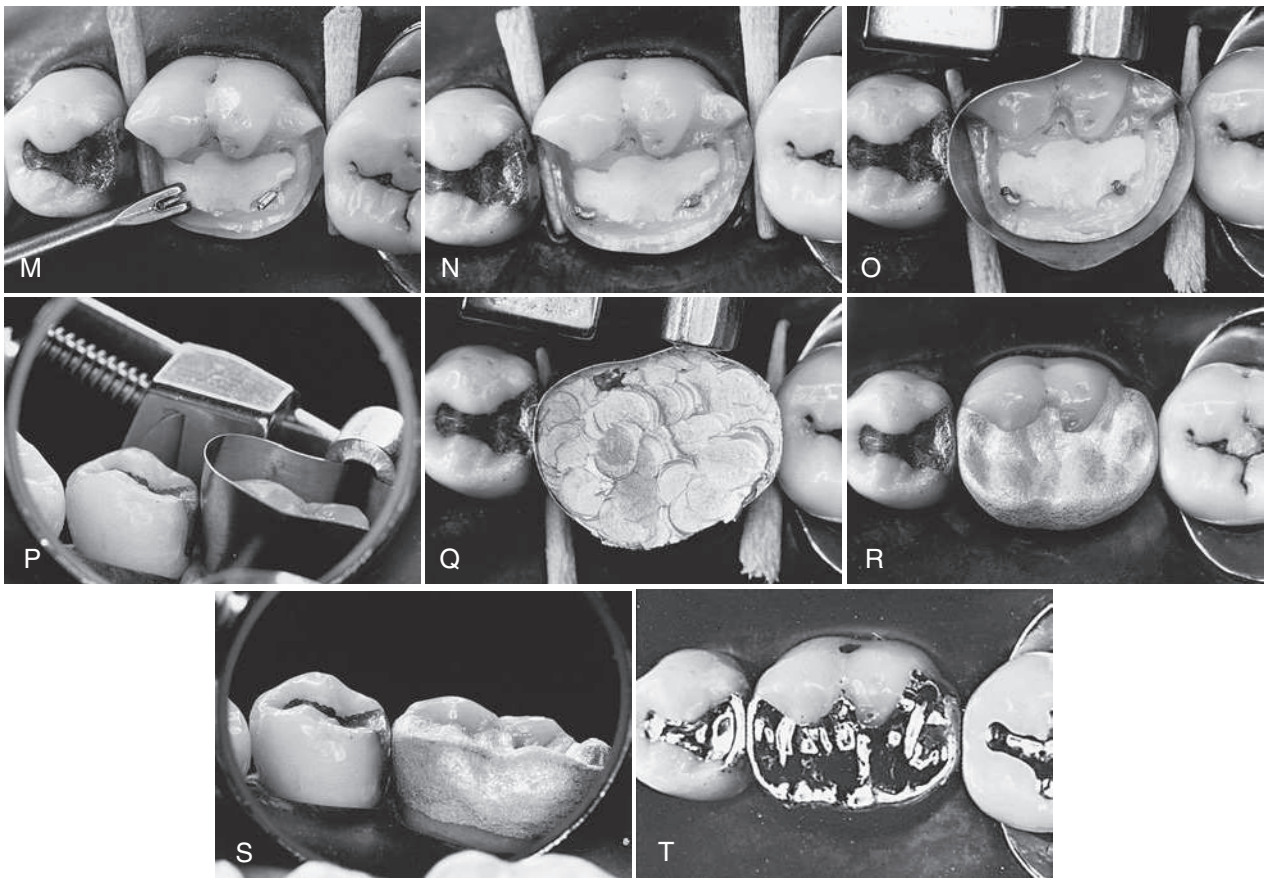
Universal Matrix

The Tofflemire retainer and band may be used successfully for most amalgam restorations (Fig. 10.200). Use of the Tofflemire retainer requires sufficient tooth structure to retain the band after it is applied. Even when the Tofflemire retainer is placed appropriately an opening may remain next to an area of the preparation tooth structure. In this case a closed system may be developed as illustrated in Fig. 10.201. A strip of matrix material that is long enough to extend from the mesial to the distal corners of the tooth is prepared. The strip must extend into these corners sufficiently that the band, when tight, holds the strip in position. Also it must not extend into the proximal areas, or a ledge would result in the restoration contour (see Fig. 10.201F). The Tofflemire retainer is loosened one half turn, and the strip of matrix material is inserted next to the opening between the matrix band and the tooth. The retainer is then tightened and the matrix is completed. Sometimes it is helpful to place a small amount of rigid material (hard-setting polyvinyl siloxane [PVS] or compound or light-cured flowable resin) between the strip and the open aspect of the band retainer to stabilize and support the strip (see Fig. 10.201G and H).

When little tooth structure remains and deep gingival margins are present, the Tofflemire matrix may not function successfully, and the AutoMatrix system (DENTSPLY Caulk, Milford, DE) may be an alternative method (Fig. 10.202).



• **Fig. 10.200** A, Mandibular first molar with fractured distolingual cusp. B, Insertion of wedges. C, Initial tooth preparation. D and E, Removal of any soft dentin and/or any remaining old restorative material as indicated. F, Application of a liner and a base (if necessary). G, Preparation of pilot holes. H, Alignment of the twist drill with the external surface of the tooth. I, Preparation of pinholes. J, Insertion of Link pins with a slow-speed handpiece. K, Depth-limiting shoulder *(arrowhead)* of inserted Link Plus pin. L, Use of a No. ¼ bur to shorten pins.



• **Fig. 10.200, cont'd** M, Bending pins (if necessary) with a bending tool. N, Final tooth preparation. O, Tofflemire retainer and matrix band applied to the prepared tooth. P, Reflecting light to evaluate the proximal area of the matrix band. Q, Preparation overfilled. R, Restoration carved. S, Reflecting light to evaluate the adequacy of the proximal contact and contour. T, Restoration polished.

AutoMatrix

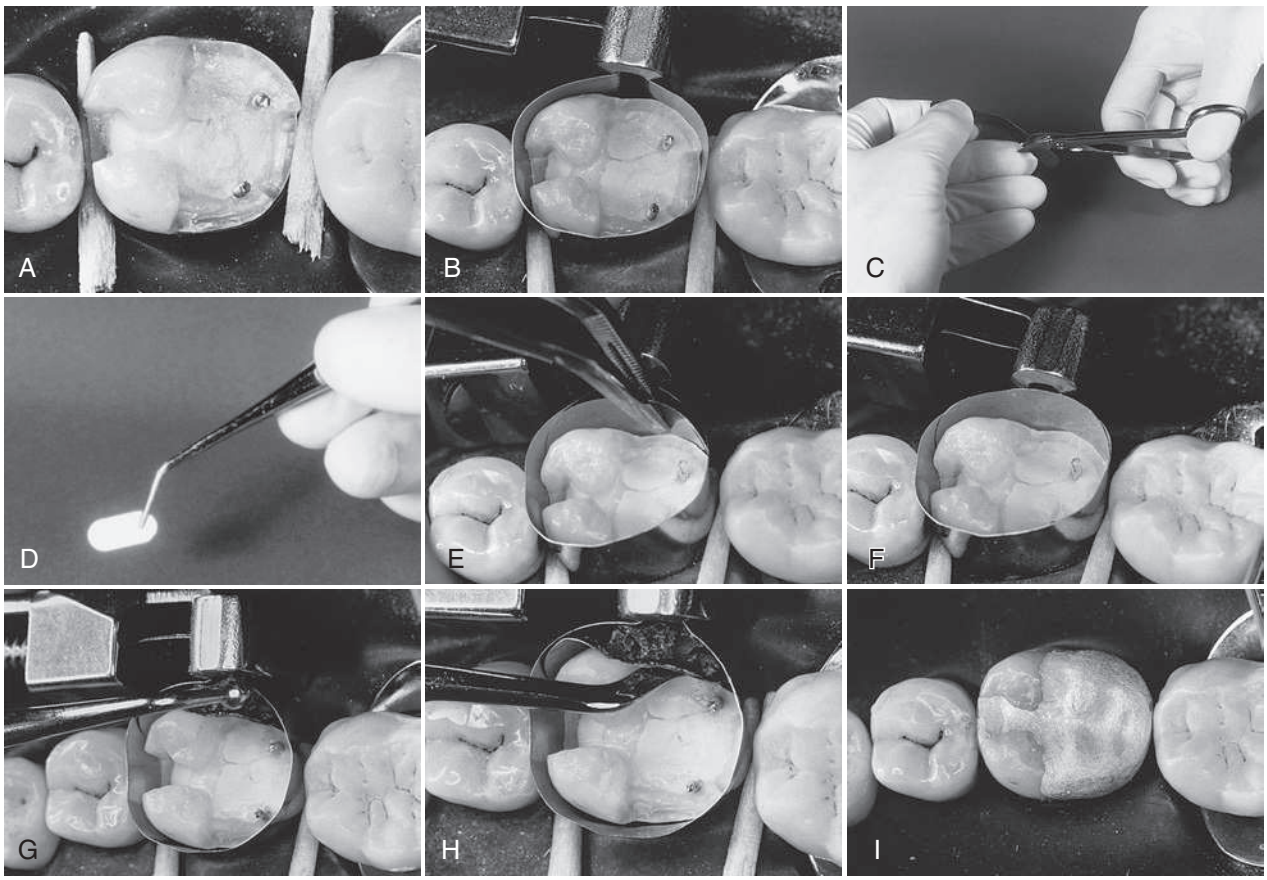
The AutoMatrix is a retainerless matrix system designed for any tooth regardless of its circumference and height. AutoMatrix bands are supplied in three widths: (1) 4.76 mm, (2) 6.35 mm, and (3) 7.94 mm. The medium band is available in two thicknesses (0.038 mm and 0.05 mm). The 4.76, 6.35, and 7.94 mm band widths are available in the 0.05 mm thickness only. Advantages of this system include (1) convenience, (2) improved visibility because of absence of a retainer, and (3) ability to place the autolock loop on the facial or lingual surface of the tooth. Disadvantages of this system are that (1) the band is flat and difficult to burnish and is sometimes unstable even when wedges are in place, and (2) development of proper proximal contours and contacts may be difficult. Use of the AutoMatrix system is illustrated in Fig. 10.203.

Regardless of the type of matrix system used, the matrix must be stable during condensation of the amalgam. The matrix should extend beyond the gingival margins of the preparation enough to provide support for the matrix and to permit appropriate wedge stabilization. The matrix should extend occlusally beyond the marginal ridge of the adjacent tooth by 1 to 2 mm. If the matrix for a complex amalgam restoration is not stable during condensation, a homogeneous restoration will not be developed. An improperly condensed amalgam may disintegrate when the matrix is removed, or eventually fail because of lack of sufficient strength secondary to internal voids or fractures. Matrix stability during condensation is especially important for slot-retained amalgam restorations. If the matrix is not secure during condensation, it may slip out of position causing separation of the main bulk of the amalgam from its retentive slots or pins.

Insertion, Contouring, and Finishing of Amalgam

A high-copper alloy is strongly recommended for the complex amalgam restoration because of excellent clinical performance and high early compressive strengths.[183-185] Spherical alloys have a higher early strength than admixed alloys, and spherical alloys may be condensed more quickly and with less pressure so as to ensure good adaptation in slots and around the pins. However, proximal contacts may be easier to achieve with admixed alloys because of their condensability. Alloys that provide extended working time are necessary to allow adequate time for condensation, removal of the matrix band, and final carving. Because complex amalgam restorations usually are quite large, a slow-set or medium-set amalgam may be selected to provide more time for the carving and adjustment of the restoration.

Regardless of the type of alloy chosen, care must be taken to condense the amalgam thoroughly in and around the retentive features of the preparation. If a mix of amalgam becomes dry or crumbly, a new



• **Fig. 10.201** Technique for closing the open space of the Tofflemire matrix system. A, Tooth preparation with wedges in place. B, Open aspect of the matrix band next to the prepared tooth structure. C and D, Cutting an appropriate length of the matrix material. E, Insertion of a strip of the matrix material. F, Closed matrix system. G and H, Placement of the rigid supporting material between the strip and the matrix band, and contouring, if necessary. I, Restoration carved.

mix is triturated immediately. Placement of a whole freshly mixed mass of amalgam (instead of using the amalgam carrier to place increments of amalgam) increases the rate of completely filling the defect. Special care must be taken when this technique is used to ensure the amalgam is well adapted in and around all retentive areas and at all margins.

With a complex (or any large) amalgam, carving time must be properly allocated. The time spent on carving occlusal anatomy (while the matrix is still in place) must be shortened to allow adequate time for carving the more inaccessible gingival margins and the proximal and axial contours (after the matrix is removed). The bulk of excess amalgam on the occlusal surface is removed and the initial anatomic contours are rapidly developed, especially the marginal ridge heights and cuspal inclines, with a discoid carver and/or Hollenback carver. The occlusal embrasures are defined by running the tine of an explorer against the internal aspect of the matrix band. Appropriate marginal ridge heights and embrasures reduce the potential of fracturing the marginal ridge when the matrix is removed.

Matrix removal is crucial when completing complex amalgam restorations, especially slot-retained restorations.[172] If the matrix is removed prematurely, the newly placed restoration may fracture immediately adjacent to the areas where amalgam has been condensed into the slot(s). Any rigid material supporting the matrix is removed with an explorer or a Black spoon. Tofflemire matrices are removed first by loosening and removing the retainer while the wedges are still in place. Leaving the wedges in place may help prevent fracturing the freshly condensed amalgam. It may be beneficial to place a fingertip on the occlusal surface of the restored tooth to stabilize the matrix while loosening and removing the retainer from the band. This is to reduce the risk of fracturing the freshly inserted amalgam while applying the force necessary to loosen the retainer. The matrix is removed by sliding each end of the band in an oblique direction (i.e., moving the band facially or lingually while simultaneously moving it in an occlusal direction). Moving the band obliquely toward the occlusal surface minimizes the possibility of fracturing the marginal ridge. Preferably the matrix band should be removed in the same direction as the wedge placement to prevent dislodging the wedges. AutoMatrix bands are removed by using the system's instruments and, after the band is open, by the same technique described for Tofflemire-retained matrices. Carving of the restoration is then continued (see Figs. 10.200R and 10.203N).

Wedges are then removed, and the gingival excess on the proximal surface is removed with an amalgam knife or explorer. Facial and lingual contours are developed with a Hollenback carver, amalgam knife, or explorer to complete the carving (see Figs. 10.200R and S and 10.201I). Appropriately shaped rotary instruments are used

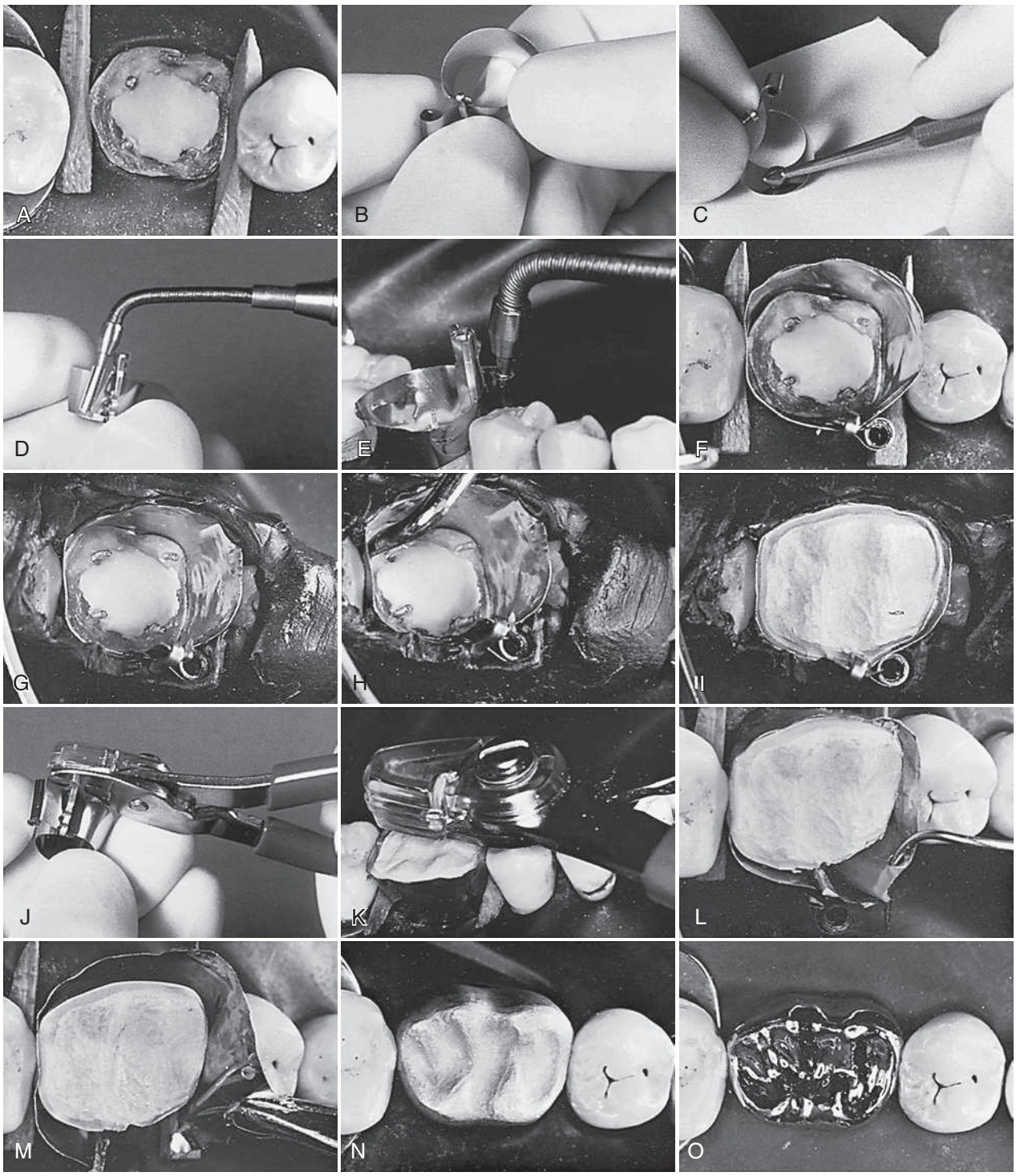


• **Fig. 10.202** A, AutoMatrix retainerless matrix system. B, AutoMatrix band. C, Automate II tightening device. D, Shielded snippers. (A, Courtesy Dentsply Caulk, Milford, DE.)

to complete the occlusal contouring if amalgam has become so hard that the force needed to carve with hand instruments might fracture portions of the restoration.

Each proximal contact is evaluated by using a mirror occlusally and lingually to ensure that no light can be reflected between the restoration and the adjacent tooth at the level of the proximal contact (see Fig. 10.200S). When the proper proximal contour or contact is not achieved in a large, complex restoration, it may be possible to prepare a conservative "ideal" two-surface tooth preparation within the initial amalgam to restore the proper proximal surface contours. Amalgam forming the walls of this "ideal" preparation must have sufficient bulk to prevent future fracture.

The rubber dam is removed and the occlusal surface of the amalgam is adjusted to obtain appropriate occlusal contacts. Thin, unwaxed dental floss may be carefully passed through the proximal contacts one time to remove amalgam shavings and to smooth the proximal surface of amalgam. Amalgam excess and loose particles are removed from the gingival sulcus by moving the floss occlusogingivally and faciolingually. The patient should be cautioned not to chew with the restoration for 24 hours. Fast-setting high-copper amalgam may be prepared for an indirect restoration within 30 to 45 minutes after insertion of the foundation. Further finishing and polishing of the complex amalgam may be accomplished, if desired, as early as 24 hours after placement (see Fig. 10.200T).



• **Fig. 10.203** Application of AutoMatrix for developing a pin-retained amalgam crown on the mandibular first molar. A, Tooth preparation with wedges in place. B, Enlargement of the circumference of the band, if necessary. C, Burnishing the band with an egg-shaped burnisher. D–F, Placement of the band around the tooth, tightening with an Automate II tightening device, and setting wedges firmly in place. G, Application of the green compound. H, Contouring of the band with the back of a warm Black spoon excavator so the softened compound will allow matrix deformation. I, Overfilling the preparation and carving the occlusal aspect. J and K, Use of shielded snippers to cut an autolock loop. L, Separating the band with an explorer. M, Removing the band in an oblique direction (facially with some occlusal vector). N, Restoration carved. O, Restoration polished.

Summary

Dental amalgam remains a predictable, cost-effective, and safe means for the restoration of posterior (and some anterior) teeth that are missing various amounts of tooth structure. The design of a successful tooth preparation must be based on the material properties of dental amalgam and the successful use of primary and secondary means of restoration retention. Complex preparations use retention grooves/coves in remaining vertical walls as well as slots, pins, and customized matrix designs to effect successful restorations. Restoration of normal anatomic contours and therefore normal clinical function may be readily accomplished with dental amalgam.

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